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TECHNICAL REPORT

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Summary of Rel-19 Work Items
(3GPP TR 21.919 version 19.0.0 Release 19)



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Foreword

This Technical Report has been produced by the 3rd Generation Partnership Project (3GPP).

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Introduction

The present document provides a summary of each and every 3GPP Release 19 Feature, and more generally of all Work Items for which a summary has been agreed to be provided. By default, the following items are not summurised here: studies, testing items, spectrum-related items, protocol enhancements.

These summaries are based on the inputs issued by the Work Item Rapporteurs, sometimes modified by the Work Plan manager to ensure overall consistency of this document. The original inputs can be retrieved as temporary document (tdoc), and/or in the folder https://ftp.3gpp.org/Information/WORK_PLAN/Description_Releases/inputs/Inputs_Rel_19. The original tdoc number (and/or file name) is provided just below the table of each clause.

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1 Scope

The present document provides a summary of each Release 19 Feature or, whenever needed, of each significant Work Item.

The information provided in the present document is limited to an overview of each Feature, explaining briefly its purpose and the main lines of the system's behaviour to execute the Feature.

2 References

The following documents contain provisions which, through reference in this text, constitute provisions of the present document.

- References are either specific (identified by date of publication, edition number, version number, etc.) or non-specific.
- For a specific reference, subsequent revisions do not apply.
- For a non-specific reference, the latest version applies. In the case of a reference to a 3GPP document (including a GSM document), a non-specific reference implicitly refers to the latest version of that document *in the same Release as the present document*.

[1] TR 21.905: "Vocabulary for 3GPP Specifications".

NOTE: Due to the specificity of the present document, consisting in a collection of independent summaries, the references are given at the end of each clause rather than in this clause.

3 Definitions of terms, symbols and abbreviations

3.1 Terms

For the purposes of the present document, the terms given in TR 21.905 [1] and the following apply. A term defined in the present document takes precedence over the definition of the same term, if any, in TR 21.905 [1].

3.2 Symbols

Void.

3.3 Abbreviations

For the purposes of the present document, the abbreviations given in TR 21.905 [1] and the following apply.
An abbreviation defined in the present document takes precedence over the definition of the same abbreviation, if any, in TR 21.905 [1]. Abbreviations specific to a given clause are provided in the clause they appear.

KPI	Key Performance Indicator
Rel	3GPP Release

4 Rel-19 Executive Summary

Rel-19 consists almost exclusively in improvements of Features introduced in previous Releases. Most aspects have been improved, such as the ones covering:

- Satellite (5GSAT) Phase 3, NTN Phase 3, UAS, Aerial. It includes enhancements related to NTN IoT support, positioning, new NTN bands, NTN PWS, etc.

- IoT and RedCap UE
- Ambient power-enabled IoT (new)
- Artificial Intelligence (AI)/Machine Learning (ML), including “Network for AI” and “AI for Network”.
- Energy Efficiency, Energy Saving
- Verticals and Non-Public Network (NPN)
- Communication services
- XR, AR, Metaverse, Edge Computing
- RAN improvements, including NR and LTE enhancements. It covers topics such as topology (femtocell, sidelink,…), MIMO, new frequency bands, etc.
- CN improvements, including on slicing, SBA, QoS, Multi-Access, etc.
- Mission Critical, eCall, Emergency
- Etc.

The structure of this document – and particularly the ordering and grouping of Features- is not performed by 3GPP but provided by the 3GPP Work Plan Manager, as to ease the presentation of 3GPP’s results. It does not reflect any relative “importance” of the different Features.

5 Rel-19 Satellite (5GSAT), NTN, UAS, Aerial

5.1 Overall and IoT-related items

5.1.1 Non-Terrestrial Networks (NTN) for NR Phase 3

1020097	**Non-Terrestrial Networks (NTN) for NR Phase 3**	NR_NTN_Ph3		RP-251954	Thales
1021097	**Core part: Non-Terrestrial Networks (NTN) for NR Phase 3**	NR_NTN_Ph3-Core	R2	RP-251954	Nicolas Chuberre, Thales
1022097	**Perf. part: Non-Terrestrial Networks (NTN) for NR Phase 3**	NR_NTN_Ph3-Perf	R4	RP-251954	Nicolas Chuberre, Thales

Summary based on the input provided by Thales and CATT in RP-252886.

Non-Terrestrial Network (NTN) support has been progressively introduced across 3GPP releases.

Release 17 defined the initial NR-NTN framework, with a primary focus on transparent-payload Low Earth Orbit (LEO) and Geostationary Orbit (GEO) satellites, while Release 18 extended this framework by adding support for regenerative payloads, mobility management, and coexistence with terrestrial networks.

This Work Item now specifies further NG-RAN enhancements targeting several aspects: **improved downlink performance** for handset-type terminals under NTN constraints such as limited payload power, large satellite footprints, and feeder-link bandwidth limitations; **improved uplink capacity** through multiplexing techniques**; support for multicast and broadcast services** when the intended service area is smaller than a Uu cell coverage area; **support for regenerative payload** architectures with a full gNB hosted on board the satellite; **and support for Reduced Capability (RedCap) and enhanced RedCap devices** operating in FR1 NR-NTN bands to enable wider-band services while maintaining low UE complexity.

To improve NR downlink coverage in NTN, the WI introduces enhancements for FR1-NTN operation, including repetitions of PDCCH in the common search space, repetitions of PDSCH carrying Msg4, and repetitions of PDSCH carrying SIB1 within 20 ms, as well as extended SS/PBCH block periodicity up to 160 ms for initial cell selection in both FR1-NTN and FR2-NTN bands.

Uplink capacity enhancements are achieved through the use of inter-slot orthogonal cover codes for PUSCH transmission, enabling multiplexing of multiple UEs.

For broadcast support, enhanced SIB signalling and core-to-NG-RAN signalling are introduced to indicate the intended service area when the satellite footprint exceeds the service coverage.

Regenerative payload support is enabled by adding gNB-on-board support in TS 38.300 and introducing the NG removal procedure.

Finally, enhancements are specified for RedCap and enhanced RedCap UEs to address timing-advance mismatches and half-duplex collision scenarios in paired spectrum, covering both semi-static and dynamically scheduled uplink and downlink transmissions.

References

List of related CRs: select "TSG Status = Approved" in:
https://portal.3gpp.org/ChangeRequests.aspx?q=1&workitem=1020097,1021097,1022097

5.1.2 Satellite access Phase 3

6.2.1 Main aspects

1020058	**Satellite access Phase 3**	5GSAT_Ph3		SP-230516	Thierry Bérisot, Novamint
960016	*Study on satellite access - Phase 3*	FS_5GSAT_Ph3	S1	SP-220679	Thierry Bérisot, Novamint
1000024	**Stage 1 of Satellite access Phase 3**	5GSAT_Ph3	S1	SP-230516	Thierry Bérisot, Novamint
1010033	*Study on Integration of satellite components in the 5G architecture Phase 3*	FS_5GSAT_Ph3_ARCH	S2	SP-231199	Jean-Yves Fine (Thales)
1020053	*Study on application enablement for Satellite access enabled 5G Services*	FS_5GSAT_Ph3_App	S6	SP-240296	Basavaraj (Basu) Pattan, Samsung
1040027	**Integration of satellite components in the 5G architecture Phase 3**	5GSAT_Ph3-ARC	S2	SP-241386	Hucheng Wang (CATT)
1040077	**Application enablement for satellite access Phase 3**	5GSAT_Ph3_App	S6	SP-241690	Basavaraj (Basu) Pattan, Samsung
1060023	**CT aspects for application enablement for satellite access Phase 3**	5GSAT_Ph3_App		CP-243071	Narendranath Durga Tangudu, Samsung
1060032	CT1 aspects for application enablement for satellite access Phase 3	5GSAT_Ph3_App	C1	CP-243071	Narendranath Durga Tangudu, Samsung
1060033	CT3 aspects for application enablement for satellite access Phase 3	5GSAT_Ph3_App	C3	CP-243071	Narendranath Durga Tangudu, Samsung
1050019	**CT aspects of integration of satellite components in the 5G architecture Phase 3**	5GSAT_Ph3_ARCH		CP-250134	Zhao, Xiaoxue (CATT)
1050041	CT1 aspects of integration of satellite components in the 5G architecture Phase 3	5GSAT_Ph3_ARCH	C1	CP-250134	Zhao, Xiaoxue (CATT)
1050042	CT3 aspects of integration of satellite components in the 5G architecture Phase 3	5GSAT_Ph3_ARCH	C3	CP-250134	Zhao, Xiaoxue (CATT)
1050043	CT4 aspects of integration of satellite components in the 5G architecture Phase 3	5GSAT_Ph3_ARCH	C4	CP-250134	Zhao, Xiaoxue (CATT)

Summary based on the input provided by Novamint and CATT by e-mail, latest version received Fri 03/04/2026 04:10, file "Summary_Update_1020058_V2.1".

The integration of satellite access and the development of Non-Terrestrial Network (NTN) as a 3GPP technology have been fundamental for 5G allowing to democratize satellite connectivity by making it a native part of the mobile network.

For satellite access, the objectives of the Rel-19 were to address new capabilities considering the technical evolutions of the satellite systems as well to be able to support the deployments of satellite access systems and networks in line with the growing market demand for IoT and broadband services by satellite.

Stage 1

Following up on Rel-19 study on Satellite Access Phase 3 (TR 22.865 [1]), the Rel-19 work item ([2]) introduces the following new capabilities related to the 5G system with satellite access:

- **Store and Forward (S&F) Satellite operation** for delay-tolerant communication services: S&F Satellite operation is an operation mode of a 5G system with satellite-access, where the 5G system can provide some level of service (in storing and forwarding the data) when satellite connectivity is intermittently/temporarily unavailable, e.g. to provide communication service for UEs under satellite coverage without a simultaneous active feeder link connection to the ground segment. This is particularly relevant for delay-tolerant IoT services via NGSO space segment.
- **UE-Satellite-UE communication**: In some scenarios, UEs need to communicate using satellite access without going to the ground network to avoid long delays and limited data rate as well as reducing the consumption of backhaul resources.
- **GNSS independent operation**: This would allow to provide satellite access to UEs without GNSS receiver or with no access to GNSS services.
- **Positioning enhancements** for satellite access: 3GPP positioning methods are needed in some scenarios for UEs using only satellite access.

In addition, the work item addressed further improvements for LAN using satellite access and information collection via satellite connections.

The new normative service requirements for satellite resulting of the work item were added into TS 22.261 ([3]). In addition, the work item conducted that all requirements related to satellite access, the already existing ones since Rel-16 and the new ones, were regrouped in one dedicated section on satellite which the objective to facilitate the work from stage 2 and stage 3 in the future.

This resulted in work conducted on satellite access in Rel-19 in SA2, SA6, SA3, SA5, CT1, CT3, CT4 as well as RAN2, RAN3 on NTN.

Stage 2

The work conducted in SA1 on satellite access resulted in stage 2 Rel-19 work on the architecture and the application enabler and mission critical specifications in order to support satellite access.

Architecture (SA2):

For the architecture, the Study on Integration of satellite components in the 5G architecture Phase 3, with results and conclusions documented in TR23.700-29 ([3]), covered the following aspects:

- **Regenerative payloads**, i.e.: embedding RAN and CN 3GPP functions within the satellite, are being considered by SA1 in Rel-19 for example to support store and forward satellite operation (e.g. for mMTC services) or to support Inter Satellite Links (e.g. for eMBB services).
- **S&F Satellite Operation**, very relevant for IoT use cases, requiring EPC enhancement.
- **UE-Satellite-UE communication** requiring 5GS enhancements.

The work conducted on Regenerative payload was referring to have the satellite payload with on-board processing capability, allowing e.g. to demodulate the received signal and then re-modulating and transmitting the signal on another link – typically a feeder link or an ISL as previous releases (Rel-17 and Rel-18) were assuming a transparent mode satellite access the satellite payload relays the signal without digitally processing it, acting as a repeater (only filtering, frequency conversion and amplification, no demodulation or modulation is done on board). The transparent mode is widely adopted in GEO satellites.

Introducing Regenerative payload in Rel-19 was allowing to provide new types of satellite services and new capabilities, in particular for Non-Geostationary Orbiting (NGSO) satellites deployments, such as S&F Satellite operation and UE-Satellite-UE communication.

The work item on Architecture on Integration of satellite components in the 5G architecture Phase 3 ([5]) resulted in changes such as introducing the new features S&F and UE-Sat-UE communication in [6] and updates or enhancements of procedures and functions in [7], [8], [9], [10], [11].

For **S&F Satellite operation**, it led to the description of two specific architectures as described in TS 23.401 ([11]):

- split MME architecture with the following principles:

1) eNB is onboard the satellite.
2) MME is split into two functions:

a) MME-onboard: the MME part which is onboard the satellite. MME-onboard is in charge of (1) handling the S1 interface with the onboard eNB and (2) handling the NAS protocol signalling from/to UEs via the onboard eNB.

b) MME-ground: the MME part which is on the ground network. MME-ground is in charge of handling the rest of interfaces towards other CN functions (e.g. S6a towards HSS, SGd towards SMS-GMSC/IWMSC /SMS Router, T6a towards SCEF, T6ai towards IWF-SCEF, S11 towards SGW).

- full EPC in satellite architecture with the following principles:

1) Each satellite contains the functionally of an eNB plus a full EPC that can include an MME, SGW, PGW, HSS, E-SMLC, SMSC etc.
2) Each satellite further includes an endpoint proxy function that emulates the behaviour of a real endpoint (e.g. an AF) from the perspective of a UE. There is also store and forward functionality on the ground that can be part of, or connected to, an NTN Gateway and that contains the proxy functionality.

It should be noted that this architecture does not support roaming.

In addition, SA2 introduced the S&F Monitoring List to indicate the satellite(s) that the UE may (re)attempt NAS procedures or receive MT data from.

For **UE-Satellite-UE communication**, the reference architecture for support of UE-Satellite-UE communication is described in TS 23.228 ([10]) with the following key aspects:
- To support UE-Satellite-UE communication for IMS services, network functions (e.g. UPF) transferring IMS user plane traffic in the 5G system shall be deployed onboard the satellites with regenerative payload that provide service link to the two UEs. It should be noted that in Release 19 only IMS voice/video services between two UEs belonging to same PLMN and in non-roaming scenario are considered.
- To keep the UE's IP address for IMS services unchanged, a UPF on the ground shall be selected during PDU Session establishment procedure as the PSA UPF for the duration of the IMS PDU session. The UPF on board the satellite shall only be selected as an UL CL/BP and local PSA to route IMS user plane traffic. In Release 19, the type of serving satellite supporting UE-Satellite-UE communication can only be LEO or MEO.

Application enabler and mission critical (SA6):

For the application enabler and mission critical specifications' part, Release 19 was the first release SA6 addressed satellite access. Therefore, SA6 conducted a complete study to the impacts of satellite access on the application enabler and mission critical and aligned with SA2 architecture.

The following aspects of the satellite access have been studied and captured in TR 23.700-01 ([12]):
- Use of satellite access characteristics to optimise application layer enablement behaviour
- Impacts of usage of satellite communication for Mission Critical (MC) services
- Discontinuous satellite coverage: Support the predictable discontinuous /intermittent connectivity patterns between UEs and AS/AFs for IoT services with satellite access e.g. asset tracking, critical infra-asset monitoring delivered through low density NGSO satellite constellations.
- Store and Forward Satellite operation (when no simultaneous active feeder link connection to the ground segment),
- UE-Satellite-UE communication for IMS services including MMTEL and for Mission Critical communications,
- Support for edge computing on board the satellite.

From the study, there has been no conclusion to progress UE-Satellite-UE communication aspect to the normative from SA6 perspective.

The other aspects have been addressed in the normative work on application enablement for satellite enabled 5G services ([13]) resulting in enhancements to support satellite access in several specifications ([14], [15], [16], [17], [18]).

Stage 3

General aspects

As a result of the work conducted in stage 2 by SA2 and the conclusions for "Support of Store and Forward satellite operation" and for "Support of UE-Satellite-UE communication", a CT1 work item ([19]) was initiated to implement the stage-2 requirements and address the stage 3 protocol enhancement for support of integration of satellite components in the 5G architecture Phase 3 on the following aspects:

- CT1:

For supporting Store and Forward Satellite operation in EPS:
a) update to the NAS procedures to support S&F operation in TS 24.301 ([20]).
b) update to the short message services due to S&F operation in TS 23.040 ([21]).
c) update to the AT commands in TS 27.007 ([22]).
For supporting UE-Satellite-UE communication in 5GS:

a) update to the Proxy CSCF (P-CSCF) procedure for the IMS-Access Gateway (IMS-AGW) (re-)selection procedure to support optimized media routing in TS 24.229 ([23])
b) update application server (AS) procedure due to early media handling in TS 24.229 ([23])

- CT3:
For supporting of Store and Forward Satellite operation in EPS:
a) update to the SCEF northbound API (e.g., MonitoringEvent API) at T8 reference point to expose S&F information to SCS/AS in TS 29.122 ([24])
For supporting of UE-Satellite-UE communication in 5GS:
a) update to PCF services and Rx/N5 interface to support UE-SAT-UE communication in TS 29.214 ([25])
b) update to SMF event exposure service to support UE-SAT-UE communication in TS 29.508, TS 29.512, TS 29.513, TS 29.514 ([26], [27], [28] and [29])

- CT4:
For supporting of Store and Forward satellite operation in EPS:
a) update to T6a, T7 and S6t interfaces to support the exposure of S&F information in TS 29.128 and in TS 29.336 ([30] and [31]).
b) update to the S6a interface to support new parameters (e.g., Request Time) for S&F operation in TS 29.272 ([32]).
c) define new diameter AVPs to support the exposure of S&F information in TS 29.230([33])
For supporting of UE-Satellite-UE communication in 5GS:
a) update to the AMF to support UE-SAT-UE communication in TS 29.502 ([34])

Stage 3 specific work and APIs

In addition, a CT3 work item ("CT aspects for application enablement for satellite access Phase 3" [35]) was conducted to define stage-3 protocol aspects and related APIs for enabling satellite applications based upon the normative Rel-19 stage 2 specification developed by SA6 and SA3 WG. The work item specified the following aspects:

- CT1:
1) Protocol enhancements for EDGE-1, EDGE-4 reference points (e.g. Edge computing on-board the satellite) in TS 24.558 ([36]).
2) Protocol enhancements for SEAL-Uu reference point for:
a) SEAL Configuration Management, Network Resource Management and Location Management services in TS 24.546, 24.546, TS 24.548, TS 24.545 ([37], [38], [39]).
b) SEAL DD management (e.g. minimise service interruption) in TS 24.543 ([40]).
3) Protocol enhancements to Mission Critical Services via satellite in TS 24.379, TS 24.282, TS 24.281 ([41], [42], [43]).

- CT3:
1) APIs enhancements for EDGE-3, EDGE-6 reference points (e.g. Edge computing on-board the satellite, Eecs_EESRegistration) in TS 29.558 ([44]).
2) APIs enhancements and new APIs for Store and Forward, configuration management service for satellite access with discontinuous coverage, ADAE and ADRF service for UE RAT connectivity analytics, over SEAL-S reference point in TS 29.549 ([45]).
3) Enhancements to Service Enabler Architecture Layer for Verticals (SEAL); SEAL Data Delivery (SEALDD) Server Services to enable Store and Forward transmission in TS 29.548 ([46]).

References

List of related CRs: select "TSG Status = Approved" in:
https://portal.3gpp.org/ChangeRequests.aspx?q=1&workitem=1020058;960016;1000024;1010033;1020053;1040027;1040077;1060023;1060032;1060033;1050019;1050041;1050042;1050043;1020044;1060065;1040014;1070014

[1] TS 22.865: “Study on satellite access - Phase 3”
[2] SP-230516: “New WID on Satellite access - Phase 3”
[3] TS 22.261: “Service requirements for the 5G system”
[4] TR 23.700-29: “Study on integration of satellite components in the 5G architecture; Phase 3.
[5] SP-241386: Integration of satellite components in the 5G architecture Phase 3”
[6] TS 23.501: “System architecture for the 5G System (5GS)”.
[7] TS 23.502: “Procedures for the 5G System (5GS)”.
[8] TS 23.503: “Policy and charging control framework for the 5G System (5GS)”.
[9] TS 23.682: “Architecture enhancements to facilitate communications with packet data networks and applications”.
[10] TS 23.228: “IP Multimedia Subsystem (IMS)”.

[11] TS 23.401: “General Packet Radio Service (GPRS) enhancements for Evolved Universal Terrestrial Radio Access Network (E-UTRAN) access.”
[12] TR 23.700-01: “Study on application enablement for Satellite access enabled 5G Services.
[13] SP-241690: Revised WID on application enablement for satellite access Phase 3”
[14] TS 23.558: “Architecture for enabling Edge Applications”
[15] TS 23.433: “Service Enabler Architecture Layer for Verticals (SEAL); Data Delivery enabler for vertical applications”
[16] TS 23.434: “Service Enabler Architecture Layer for Verticals (SEAL) - Functional architecture and information flows”.
[17] TS 23.436: “Functional architecture and information flows for Application Data Analytics Enablement Service”
[18] TS 23.289: “Mission Critical services over 5G System”
[19] CP-250134: “Revised WID on CT aspects of integration of satellite components in the 5G architecture Phase 3”
[20] TS 24.301: “Non-Access-Stratum (NAS) protocol for Evolved Packet System (EPS)
[21] TS 23.040: “Technical realization of the Short Message Service (SMS)”
[22] TS 27.007: “AT command set for User Equipment (UE)”
[23] TS 24.229: “IP multimedia call control protocol based on Session Initiation Protocol (SIP) and Session Description Protocol (SDP); Stage 3”
[24] TS 29.122: “T8 reference point for Northbound APIs”
[25] TS 29.214: “Policy and charging control over Rx reference point”
[26] TS 29.508: “5G System; Session Management Event Exposure Service; Stage 3”
[27] TS 29.512: “5G System; Session Management Policy Control Service; Stage 3”
[28] TS 29.513: “5G System; Policy and Charging Control signalling flows and QoS parameter mapping; Stage 3”
[29] TS 29.514: “5G System; Policy Authorization Service; Stage 3”
[30] TS 29.128: “Mobility Management Entity (MME) and Serving GPRS Support Node (SGSN) interfaces for interworking with packet data networks and applications”
[31] TS 29.336: “Home Subscriber Server (HSS) diameter interfaces for interworking with packet data networks and applications”.
[32] TS 29.272: “Evolved Packet System (EPS); Mobility Management Entity (MME) and Serving GPRS Support Node (SGSN) related interfaces based on Diameter protocol”
[33] TS 29.230: “Diameter applications; specific codes and identifiers”.
[34] TS 29.502: “5G System; Session Management Services; Stage 3”.
[35] CP-243071: “New WID on CT aspects for application enablement for satellite access Phase 3”
[36] TS 24.558: “Enabling Edge Applications; Protocol specification”
[37] TS 24.546: “Configuration management - Service Enabler Architecture Layer for Verticals (SEAL); Protocol specification”
[38] TS 24.548: “Network Resource Management - Service Enabler Architecture Layer for Verticals (SEAL); Protocol specification”
[39] TS 24.545: “Location Management - Service Enabler Architecture Layer for Verticals (SEAL); Protocol specification”
[40] TS 24.543: “Data delivery management - Service Enabler Architecture Layer for Verticals (SEAL); Protocol Specification”
[41] TS 24.379: “Mission Critical Push To Talk (MCPTT) call control; Protocol specification”
[42] TS 24.282: “Mission Critical Data (MCData) signalling control; Protocol specification”
[43] TS 24.281: “Mission Critical Video (MCVideo) signalling control; Protocol specification”
[44] TS 29.558: “Enabling Edge Applications; Application Programming Interface (API) specification; Stage 3”
[45] TS 29.549: “Service Enabler Architecture Layer for Verticals (SEAL); Application Programming Interface (API) specification; Stage 3”
[46] TS 29.548: “Service Enabler Architecture Layer for Verticals (SEAL); SEAL Data Delivery (SEALDD) Server Services; Stage 3”

6.2.2 Security Aspects of 5G Satellite Access Phase 3

1020044	*Study on Security Aspects of 5G Satellite Access Phase 3*	FS_5GSAT_Ph3_SEC	S3	SP-240232	Zhou Wei, CATT
1060065	**Security Aspects of 5G Satellite Access Phase 3**	5GSAT_Ph3_SEC	S3	SP-241780	Wei Zhou, CATT

This Work Item specifies the security aspects of Store and Forward Satellite operation, namely:

- Security protection in Store and Forward Satellite operation in EPS. Both split MME architecture and full EPC in satellite architecture are supported.

- Privacy protection in Store and Forward Satellite operation in EPS.

The security aspects of Store and Forward Satellite operation are specified in clause Annex N of TS 33.401 [1]. The general features of Store and Forward Satellite operation are described in TS 23.401 [2].

There are two example deployment options for Store and Forward Satellite operation given in Annex O of TS 23.401 [2], i.e. Split MME architecture and Full EPC in each satellite. In both cases, regular LTE procedures shall be used to provide the security between the UE and network. The security of communications between the proxies on satellite and the ground station(s) is out of 3GPP scope.

For Security aspects of Split MME architecture, the MME-ground is responsible for the selection and provisioning of MME-onboard with the necessary information (e.g., Authentication Vector) to perform or finish an authentication procedure. The NAS security is terminated on the MME-onboard, and the MME-ground ensures that the latest NAS security context of the UE, or an Authentication Vector, is available at the MME-onboard. The distribution and synchronization of the latest NAS security context between the MME-ground and MME-onboard is out of the scope of 3GPP. (D)DoS attack in the Split MME deployment is left to implementation.

For Security aspects of Full EPC in each satellite, the security credentials required for the mutual authentication between the UE and network are stored in the HSS onboard the satellite. To enable having only a satellite-specific subscriber key stored in the satellite, IOPS-based keying method described in Annex F of TS 33.401 [1] is used to limit the impact of exposing long-term keys in the HSS onboard the satellite.

References

Related CRs: set "TSG Status = Approved" in:

https://portal.3gpp.org/ChangeRequests.aspx?q=1&workitem=1060065

[1] TS 33.401: "3GPP System Architecture Evolution: Security Architecture".
[2] TS 23.401: "General Packet Radio Service (GPRS) enhancements for Evolved Universal Terrestrial Radio Access Network (E-UTRAN) access".

6.2.3 Charging aspects of satellite access Phase 3

1040014	*Study on charging aspects of satellite access Phase 3*	FS_5GSAT_Ph3_CH	S5	SP-240980	Yingying Liu, CATT,
1070014	**Charging aspects of satellite access Phase 3**	5GSAT_Ph3-CH	S5	SP-250365	Yingying Liu, CATT

Summary based on the input provided by CATT in SP-250365.

This Work Item specifies the Charging aspects of satellite access Phase 3. TR 28.846 [1] has identified the business scenarios, possible charging scenarios and requirements, and developed and evaluated the potential solutions. This Work Item is the normative work Rel-19 based on the recommendations.

Based on the conclusions documented in the TR 28.846 [1], the charging aspects of satellite access Phase 3 is specified in the following aspects:

- Support of Store and Forward satellite operation with the enhancement of the CP CIoT, UP CIoT and SMS service in TS 32.240[2], TS 32.251[3] and TS 32.298[7].
- Support UE-satellite-UE communication with the enhancement of the IMS charging in TS 32.260[4], TS 32.291[6] and TS 32.298[7].
- Enhance the roaming charging principle to support transparent satellite access in TS 32.255[5].
- Enhance the charging principle for SSP to perform wholesale to MNO in TS 32.257[8].
- Support of MVNO which provides satellite communication services in TS 32.255[5].

References

Related CRs: set "TSG Status = Approved" in:

https://portal.3gpp.org/ChangeRequests.aspx?q=1&workitem=1070014

[1] TS 28.846: " Study on charging aspects of satellite access Phase 3".
[2] TS 32.240: "Telecommunication management; Charging management; Charging Architecture and Principles".
[3] TS 32.251: "Packet Switched (PS) domain charging ".
[4] TS 32.260: "IP Multimedia Subsystem (IMS) charging ".
[5] TS 32.255: " Telecommunication management; Charging management; 5G data connectivity domain charging; Stage 2".
[6] TS 32.291: "Telecommunication management; Charging management; 5G system; Charging service, stage 3".
[7] TS 32.298: "Telecommunication management; Charging management; Charging Data Record (CDR) parameter description".
[8] TS 32.257: "Edge computing domain charging".

5.1.3 Non-Terrestrial Networks (NTN) for Internet of Things (IoT) Phase 3 (for LTE)

1020096	**Non-Terrestrial Networks (NTN) for Internet of Things (IoT) Phase 3 (for LTE)**	IoT_NTN_Ph3		RP-252504	Mediatek
1021096	**Core part: Non-Terrestrial Networks (NTN) for Internet of Things (IoT) Phase 3**	IoT_NTN_Ph3-Core	R2	RP-252504	Mediatek
1032087	**Perf. part: Non-Terrestrial Networks (NTN) for Internet of Things (IoT) Phase 3**	IoT_NTN_Ph3-Perf	R4	RP-252504	Mediatek

Summary based on the input provided by MediaTek Inc. in RP-252505.

This Work Item introduces enhancements in three areas: Store-and-Forward (S&F) satellite operation using a full eNB as a regenerative payload; uplink capacity improvements; and support for broadcasting public warning system (PWS) messages for NB-IoT.

For S&F satellite operation, the work enables the network to distinguish between legacy UEs and Rel-19 S&F-capable UEs by introducing a new barring mechanism, allowing S&F-capable devices to be served while legacy devices are barred when needed. New system information indications are defined to inform UEs whether the serving and neighbouring cells are operating in S&F mode and to signal the transition timing between S&F and normal operation. In addition, an S1 Removal procedure is introduced to support the operation of a full eNB acting as a regenerative satellite payload.

To enhance uplink capacity, the work item introduces orthogonal cover code (OCC) support for NPUSCH format 1, enabling multiplexing of multiple UEs on shared uplink resources. Symbol-level OCC is supported for 3.75 kHz subcarrier spacing, while slot-level OCC is supported for 15 kHz subcarrier spacing, along with the definition of OCC-based reference signals. In connected mode, OCC usage is enabled through UE-specific RRC signalling, with dynamic activation and deactivation controlled by DCI. The work also reduces signalling overhead for early data transmission by introducing a contention-based Msg3 early data transmission procedure that allows UEs in RRC_IDLE to transmit data without random access preambles or responses. Support for diversity slotted ALOHA is added to allow multiple Msg3 replicas, system information provides the necessary configuration parameters, and a new MAC PDU format for Msg4 enables multiplexing of multiple UEs in a single transmission.

For NB-IoT public warning services, the work item introduces new system information blocks to support primary and secondary ETWS notifications and CMAS notifications. Paging and direct indication mechanisms are enhanced to inform UEs about PWS reception, and the concept of an acceptable cell is introduced so that NB-IoT UEs can camp on a cell solely for receiving PWS broadcasts even if the cell does not belong to the UE’s subscribed PLMN.

References

List of related CRs: select "TSG Status = Approved" in:
https://portal.3gpp.org/ChangeRequests.aspx?q=1&workitem=1020096,1021096,1032087

5.1.4 Introduction of IoT-NTN TDD mode

1050123	**Introduction of IoT-NTN TDD mode**	IoT_NTN_TDD		RP-252935	Iridium
1051123	**Core part: Introduction of IoT-NTN TDD mode**	IoT_NTN_TDD-Core	R1	RP-252935	Iridium
1052123	**Perf. part: Introduction of IoT-NTN TDD mode**	IoT_NTN_TDD-Perf	R4	RP-252935	Iridium

Summary based on the input provided by Iridium Satellite LLC in RP-252872.

This Work Item introduces a new mode of operation for NB-IoT NTN that enables half-duplex use of radio resources, allowing time-division duplex (TDD) operation in both the Satellite Access Node (SAN) and the IoT NTN UE.

By supporting TDD, this mode extends 3GPP NB-IoT NTN operation to additional non-geostationary satellite (NGSO) systems. In earlier releases, including the Rel-17 work item LTE_NBIOT_eMTC_NTN, the Rel-18 work item IoT_NTN_enh, and the Rele-19 work item IoT_NTN_Ph3, NB-IoT NTN was specified exclusively for frequency-division duplex (FDD) operation; this work item is the first to enable NB-IoT NTN operation in TDD spectrum.

The IoT NTN TDD mode is designed to reuse common principles from existing NB-IoT FDD NTN operation and targets the Mobile-Satellite Service (MSS)-allocated spectrum in the 1616–1626.5 MHz band. The intended deployment is standalone, supporting both anchor and non-anchor carriers.

Frame Structure Type-1 is applied, where uplink and downlink transmissions are separated in the time domain and consist of eight contiguous downlink subframes followed by eight contiguous uplink subframes, separated by a fixed guard period of 50 ms, with this pattern repeating every nine radio frames. Compared to FDD operation, most physical-layer transmissions in IoT NTN TDD are deferred to the next available downlink (D) or uplink (U) subframe group, reflecting the half-duplex nature of the mode. See figure 1 below.

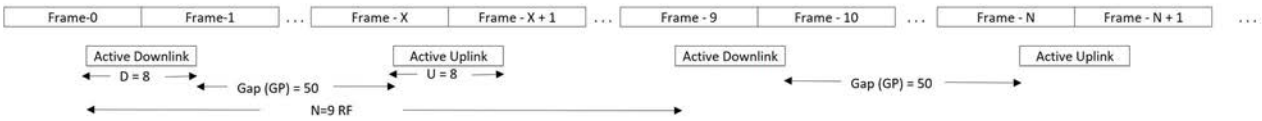


Figure 1. IoT NTN TDD mode with Half Duplex FDD frame structure

References

List of related CRs: select "TSG Status = Approved" in:
https://portal.3gpp.org/ChangeRequests.aspx?q=1&workitem=1050123,1051123,1052123

[1] Last status report: RP-252329

5.1.5 Enhanced requirements and test methodology for NR NTN and IoT NTN

1030086	**Enhanced requirements and test methodology for NR NTN and IoT NTN**	NR_IoT_NTN_req_test_enh		RP-252076	Samsung
1031086	**Core part: Enhanced requirements and conductive test methodology for NR NTN and IoT NTN**	NR_IoT_NTN_req_test_enh-Core	R4	RP-252076	Samsung
1032086	**Perf. Part: Enhanced requirements and conductive test methodology for NR NTN and IoT NTN**	NR_IoT_NTN_req_test_enh-Perf	R4	RP-252076	Samsung
1090063	**UE Conformance - Enhanced requirements and conductive test methodology for NR NTN and IoT NTN**	NR_IoT_NTN_req_test_enh-UEConTest	R5	RP-252575	Xiaomi

Summary based on the input provided by Samsung and Xiaomi in RP-252075.

This WI focues on two main aspects for further enhancement on NR-NTN and IoT-NTN in RAN4. One is to enable both NR-NTN and IoT-NTN high power UE (HPUE) to transmit with higher power in FR1-NTN bands, by conducting and concluding co-existence studies and specifying UE RF requirements. The other one is to enable NR-NTN to operate with less than 5MHz channel bandwidth, by specifying corresponding system parameters, UE RF requirements, RRM requirements and satellite-access node (SAN) requirements for these channel bandwidths.

Core: High power UE (HPUE) for NTN

In this WI, in order to protect the operators in adjacent channel from this HPUE operation, the co-existence studies were performed and concluded for between NR-NTN/IoT-NTN and NR-TN network for the UEs transmitting in 26dBm for Power class (PC) 2, 29dBm for PC1.5 and 31dBm in PC1 in FR1-NTN bands. The co-ex studies were performed by system level simulations from different sourcing companies, and the simulations followed the same methodologies and deployment assumptions as Rel-17 NR-NTN co-ex studies in same FR1-NTN bands.

With the observations and conclusions from the co-ex studies, after discussion, the adjacent channel leakage ratio (ACLR) for NR-NTN HPUE were concluded as 31dBc and 34dBc for PC2 and PC1 respectively; and ACLR for NB-IoT based IoT-NTN were concluded as 37 dBc for both PC2 and PC1; and ALCR for eMTC based IoT-NTN was concluded as 31 dBc for PC2.

For the NR-NTN HPUE RF requirements, the power class 2 and power class 1 for NR-NTN band n256 and n255 are introduced. For the IoT-NTN HPUE RF requirements, the power class 2 and power class 1 for NB-IoT based IoT-NTN band 256 and 255 are introduced; and power class 2 for eMTC based IoT-NTN band 256 and 255 are introduced in this release. The relevant HPUE requirements for these power classes and bands are specified, including maximum output power (MOP) and associated tolerances, maximum output power reduction (MPR) to meet emission requirements and ACLR for these power classes; additional maximum output power (A-MPR) to meet additional emission requirements and reference sensitivity degradation (RSD) for these newly introduced bands.

Core: Less than 5MHz for NTN

In this WI, in order to enable less than 5MHz channel bandwidth operations for NR-NTN, the system parameters, UE RF requirements, RRM requirements and satellite-access node (SAN) requirements are enhanced from existing FR1-NTN requirements.

For the system parameters, 3MHz transmission bandwidth is introduced to the maximum transmission bandwidth configuration; minimum guard band is specified to 3MHz UE channel bandwidth; GSCN and sync raster are enhanced with 3MHz channel bandwidth; and this UE channel bandwidth 3MHz is introduced to FR1-NTN bands n256, n255 and n254 in this release.

For UE RF requirements, the enhancements are made to almost all Tx and Rx requirements that related to UE channel bandwidth. The impacts include transmitter power, output power dynamics, output RF spectrum emissions in Tx parts, and reference sensitivity, maximum input level, adjacent channel selectivity, blocking characteristics and spurious emissions in Rx parts. And the changes are made to introduce relevant requirements to 3MHz channel bandwidth.

For RRM requirements, the enhancements are made to handover, conditional handover, satellite switching with re-sync, link recovery, intra-frequency measurements, inter-frequency measurements procedures. The key enhancements of these changes are made to incorporate the different SSB for less than 5MHz channel bandwidth.

For SAN RF requirements, the enhancements are also made to introduce missing requirements for 3MHz channel bandwidths in most of Tx and Rx requirements. The impacts include the reference sensitivity level, dynamic range, in-band sensitivity and blocking, in-channel selectivity, OTA output power dynamics, OTA transmitted signal quality, OTA unwanted emissions, OTA sensitivity, OTA reference sensitivity level, OTA dynamic range, OTA in-band selectivity and blocking, OTA in-channel selectivity and associated reference channels for Rx requirements and dynamic ranges.

Performance parts

This WI also includes the RF, RRM, demodulation performance requirements work by introducing new NGSO channel model, and also the SAN conformance and RRM conformance work by introducing less than 5MHz channel bandwidth. These works are still under discussion, and will be further completed in the performance phase of this WI.

References

List of related CRs: select "TSG Status = Approved" in:
https://portal.3gpp.org/ChangeRequests.aspx?q=1&workitem=1030086,1031086,1032086,1090063

[1] [1] RP-252076, Revised WID on Enhanced requirements and conductive test methodology for NR NTN and IoT NTN
[2] RP-252074, Status Report to TSG: Enhanced requirements and conductive test methodology for NR NTN and IoT NTN

5.2 Positionning-related items

5.2.1 On-demand broadcast of GNSS assistance data

1050078	**On-demand broadcast of GNSS assistance data**	TEI19_OBGAD		SP-240487	Aihua Li, China Mobile

1030010	**Stage 2 of On-demand broadcast of GNSS assistance data**	TEI19_OBGAD	S2	SP-240487	Aihua Li, China Mobile
1050003	**CT aspects of On-demand broadcast of GNSS assistance data**	TEI19_OBGAD	C4	CP-242023	Rong Wang, China Mobile
1050079	CT4 aspects of On-demand broadcast of GNSS assistance data	TEI19_OBGAD	C4	CP-242023	Rong Wang, China Mobile

Summary based on the input provided by China Mobile by e-mail received Tue 31/03/2026 14:17, file "21919-110-revision".

This work item introduces an on-demand mechanism for broadcasting GNSS assistance data between the LMF and AMF: based on information received from the AMF and/or UEs, the LMF decides whether to initiate a broadcast.

The GNSS assistance data broadcast procedure is enhanced by adding an optional subscription and notification interaction between the AMF and LMF. Additionally, optional logic is introduced in the LMF to track how many UEs have requested Real-Time Kinematic (RTK) assistance data (specifically, which RTK AD or posSIB) using Mobile-Originated Location Request (MO-LR) within a given area.

RTK is a high-precision positioning technique that improves standard GNSS accuracy from meter level to centimeter level by using real-time correction data. RTK AD (RTK Assistance Data) refers to GNSS assistance data required by a UE to perform RTK positioning. A posSIB is a Positioning System Information Block, i.e. a broadcast RRC container used to deliver positioning assistance data to UEs. RTK Assistance Data can be mapped to specific posSIB types.

References

List of related CRs: select "TSG Status = Approved" in:
https://portal.3gpp.org/ChangeRequests.aspx?q=1&workitem=1050078;1030010;1050003;1050079

[1] TS 23.501: “System architecture for the 5G System (5GS)
[2] TS 23.273: ”5G System (5GS) Location Services (LCS); Stage 2”

5.2.2 Introduction of BDS (BeiDou Navigation Satellite System) B2b Signal in A-GNSS for LTE and NR

1050121	**Introduction of BDS (BeiDou Navigation Satellite System) B2b Signal in A-GNSS for LTE and NR**	LCS_BDS_B2b_LTE_NR		RP-252083	CAICT
1051121	**Core part: Introduction of BDS (BeiDou Navigation Satellite System) B2b Signal in A-GNSS for LTE and NR**	LCS_BDS_B2b_LTE_NR-Core	R2	RP-252083	CAICT
1052121	**Perf. part: Introduction of BDS (BeiDou Navigation Satellite System) B2b Signal in A-GNSS for LTE and NR**	LCS_BDS_B2b_LTE_NR-Perf	R4	RP-252083	CAICT

Summary based on the input provided by CAICT and CATT in RP-252107.

This WI introduces the B2b signal of BDS in A-GNSS in both E-UTRA and NR systems to increase the positioning accuracy.

The BDS B2b signal was supported in A-GNSS for both E-UTRA and NR, including corresponding ICD as reference in stage 2 and stage 3 TSs and corresponding assistant information in dedicated signaling and performance requirements were specified.

References

List of related CRs: select "TSG Status = Approved" in:
https://portal.3gpp.org/ChangeRequests.aspx?q=1&workitem=1050121,1051121,1052121

[1] RP-252108 Status Report for WI: Introduction of BDS (BeiDou Navigation Satellite System) B2b Signal in A-GNSS for LTE and NR CAICT, CATT

5.2.3 Introduction of A-GNSS support for NavIC (Navigation with Indian Constellation) L1 SPS (Standard Positioning Service) in NR & LTE

1050122	**Introduction of A-GNSS support for NavIC (Navigation with Indian Constellation) L1 SPS (Standard Positioning Service) in NR & LTE**	LCS_NAVIC_L1_SPS_NR_LTE		RP-251552	Reliance Jio
1051122	**Core part: Introduction of A-GNSS support for NavIC (Navigation with Indian Constellation) L1 SPS (Standard Positioning Service) in NR & LTE**	LCS_NAVIC_L1_SPS_NR_LTE-Core	R2	RP-251552	Reliance Jio
1052122	**Perf. part: Introduction of A-GNSS support for NavIC (Navigation with Indian Constellation) L1 SPS (Standard Positioning Service) in NR & LTE**	LCS_NAVIC_L1_SPS_NR_LTE-Perf	R4	RP-251552	Reliance Jio

Summary based on the input provided by Reliance Jio in RP-252828.

This Work Item introduces support for NavIC L1 band SPS–specific assistance data, enabling UE-based and UE-assisted A-GNSS positioning in both NG-RAN and E-UTRAN across control-plane and user-plane positioning.

Prior to this work, A-GNSS support in NR and LTE was limited to NavIC L5 band SPS, alongside GPS, Galileo, GLONASS, BDS, SBAS, and QZSS constellations. The addition of NavIC L1 SPS assistance data improves positioning accuracy and time-to-fix, enhances receiver sensitivity, and reduces UE power consumption for devices supporting NavIC RNSS.

The key updates introduced by this WI are :

- support for NavIC satellite L1 SPS system, added to the A-GNSS specifications by assignment of new ‘GNSS Signal ID’, and ‘GNSS Time ID’, together with ASN.1 extensions to existing assistance data elements.
- NavIC L1 specific New Clock Model, Orbit Model, Almanac Model, UTC model update, Differential corrections and Ionospheric Model have been added to existing assistance data elements.
- The GNSS-GenericAssistData used by the location server to aid data for a specific GNSS has been updated by introducing NavIC satellite system specific Differential corrections.

References

List of related CRs: select "TSG Status = Approved" in:
https://portal.3gpp.org/ChangeRequests.aspx?q=1&workitem=1050122,1051122,1052122

[1] TS 37.355: LTE Positioning Protocol (LPP); Protocol specification
[2] RP-251552: Revised WID LCS_NAVIC_L1_SPS_NR_LTE, Reliance Jio
[3] RP-252826: Introduction of A-GNSS support for NavIC L1 SPS in NR and LTE, Reliance Jio
[4] RP-251554: SR-Introduction of A-GNSS support for NavIC L1 SPS in NR and LTE

5.3 Other aspects (non band-related)

5.3.1 Support for PWS in Satellite E-UTRAN and Satellite NG-RAN

1070009	**Support for PWS in Satellite E-UTRAN and Satellite NG-RAN**	PWS_NTN		CP-251222	Amer Catovic
1070049	**CT1 aspects of Support for PWS in Satellite E-UTRAN and Satellite NG-RAN**	PWS_NTN	C1	CP-251222	Amer Catovic

Summary based on the input provided by Qualcomm by e-mail received Mon 16/03/2026 21:54, file "Summary_1070009_1070049".

This WI introduces support for PWS over Satellite E-UTRAN and Satellite NG-RAN. More precisely, the following aspects have been introduced:

- PWS over Satellite E-UTRAN
- ETWS geo-fencing
- duplicate detection when PLMN uses shared MCC

- PWS over Satellite NG-RAN (introduced from Rel-17)

References

List of related CRs: select "TSG Status = Approved" in:
https://portal.3gpp.org/ChangeRequests.aspx?q=1&workitem=1070009,1070049

5.3.2 Enhancements for Air-To-Ground (ATG) network for NR

1030083	**Enhancements for Air-to-ground network for NR**	NR_ATG_enh		RP-243285	CMCC
1031083	**Core part: Enhancements for Air-to-ground network for NR**	NR_ATG_enh-Core	R4	RP-243285	CMCC
1032083	**Perf. part: Enhancements for Air-to-ground network for NR**	NR_ATG_enh-Perf	R4	RP-243285	CMCC

Summary based on the input provided by CMCC in RP-252603.

This targets two main aspects: enhance throughput to meet ATG capacity demand; and fully utilize operators' spectrum especially for the fragment spectrum. To achieve these two objectives, the RF and RRM requirements are specified for ATG UE to support DL CA and MIMO.

Core parts

In order to help operators fully utilize spectrum and enhance throughput for ATG, RF and RRM core requirements for ATG supporting DL CA, and RF requirements for ATG supporting UL MIMO are specified. For ATG supporting DL CA, this WI focus on the scenario of intra-band co-located (Example band combination: CA_n79C) and inter-band co-located DL CA (Example band combination: CA_n3A-n39A). For ATG supporting UL MIMO, the scenario of 2TX for single CC for ATG UE is studied.

For the NR ATG RF part, the requirement for DL CA and UL/DL MIMO are introduced. For ATG supporting DL CA, a complete set of Rx RF requirements—including Reference sensitivity, Maximum input level, ACS, Blocking characteristics, and Spurious emissions have been defined in TS38.101-1. In addition to supporting CA_n79C and CA_n3A-n39A with UL on n39, and after fully considering ATG UE architecture and spectrum utilization, RF requirements for CA_n3A-n39A with UL on n3 have also been specified. For ATG supporting MIMO, a complete set of Tx RF requirements, including Transmitter power, Output power dynamics, Transmit signal quality, Output RF spectrum emissions have been specified for ATG UE, and the TAE requirements have been specified for ATG BS.

For RRM requirements, a complete set of requirements for DL CA and UL/DL MIMO are introduced in TS 38.133, including MRTD, interruption, SCell activation and deactivation delay, Link Recovery procedures for SCell, Active BWP switch delay on multiple CC, measurement requirement for CC with deactivated SCell. Meanwhile, considering the UE antenna type on PCC and SCC, and beam implementation when UE capable of *antennaArrayType-r18* on both PCC and SCC, a set of requirements are updated in TS 38.133, including RLM, Link Recovery, intra-frequency measurement, inter-frequency measurement without gap, L1-RSRP, L1-SINR and CSI-RS based measurement requirements.

Performance parts

This WI also includes the RRM performance and demodulation requirements for intra-band co-located and inter-band co-located DL CA, the demodulation requirements for support of UL MIMO with 2TX for single CC for UE with dual-polarization antenna, the demodulation requirements for support of DL MIMO for UE with dual-polarization antenna, and the conformance testing and conformance requirements for BS.

References

List of related CRs: select "TSG Status = Approved" in:
https://portal.3gpp.org/ChangeRequests.aspx?q=1&workitem=1030083,1031083,1032083

[1] RP-252602, Status Report for Enhancements for Air-to-ground network for NR

5.3.3 Inter-RAT mode mobility support from E-UTRAN TN to NR NTN

1030076	**Inter-RAT mode mobility support from E-UTRAN TN to NR NTN**	LTE_TN_NR_NTN_mob		RP-251974	Thales
1031076	**Core part: Inter-RAT mode mobility support from E-UTRAN TN to NR NTN**	LTE_TN_NR_NTN_mob-Core	R2	RP-251974	Thales

Summary based on the input provided by Thales and CATT in RP-252846.

This Work Item introduces support for LTE terrestrial-network to NR non-terrestrial-network idle-mode mobility.

The feature enables **idle-mode mobility** based on cell reselection from E-UTRA TN to NR NTN, where the E-UTRA TN cell provides information about neighbouring NR NTN satellite cells through system information. The scope of this work item impacts RAN2 only and applies to LTE specifications, with the Stage-2 description defined in TS 36.300, RRC signalling specified in TS 36.331, and UE capabilities defined in TS 36.306.

The enhancements focus on enabling idle-mode cell reselection from E-UTRA TN to NR NTN by allowing E-UTRA TN cells to **broadcast NR neighbour satellite assistance information** via SIB33, to indicate satellite ID lists per frequency in SIB24 for associating NR satellite assistance information during reselection, and to **support redirection from LTE TN to NR NTN**. In addition, **E-UTRAN cells broadcasting NR satellite assistance** information in SIB33 are enhanced to configure a dedicated priority for an NR NTN frequency, further facilitating efficient idle-mode mobility toward NR NTN.

References

List of related CRs: select "TSG Status = Approved" in:
https://portal.3gpp.org/ChangeRequests.aspx?q=1&workitem=1030076,1031076

5.3.4 Uncrewed Aerial System Phase 3

1020059	**Uncrewed Aerial System Phase 3**	UAS_Ph3		SP-230518	LaeYoung Kim (LG Electronics)
960017	*Study on (Stage 1 of) UAV Phase 3*	FS_UAV_Ph3	S1	SP-220954	Pengtai Qin , China Mobile
1020069	*Study on (Stage 2 of) Phase 3 for UAS, UAV and UAM*	FS_UAS_Ph3	S2	SP-231801	LaeYoung Kim (LG Electronics)
1000032	**Stage 1 of Uncrewed Aerial System Phase 3**	UAS_Ph3	S1	SP-230518	Pengtai Qin , China Mobile
1040037	**(Stage 2 of) UAS, UAV and UAM Phase 3**	UAS_Ph3	S2	SP-241287	LaeYoung Kim (LG Electronics)
1050013	**CT aspects of UAS, UAV and UAM Phase 3**	UAS_Ph3		CP-251136	Sunhee Kim, LG Electronics
1050091	CT1 aspects of UAS, UAV and UAM Phase 3	UAS_Ph3	C1	CP-251136	Sunhee Kim, LG Electronics
1050092	CT3 aspects of UAS, UAV and UAM Phase 3	UAS_Ph3	C3	CP-251136	Sunhee Kim, LG Electronics
1060047	CT4 aspects of UAS, UAV and UAM Phase 3	UAS_Ph3	C4	CP-251136	Sunhee Kim, LG Electronics
1060058	**Security of Uncrewed Aerial System Phase 3**	UAS_Ph3_Sec	S3	SP-241773	Markus Hanhisalo, Ericsson
1070017	**Charging for Uncrewed Aerial Systems Phase 3**	UAS_Ph3-CH	S5	SP-250512	Jia Dong, China Mobile
1010008	**Application Architecture for UAS applications Phase 3**	UASAPP_Ph3	S6	SP-240741	Roy, Michel, InterDigital
1040043	**CT Aspects of Application Layer Support for Uncrewed Aerial Systems (UAS), Phase 3**	UASAPP_Ph3		CP-241281	Taimoor Abbas, InterDigital
1040069	CT1 Aspects of Application Layer Support for Uncrewed Aerial Systems (UAS), Phase 3	UASAPP_Ph3	C1	CP-241281	Taimoor Abbas, InterDigital
1040070	CT3 Aspects of Application Layer Support for Uncrewed Aerial Systems (UAS), Phase 3	UASAPP_Ph3	C3	CP-241281	Taimoor Abbas, InterDigital
1030033	**UAS security enhancements**	FS_UAS_Ph3_Sec	S3	SP-240504	Markus Hanhisalo,

1040017	*Study on Charging aspects of Uncrewed Aerial Systems*	FS_UAS_CH	S5	SP-240983	Jia Dong, China Mobile,

Summary based, for the main aspects, on the input provided by LG Electronics in SP-251129, and, for the UASAPP_Ph3 aspects, by InterDigital by e-mail received Mon 30/03/2026 21:06, file "Addition_of_UASAPP_Ph3_to_TR 21.919"

This Work Item mainly specifies architecture and system level enhancement to support the following scenarios and requirements for UAV (Uncrewed Aerial Vehicle) and UAM (Urban Air Mobility):
- Enhance NEF services to support service exposure and interactions between MNOs and UTM functions for support the following:
- Pre-mission flight planning and in-mission flight monitoring for UAVs.
- Command and Control (C2) communication reliability [the communication link and operational control channel used to manage and direct a UAV during flight].
- Multiple USS serving different geographical areas corresponding the UAV flight path, i.e. changeover of the UAV from one USS to another USS.
- Enable network assisted/ground-based mechanism for DAA (Detect and Avoid)
- Support NTZs (No-Transmit Zones) for UAVs

Description, procedures and protocols related to architecture and system level enhancement for UAS (Uncrewed Aerial System) services are specified as below:
- Pre-flight Planning and In-flight Monitoring for UAVs as defined in TS 23.256 [1]
- Multiple USS serving different geographical areas as defined in TS 23.256 [1]
- Altitude reporting for aerial UEs as defined in TS 23.256 [1]
- Support of Network-Based/Assisted DAA as defined in TS 23.256 [1]
- Support of NTZs as defined in TS 23.256 [1]
- Methods to support C2 communication reliability as Informative Annex as defined in TS 23.256 [1]
- Updates to the existing Analytics, e.g. QoS Sustainability Analytics as defined in TS 23.288 [2]
- Updates to enhance NEF services as defined in TS 23.502 [3]
- Support of multiple USS serving different geographical areas corresponding to the UAV flight path as defined in TS 24.301 [4] and TS 24.501 [5]
- Support of enforcement of the NTZ assistance information by the UE as defined in TS 24.301 [4] and TS 24.501 [5]
- introduce new AT command +CNTZAI as defined in TS 27.007 [6]
- USS service to support pre-mission flight planning assistance service, in-mission flight monitoring service, multiple-USS service (e.g. USS/UTM changeover procedure for an UAV) and network-assisted DAA service as defined in TS 29.255 [7]
- Enhancements to NEF (UAS-NF) southbound service in Nnef_Authentication API as defined in TS 29.256 [8]
- Update to UDM services for expected UE behavior provisioning, e.g. for new elements for assigned trajectory as defined in TS 29.503 [9]
- Updates to the AMF Event Exposure service for new event filter includes new elements for assigned trajectory as defined in TS 29.518 [10]
- Updates NWDAF interface to enhance the inputs of Movement Behavior Analytics with height information and QoS Sustainability Analytics with 3D location waypoints as defined in TS 29.520 [11]
- Updates to the NEF northbound and USS interface to support pre-mission flight planning assistance service, in-mission flight monitoring service, multiple-USS service (e.g. USS/UTM changeover procedure for an UAV) and network-assisted DAA service as defined in TS 29.522 [12]
- Updates to the NEF northbound to support the exposure of NWDAF Analytics service of the list of UEs in QoS Sustainability and Movement Behavior Analyses, and TTC (Time To Collision) prediction in Relative Proximity Analytics as defined in TS 29.522 [12]
- Support of real time UAV flight path monitoring assistance as defined in TS 23.255 [14], TS 24.257 [15] and TS 29.257 [16]
- Support of UAS provided flight routes as defined in TS 23.255 [14], TS 24.257 [15] and TS 29.257 [16]
- Support of UAE layer for NTZ as defined in TS 23.255 [14], TS 24.257 [15] and TS 29.257 [16]
- Enhancements to C2 Communication mode selection and switching as defined in TS 23.255 [14], TS 24.257 [15] and TS 29.257 [16]
- Enhancements to UAE layer support for DAA services as defined in TS 23.255 [14], TS 24.257 [15] and TS 29.257 [16]

Regarding security enhancements to NEF service for supporting the scenario of multiple USS serving different geographical area corresponding to the UAV flight path, support a changeover from one USS to another USS for UAV assisted by 5GS is specified in TS 33.256 [13].

References

List of related CRs: select "TSG Status = Approved" in:
https://portal.3gpp.org/ChangeRequests.aspx?q=1&workitem=1020059,960017,1020069,1000032,1040037,1050013,1050091,1050092,1060047,1060058,1070017,1010008,1040043,1040069,1040070,1030033,960018,1040017,1020013

[1] TS 23.256: "Support of Uncrewed Aerial Systems (UAS) connectivity, identification and tracking; Stage 2".
[2] TS 23.288: "Architecture enhancements for 5G System (5GS) to support network data analytics services".
[3] TS 23.502: "Procedures for the 5G System (5GS); Stage 2".
[4] TS 24.301: "Non-Access-Stratum (NAS) protocol for Evolved Packet System (EPS); Stage 3".
[5] TS 24.501: "Non-Access-Stratum (NAS) protocol for 5G System (5GS); Stage 3;".
[6] TS 27.007: "AT command set for User Equipment (UE)".
[7] TS 29.255: "Uncrewed Aerial System Service Supplier (USS) Services; Stage 3".
[8] TS 29.256: "Uncrewed Aerial Systems Network Function (UAS-NF); Aerial Management Services; Stage 3".
[9] TS 29.503: "5G System; Unified Data Management Services; Stage 3".
[10] TS 29.518: "5G System; Access and Mobility Management Services; Stage 3".
[11] TS 29.520: "5G System; Network Data Analytics Services; Stage 3".
[12] TS 29.522: "5G System; Network Exposure Function Northbound APIs; Stage 3".
[13] TS 33.256: "Security aspects of Uncrewed Aerial Systems (UAS)".
[14] TS 23.255 " Application layer support for Uncrewed Aerial System (UAS); Functional architecture and information flows".
[15] TS 24.257 " Uncrewed Aerial System (UAS) Application Enabler (UAE) layer; Protocol aspects; Stage 3".
[16] TS 29.257 " Application layer support for Uncrewed Aerial System (UAS); UAS Application Enabler (UAE) Server Services; Stage 3".

5.3.5 Management Aspects of Rel-18's NTN Phase 2

1060016	**Management Aspects of NTN Phase 2**	NTN_OAM_Ph2	S5	SP-250514	MingRui Sun (China Unicom)
1020013	*Study on Management Aspects of NTN Phase 2*	FS_NTN_OAM_Ph2	S5	SP-231733	MingRui Sun (China Unicom)

Summary based on the input provided by China Unicom in SP-260238.

This work item specifies the management aspects of supporting IoT and NR over non-terrestrial networks (NTN), taking into account satellite-specific capabilities such as Store-and-Forward operation and direct UE-Satellite-UE communication enabled by regenerative payload architectures and operation without having to use the terrestrial network.

The scope covers all NTN cell types, including moving, quasi-earth-fixed, and fixed cells, and defines management connections and associations between satellite and ground systems, encompassing radio nodes, the core network, and management systems. It also enhances management support for mobility coordination between NTN and terrestrial networks, as well as between different NTNs, in order to improve service continuity.

In addition, the work item specifies management requirements, use cases, and solutions for these satellite-specific operation modes and communication scenarios, and addresses the management of secure connections in NTN environments.

Management aspects of supporting IoT and NR over non-terrestrial networks (NTN) are specified in TS 28.541 [1] and TS 28.552 [2],TS 28.554 [3], TS 28.657 [4], TS 28.658 [5], TS 28.659 [6], TS 28.707 [7], TS 28.708 [8], TS 28.709 [9], TS 28.540 [10], and TS 28.530 [11].

References

List of related CRs: select "TSG Status = Approved" in:
https://portal.3gpp.org/ChangeRequests.aspx?q=1&workitem=1060016

[1] TS 28.541: " Management and orchestration; 5G Network Resource Model (NRM); Stage 2 and stage 3".
[2] TS 28.552: " Management and orchestration; 5G performance measurements".
[3] TS 28.554: " Management and orchestration; 5G end to end Key Performance Indicators (KPI)".

[4] TS 28.657: " Telecommunication management; Evolved Universal Terrestrial Radio Access Network (E-UTRAN) Network Resource Model (NRM) Integration Reference Point (IRP); Requirements".
[5] TS 28.658: " Telecommunication management; Evolved Universal Terrestrial Radio Access Network (E-UTRAN) Network Resource Model (NRM) Integration Reference Point (IRP); Information Service (IS)".
[6] TS 28.659: " Telecommunication management; Evolved Universal Terrestrial Radio Access Network (E-UTRAN) Network Resource Model (NRM) Integration Reference Point (IRP); Solution Set (SS) definitions".
[7] TS 28.707: " Telecommunication management; Evolved Packet Core (EPC) Network Resource Model (NRM) Integration Reference Point (IRP); Requirements".
[8] TS 28.708: " Telecommunication management; Evolved Packet Core (EPC) Network Resource Model (NRM) Integration Reference Point (IRP); Information Service (IS)".
[9] TS 28.709: " Telecommunication management; Evolved Packet Core (EPC) Network Resource Model (NRM) Integration Reference Point (IRP); Solution Set (SS) definitions".
[10] TS 28.540: " Management and orchestration; 5G Network Resource Model (NRM); Stage 1".
[11] TS TS 28.530: " Management and orchestration; Concepts, use cases and requirements".

5.4 Band-related items

5.4.1 New LTE band for 5G broadcast for region 3 utilizing a geosynchronous satellite

1070064	**New LTE band for 5G broadcast for region 3 utilizing a geosynchronous satellite**	LTE_band_5G_bcast_GSO	R4	RP-251419	Astrum Mobile
1071064	**Core part: New LTE band for 5G broadcast for region 3 utilizing a geosynchronous satellite**	LTE_band_5G_bcast_GSO-Core	R4	RP-251419	Astrum Mobile
1072064	**Perf. part: New LTE band for 5G broadcast for region 3 utilizing a geosynchronous satellite**	LTE_band_5G_bcast_GSO-Perf	R4	RP-251419	Astrum Mobile

Summary based on the input provided by Astrum Mobile in RP-252885.

This WI specifies a new SDO (Standalone Downlink Only) band for LTE-based 5G Broadcast operation in 1467 to 1492 MHz in ITU's Region 3 for GSO (Geosynchronous Orbit) communication, based on E-UTRA operation over NTN (Non-Terrestrial Network).

UE RF requirements have been defined for 10 MHz PMCH (Physical Multicast Channel) in TS 36.102 (E-UTRA UE radio transmission and reception for satellite access). At the network side, related SAN (Satellite Access Node) RF requirements for 10 MHz PMCH have been defined in TS 36.108 (E-UTRA Satellite Access Node radio transmission and reception), considering applicable regulatory requirements based on WRC-19 Resolution 761.

References

List of related CRs: select "TSG Status = Approved" in:
https://portal.3gpp.org/ChangeRequests.aspx?q=1&workitem=1070064,1071064,1072064

5.4.2 Other items

Note: no summaries are provided for band-realted items.

1040135	**Introduction of Ku bands for NR NTN**	NR_NTN_Ku_bands		RP-252911	Intelsat
1050127	**Introduction of additional operating NR bands for HAPS (High Altitude Platform Station)**	NR_HAPS_bands		RP-242891	SoftBank
1040129	**Introduction of another NR NTN S-band (MSS band 2000-2020 MHz UL and 2180-2200 MHz DL)**	NR_NTN_Sband		RP-250488	Qualcomm
1040134	**New NR NTN bands to support Extended L-band and combined MSS L-band and Extended L-band ranges**	NR_NTN_combined Lband		RP-252893	Aalyria

1040130	**Introduction of another IoT-NTN S-band (MSS band 2000-2020 MHz UL and 2180-2200 MHz DL)**	IoT_NTN_FDD_S_band		RP-251820	Mediatek

6 Rel-19 Internet of Things (IoT) and Reduced Capability (RedCap) UE

6.1 NAS layer overhead reduction for data transfer using CP CIoT

1060018	**NAS layer overhead reduction for data transfer using CP CIoT**	NORDAT_CP	C1	CP-252195	Amer Catovic

Summary based on the input provided by Qualcom's e-mail received Mon 16/03/2026 21:54, file "Summary_1060018".

This Work Item defines a new NAS protocol message, namely EMM TRANSPORT, for transporting user data, SMS, and location information over NAS in both idle and connected modes.

It replaces the legacy CONTROL PLANE SERVICE REQUEST message. It contains only two bytes of NAS protocol overhead compared to 8 bytes for the legacy message.

This is a protocol optimization without impact on stage 2 procedures or on the overall protocol functionality.

References

List of related CRs: select "TSG Status = Approved" in:
https://portal.3gpp.org/ChangeRequests.aspx?q=1&workitem=1060018

6.2 Management Aspects of RedCap features

1060008	**Management Aspects of RedCap features**	NR_RedCap_OAM	S5	SP-241941	Zhen Xing (China Unicom)
1020017	*Study on management aspects of RedCap feature*	FS_NR_RedCap_OAM	S5	SP-231734	Zhen Xing (China Unicom)

Summary based on the input provided by China Unicom in SP-260235.

This work item specifies the management aspects of RedCap features, including enhancements to Network Resource Model (NRM) and Performance Measurements (PM).

The enhancements to NRM are to specify the configuration management to control the RedCap UEs' access to communication service and to specify the configuration management to support Multi-Initial Bandwidth Part (BWP) in co-existence scenario. A Bandwidth Part is a configured portion of the NR carrier bandwidth used by a UE at a given time for uplink or downlink operation.

The enhancements to PM are to add UE type as filter to support the performance measurements such as RedCap RRC connection/inactive number, RedCap PRB usage, etc.

The NRM enhancements are specified in TS 28.540 [1] and TS 28.541 [2], and the corresponding changes on YAML are updated on Forge. The PM enhancements are specified in TS 28.552 [3].

References

List of related CRs: select "TSG Status = Approved" in:
https://portal.3gpp.org/ChangeRequests.aspx?q=1&workitem=1060008

[1] TS 28.540: "Management and orchestration; 5G Network Resource Model (NRM); Stage 1"

[2] TS 28.541: "Management and orchestration; 5G Network Resource Model (NRM); Stage 2 and stage 3"
[3] TS 28.552: “Management and orchestration; 5G performance measurements”

6.3 See also

NTN for IoT Phase 3 (for LTE) (1020096, IoT_NTN_Ph3); "NR power class 2 RedCap (Reduced Capability) UE in FR1" (no summary)

Introduction of IoT-NTN TDD mode (1050123, IoT_NTN_TDD)

Enhanced requirements and test methodology for NR NTN and IoT NTN (1030086, NR_IoT_NTN_req_test_enh)

Introduction of another IoT-NTN S-band (MSS band 2000-2020 MHz UL and 2180-2200 MHz DL) (1040130, IoT_NTN_FDD_S_band)

7 Ambient power-enabled Internet of Things (IoT)

7.1 Ambient power-enabled Internet of Things (IoT) (SA and CT)

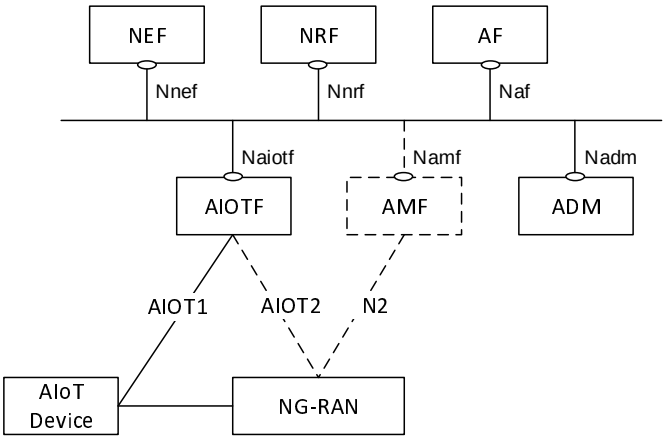
1020055	**Ambient power-enabled Internet of Things**	AmbientIoT		SP-231403	Runze Zhou (Huawei)
950004	*Study on Ambient power-enabled Internet of Things*	FS_AmbientIoT	S1	SP-220085	Xu, weijie, OPPO
1020071	*Study on Architecture support of Ambient power-enabled Internet of Things*	FS_AmbientIoT	S2	SP-240969	Runze Zhou (Huawei)
1020030	**Stage 1 of Ambient power-enabled Internet of Things**	AmbientIoT	S1	SP-231403	Xu, Weijie, OPPO
1070010	**Architecture support of Ambient power-enabled Internet of Things**	AmbientIoT-ARC	S2	SP-250377	Runze Zhou (Huawei)
1080021	**CT aspects of Ambient power-enabled Internet of Things**	AmbientIoT-CT		CP-252196	Wass, Mikael, Huawei
1080030	CT1 aspects of Ambient power-enabled Internet of Things	AmbientIoT-CT	C1	CP-252196	Wass, Mikael, Huawei
1080031	CT3 aspects of Ambient power-enabled Internet of Things	AmbientIoT-CT	C3	CP-252196	Wass, Mikael, Huawei
1080032	CT4 aspects of Ambient power-enabled Internet of Things	AmbientIoT-CT	C4	CP-252196	Wass, Mikael, Huawei
1020085	*Study on solutions for Ambient IoT (Internet of Things) in NR*	FS_Ambient_IoT_solutions	R1	RP-240826	Xiaodong Shen, CMCC
1030031	*Study on Security Aspect of Ambient IoT Services in 5G*	FS_Ambient_IoT_Sec	S3	SP-240506	Marcus Wong, Oppo
1070018	**Charging for Ambient power-enabled Internet of Things**	AmbientIoT-CH	S5	SP-250369	YiMeng, DENG, Huawei
1070022	**Security Aspects of Ambient IoT Services in 5G for Isolated Private Networks**	AmbientIoT-SEC	S3	SP-251196	Marcus Wong, Oppo

Summary provided by TR 21.919 Rapporteur.

Ambient IoT (AIoT) is a 5G System capability designed to support extremely low-power IoT devices that operate on harvested ambient energy rather than conventional batteries. An AIoT Device has very limited energy storage (or is battery-less) and may harvest power from sources such as radio waves, light, motion, or heat.

AIoT is targeted at isolated private network deployments (e.g., SNPN) and provides a controlled way to identify AIoT devices, discover them in a given area, and exchange application data with them despite their intermittent, energy-constrained operation.

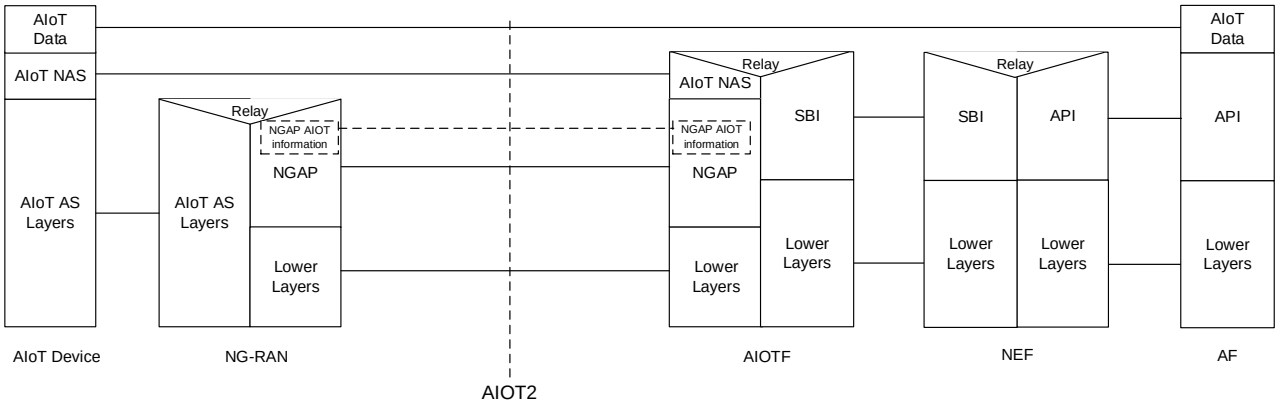
The architecture is described below:



System Architecture for AIoT (Non-roaming case)

It introduces the Ambient IoT Function (AIOTF) and supports AIoT readers within NG-RAN (gNB) to reach AIoT Devices, either through direct connectivity between AIOTF and NG-RAN or indirect connectivity via an AMF. Application Functions (AFs) access AIoT capabilities either directly (trusted AF) or through the Network Exposure Function (NEF) for authorization and exposure. Two core service types are defined: an Inventory service to discover AIoT Devices, and a Command service (Read, Write, and Permanent Disable) to retrieve or update device user memory or permanently disable RF transmission. The system also defines identifiers (including globally unique AIoT Device Permanent Identifiers and optional temporary identifiers for privacy), area concepts for targeting operations, and mechanisms for assistance information and aggregation so that NG-RAN and AIOTF can efficiently handle responses from potentially many devices, with optional reporting of device location at reader granularity when supported by policy and configuration.

The protocol stacks is as follows:



Protocol stack between the AF and the AIoT device (for NG-RAN Direct Connectivity)

The following protocols are introduced:
- AIoT Data: It is the application data exchanged between the AIoT Device and the AF.
- AIoT NAS: The NAS protocol between AIoT Device and the AIOTF.
- AIoT AS: It is between the AIoT Device and the NG-RAN as specified in TS 38.300 [5].
- NGAP: Application Layer Protocol between the AIOTF and the NG-RAN as defined in TS 38.413 [10].
- NGAP AIoT Information: It is the subset of NGAP information and is included in the NGAP messages over AIOT2 reference point.

Charging aspects (from Huawei in SP-251554., SP-251554): The charging for using Ambient IoT Services can be per API basis, and is supported in the northbound API exposure charging, as specified in TS 32.254 [3]. The A-IoT Function (AIOTF) is specified as one of the API Target Network Function in the NEF API Charging Information, see TS 32.254 [3], TS 32.291 [4] and TS 32.298 [5].

The Security aspects have been covered by the work item "Security Aspects of Ambient IoT Services in 5G for Isolated Private Networks"

References

List of related CRs: select "TSG Status = Approved" in:
https://portal.3gpp.org/ChangeRequests.aspx?q=1&workitem=1020055;950004;1020071;1020030;1070010;1080021;1080030;1080031;1080032;1020085;1030031;1070018;1070022

[1] TS 23.369: "Architecture support of Ambient power-enabled Internet of Things"
[2] TS 23.501: “System architecture for the 5G System (5GS)”
[3] TS 32.254: " Telecommunication management; Charging management; Exposure function Northbound Application Program Interfaces (APIs) charging ".
[4] TS 32.291: "Telecommunication management; Charging management; 5G system, charging service; Stage 3".
[5] TS 32.298: "Telecommunication management; Charging management; Charging Data Record (CDR) parameter description".
[6] TS 32.240: "Telecommunication management; Charging management; Charging architecture and principles".

7.2 Solutions for Ambient IoT (Internet of Things) in NR

1060084	**Solutions for Ambient IoT (Internet of Things) in NR**	Ambient_IoT_Solutions		RP-252105	CMCC
1061084	**Core part: Solutions for Ambient IoT (Internet of Things) in NR**	Ambient_IoT_Solutions-Core	R1	RP-250796	CMCC
1062084	**Perf. part: Solutions for Ambient IoT (Internet of Things) in NR**	Ambient_IoT_Solutions-Perf	R4	RP-250796	CMCC

Summary based on the input provided by CMCC, Huawei and T-Mobile USA Inc in RP-252928.

Note: this is a summary of RP-252928. RP-252928 provides more information.

This Work Item defines the Ambient IoT framework in NR , standardizing radio access support for **passive, ultra-low-power IoT devices** that operate without batteries, or with minimal energy storage, by harvesting ambient energy from the environment. These devices are designed with extremely low complexity and small form factors, enabling power consumption orders of magnitude lower than existing LPWA technologies such as NB-IoT and eMTC, and supporting use cases that cannot be efficiently addressed by current 3GPP IoT solutions.

The specification focuses on **Ambient IoT Device 1**, targeting approximately **1 µW peak power consumption** and relying on **backscatter-based device-to-reader communication** using an externally provided carrier wave. The scope concentrates on representative indoor use cases, including **inventory tracking and command-based interactions**, assuming a deployment where devices communicate directly with an indoor base-station reader. Operation is defined in **FR1 licensed FDD spectrum**, with reader-to-device signaling in downlink and backscattered device-to-reader transmission in uplink, supporting both device-terminated and device-originated triggered traffic.

To accommodate the extreme device constraints, the work item specifies a **highly simplified radio interface**, limited to PHY and MAC layers, using lightweight modulation, minimal coding, and simple synchronization mechanisms. Reader-to-device communication employs OOK-based waveforms with Manchester encoding, while device-to-reader communication supports OOK or BPSK backscatter with basic error protection. Ambient IoT is integrated into the existing 5G system architecture by supporting the reader function within the gNB and introducing a dedicated **A-IoT Function (AIOTF)** in the 5G Core to manage device access, signaling, and application interaction.

References

List of related CRs: select "TSG Status = Approved" in:
https://portal.3gpp.org/ChangeRequests.aspx?q=1&workitem=1060084,1061084,1062084

[1] TR 38.848, Study on Ambient IoT (Internet of Things) in RAN (Release 18), v18.0.0.

8 Rel-19 Artificial Intelligence (AI)/Machine Learning (ML)

8.1 AI for network (using AI to optimize the system)

8.1.1 Artificial Intelligence (AI)/Machine Learning (ML) for NR air interface

1020093	Artificial Intelligence (AI)/Machine Learning (ML) for NR air interface	NR_AIML_air		RP-252930	Qualcomm
1021093	Core part: Artificial Intelligence (AI)/Machine Learning (ML) for NR air interface	NR_AIML_air-Core	R1	RP-252930	
1022093	Perf. part: Artificial Intelligence (AI)/Machine Learning (ML) for NR air interface	NR_AIML_air-Perf	R4	RP-252930	

Summary based on the input provided by Qualcomm in RP-253362.

The application of Artificial Intelligence (AI) and Machine Learning (ML) techniques to the air interface of NR was studied in the course of Rel-18 under the WI code FS_NR_AIML_air [1] where a number of uses cases were identified as promising for their anticipated performance gains over non-AI/ML based techniques. The Rel-18 study also identified the general framework which could be used to specify different use cases in the future. The Technical Report in 38.843 [2] summarizes the findings from the Rel-18 studies.

This Rel-19 Feature provides the normative support for a general framework of one-sided (at the gNB or UE sides) AI/ML models as well as provides the normative support for the recommended use cases in the Rel-18 study, namely beam management, positioning enhancements and CSI prediction. The last Status Report (SR) for this feature can be found in [4].

In parallel to this feature, some studies unfinished during Rel-18 were further studied in the feature FS_NR_AIML_air_Ph2 [5]. The TR 38.843 [2] was further updated in Rel-19 with further evaluations carried out during the Rel-19 features as well as with further findings on two-sided AI/ML models for CSI compression and UE data collection.

A follow-up Rel-20 feature [6] is providing normative support for two-sided AI/ML models for CSI compression. In addition, the specification of a standardized data collection mechanism for UE training data is being further discussed.

The outcomes of this Rel-19 feature can be categorized as:

General framework for one-sided AI/ML models:

- Signalling and protocol aspects of Life Cycle Management (LCM) enabling functionality and model selection, activation, deactivation, switching, fallback

- Necessary signalling/mechanism(s) for LCM to facilitate model training, inference, performance monitoring, data collection (except for the purpose of CN/OAM/OTT collection of UE-sided model training data) for both UE-sided and NW-sided models

- Signalling mechanism of applicable functionalities/models

Use case specific enhancements:

- Beam management - DL Tx beam prediction for both UE-sided model and NW-sided model, encompassing Spatial-domain DL Tx beam prediction for Set A of beams based on measurement results of Set B of beams (“BM-Case1”) and Temporal DL Tx beam prediction for Set A of beams based on the historic measurement results of Set B of beams (“BM-Case2”)

- Positioning accuracy enhancements, encompassing Direct AI/ML positioning: (Case 1: UE-based positioning with UE-side model, direct AI/ML positioning; Case 3b: NG-RAN node assisted positioning with LMF-side model, direct AI/ML positioning) and AI/ML assisted positioning (Case 3a: NG-RAN node assisted positioning with gNB-side model, AI/ML assisted positioning)

- CSI prediction (UE-sided model), assuming functionality-based LCM

Core requirements for the above three use cases for AI/ML LCM procedures, including performance monitoring, and UE features. Note that offline training is assumed for the purpose of this feature.

References

List of related CRs: select "TSG Status = Approved" in:
https://portal.3gpp.org/ChangeRequests.aspx?q=1&workitem=1020093,1021093,1022093

[1] RP-221348, “Revised SID: Study on Artificial Intelligence (AI)/Machine Learning (ML) for NR Air Interface”.
[2] TR 38.843, “Study on Artificial Intelligence (AI)/Machine Learning (ML) for NR air interface”, available at https://www.3gpp.org/ftp/Specs/archive/38_series/38.843/38843-j00.zip
[3] RP-252930, “Revised WID: Artificial Intelligence (AI)/Machine Learning (ML) for NR Air Interface”
[4] RP-253361, “Status Report for AI/ML for NR Air Interface”
[5] RP-250308, “Revised SID: Study on Artificial Intelligence (AI)/Machine Learning (ML) for NR air interface Phase 2”.
[6] RP-252445, “Revised WID: Artificial Intelligence (AI)/Machine Learning (ML) for NR air interface Phase 2”

8.1.2 Enhancements for Artificial Intelligence (AI)/Machine Learning (ML) for NG-RAN

1050124	**Enhancements for Artificial Intelligence (AI)/Machine Learning (ML) for NG-RAN**	NR_AIML_NGRAN_enh		RP-251245	ZTE
1051124	**Core part: Enhancements for Artificial Intelligence (AI)/Machine Learning (ML) for NG-RAN**	NR_AIML_NGRAN_enh-Core	R3	RP-251245	ZTE
1020083	*Study on enhancements for Artificial Intelligence (AI)/Machine Learning (ML) for NG-RAN*	FS_NR_AIML_NGRAN_enh	R3	RP-240323	ZTE

Summary based on the input provided by ZTE Corporation and NEC in RP-252149.

This WI specifies new AI/ML-based use cases (i.e., Network Slicing and Coverage and Capacity Optimization) and to introduce further enhancements to finalize the Rel-18 leftovers based on the conclusions captured in TR 38.743.The following aspects are specified.

AI/ML enabled Network Slicing: The feature enables the data collection requesting and reporting between gNBs to support AI/ML enabled Network Slicing. The predicted radio resource status per slice, predicted slice available capacity and per slice UE performance can be configured to be collected and measured by a gNB by using the Data Collection Reporting Initiation procedure, while the actual reporting is performed through the Data Collection Reporting procedure.

AI/ML enabled Coverage and Capacity Optimization (CCO): The feature enables that the gNB derives predicted CCO issues and the corresponding future coverage status for its affected cells and beams, and notify its neighbour gNBs about the future coverage state changes together with modification cause and time information through NG-RAN Node Configuration Update procedure. In case of split architecture, a gNB-DU may receive the predicted CCO assistance information including a predicted CCO issue, the predicted affected cells and beams and the time when the predicted CCO issue will happen. The gNB-DU may indicate to the gNB-CU a future coverage configuration for its cells and beams; a gNB-DU may also receive the future coverage configurations of a list of cells and beams not served by the gNB-DU with their corresponding future coverage cause and the time of application of the future coverage configuration. In response, the gNB-DU may indicate to the gNB-CU a future coverage configuration for its cells and beams, and an associated time and cause.

Also, several aspects which remained unsolved in Rel-18 have been covered. These are the ones specified below.

Split Architecture: The feature enables the support of data collection in split architecture. With respect to Network Energy Saving, the gNB-DU can report the Energy Cost (EC) to gNB-CU upon request through the Resource Status Reporting Initiation procedure, and the Resource Status Reporting procedure. Regarding the UE performance feedback, Packet Delay measurement collection for UE performance from gNB-DU is reported to gNB-CU-UP through the ASSISTANCE INFORMATION DATA frame. PDCP Average UE DL/UL packet loss from gNB-CU-UP is configured and reported through the Data Collection Reporting Initiation procedure and the Data Collection Reporting procedure.

Mobility Optimization for NR-DC: The feature enables that the Master Node (MN) configures the collection and reporting of UE performance measurements from the Secondary Node (SN) after successful SN addition or SN change. The MN can request the UE performance feedback from the SN through the Data Collection Reporting Initiation procedure, while the actual reporting from SN is through the Data Collection Reporting procedure.

Continuous MDT Collection: The feature enables the measurement continuity to ensure that the same UE collecting MDT measurements during the same RRC state and across different RRC states, and trace correlation to ensure that the TCE which eventually receives the MDT reports can associate the received logged and immediate MDT measurements to a continuous data collection period from the same UE. OAM configures management-based MDT for continuous MDT towards the NG-RAN and provides specific information to NG-RAN for NG-RAN to identify continuous MDT.

References

List of related CRs: select "TSG Status = Approved" in:
https://portal.3gpp.org/ChangeRequests.aspx?q=1&workitem=1050124,1051124

[1] RP 251245, Revised WID: Enhancements for Artificial Intelligence (AI)/Machine Learning (ML) for NG-RAN, ZTE Corporation, NEC
[2] RP-252150, Status report for Enhancements for Artificial Intelligence (AI)/Machine Learning (ML) for NG-RAN, ZTE Corporation, NEC

8.1.3 Core Network Enhanced Support for Artificial Intelligence (AI)/Machine Learning (ML)

1050047	**Core Network Enhanced Support for Artificial Intelligence (AI)/Machine Learning (ML)**	AIML_CN		SP-240991	Xiaobo Wu, vivo
1040033	**Stage 2 of Core Network Enhanced Support for Artificial Intelligence (AI)/Machine Learning (ML)**	AIML_CN	S2	SP-240991	Xiaobo Wu, vivo
1050014	**CT aspects of Core Network Enhanced Support for Artificial Intelligence (AI)/Machine Learning (ML)**	AIML_CN		CP-251283	Yizhong ZHANG, vivo
1050048	CT3 aspects of Core Network Enhanced Support for Artificial Intelligence (AI)/Machine Learning (ML)	AIML_CN	C3	CP-251283	Yizhong ZHANG, vivo
1050049	CT4 aspects of Core Network Enhanced Support for Artificial Intelligence (AI)/Machine Learning (ML)	AIML_CN	C4	CP-251283	Yizhong ZHANG, vivo
1060059	**Security aspects of Core Network Enhanced Support for AIML**	AIML_CN_SEC	S3	SP-241957	Chang Liu, China Mobile
1020068	*Study on Core Network Enhanced Support for Artificial Intelligence (AI)/Machine Learning (ML)*	FS_AIML_CN	S2	SP-241936	Xiaobo Wu, vivo
1030035	*Study on security aspects of Core Network Enhanced Support for AIML*	FS_AIML_CN_SEC	S3	SP-240509	Chang Liu,
1020007	*Study on AI/ML management - phase 2*	FS_AIML_MGT_Ph2	S5	SP-241567	Yizhi Yao (Intel)
1010005	*Study on application layer support for AI/ML services*	FS_AIMLAPP	S6	SP-231182	Pateromichelakis, Emmanouil (Manos)

Summary based on the input provided by Vivo by e-mail received Tue 31/03/2026 11:59, file "Summary_1050047 for Rel-19 Core Network Enhanced Support for AIML_v0".

This Work Item covers the work related to the Core Network Enhanced Support for Artificial Intelligence (AI)/Machine Learning (ML) in 5GC.

The convergence of communication network and Artificial Intelligence (AI) technology is progressing based on work done in previous releases. For Rel-19, it is proposed to expand the scope of network AI services to leverage AI/ML technologies to enable 5GC and Air interface Intelligence by providing network automation and improving the efficiency of the 5G network architecture.

The outputs of the corresponding stage-2 study phase FS_AIML_CN are documented in the TR 23.700-84 [1].

The Feature enhances the following aspects:
- Enhancements to LCS to support Direct AI/ML based Positioning
- 5GC Support for Vertical Federated Learning
- NWDAF-assisted policy control and QoS enhancement
- NWDAF enhancements to support network abnormal behaviours (i.e. signalling storm) mitigation and prevention

by introducing the following set of capabilities:

Enhancements to LCS to support Direct AI/ML based Positioning: LMF-based AI/ML positioning enables UE location estimation by leveraging machine learning models within the LMF as described in the TS 23.273 [2]. The LMF collects input data from NG-RAN for the LMF-based AI/ML Positioning to perform location calculation and provide the location to the consumer. The LMF can either train ML models locally or obtain them from NWDAF. Either LMF or NWDAF containing MTLF may perform performance monitoring for LMF-based AI/ML Positioning.

5GC Support for Vertical Federated Learning: Vertical Federated learning is a machine learning technique working without exchanging/sharing of local data set, while maintaining some level of coordination amongst VFL participants, i.e. VFL Server and VFL Client, when training and inference are performed on local ML Models, wherein the local data set in different VFL Participant for local model training have different feature spaces for the same samples (e.g. UE IDs) as described in the TS 23.288 [3]. Vertical Federated Learning may involve multiple NWDAFs and AFs. For Vertical Federated Learning, there may be one NWDAF or one AF acting as a VFL server and one or multiple NWDAF(s) and/or one or multiple AF(s) acting as VFL Client(s).

NWDAF-assisted policy control and QoS enhancement: The QoS and policy assistance analytics enables NWDAF to provide predictions of the QoE associated to a set of QoS parameters as described in the TS 23.288 [3]. When the service consumer needs assistance information to determine QoS or policy, it may send a request to the NWDAF with one or multiple sets of QoS parameters and optionally a requested QoE. The NWDAF provides the candidate set(s) of QoS parameters and the corresponding predicted QoE that can be achieved by applying the candidate QoS parameters set(s). The candidate QoS parameters set(s) provided by the NWDAF are within the sets of QoS parameters provided by the consumer (i.e. PCF). The NWDAF shall only include the parameter(s) and the corresponding value(s) that are provided by the consumer request (i.e. PCF) in the QoS parameter set(s).

NWDAF enhancements to support network abnormal behaviours (i.e. signalling storm) mitigation and prevention: The signalling storm analytics enables NWDAF to provide analytics information (statistics or predictions) regarding the abnormal signalling rate condition as described in the TS 23.288 [3]. An abnormal signalling rate condition is when signalling rate is experiencing an abnormal growth. The abnormal signalling growth may indicate a potential signalling storm due to a significant deviation from the signalling level in the normal conditions. The analytics of signalling storm provides the identification of whether the signalling storm is due to massive signalling from NF(s) or from UEs based on the analytics consumer request.

References

List of related CRs: select "TSG Status = Approved" in:
https://portal.3gpp.org/ChangeRequests.aspx?q=1&workitem=1050047;1040033;1050014;1050048;1050049;1060059

[1] TR 23.700-84: " Study on Core Network Enhanced Support for Artificial Intelligence (AI)/Machine Learning (ML)"
[2] TS 23.273: "5G System (5GS) Location Services (LCS); Stage 2".
[3] TS 23.288: "Architecture enhancements for 5G System (5GS) to support network data analytics services".

8.1.4 AI/ML Management Phase 2

1060011	AI/ML Management Phase 2	AIML_MGT_Ph2	S5	SP-251197	Hassan Al-Kanani (NEC)
1020007	*Study on AI/ML management - phase 2*	FS_AIML_MGT_Ph2	S5	SP-241567	Hassan Al-Kanani (NEC) Yizhi Yao (Intel)

Summary based on the input provided by NEC in SP-260231

Building upon the accomplishments of Release 18 specifications, the Release 19 AI/ML Management Phase 2 introduced the following enhancements related to AI/ML lifecycle management (LCM).

ML training enhancements:

Pre-specialized training was introduced, enabling ML models to be trained on datasets not specific to any single inference type. This allows the resulting model to be adapted (via fine-tuning) to support one or more inference types,

such as MDA types in MDA, analytics types in NWDAF, NG-RAN AI/ML-supported functions, or vendor-specific extensions.

Fine-tuning was introduced to support narrowing the inference scope of a pre-specialized trained ML model to a single inference type, producing a new ML model. Retraining was clarified to maintain the same inference type, distinguishing it from fine-tuning.

ML model training for multiple contexts enables grouping ML model instances with similar operational characteristics into clusters, allowing them to be trained from a shared baseline model rather than independently. Clustering is based on criteria such as task type, performance metrics, and training or inference time, improving training efficiency and supporting adaptive retraining across related contexts. Model diversity preferences may also be considered to allow variation in model types within a cluster.

Federated learning and Distributed training support:

Distributed training management capabilities were introduced, allowing training workloads to be split across multiple ML training functions. Consumers can specify distributed training expectations including expected training time, data split indication, and suggested training node list.

Federated learning (FL) support was introduced, including modelling of FL Server and FL Client roles within the ML training function. Consumers can specify FL training requirements and client selection criteria, while the FL Server provides per-client training reports to ensure visibility of client participation and performance.

Reinforcement learning (RL) support: Reinforcement learning support was introduced, enabling management of models trained via agent–environment interactions using reward-based optimization. RL training can be performed in offline or online modes. Consumers can specify RL requirements including RL environment type (simulation or real network), RL environment scope (area, time window, managed entities), and RL performance requirements (e.g., tolerable degradation thresholds). Exploration in RL is supported with configurable ranges and fallback actions.

Other AI/ML lifecycle management enhancements:

AI/ML inference emulation was enhanced with the ability to select an emulation environment, enabling evaluation of ML model inference performance under specified runtime contexts prior to deployment. Configuration parameters such as emulation scope and time can be defined.

ML model loading relationships were updated to support multiplicity, allowing multiple ML models to be loaded via a single ML model loading request and an associated ML model loading process.

Training data management enhancements: Training data statistical properties were introduced to represent key statistical characteristics of training datasets, including uniformly distributed training data and training data with or without outliers.

AI/ML Explainability: AI/ML inference explainability to support AI/ML inference explainability. The management system can report explanations of inference results to authorised consumers, enabling understanding why a specific output was produced by an ML model. Explanations focus on critical features in the training or inference data that influenced the output, supporting informed operational decisions, such as deactivating an inference or reverting to a previous model version. The InferenceOutput data type includes inferenceExplanationInf to indicate the inference explanation information of the ML model inference results e.g., the critical features in the training or inference data.

Monitoring and feedback enhancements: Performance monitoring of ML models in live networks was introduced, enabling identification of underperforming models and their impact on network KPIs.

The Release 19 enhancements collectively extend the Release 18 AI/ML management framework by introducing new training modes (including pre-specialized, fine-tuning, and multi-context training), improving lifecycle control (including emulation environment selection and model loading multiplicity), expanding data handling capabilities (including statistical properties and clustering), and adding support for Distributed training, Federated, and Reinforcement learning with enhanced configuration and reporting mechanisms.

References

List of related CRs: select "TSG Status = Approved" in:
https://portal.3gpp.org/ChangeRequests.aspx?q=1&workitem=1060011

[1] TS 28.105: “Management and orchestration; Artificial Intelligence/ Machine Learning (AI/ML) management”

8.2 Network for AI (transport of AI/ML-related data)

8.2.1 Void

1000030	**AI/ML Model Transfer Phase 2**	AIML_MT_Ph2	S1	SP-230514	Yang Xu, OPPO

This section was covereing "AI/ML Model Transfer Phase 2" (UID 1000030) but it was clarified that only Stage 1 was defined in Rel-19, so the Stage 1 was moved to the next Release, as to keep consistency between all Stages.

8.2.2 Application enablement for AI/ML services

1050044	**Application enablement for AI/ML services**	AIML_App		SP-241008	Pateromichelakis, Emmanouil (Manos),
1040075	**Stage 2 of Application enablement for AI/ML services**	AIML_App	S6	SP-241008	Pateromichelakis, Emmanouil (Manos),
1050024	**CT aspects of Application enablement for AI/ML services**	AIML_App		CP-243310	Atarius, Roozbeh (Lenovo)
1050045	CT1 aspects of Application enablement for AI/ML services	AIML_App	C1	CP-243310	Atarius, Roozbeh (Lenovo)
1050046	CT3 aspects of Application enablement for AI/ML services	AIML_App	C3	CP-243310	Atarius, Roozbeh (Lenovo)

Summary based on the input provided by Lenovo in SP-251511.

Vertical-specific and edge applications are expected to be major users of 3GPP AI/ML support services. This Work Item introduces a new application-layer feature to enhance and optimize AIML service enablement.

It provides support for ML model lifecycle operations—such as distribution, training, and inference—in alignment with SA2 system architecture, and extends existing analytics capabilities (e.g., SEAL ADAES) using AI/ML services.

In Rel-19, SA6 specified the AIML Enablement (AIMLE) architecture and a new ML repository in TS 23.482 [1], along with ML-enabled analytics enhancements in TS 23.436 [2].

Within the SEAL framework, AIMLE consists of an AIMLE server offering exposure of AIML functions—including federated learning management, split AI/ML support, and ML model-related operations—an AIMLE client entity, and an ML repository serving as both a registry and a model-information store. AIMLE plays a key role in enabling ML model lifecycle operations by mediating interactions between application entities that perform ML tasks. Its capabilities include ML-model support (retrieval, discovery, storage), ML-operation support (VFL/HFL, transfer learning, split AI/ML, data management, FL assistance), and AIMLE-client support (registration, discovery, selection, monitoring), with the detailed procedures documented in TS 23.482.

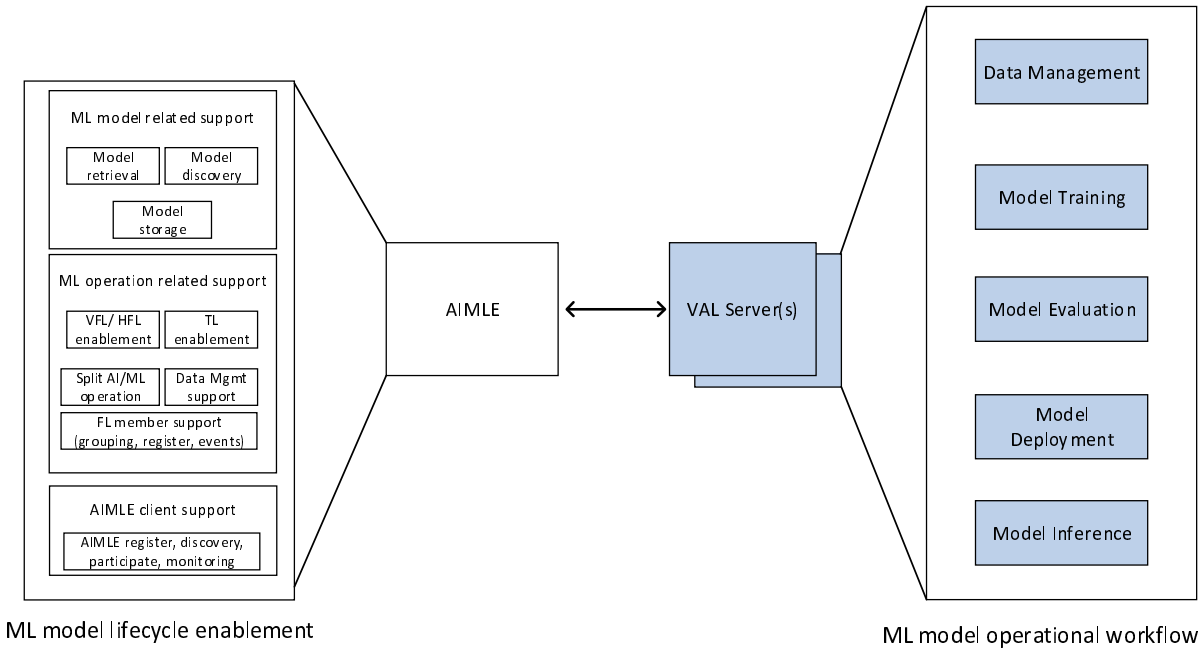


Figure 1: ML model Lifecycle Enablement at AIMLE

References

List of related CRs: select "TSG Status = Approved" in:
https://portal.3gpp.org/ChangeRequests.aspx?q=1&workitem=1050044,1040075,1050024,1050045,1050046

[1] TS 23.482: “Functional architecture and information flows for AIML Enablement Service”
[2] TR 23.436: “Functional architecture and information flows for Application Data Analytics Enablement Service”

8.2.3 Protocol for AI Data Collection from UPF

1040005	**Protocol for AI Data Collection from UPF**	PAIDC_UPF			Huang Zhenning, China Mobile
1040999	*Study on Protocol for AI Data Collection from UPF*	FS_PAIDC_UPF	C4	CP-251039	Huang Zhenning, China Mobile
1090001	**CT aspects of PAIDC_UPF**	PAIDC_UPF		CP-252243	Wang Rong, China Mobile
1090036	CT3 aspects of PAIDC_UPF	PAIDC_UPF	C3	CP-252243	Wang Rong, China Mobile
1090037	CT4 aspects of PAIDC_UPF	PAIDC_UPF	C4	CP-252243	Wang Rong, China Mobile

Summary based on the input provided by China Mobile in CP-260159.

The 5G service-based architecture (SBA) enables flexible information exchange through standardized service-based interfaces (SBIs). This provides a common framework for AI/ML functions such as the Network Data Analytics Function (NWDAF) to collect data from multiple network sources. (NWDAF is a 5G CN function that gathers data from network functions and other sources, analyzes it, and provides analytics and predictions to consumer functions to support network automation and optimization.)

As AI/ML usage grows for network automation and analytics, there is an increasing need for more efficient protocols to collect data over SBIs from both user-plane and control-plane functions. This is particularly important for scenarios involving high data volumes or frequent data transfers, where scalability and responsiveness must be ensured without negatively impacting network performance.

This work item specifies enhancements that allow an NF service consumer to create UPF-related event exposure subscriptions with reduced reporting instructions and for the UPF to skip generating event reports in accordance with

those instructions. It also enables bundling of event reports originating from different UPF event exposure subscriptions. In addition, it allows an NF service consumer to request inclusion of RAT type information in UPF event reports and to specify the RAT type or types for which event reports are required, limited to PDU sessions of UEs camping on those RAT types.

References

List of related CRs: select "TSG Status = Approved" in:
https://portal.3gpp.org/ChangeRequests.aspx?q=1&workitem=1040005,1090001,1090036,1090037

[1] TS 29.208: "5G System; Session Management Event Exposure Service; Stage 3"
[2] TS 29.564: "5G System; User Plane Function Services; Stage 3"

8.3 Other AI/ML aspects

8.3.1 Study on 3GPP AI/ML Consistency Alignment

1040084	*Study on 3GPP AI/ML Consistency Alignment*	FS_AIML_CAL	SP	SP-250745	Johannes Achter, Magenta

Summary based on the input provided by vivo/Deutsche Telekom in SP-260239.

Note that Studies are by default not meant to be summarized in this document.

This study item covers the work related to 3GPP AI/ML consistency alignment in 5GC.

The FS_AIML_CAL study investigates ongoing AI/ML work in TSG CT, TSG RAN and TSG SA Working Groups, identifies instances of any potential misalignment and/or inconsistencies and provides information on any potential outcome to the respective WGs to resolve any issues with appropriate SA-level co-ordination as necessary.

The study is led by TSG SA in close collaboration and inputs from TSG CT and TSG RAN. The study item does not impact studies and normative work for AI/ML across all SA/RAN/CT WGs for Rel-19.

This study investigates and identifies AI/ML related activities of all working groups of Rel-18 features and Rel-19 studies and work items in TSG CT, TSG RAN and TSG SA Working Groups, identifies any potential misalignments and inconsistencies for AI/ML terminology across 3GPP, and AI/ML related features.

The study confirms that AI/ML-related terminology across 3GPP WGs is broadly aligned at the conceptual level but shows differences in emphasis, scope, and granularity. Most variations are intentional and linked to domain requirements and the relevant specific use cases, rather than contradictions. The analysis of AI/ML terminology revealed that several AI/ML terms were overlapping across Working Groups.

A set of unified terminologies has been developed during the study, providing consistent definitions for key concepts such as ML model, ML model training, ML model inference, ML model lifecycle management, Functionality-based lifecycle management, Federated Learning (including Horizontal and Vertical), Transfer Learning.

The evaluation of AIML-related terminology yielded the following observation:
- No critical inconsistencies in terminology have been identified that would block cross-domain AI/ML lifecycle management.
- Differences mainly reflect WG scope and perspective, not fundamental conflicts.

The analysis of AI/ML related features has identified services for AI/ML model training, AI/ML model inference and AI/ML performance are specified across the SA WG2, SA WG5 and SA WG6 Working Groups.

Additionally, analysis of AI/ML related features has also identified that services for data collection for AI/ML are specified across the SA WG2, SA WG6 and RAN WG3 Working Groups.

No detailed analysis was conducted on potential misalignment, inconsistencies or overlap between these services.

The study has identified a wide range of AI/ML-related activities across 3GPP working groups, including lifecycle management (LCM) functionalities, data handling mechanisms and terminology usage. These variations largely reflect justified functional scopes, architectural focuses and abstraction levels per WG. Overall, AI/ML-related terminology and lifecycle concepts are conceptually aligned, with differences arising primarily from WG-specific perspectives rather

than from contradictory foundations. No critical inconsistencies were identified that would block cross-WG coordination or AI/ML lifecycle management.

To support effective TSG SA-level coordination and minimise risks of ambiguity or fragmentation, the following conclusions are drawn:
- A shared understanding of key AI/ML lifecycle concepts across WGs facilitates interoperability and semantic clarity.
- Semantic alignment in terminology is encouraged where it improves cross-specification integration, while respecting justified domain-specific definitions.
- Differences in terminology or treatment of AI/ML concepts should not be considered inconsistencies unless they create ambiguity or interoperability concerns.
- Unified AI/ML-related terminology identified in this study will inform proposed CRs towards TR 21.905 [1], establishing a common reference terminology.
- TSG SA-level guidance is intended as a reference to support convergence, particularly for AI/ML lifecycle elements evaluated in clause 7 (e.g., training, re-training/update, testing, pre-specialised training/fine-tuning, activation/deactivation, inference emulation), while allowing justified WG-specific adaptations.

"AI/ML" may be used as a general collective reference when referring to AI- or ML-based functionality without needing a distinction between the two. This is not a mandatory replacement for "AI" or "ML" and these terms may still be used independently where technical clarity requires differentiation.

The term "output" is proposed as unified term for e.g. decision or prediction or statistic or recommendation.

The outputs of the TSG SA level study are documented in the TR 22.850 [1].

Based on the outputs of the study, the consistent terminologies and abbreviations to be used across all SA/RAN/CT WGs are documented in a CR SP-251666 to TR 21.905 [2].

TSG SA#110 has sent an LS-out SP-251699 to inform about the completion of this study to all SA/RAN/CT WGs.

References

List of related CRs: select "TSG Status = Approved" in:
https://portal.3gpp.org/ChangeRequests.aspx?q=1&workitem=1040084

[1] TR 22.850: "Study on 3GPP AI/ML Consistency Alignment"
[2] TR 21.905: "Vocabulary for 3GPP Specifications"

9 Rel-19 Energy Efficiency, Energy Saving

9.1 Enhancements of Network energy savings for NR

1020095	**Enhancements of Network energy savings for NR**	Netw_Energy_NR_enh		RP-252332	Ericsson
1021095	**Core part: Enhancements of Network energy savings for NR**	Netw_Energy_NR_enh-Core	R1	RP-252332	Ajit Nimbalker, Ericsson
1022095	**Perf. part: Enhancements of Network energy savings for NR**	Netw_Energy_NR_enh-Perf	R4	RP-252332	Ajit Nimbalker, Ericsson
1030004	*Study on Media enerGy consumption exposuRE and EvaluatioN framework*	FS_MediaEnergyGREEN	S4	SP-240481	Julien Lemotheux, Orange,

Summary based on the input provided by Ericsson and Apple in RP-252333

Energy performance continues to be a critical part of network operations and energy savings techniques are important in reducing operation costs while maintaining high network quality. Following up on Rel-18 study and work items on network energy savings ([1][2]), the Rel-19 work item ([3]) introduces the following new functionalities for network energy savings in NR:
- On-demand SSB operation for SCells
- On-demand SIB1 for UEs in idle/inactive mode
- Adaptation of common signal/channel transmissions

Below is a brief description of the functionalities introduced in the Rel-19 network energy savings enhancements work item.

On-demand SSB operation for SCells: Network energy consumption on SCells can be reduced by providing SSBs on demand. On-demand SSB operation for an SCell is introduced for both of the following cases: 1) no always-on SSB on the SCell, and 2) always-on SSB on the SCell. The on-demand SSBs can be configured by higher layers, and the on-demand SSB transmissions can be activated, deactivated or adapted using RRC-based and/or MAC-CE based signalling from the network. The center frequency locations of always-on SSB and on-demand SSB can be same or different. Both L1 and L3 measurements based on on-demand SSBs are supported. The on-demand SSBs are not cell-defining.

On-demand SIB1: Energy consumption due to periodic SIB1 transmissions can be reduced by transmitting SIB1 on demand. A UE can transmit an uplink wakeup signal based on a SIB1 request configuration to request SIB1 transmission from a gNB. A cell can transmit SIB1 request configuration(s) of one or multiple cells including its own SIB1 request configuration. The uplink wakeup signal is based on PRACH. Upon transmission of the uplink wakeup signal, the UE monitors for RAR and the UE can receive SIB1 in a time window. Inter NG-RAN node signalling is introduced for the exchange of the SIB1 request configurations.

Adaptation of common signal/channel transmissions:

For an SCell, SSB periodicity adaptation (e.g. between sparse and dense periodicities) can facilitate network energy savings. A UE can be configured with up to two additional SSB periodicities and corresponding frame-level/half-frame offsets for an SCell. Group-common DCI 2_9 signalling is introduced to indicate the applicable SSB periodicity value for an SCell, and the indication is also used for indicating the applicable SMTC from a set of SMTCs configured for the SCell. SSB periodicity adaptation is supported for SSBs that are not cell-defining.

Adaptation of PRACH in time domain (e.g. between sparse and dense random-access occasions in time domain) can facilitate network energy savings. A gNB can configure additional RACH resources in addition to legacy RACH resources in a cell and the availability of the additional RACH resources can be indicated using DCI 1_0 with P-RNTI and a configured validity duration or using DCI 1_0 with C-RNTI for PDCCH order. A PRACH subset mask is also introduced to allow selection of a subset of the additional RACH resources. Adaptation of PRACH is supported for both idle/inactive and connected mode operation.

Adaptation of paging by confining paging occasions in time domain can facilitate network energy savings. The value range for parameter N is extended to make it possible to have an increased interval between PFs. The value range for the number of paging occasions within one paging frame is increased to compensate for the decrease in the number of PFs. UEs supporting paging adaption and PEI can monitor PEI according to the additional PEI configuration, if configured.

References

List of related CRs: select "TSG Status = Approved" in:
https://portal.3gpp.org/ChangeRequests.aspx?q=1&workitem=1020095,1021095,1022095

[1] TR 38.864, Study on network energy savings for NR, March 2023
[2] RP-230566, WID revision: Network energy savings for NR, RAN#99, March 2023
[3] RP-251678, Revised WID on enhancements of network energy savings for NR, RAN#108, June 2025
[4] RP-252331, Status report for enhancements of network energy savings for NR, Ericsson, Apple, RAN#109, Sep 2025

9.2 Low-power wake-up signal and receiver for NR (LP-WUS/WUR)

1020094	**Low-power wake-up signal and receiver for NR (LP-WUS/WUR)**	NR_LPWUS		RP-251200	vivo
1021094	**Core part: Low-power wake-up signal and receiver for NR (LP-WUS/WUR)**	NR_LPWUS-Core	R1	RP-251200	Xueming Pan, vivo
1022094	**Perf. part: Low-power wake-up signal and receiver for NR (LP-WUS/WUR)**	NR_LPWUS-Perf	R4	RP-251200	Xueming Pan, vivo

Summary based on the input provided by vivo, NTT DOCOMO in RP-252009

UE power saving and device battery life are important for user experience. In Rel-18 study item for "Study on low-power wake up signal and receiver for NR", the low-power wake-up signal and receiver (LP-WUS/WUR), including its power saving benefit, coverage, system impact and design aspects were studied. From the study it was concluded that LP-WUS/WUR can be significantly beneficial for UE operations in IDLE/INACTIVE and CONNECTED modes, and across different traffic types. Based on the study outcome [1], a Rel-19 work item on “Low-power wake-up signal and receiver for NR (LP-WUS/WUR)” was approved with following objectives [2]:

- Specify an LP-WUS design commonly applicable to both IDLE/INACTIVE and CONNECTED modes, including OOK based LP-WUS with overlaid OFDM sequence(s) over OOK symbols
- For IDLE/INACTIVE mode, specify
- Procedures and configurations of LP-WUS indicating paging monitoring triggered by LP-WUS
- Periodic LP-SS for synchronization and RRM measurement for serving cell
- Further RRM relaxation of UE MR for both serving and neighbor cell measurements, and UE serving cell RRM measurement offloaded from MR to LP-WUR, including the necessary conditions
- For CONNECTED mode, specify procedures to allow UE MR PDCCH monitoring triggered by LP-WUS
- Specify the necessary RAN4 core RF and RRM requirement(s) for the feature

LP-WUS and LP-SS signal design

LP-WUS is designed to be commonly appliable to IDLE/INACTVIE and CONNECTED mode operations. A LP-WUS is transmitted based on OOK and overlaid OFDM sequence(s) over OOK ON symbols, and can carry up to 5 information bits and one codepoint out of up to 32 codepoints. A UE supports detection of LP-WUS information carried by OOK or overlaid OFDM sequences, or both, subject to UE capability. For LP-WUS operation in IDLE/INACTIVE modes, the same information is delivered by OOK and overlaid OFDM sequences. For LP-WUS, the number of OOK symbols within an OFDM symbol can be configured as 1, 2 or 4. For IDLE/INACTIVE mode operation, a UE monitors two codepoints for LP-WUS. For CONNECED mode operation, a UE can be configured to monitor up to 2,4,6 or 8 codepoints for LP-WUS, subject to UE capability.

For LP-WUS operation in IDLE/INACTIVE modes, LP-SS is supported for UE LR (Low-power wake-up Receiver) to maintain synchronization and perform serving cell RRM measurements. LP-SS transmission is based on OOK with or without overlaid OFDM sequence. For UE capable of detection of LP-WUS information carried by OOK, LP-SS is required to maintain synchronization and perform serving cell RRM measurements. For UE capable of detecting overlaid OFDM sequence by LR, PSS/SSS can be used for UE LR to maintain synchronization and perform serving cell RRM measurements while using LP-SS for these functions is optional. LP-SS is not supported for RRC_CONNECTED mode operations.

LP-WUS operation for IDLE/INACTIVE mode

In IDLE/INACTIVE mode operation, LP-WUS is used to trigger UE paging monitoring. LP-WUS carries subgroup indication and up to 31 subgroups can be configured for LP-WUS. Either UE ID based subgrouping or CN based subgrouping can be configured. In addition to monitoring a codepoint associated with its subgroup ID, a UE configured with LP-WUS monitoring also monitors a common codepoint associated with all subgroups in a PO, if applicable.

In IDLE/INACTIVE mode operations, 3 candidate values for wake-up delay are supported for UE to report via capability signalling, where wake-up delay is defined as the minimum time gap between the LP-WUS reception and Main Receiver (MR) to start paging PDCCH monitoring. For IDLE/INACTIVE mode operation, the frequency resource of LP-WUS and LP-SS can be configured within or outside initial DL BWP in the carrier where the UE monitors paging.

In IDLE/INACTIVE mode operation, UE power saving can also be achieved by allowing UEs supporting LP-WUS to relax serving cell measurements on MR, further neighbour cell measurements relaxation on MR and/or offload serving cell measurements from MR to LR. Entry conditions for serving cell and further neighbour cell measurement relaxation and/or offloading serving cell measurements from MR to LR are based on MR and optionally LR measurements. Exit conditions are based on LR measurements.

LP-WUS operation for CONNECTED mode

In CONNECTED mode operation, LP-WUS is used to trigger UE PDCCH monitoring with C-DRX configuration. LP-WUS carries codepoint indication where a UE can be configured to monitor one or multiple codepoints. LP-WUS is monitored outside active-time. If LP-WUS is detected, based on configuration, the UE shall start the on-duration timer or lpwus-PDCCH-MonitoringTimer to start PDCCH monitoring and enter active-time. If the UE is configured to start on-duration timer after LP-WUS reception, the UE monitors LP-WUS at occasion(s) at a configured offset before the on-duration, and UE does not monitor LP-WUS when short DRX cycle is used. If the UE is configured to start lpwus-PDCCH-MonitoringTimer after LP-WUS reception, the UE monitors LP-WUS at occasion(s) according to the

configured periodicity and offset which can be same or different from the periodicity and offset configured for C-DRX cycle, and UE monitors LP-WUS regardless of long or short DRX cycle is being used.

In CONNECTED mode operation, 3 candidate values for minimum time gap are supported for UE to report via capability signalling, where the minimum time gap is between the LP-WUS reception and MR to start PDCCH monitoring. gNB configures the time offset between LP-WUS monitoring and the corresponding PDCCH monitoring. UAI can be used to indicate the UE preferred time offset. In CONNECTED mode operation, the frequency resource of LP-WUS can be configured within or outside the UE active DL BWP, where the support of LP-WUS monitoring outside active DL BWP is optional.

In CONNECTED mode operation, UE is not required to support simultaneous LR and MR operations, where LR operations is the UE operations for LP-WUS monitoring, MR operation is the UE operations for all other NR signals/channels transmissions/receptions.

References

List of related CRs: select "TSG Status = Approved" in:
https://portal.3gpp.org/ChangeRequests.aspx?q=1&workitem=1020094,1021094,1022094

[1] TR 38.869, Study on low-power wake up signal and receiver for NR.
[2] RP-241824, Revised WID: Low-power wake-up signal and receiver for NR (LP-WUS/WUR)
[3] RP-252008, Status report for WI: Low-power wake-up signal and receiver for NR (LP-WUS/WUR); rapporteurs: vivo, NTT DOCOMOReferences

9.3 Energy Efficiency as Service Criteria

1020060	**Energy Efficiency as Service Criteria**	**EnergyServ**	S2	SP-250069	Primary Jungshin Park (Samsung)
960019	*Study on Energy Efficiency as service criteria*	FS_EnergyServ	S1	SP-230235	Xiaonan Shi, CMCC
1000033	**Stage 1 of Energy Efficiency as Service Criteria**	EnergyServ	S1	SP-230520	Xiaonan Shi, CMCC
1010029	*Study on Energy Efficiency and Energy Saving*	FS_EnergySys	S2	SP-231391	Jungshin Park (Samsung)
1050112	**Energy Efficiency and Energy Saving**	EnergySys	S2	SP-250069	Primary Jungshin Park (Samsung)
1070002	**CT aspects of energy efficiency and energy saving**	EnergySys		CP-250247	Suh, Kyungjoo(Grace), Samsung
1070034	CT3 aspects of energy efficiency and energy saving	EnergySys	C3	CP-250247	Suh, Kyungjoo(Grace), Samsung
1070035	CT4 aspects of energy efficiency and energy saving	EnergySys	C4	CP-250247	Suh, Kyungjoo(Grace), Samsung
1030038	*Study on security aspects of energy saving in 5G*	FS_Energy_Sec	S3	SP-240512	Bo Bjerrum
1040018	**Charging Aspects for Energy Efficiency of Network Slice**	EnergySys_CH	S5	SP-250124	YiMeng, DENG, Huawei,
1060007	**Energy efficiency and energy saving aspects of 5G networks and services**	Energy_OAM_Ph3	S5	SP-251274	Jean-Michel Cornily (Huawei)
1020021	*Study on energy efficiency and energy saving aspects of 5G networks and services*	FS_Energy_OAM_Ph3	S5	SP-241568	Jean-Michel Cornily (Huawei)

Main aspects: Summary based on the input provided by Samsung by e-mail received Mon 30/03/2026 07:56, file "Summary_1020060".
Charging aspects: from Huawei, SP-251555.
OAM aspects: from Huawei and Samsung, e-mail from Huawei received Thu 26/03/2026 03:28, file "Summary_1060007 Energy efficiency and energy saving aspects of 5G networks and services_final".

Main aspects:

The goal of energy usage control as service criteria is to supervise services in an energy-aware manner, ensuring the services are offered as intended by service providers, network operators or subscribers, with energy constraints and no performance consequences.

Stage 1 requirements have been identified [1] that promise increased control over energy usage to achieve service objectives for mobile network operators, service providers and their customers. Examples of these functional requirements include the ability to control energy usage based on the operator policies applying to services provided for a UE or group of UEs.

5G system enhancements to improve energy efficiency and to support energy saving in the network is specified [2][3][4]:
- Energy Information Function (EIF) is defined to support the collection and calculation of energy consumption in the network and provide the exposure of network energy related information towards authorized consumers in the granularities of per-UE, per-UE-per-application, per-UE-per-PDU session and/or per-UE-per-QoS flow
- The UE Subscription Data in UDM/UDR is extended to include energy saving indication information per UE that indicates the applicability of network energy saving strategies for the UE
- The PCF may use the energy saving indication information to make policy decisions

Specifications for management system are extended to support improved energy efficiency in the network:
- TS 28.310 is extended for NRM to support energy supply, energy source, carbon emission and renewable energy related information in network
- Multi-dimensional network energy efficiency metrics is introduced, i.e. a new KPI of energy efficiency based on cell availability and network quality was introduced in TS 28.554. Also, a new KPI for gNB estimated carbon emission was introduced in TS 28.554
- TS 28.541 is extended for the NRM to support provisioning of network slices depending on the percentage of renewable energy usage in the network

Network slice performance and management charging [9][10][11][12][13] are extended to include energy-related information by adding estimated energy consumption and energy efficiency to charging specifications, APIs, and CDRs.

Charging aspects (UID 1040018)**:**

Network slice performance and management charging are extended to include energy-related information by adding estimated energy consumption and energy efficiency to charging specifications, APIs, and CDRs.

OAM aspects (UIDs 1020021 and 1060007)**:**

The OAM aspects consist in implementing the recommendations made in TR 28.880 [23] (resulting from FS_Energy_OAM_Ph3, UID 1020021) into normative specifications.

The normative part (Energy_OAM_Ph3) provides:
1 Energy supply, energy source, carbon emission and renewable energy information related to managed entities such as gNBs and 5GC NFs.
2 Multi-dimensional Energy efficiency (EE) metrics, including EE evaluated from network data traffic dimension; from network availability dimension and from network quality dimension.
3 Provisioning of network slices as per the requested minimum percentage of renewable energy usage.
4 Energy saving optimization for multi-carrier RAN scenarios use case.
5 KPIs for gNB estimated carbon emission and NG-RAN estimated carbon emission.

References:
List of related CRs: select "TSG Status = Approved" in:
https://portal.3gpp.org/ChangeRequests.aspx?q=1&workitem=1020060;960019;1000033;1040018;1010029;1050112;1070002;1070034;1070035;1020021;1030038;1060007

[1] TS 22.261: “Service requirements for the 5G system”
[2] TS 23.501: “System architecture for the 5G System (5GS)”
[3] TS 23.502: “Procedures for the 5G System (5GS)”
[4] TS 23.503: “Policy and charging control framework for the 5G System (5GS); Stage 2”
[5] TS 29.566: “5G System; Energy Information Function Services; Stage 3”
[6] TS 28.310: “Management and orchestration; Energy efficiency of 5G”
[7] TS 28.554: “Management and orchestration; 5G end to end Key Performance Indicators (KPI)”
[8] TS 28.541: “Management and orchestration; 5G Network Resource Model (NRM); Stage 2 and stage 3”
[9] TS 32.240: "Telecommunication management; Charging management; Charging architecture and principles".
[10] TS 28.201: "Charging management; Network slice performance and analytics charging in the 5G System (5GS); Stage 2"
[11] TS 28.202: "Charging management; Network slice management charging in the 5G System (5GS); Stage 2"

[12] TS 32.291: "Telecommunication management; Charging management; 5G system, charging service; Stage 3"
[13] TS 32.298: "Telecommunication management; Charging management; Charging Data Record (CDR) parameter description"
[14] TS 32.240: "Telecommunication management; Charging management; Charging architecture and principles".
[15] TS 28.201: "Charging management; Network slice performance and analytics charging in the 5G System (5GS); Stage 2".
[16] TS 28.202: "Charging management; Network slice management charging in the 5G System (5GS); Stage 2".
[17] TS 32.291: "Telecommunication management; Charging management; 5G system, charging service; Stage 3".
[18] TS 32.298: "Telecommunication management; Charging management; Charging Data Record (CDR) parameter description".
[19] TS 28.310: "Management and orchestration; Energy efficiency of 5G"
[20] TS 28.541: "Management and orchestration; 5G Network Resource Model (NRM); Stage 2 and stage 3"
[21] TS 28.554: "Management and orchestration; 5G end to end Key Performance Indicators (KPI)"
[22] TS 28.623: "Telecommunication management; Generic Network Resource Model (NRM) Integration Reference Point (IRP); Solution Set (SS) definitions"
[23] TR 28.880 "Study on energy efficiency and energy saving aspects of 5G networks and services"

10 Rel-19 Verticals and Non Public Network

10.1 Rel-19 Enhancements of 3GPP Northbound and Application Layer Interfaces and APIs

1040009	**Rel-19 Enhancements of 3GPP Northbound and Application Layer Interfaces and APIs**	NBI19		CP-243079	Abdessamad El Moatamid, Huawei
1040060	**CT1 aspects of Rel-19 Enhancements of 3GPP Northbound and Application Layer Interfaces and APIs**	NBI19	C1	CP-243079	Abdessamad El Moatamid, Huawei
1040061	**CT3 aspects of Rel-19 Enhancements of 3GPP Northbound and Application Layer Interfaces and APIs**	NBI19	C3	CP-243079	Abdessamad El Moatamid, Huawei

Summary based on the input provided by Huawei in CP-253217

The 3GPP Network Capability Exposure is a framework defined to expose 3GPP mobile network capabilities to applications and application providers/developers so as to enable their easy, efficient, scalable and lucrative monetization by Mobile Network Operators (MNOs) via:

- The 3GPP Northbound Interfaces and APIs enabling external entities and third-party Application Servers/Functions to access a set of exposed 3GPP network services and capabilities in a secure and controlled manner (via the SCEF/NEF).

- The 3GPP Application Enabler Layer APIs, including set of abstraction layers that are common or application customized/specialized allowing the support of various applications/services (e.g., CAPIF, V2XAPP, UASAPP, EDGEAPP, SEAL, MSGin5G, PINAPP, etc.) over 3GPP networks and ensure the efficient use and deployment of these applications/services over 3GPP network.

The 3GPP Network Capability Exposure interfaces and APIs, which are mainly consumed by various external entities including 3rd Party Application Servers/Functions, should be continuously maintained, enhanced and optimized.

Therefore, the main work carried out was to introduce pure stage 3 (i.e., there are no related stage 1 nor stage 2 requirements) technical improvements and enhancements, i.e., improvement of the overall efficiency, reliability and flexibility, enhancement of the signalling efficiency, consolidation of the common protocol aspects, alignment with the 5GC service based interface design principles and guidelines whenever this is identified as relevant, corrections/changes missed in the previous 3GPP Releases, etc., especially that such enhancements may not be covered by the other more dedicated and functionality-driven work items.

The continuous maintenance and enhancement of the NBI TS template/skeleton for technical specifications documenting 3GPP Network Capability Exposure interfaces/APIs was also part of the activities carried out under this work item, with the objective to enhance the quality of the produced 3GPP specifications for Network Capability Exposure.

References

List of related CRs: select "TSG Status = Approved" in:
https://portal.3gpp.org/ChangeRequests.aspx?q=1&workitem=1040009,1040060,1040061

[1] TS 24.558: "Enabling Edge Applications; Protocol specification"
[2] TS 24.559: "Application Data Analytics Enablement Service (ADAES); Stage 3"
[3] TS 29.122: "T8 reference point for Northbound APIs"
[4] TS 29.222: "Common API Framework for 3GPP Northbound APIs"
[5] TS 29.255: "Uncrewed Aerial System Service Supplier (USS) Services; Stage 3"
[6] TS 29.257: "Application layer support for Uncrewed Aerial System (UAS); UAS Application Enabler (UAE) Server Services; Stage 3"
[7] TS 29.343: "Proximity-services (ProSe) function to ProSe application server aspects (PC2); Stage 3"
[8] TS 29.435: "Service Enabler Architecture Layer for Verticals (SEAL); Network Slice Capability Enablement (NSCE) Server Services"
[9] TS 29.486: "V2X Application Enabler (VAE) Services; Stage 3"
[10] TS 29.522: "5G System; Network Exposure Function Northbound APIs; Stage 3"
[11] TS 29.538: "Enabling MSGin5G Service; Application Programming Interfaces (API) specification; Stage 3"
[12] TS 29.548: "Service Enabler Architecture Layer for Verticals (SEAL); SEAL Data Delivery (SEALDD) Server Services; Stage 3"
[13] TS 29.549: "Service Enabler Architecture Layer for Verticals (SEAL); Application Programming Interface (API) specification; Stage 3"
[14] TS 29.558: "Enabling Edge Applications; Application Programming Interface (API) specification; Stage 3"
[15] TS 29.583: "Application layer support for Personal IoT Network (PINAPP); Personal IoT Network (PIN) server services"

10.2 SEAL DD (Data Delivery) Phase 2

1050071	**SEAL DD (Data Delivery) Phase 2**	SEALDD_Ph2		SP-230989	Cuili Ge, Huawei
1010006	**Stage 2 of SEAL DD (Data Delivery) Phase 2**	SEALDD_Ph2	S6	SP-230989	Cuili Ge, Huawei
1050017	**CT aspects of SEAL DD (Data Delivery) Phase 2**	SEALDD_Ph2		CP-243261	Herrero Veron, Christian (Huawei)
1050072	CT1 aspects of SEAL DD (Data Delivery) Phase 2	SEALDD_Ph2	C1	CP-243261	Herrero Veron, Christian (Huawei)
1050073	CT3 aspects of SEAL DD (Data Delivery) Phase 2	SEALDD_Ph2	C3	CP-243261	Herrero Veron, Christian (Huawei)

Summary based on the input provided by Huawei in SP-251504.

In Rel-18, 3GPP introduced the Service Enabler Architecture Layer for Verticals Data Delivery (SEALDD) to provide vertical applications with simple and efficient support for 5G data transmission (see TR 21.918 [3], clause 10.6).

Rel-19 SEALDD Phase 2 introduces several enhancements, including support for SEALDD connections over Non-3GPP access and seamless switching with quality monitoring, improved BDT configuration efficiency, XR device capability reporting for better bandwidth control, client-initiated connection establishment, connection-status subscription and notification, congestion control using L4S exposure and client feedback, and enabler-layer uplink data delivery control using network-exposed BAT and periodicity information.

References

List of related CRs: select "TSG Status = Approved" in:
https://portal.3gpp.org/ChangeRequests.aspx?q=1&workitem=1050071,1010006,1050017,1050072,1050073

[1] TS 23.433: "Service Enabler Architecture Layer for Verticals (SEAL); Data Delivery enabler for vertical applications, stage 2"

[2] TS 23.434: "Service Enabler Architecture Layer for Verticals (SEAL); Functional architecture and information flows, stage 2"
[3] TR 21.917: "Release 17 Description; Summary of Rel-17 Work Items"
[5] TS 33.434: "Security aspects of Service Enabler Architecture Layer (SEAL) for verticals"
[6] TS 24.543: "Data delivery management - Service Enabler Architecture Layer for Verticals (SEAL); Protocol Specification; "
[7] TS 29.548: "Service Enabler Architecture Layer for Verticals (SEAL); SEAL Data Delivery (SEALDD) Server Services; Stage 3"

10.3 Common Application Programming Interface (API) Framework (CAPIF) Phase 3

1060040	**Common API Framework (CAPIF) Phase 3**	CAPIF_Ph3		SP-241381	Niranth Amogh, Nokia
1020062	*Study on CAPIF Phase 3*	FS_CAPIF_Ph3	S6	SP-231741	Preciado Rojas, Diego, Nokia
1050035	**Stage 2 of Common API Framework (CAPIF) Phase 3**	CAPIF_Ph3	S6	SP-241694	Niranth Amogh, Nokia
1060026	**CT aspects of Common API Framework (CAPIF) Phase 3**	CAPIF_Ph3		CP-252059	Paul, Bhaskar, Nokia
1060041	CT3 aspects of Common API Framework (CAPIF) Phase 3	CAPIF_Ph3	C3	CP-252059	Paul, Bhaskar, Nokia
1040025	*Study on security aspects of CAPIF Phase 3*	FS_CAPIF_Ph3-sec	S3	SP-240654	Henry Leung (Xiaomi)
1060061	**Security aspects of CAPIF Phase3**	CAPIF_Ph3_sec	S3	SP-250076	1st Henry Leung (Xiaomi),
1040015	*Study on Charging Aspects of CAPIF*	FS_CAPIF_Ph2_CH	S5	SP-240981	João Rodrigues, Nokia,
1070015	**Charging aspects of CAPIF**	CH_CAPIF_EXT	S5	SP-250820	João Rodrigues, Nokia
1080014	*Study on Charging Aspects of CAPIF phase 3*	FS_CAPIF_Ph3_CH	S5	SP-250869	ur: João Rodrigues (Nokia),
1080019	**Converged Charging aspects of CAPIF Phase 3**	CAPIF_Ph3_con-CH	S5	SP-251027	
1060009	**Enhanced OAM for management service exposure to external consumers through CAPIF**	MExpo	S5	SP-241942	Winnie Nakimuli (Nokia)

Summary based on the input provided by Nokia in SP-260256.

3GPP specified the CAPIF (Common API Framework) in Rel-15 to provide a framework for accessing 3GPP northbound APIs. In Rel-18 CAPIF was enhanced with RNAA (Resource owner-aware Northbound API Access) feature to specify the authorization of API invokers when access to protected resources (via API invocations) are granted by resource owners (e.g., UE users or an MNO subscribers).

In Rel-19, CAPIF is enhanced for a series of key aspects specified in TS 23.222 [1] presented below.

Finer granularity-based authorization to access service API: Until Rel-18, the authorization to access service API was governed by high level information related to the service API (like service API identifier). In Rel-19, the access to service API is governed by more granular levels like service API operation level or resource level.

Support for obtaining collective authorization from a resource owner: In Rel-18, CAPIF enables to authorize an API invoker to invoke a service API based on the authorization information from the resource owner given before the API invocation. In Rel-19, an extension is supported in CAPIF where authorization information from the resource owner relating to multiple API invokers and service APIs can be obtained in a single interaction.

Support for UE-deployed API invoker accessing other UEs' resources of a group and other UE's resources: CAPIF enables the support for an API invoker on a UE to access the services which expose resources of other UEs of a group or resources of other UEs. In case of group, the UEs should belong to the same group and the same Application layer service provider (e.g in IoT case) or the same user and all of them belong to the same CAPIF provider domain.

Support for CAPIF Core Function (CCF) to obtain resource owner authorization: In Rel-18, the support for CAPIF-8 procedures between CCF and Resource Owner Function (ROF) to obtain resource owner authorization was not specified. In Rel-19, CAPIF-8 procedure is specified which enables interactions between CCF and ROF for authorization information.

Revoking resource owner authorization: CAPIF enables revocation of resource owner authorization which can be triggered by ROF towards the CCF. This CAPIF-8 procedure enables ROF to revoke resource owner authorizations provided previously. When revoked, all affected entities (AEF, API invoker) are notified.

Open Discover service APIs: Until Rel-18, the discover service API operation could only be performed by an Onboarded API invoker. In Rel-19, open discovery operation allows a requestor entity to discover service API information about the available set of APIs offered by CCF before onboarding.

Enhancements to CAPIF Interconnection: CAPIF interconnection feature was enhanced to include service API unpublish, retrieve service APIs, update service APIs, authorization for service API access (API invoker and AEF belong to different CAPIF provider domain), revoking API invoker authorization, obtain access control, obtain security information operations.

Enhanced security features for CAPIF_Ph3

The security aspects for CAPIF are specified in TS 33.122 [3].

The interactions on CAPIF-8 (for resource owner authorization and revocation messages) are specified to be integrity protected, confidentiality protected and protected from replay attacks using TLS. Authentication of CCF is based on certificates and authentication of ROF is left for implementation.

The security model for the following CAPIF interconnection architecture over CAPIF-6/6e is specified.

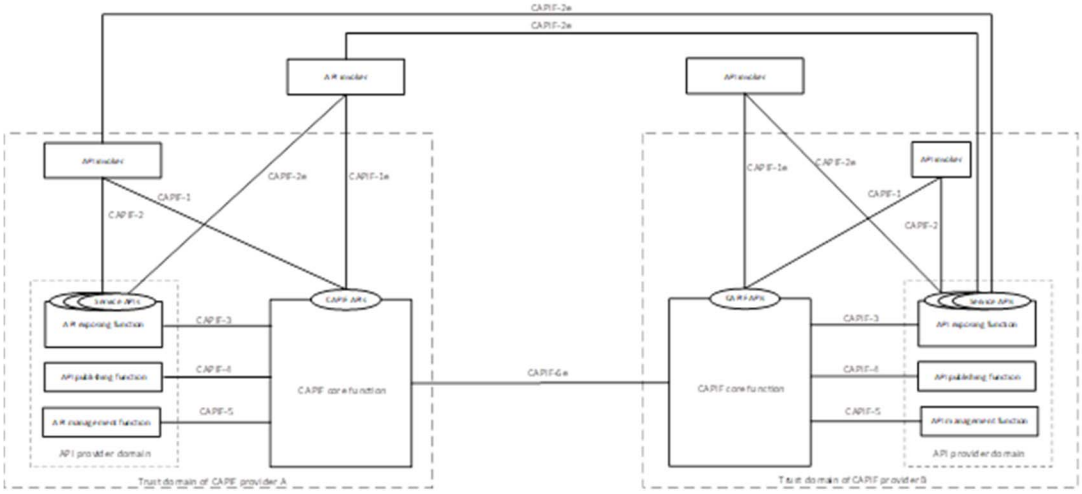


Figure 1: Functional security model for CAPIF interconnection architecture

For API invoker onboarded to CCF B is to access AEF registered to CCF A, the security method negotiation between CCF A and CCF B is supported over CAPIF-6/6e. TLS-PSK, PKI, TLS with OAuth Token are supported for authentication and authorization procedures between API invoker (CAPIF Provider B) and AEF (CAPIF Provider A).

Authorization for nested API invocation (from first API exposing function to second API exposing function and so on) is supported. The token exchange procedure specified in IETF RFC 8693 [4] is utilized to perform authorization of API invocation triggered towards the second API exposing function, where the access token of the API invoker to be used towards first API exposing function is used as the subject token.

CAPIF related converged charging

The charging aspects for CAPIF are specified in TS 32.254 [6].

When CAPIF is used, northbound API converged charging is supported for CAPIF APIs (exposed over CAPIF-1/, CAPIF-3, CAPIF-4, CAPIF-5) and service APIs (e.g. NEF) which are exposed via CAPIF-2. The following is the architecture:

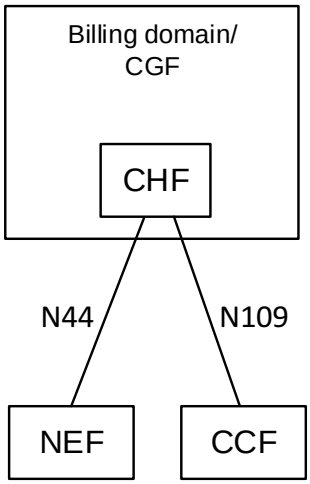


Figure 2: Northbound API converged charging architecture (CAPIF)

For converged charging, the CCF supports collecting charging information like invocations/notifications count of the northbound APIs, identification of the AF and the associated northbound API invocation/notification, timestamp of the northbound API invocation/notification, northbound API related information, e.g. location. The CCF performs convergent charging by interacting with CHF, for charging data related to Northbound API access.

Protocol aspects of CAPIF: The open API specification for CAPIF interfaces as specified in TS 29.222 [8].

A study on CAPIF Phase 3 capturing the key issues and solutions is captured in TR 23.700-22 [2]. A related study on security aspects is captured in TR 33.700-22 [5]. A related study on charging aspects is captured in TR 28.849 [7].

References

List of related CRs: select "TSG Status = Approved" in:
https://portal.3gpp.org/ChangeRequests.aspx?q=1&workitem=1060040;1050035;1060026;1060041;1060061;1070015;1080019;1060009

[1] TS 23.222: “Common API Framework for 3GPP Northbound APIs”
[2] TR 23.700-22: “Study on CAPIF Phase 3”
[3] TS 33.122: "Security Aspects of Common API Framework for 3GPP Northbound APIs"
[4] IETF RFC 8693: "OAuth 2.0 Token Exchange"
[5] TR 33.700-22: “Study on security aspects of Common API Framework (CAPIF) Phase3”
[6] TS 32.254: "Telecommunication management; Charging management; Exposure function northbound Application Programming Interfaces (APIs) charging”
[7] TR 28.849: “Study on charging aspects of Common API Framework for Northbound APIs (CAPIF) phase 2”
[8] TS 29.222: “Common API Framework for 3GPP Northbound APIs”

10.4 Void

Was "Enhanced OAM for management service exposure to external consumers through CAPIF", now part of 10.3.

10.5 Non-Public Network (NPN) security

1000027	Non-Public Network (NPN) security considerations	SecNPN	S1	SP-230523	Yinglin Chen, China Telecom
1020039	*Study on security for PLMN hosting a NPN*	PLMNNPN_SEC	S3	SP-231786	Jun Shen
1060067	Security for PLMN hosting a NPN	PLMNNPN	S3	SP-241782	Jun Shen, China Telecom

Summary based on the input provided by China Telecom in SP-260227.

Note that this feature is referred to by two different names: at Stage 1 it is called “Non-Public Network (NPN) security considerations”, while at Stages 2 and 3 it is referred to as “Security for a PLMN hosting an NPN.”

NPN customers may request that dedicated network functions be deployed on customer premises. Deploying such dedicated network functions within customer premises can introduce security risks to the public land mobile network (PLMN) that hosts the NPN. The objective of this feature is therefore to ensure that the security of the hosting PLMN is not compromised when supporting an NPN. The corresponding service requirements are specified in TS 22.261[1].

To address these risks, one or more proxy entities are recommended at the boundary between the PLMN and the PNI-NPN operational domains. These proxy entities provide security capabilities such as transport-layer protection, authorization, topology hiding at IP and application levels, message inspection and filtering, and handling of malformed messages. In addition, the proxy entities may enforce policy checks to control information exchange between the PLMN and PNI-NPN operational domains (PNI-NPN being the Public Network Integrated Non-Public Network).

Dedicated network functions within the PNI-NPN operational domain are also provisioned with appropriate DNS security configurations. These security enhancements are specified in TS 33.501[2].

References

List of related CRs: select "TSG Status = Approved" in:
https://portal.3gpp.org/ChangeRequests.aspx?q=1&workitem=1000027;1020039;1060067

[1] TS 22.261: "Service requirements for the 5G system; Stage 1".
[2] TS 33.501: “Security architecture and procedures for 5G system”.

10.6 Interconnect of Standalone Non-Public Network (SNPN)

1030041	**Interconnect of SNPN**	ISN	S1	SP-240194	Thierry Bérisot, Novamint,
990053	*Study on Interconnect of SNPN*	FS_ISN	S1	SP-230236	Thierry Bérisot, Novamint

Summary based on the input provided by Novamint by e-mail received Tue 31/03/2026 22:08, file "Summary_1030041".

3GPP has defined, as part of 5G, the concept of Non-Public Network (NPN) as a network that is intended for non-public use in TS 22.261 ([2]), including requirements related to multiple subscriptions and access to multiple non-public networks and corresponding services.

There are two large categories of NPN deployment:

- PNI-NPN (Public Network Integrated NPN), referring to an NPN that is deployed with the support of a Public Land Mobile Network (PLMN).
- SNPN (Standalone Non-Public Network) referring to an NPN that does not rely on network functions provided by a PLMN

In many sectors, verticals have deployed a Standalone Non-Public Networks (SNPN) in ports, airports, mainports, utilities, railway, etc. to support private network in areas not covered by MNOs and for which they have their own licenced spectrum.

This is leading to the emergence of many new use cases where the verticals are confronted with aspects related to the interconnect of SNPNs with different owners, in locations where there is no public network, and where there is a need to deal with security, authorisation, onboarding, continuity of service, and resilience. In this case, the term “interconnect” refers to the relationship between two SNPNs owned by distinct administrations that allow users subscribed to one SNPN to use the mobile network access of the other SNPN.

In such use cases, there is a need to facilitate the interconnect between the SNPNs for the benefits of the end users and support of vertical services and applications.

In addition, a vast number of identity providers authenticate and authorize access to private networks inherited from past deployments of such networks for private users. New issues are appearing in terms of scalability and security aspects to support the interconnect between SNPNs and identity providers as the number of SNPNs increases.

The Rel-19 study on use cases of interconnect between SNPNs as well as interconnect of SNPNs with identity providers was done to identify new or enhanced service requirements related these issues and those use cases. The outcomes of the study were captured in TR 22.848 ([1]).

Following the study, the Rel-19 work item ([2]) introduces two aspects in TS 22.261 ([3]) as a result:

- Groups of interconnected SNPNs
- Enabling of SNPN cellular hotspots (Scalable SNPN Interconnect with dynamic connections)

Charging and billing aspects were not covered in the study and the resulting work item..

References

List of related CRs: select "TSG Status = Approved" in:
https://portal.3gpp.org/ChangeRequests.aspx?q=1&workitem=1030041

[1] TR 22.848: Study on Interconnect of SNPN
[2] SP-240194: WID on Interconnect of SNPN
[3] TS 22.261: Service requirements for the 5G system

10.7 ProSe support in NPN

1050080	ProSe support in NPN	TEI19_ProSe_NPN		SP-240122	Walter Dees, Philips International B.V.,
1030016	**Stage 2 of ProSe support in NPN**	TEI19_ProSe_NPN	S2	SP-240122	Walter Dees, Philips International B.V.,
1050021	**CT aspects of ProSe support in NPN**	TEI19_ProSe_NPN		CP-251224	Li, Mingxue, China Telecom
1050081	CT1 aspects of ProSe support in NPN	TEI19_ProSe_NPN	C1	CP-251224	Li, Mingxue, China Telecom
1060042	CT3 aspects of ProSe support in NPN	TEI19_ProSe_NPN	C3	CP-251224	Li, Mingxue, China Telecom
1050082	CT4 aspects of ProSe support in NPN	TEI19_ProSe_NPN	C4	CP-251224	Li, Mingxue, China Telecom

Summary based on the input provided by Philips International B.V. in SP-260281.

In Release 17 and 18, support for Proximity Services (ProSe) was added to 5G [1]. However, in these Releases, the architecture for ProSe was only specified for PLMNs, and not for NPNs. Without NPN support, deployment of ProSe (e.g. in factories) is hampered, and important use cases identified and requirements for ProSe would not be supported. Also other features such as ranging would be affected.

In Release 19, several enhancements are specified to support ProSe in the context of NPNs. The enhancements include:
- Expanding PLMN IDs with a Network Identifier (NID) to enable identification of SNPNs for 5G ProSe direct discovery and connection setup, 5G ProSe UE-to-Network Relay discovery and connection setup and 5G ProSe UE-to-UE Relay discovery and connection setup.
- Updating PLMN specific terminology in ProSe stage-2 specification [1] to also cover NPNs.
- Specifying restrictions on which 5G ProSe UE-to-Network relays can be selected (i.e. only from same SNPN for non-emergency services)
- Restricting the use of ProSe in SNPNs to UEs operating in SNPN access mode.
- Specifying the use of CAG for Layer-2 5G ProSe UE-to-Network Relays.
- Stage-3 details on how to support ProSe in NPNs for specific features, such as relaying PWS messages.

Some limitations include:
- Roaming architecture and Inter-SNPN operation are not supported for ProSe in NPN.
- SNPN onboarding procedure is not supported via 5G ProSe UE-to-Network Relay in NPN..

References

List of related CRs: select "TSG Status = Approved" in:
https://portal.3gpp.org/ChangeRequests.aspx?q=1&workitem=1050080;1030016;1050021;1050081;1060042;1050082

[1] TS 23.304 “Proximity based Services (ProSe) in the 5G System (5GS)”

10.8 See also

See also “Application enablement for AI/ML services (1050044, AIML_App)”

11 Rel-19 communications services

11.1 Media Messaging Enhancements

1080048	**Media Messaging Enhancements**	MeME-MED	S4	SP-250632	Thomas Stockhammer, Qualcomm Incorporated
1030005	*Study on Media Messaging*	FS_MeMe	S4	SP-240477	Thomas Stockhammer, Qualcomm Incorporated,

Summary based on the input provided by Qualcomm Incorporated in SP-251148.

TS 26.143 [2] as completed in Rel-18 defines capabilities and profiles for messaging services. In addition, the profiles permit to encapsulate real-time video into the 3GP file format using the baseline profile. The media types for the player address basic text, audio/speech, images, video, text/subtitle, 3D scenes including AR as well as simple HTML-5 presentations. For the content generator, the supported media types in this profile are basic text, audio/speech, images, video, and text.

In the Rel-19 study FS_MeME, additional stage-3 work has been recommended in clause 6.3.2 of TR 26.841 [1], in particular to support Media Service Enabler (MSE) as defined in TR 26.857 [3] as introduced in clause 5.6 and based on the conclusions in clause 5.6.4, the following aspects are addressed:
1. Connect TS 26.143 to TR 26.857 to fulfil several of the MSE concepts
2. Add a call flow to TS 26.143 aligned with clause 4.1 of TR 26.841
3. Add a set of stage-2 APIs and parameter that can be assigned to player and generator
4. Add more examples for valid content

In addition, when checking the existing capability support in TS 26.143 in context of GSMA RCS messaging, some key gaps were identified and are addressed in the work item:
1. HEIC image format with 10 bit and HDR capability is now supported
2. Permitted the use of MP4, in particular 'mp42' as the major brand, in addition to 3gpx major brands.
3. Clarified that the main 10 profile decoders can decode Main profile content.
4. Updated the related media playback and generator capabilities and the related media messaging profiles

The work item and the respective CR to TS 26.143 address the above updates.

References

List of related CRs: select "TSG Status = Approved" in:
https://portal.3gpp.org/ChangeRequests.aspx?q=1&workitem=1080048

[1] TR 26.841, "Study on Media Messaging"
[2] TS 26.143, "Messaging Media profiles"
[3] TS 26.857, "5G Media Service Enablers"

11.2 Terminal Audio quality performance and Test methods for Immersive Audio Services, Phase 2

1050113	**Terminal Audio quality performance and Test methods for Immersive Audio Services, Phase 2**	ATIAS_Ph2	S4	SP-241314	Jan Reimes, HEAD acoustics GmbH,

Summary based on the input provided by HEAD acoustics GmbH in SP-251544.

The work item ATIAS_Ph2 continued and expanded upon previous work on Terminal Audio quality performance and Test methods for Immersive Audio Services (ATIAS). The focus was on test setups/infrastructure, test methods and requirements needed to evaluate assessment of immersive audio terminals supporting IVAS codec. The work addressed gaps remaining from Rel-18 and concluded with updated specifications that are highly relevant for the industry.

ATIAS_Ph2 developed and consolidated test methods in TS 26.260 [1] and performance requirements in TS 26.261 [2] for immersive UEs across several technical areas:

Test setups and UE configurations: Clarifications on UE types, bitrate settings and simultaneous capture modes/capabilities.

Acoustic echo control for immersive communication: Initial work in this area introduced test methods evaluating acoustic echo control for immersive scenarios. Methodologies for terminal coupling loss were adapted from principles in related standards on mono telephony. Provisional performance requirements were specified for all send/receive UE types.

Headtracking and rendering capabilities: Substantial progress was achieved for evaluating headtracking performance of immersive UEs. Measurement setup, test methods and requirements involving motorized Head and Torso Simulators (HATS) were specified to quantify the relationship between head movement and associated audio scene adaptation in terms of motion-to-sound latency.

Ambient sound field transmission: Efforts in this area resulted in new procedures for testing the transparent capture of transmitted ambient sound fields. Test cases and requirements addressed loudness, frequency response and temporal characteristics. Emerging developments in IVAS_Ph2 regarding new SDP parameters to control noise suppression functionalities in the UE were taken into account [3].

Enhanced testing for single sound source: To accommodate an increased spatial resolution, the existing test methods on half-circular planar capture capabilities were extended for full-circular planar and 3D spherical capabilities. In addition, new performance requirements and objectives for SBA and MASA were introduced to accompany this.

References

List of related CRs: select "TSG Status = Approved" in:
https://portal.3gpp.org/ChangeRequests.aspx?q=1&workitem=1050113

[1] TS 26.260: "Objective test methodologies for the evaluation of immersive audio systems"
[2] TS 26.261: "Terminal audio quality performance requirements for immersive audio services"
[3] TS 26.253: "Codec for Immersive Voice and Audio Services; Detailed Algorithmic Description incl. RTP payload format and SDP parameter definitions"

11.3 EVS Codec Extension for Immersive Voice and Audio Services, Phase 2

1040021	**EVS Codec Extension for Immersive Voice and Audio Services, Phase 2**	IVAS_Codec_Ph2	S4	SP-250262	Su, Huan-yu, Huawei Technologies Co Ltd.,
1040020	*Study on Haptics in 5G Media Services*	FS_HapticsMedia	S4	SP-241121	Gaëlle Martin-Cocher, InterDigital

Summary based on the input provided by vivo in SP-251375.

This Work Item enhances the Immersive Voice and Audio Services (IVAS) codec specifications in TS 26.250, originally introduced under Rel-18's IVAS_Codec [3].

It delivers a fixed-point reference implementation, updates the conformance process with non-bit-exact testing for floating-point code, and provides a comprehensive performance characterization across all codec operating modes. Tests confirm full interoperability between the floating-point reference code (TS 26.258) and the new fixed-point code (TS 26.251).

The key results of IVAS_Codec_Ph2 are:

- TS 26.251 fixed-point reference code, functionally aligned with the floating-point code in TS 26.258 and offering comparable audio quality and complexity [6][7][8].

- Extensive codec characterization, summarized in TR 26.997, including:

  - Performance across formats, bitrates, rendering modes, DTX, PLC, JBM, and more [4][5][8].

  - Verified interoperability between TS 26.258 (floating-point) and TS 26.251 (fixed-point).

- Updated conformance testing in TS 26.252, adding new fixed-point test sequences and a non-bit-exact conformance method for floating-point implementations.

References

List of related CRs: select "TSG Status = Approved" in:
https://portal.3gpp.org/ChangeRequests.aspx?q=1&workitem=1040021

[1] (void)
[2] (void)
[3] SP-220608 EVS Codec Extension for Immersive Voice and Audio Services
[4] S4-251561 IVAS-7b Processing plan for Characterization phase, v.1.1.0.
[5] S4-252024 IVAS Permanent Document IVAS-8b: Test Plan for Characterization Phase v.1.3.0
[6] S4-251013 Verification of IVAS BASOP encoder.docx
[7] S4-242054 Verification of IVAS BASOP decoder and renderer
[8] S4-251668 IVAS Characterization Tests - Global Analysis Lab Report

11.4 5GMSG Service phase 3

1060051	5GMSG Service phase 3	5GMARCH_Ph3		SP-230781	Liu, Yue, China Mobile,
1000037	**Stage 2 of GMSG Service phase 3**	5GMARCH_Ph3	S6	SP-230781	Liu, Yue, China Mobile,
1060019	**CT aspects for enabling MSGin5G Service phase 3**	5GMARCH_Ph3		CP-251225	Liu, Yue, China Mobile
1060052	CT1 aspects for enabling MSGin5G Service phase 3	5GMARCH_Ph3	C1	CP-251225	Liu, Yue, China Mobile
1060053	CT3 aspects for enabling MSGin5G Service phase 3	5GMARCH_Ph3	C3	CP-251225	Liu, Yue, China Mobile
1030030	**Security aspects for MSGin5G Service Phase 3**	5GMARCH_SEC_Ph3	S3	SP-240516	Xiaoting Huang,

Summary based on the input provided by China Mobile by e-mail received Tue 31/03/2026 10:31, file "10.4_Summary for 5GMARCH_Ph3".

This Work Item is a further refinement and enhancement of the 5GMSG Service specified in TS 22.262 [1] based on the foundation established in Rel-18 5GMARCH_Ph2 specified in TS 23.554 [2]. It covers:

- Deferred delivery for unregistered recipients (not sender-triggered), addressing overlapping scenarios with Store and Forward;
- Procedures for adding specific Application Servers to Group Messaging;
- Delivery using 5GC QoS to meet delay requirements;
- APIs from MSGin5G Client supporting Application Clients via MSGin5G-5 reference point;
- Improved Message Topic subscription procedures between MSGin5G Servers.

Protocols are specified in TS 24.538 [3] and the APIs in TS 29.538 [4].

References

List of related CRs: select "TSG Status = Approved" in:
https://portal.3gpp.org/ChangeRequests.aspx?q=1&workitem=1060051;1000037;1060019;1060052;1060053;1030030

[1] TS 22.262: "Message Service within the 5G System".
[2] TS 23.554: "Application architecture for MSGin5G Service; Stage 2".
[3] TS 24.538: " Enabling MSGin5G Service;Protocol specification".
[4] TS 29.538: " Enabling MSGin5G Service;Application Programming Interfaces (API) specification".

11.5 Video Operating Points - Harmonization and Stereo MV-HEVC

1030002	Video Operating Points - Harmonization and Stereo MV-HEVC	VOPS	S4	SP-250818	Waqar Zia

Summary based on the input provided by Apple in SP-251503.

This Work Item consolidates video operation points and capabilities previously scattered across multiple SA4 specifications into the new TS 26.265 [1]. A key objective was to define and integrate stereoscopic encoding/decoding capabilities and operation points based on the Multiview Extensions of HEVC (MV-HEVC), in coordination with related SDOs such as MPEG. The work also included the development of a conformance framework, including tools and test vectors, particularly for MV-HEVC.

The resulting TS 26.265 specifies a set of service-independent video operation points and capabilities. It defines three main video representation formats—each based on established recommendations (ITU-R BT.709-6 and BT.2100-2) but refined through 3GPP-specific constraints:

- 3GPP High-Definition (HD)
- 3GPP High Dynamic Range (HDR)
- 3GPP Stereoscopic

The specification further introduces common bitstream constraints such as progressive-scan requirements and limitations on VUI (Video Usability Information) syntax for HEVC. It also defines several decoding capabilities for AVC and HEVC by codec, profile, tier, and level, which are combined with the formats above to form video operation points, including:

- 3GPP AVC HD
- 3GPP HEVC HD
- 3GPP HEVC HDR
- 3GPP HEVC UHD
- 3GPP HEVC Stereo

Two additional operation points with multiview support—3GPP-MV-HEVC-Main-Stereo and 3GPP-MV-HEVC-Ext-Stereo—were also specified, both requiring the 3GPP Stereoscopic format.

Annex B of TS 26.265 [1] provides the conformance assets, including video test material, reference encoder configurations, a conformance validator, and conformance bitstreams.

References

List of related CRs: select "TSG Status = Approved" in:
https://portal.3gpp.org/ChangeRequests.aspx?q=1&workitem=1030002

[1] TS 26.265: "Media Delivery: Video Capabilities and Operating Points

11.6 Advanced Media Delivery

1030006	**Advanced Media Delivery**	AMD	S4	SP-241011	Thomas Stockhammer, Qualcomm Incorporated Richard Bradbury, BBC
1030006	*Study on Advanced Media Delivery*	FS_AMD	S4	SP-241011	
1060069	Stage 2 for Advanced Media Delivery	AMD-ARCH-MED	S4	SP-241963	
1070057	Stage 3 for Advanced Media Delivery	AMD_PRO-MED	S4	SP-250265	
1080004	CT aspects of Advanced Media Delivery	AMD_PRO-MED-CT		CP-252191	Hanna Lim, Qualcomm Incorporated
1080034	CT4 aspects of Advanced Media Delivery	AMD_PRO-MED-CT	C4	CP-252191	Hanna Lim, Qualcomm Incorporated
1030007	*Study of 5G Real-time Transport Protocol Configurations, Phase 2*	FS_5G_RTP_Ph2	S4	SP-240482	Igor Curcio, Nokia Corporation

Summary based on the input provided by Qualcomm Incorporated and BBC in SP-251559, and from Qualcomm in SP-250312.

Advanced Media Delivery (AMD) is a set of 5G enhancements that significantly improves media streaming and broadcast capabilities through new architectural features, protocols, and APIs across 5GMS and MBS.

The 5G Media Streaming (5GMS) and Multicast-Broadcast Services (MBS) architectures continue to evolve to meet the growing demand for advanced mobile media delivery. TS 26.501 defines the 5GMS architecture, while TS 26.502 specifies the MBS User Service architecture. Building on the feasibility study on Advanced Media Delivery (FS_AMD) and the conclusions of TR 26.802 and TR 26.804, 3GPP initiated both Stage-2 and Stage-3 normative work to enhance media delivery capabilities in Release 19. Reference implementations for 5GMS and MBS are also being developed within 5G-MAG.

At Stage-2, several architectural enhancements are introduced. TS 26.502 is updated with support for in-session unicast repair for MBS object distribution, enabling unicast recovery of lost data during live multicast/broadcast sessions. Time synchronization mechanisms are added, offering both SIB9-based and user-domain time services. Deployment guidelines for operating MBS User Services over eMBMS are also documented to harmonize interfaces and support potential 5G Broadcast deployments. In parallel, TS 26.501 is enhanced with in-band reporting for Common Client Metadata, enabling collection of playback metrics for analytics and exposure via NWDAF. Additional updates include support for multi-access media delivery, media delivery from multiple service endpoints or CDNs, workflows for distributing encrypted and high-value content, improved QoS support including ECN marking for L4S and QoS monitoring, and procedures enabling media streaming over network slices.

Stage-3 work in SA4 builds on these architectural foundations by specifying the corresponding protocols, APIs, and implementation details. Enhancements include standardized Common Media Client Data reporting based on CTA-5004, enabling improved operational insight and quality of experience. Media delivery from multiple service endpoints is further developed to support multi-CDN operation, content steering, and Coded Multi-Source Media Format, improving reliability and latency. Multi-access media delivery is extended to support multipath protocols such as MPTCP and MPQUIC, along with configuration and reporting mechanisms. Secure distribution of encrypted and high-value content is strengthened through CPIX-based key exchange and DRM-enabled DASH workflows. QoS support is expanded with mechanisms for L4S ECN marking and dynamic policy invocation. Additional Stage-3 work addresse network slicing for differentiated media services, in-session unicast repair for MBS/MBMS sessions, and protocols for time synchronization across devices and networks.

These developments are coordinated with SA2, CT3, CT4, and external organizations such as ETSI and DASH-IF. The resulting enhancements are designed to remain compatible with existing OTT practices while enabling advanced capabilities for 5G media streaming and MBS User Services, supported by reference tools and client implementations from 5G-MAG.

References

List of related CRs: select "TSG Status = Approved" in:
https://portal.3gpp.org/ChangeRequests.aspx?q=1&workitem=1030006,1030006,1060069,1070057,1080004,1080034,1030007

[1] TS 26.501, 5G Media Streaming (5GMS); General description and architecture
[2] TS 26.502, 5G multicast-broadcast services; User service architecture
[3] TS 26.510, 5G; Media delivery; interactions and APIs for provisioning and media session handling
[4] TS 26.512, 5G Media Streaming (5GMS); Protocols
[5] TS 26.517, 5G Multicast-Broadcast User Services; Protocols and Formats
[6] TS 26.346, Multimedia Broadcast/Multicast Service (MBMS); Protocols and codecs
[7] TR 26.802, Multicast Architecture Enhancement for 5G Media Streaming
[8] TR 26.804, Study on 5G media streaming extensions
[9] ETSI TS 103 973, Coded Multisource Media Format (CMMF) for Content Distribution and Delivery
[10] ETSI TS 103 998, DASH-IF: Content Steering for DASH
[11] ETSI TS 103 799, DASH-IF Content Protection Information Exchange Format (CPIX)
[12] Consumer Technology Association (CTA) CTA-5004, Common Media Client Data (CMCD)

11.7 5G Real-time Transport Protocol Configurations, Phase 2

1060021	**5G Real-time Transport Protocol Configurations, Phase 2**	5G_RTP_Ph2	S4	SP-241961	Igor Curcio, Nokia Corporation
1030007	*Study of 5G Real-time Transport Protocol Configurations, Phase 2*	FS_5G_RTP_Ph2	S4	SP-240482	Igor Curcio, Nokia Corporation

Summary based on the input provided by Nokia in SP-251144.

This Work Item specifies the continuation work of a horizontal protocol specification in TS 26.522 that can be used by different 3GPP 5G XR services and enablers.

The origin of this work was TR 26.822 (Feasibility Study of 5G Real-time Transport Protocol Configurations, Phase 2, FS_5G_RTP_Ph2) which identified multiple aspects of normative work in Rel. 19 to support 5G/AR Real-time Communication. The study conclusions recommended doing normative specification for several key issues related to RTP and RTCP in order to better support real-time media transport for conversational services in the 5G system for both WebRTC and IMS.

This Work Item focused on optimizing the use of RTP for the transport of real-time XR media (including conversational media) and associated metadata into TS 26.522, TS 26.510 and TS 26.113.

This is the primary feature list developed as part of this work item:
- Normative work to mitigate the inaccuracy of PDU Set Size information.
- Extended the RTC Dynamic Policy API in TS 26.510 and TS 26.113 to include PDU Set Importance values for PDUs that may be treated as lone PDUs in the UPF.
- Developed guidelines for handling of lone PDUs at the UPF.
- Normative work on multiplexed RTP streams and defined guidelines in TS 26.522, TS 26.510 and TS 26.113 for RTP senders that use multiplexing.
- Normative work on burst size, time to next burst, expedited transfer indication and the definition of data burst.
- Normative work on guidelines for marking PDU Set Importance for PDU Sets that are not defined as video frames or slices.

References

List of related CRs: select "TSG Status = Approved" in:
https://portal.3gpp.org/ChangeRequests.aspx?q=1&workitem=1060021

11.8 Next Generation Real time Communication services Phase 2

1050066	**System architecture for Next Generation Real time Communication services Phase 2**	NG_RTC_Ph2		SP-240985	Yi Jiang, China Mobile
1010030	*Study on system architecture for next generation real time communication services phase 2*	FS_NG_RTC_Ph2	S2	SP-231196	Yi Jiang (China Mobile)
1040016	*Study on Charging aspects of next generation real time communication services phase 2*	FS_NG_RTC_Ph2_CH	S5	SP-240982	Chen Ai, China Mobile
1040026	**Stage 2 of System architecture for Next Generation Real time Communication services Phase 2**	NG_RTC_Ph2	S2	SP-241384	Yi Jiang, China Mobile
1050022	**CT aspects of System architecture for Next Generation Real time Communication services Phase 2**	NG_RTC_Ph2		CP-242250	Chen Xu, China Mobile
1050067	CT1 aspects of System architecture for Next Generation Real time Communication services Phase 2	NG_RTC_Ph2	C1	CP-242250	Chen Xu, China Mobile
1050068	CT3 aspects of System architecture for Next Generation Real time Communication services Phase 2	NG_RTC_Ph2	C3	CP-242250	Chen Xu, China Mobile
1050069	CT4 aspects of System architecture for Next Generation Real time Communication services Phase 2	NG_RTC_Ph2	C4	CP-242250	Chen Xu, China Mobile
1050070	CT6 aspects of System architecture for Next Generation Real time Communication services Phase 2	NG_RTC_Ph2	C6	CP-242250	Chen Xu, China Mobile
1070016	**Charging aspects of next generation real time communication services phase 2**	NG_RTC_Ph2-CH	S5	SP-250513	Chen Ai, China Mobile
1020038	*Study on the security support for the Next Generation Real Time Communication services phase 2*	FS_NG_RTC_SEC_Ph2	S3	SP-240233	Vlasios Tsiatsis
1060063	**Security support for the Next Generation Real Time Communication services Phase 2**	NG_RTC_SEC_Ph2	S3	SP-250649	Vlasios Tsiatsis, Ericsson

Summary based on the input provided by China Mobile by e-mail received Fri 27/03/2026 11:58, file "Summary_1050066", including Security Aspects provided by Ericsson.

The system architecture for the next generation real time communication services based on IMS enhancements has been specified in Rel-18. The fundamental architecture, interfaces and procedures of IMS data channel and AR communication service based on IMS data channel have been standardized.

In Rel-19, it is still required to specify enhanced features to improve the support for IMS Data Channel and to support more muti-media real time communication services using IMS Data Channel. The charging and security aspects of using IMS data channel is also needed to be enhanced.

System Architecture Aspects:
The enhancements on IMS architecture, procedures and protocols are specified in Rel-19 to support the following aspects:

- Architectural and procedural enhancement to enable trusted and untrusted DC applications to subscribe to and receive reports about IMS events;
- Enhancements to IMS architecture, interfaces and procedures to expose IMS services in the context of IMS Data Channel;
- Enhancements to IMS architecture and procedures to provide interworking with a MTSI UE supporting only legacy media types (audio/video/messaging) ;
- Enhancements to the IMS architecture and procedures to support third party identities in IMS sessions;
- Enhancements to IMS procedures to support PS Data off for services over IMS DC;
- Enhancements of IMS procedures to support standalone IMS data channel session;
- Enhancements of IMS architecture and procedures to support multiplexing and de-multiplexing DC applications over single SCTP connection;
- Enhancements of IMS architecture and procedures to support IMS based Avatar communication.

Security Aspects
The security work in Rel-19 included the following aspects:

- IMS third party identity security handling. A new clause 10 was introduced in TS 33.203 [4], which provided the security requirements, a summary of the third party specific identities feature and a reference to the related clauses in TS 23.228 [1]. This clause also included the security of the exposure of the parameter provisioning procedure introduced in TS 23.228 [1].
- Security aspects related to the IMS data channel exposure. The normative work included a new Annex Q in TS 33.328 [3] which provided a summary of the new feature, a reference to the related clauses in TS 23.228 [1] and TS 23.502 [2], a list of security requirements and a statement that existing security specification in TS 33.501 [5], clause 12 for the NEF-AF interface is reused in the procedures.
- Security aspects related to the IMS avatar communication. The normative work was based on existing security specification in TS 33.328 [3] with the addition of a few authorization principles. More specifically, the normative work included a new Annex R in TS 33.328 [3]. which provided a summary of the IMS avatar feature, a reference to the related clauses in TS 23.228 [1], a list of security requirements and the security procedures for this feature.

Charging Aspects
The charging principles, procedures have been enhanced in Rel-19 to support next generation real time communication services with the following aspects:

- Charging support of standalone IMS Data Channel sessions in TS 32.255 [6] and TS 32.260 [7].
- Charging support of DC application download and usage in TS 32.255 and TS 32.260.
- Charging support of IMS network capabilities exposure in TS 32.254 [8].
- Charging support of IMS Data Channel as a PS Data Off Exempt Service in TS 32.255.
- Charging support of avatar communication in TS 32.260.

The corresponding Open API and ASN.1 are specified in the TS 32.291 [9] and TS 32.298 [10].

References

List of related CRs: select "TSG Status = Approved" in:
https://portal.3gpp.org/ChangeRequests.aspx?q=1&workitem=1050066;1010030;1040016;1040026;1050022;1050067;1050068;1050069;1050070;1070016;1020038;1060063

[1] TS 23.228: "IP Multimedia Subsystem (IMS); Stage 2"
[2] TS 23.502: "Procedures for the 5G System (5GS)"
[3] TS 33.328: "IP Multimedia Subsystem (IMS) media plane security"
[4] TS 33.203: "3G security; Access security for IP-based services"
[5] TS 33.501: "Security architecture and procedures for 5G System"
[6] TS 32.255: "Telecommunication management; Charging management; 5G Data connectivity domain charging; stage 2".
[7] TS 32.260: "Telecommunication management;Charging management;IP Multimedia Subsystem (IMS) charging".
[8] TS 32.254: "Telecommunication management; Charging management; Exposure function Northbound Application Program Interfaces (APIs) charging".
[9] TS 32.291: "Telecommunication management; Charging management; 5G system; Charging service, stage 3".

[10] TS 32.298: "Telecommunication management; Charging management; Charging Data Record (CDR) parameter description".
[11] TS 24.186: "IMS Data Channel applications; Protocol specification"
[12] TS 24.229: "IP multimedia call control protocol based on Session Initiation Protocol (SIP) and Session Description Protocol (SDP); Stage 3"
[13] TS 29.522: "5G System; Network Exposure Function Northbound APIs; Stage 3"
[14] TS 29.165: "Inter-IMS Network to Network Interface (NNI)"
[15] TS 29.175: "IP Multimedia Subsystem (IMS) Application Server (AS) Services Stage 3"
[16] TS 29.176: "IP Multimedia Subsystems (IMS); Media Function (MF) Services; Stage 3"
[17] TS 29.562: "5G System; Home Subscriber Server (HSS) services; Stage 3"
[18] TS 29.510: "5G System; Network function repository services; Stage 3"
[19] TS 29.230: "Diameter applications; 3GPP specific codes and identifiers"
[20] TS 29.328: "IP Multimedia (IM) Subsystem Sh interface; Signalling flows and message contents"
[21] TS 29.329: "Sh interface based on the Diameter protocol; Protocol details"
[22] TS 31.102: "Characteristics of the Universal Subscriber Identity Module (USIM) application"
[23] TS 31.103: "Characteristics of the IP Multimedia Services Identity Module (ISIM) application"

11.9 Application enablement aspects for MMTel

1070036	**Application enablement aspects for MMTel**	MMTel_App		SP-241019	Liu, Yue, China Mobile
1040073	**Stage 2 of Application enablement aspects for MMTel**	MMTel_App	S6	SP-241019	Liu, Yue, China Mobile
1070008	**CT aspects of Application enablement aspects for MMTel**	MMTel_App		CP-250249	Liu, Yue, China Mobile
1070037	CT1 aspects of Application enablement aspects for MMTel	MMTel_App	C1	CP-250249	Liu, Yue, China Mobile
1070038	CT3 aspects of Application enablement aspects for MMTel	MMTel_App	C3	CP-250249	Liu, Yue, China Mobile
1000034	*Study on Service aspects for supporting the eMMTel service*	FS_eMMTelAPP	S6	SP-230779	Liu, Yue, China Mobile

Summary based on the input provided by China Mobile by e-mail received Tue 31/03/2026 10:31, file "10.9_Summary for Application enablement aspects for MMTel".

This Work Item addresses application enablement for the MMTel service, based on stage 1 requirements (TS 22.261 [1]) and architecture (TS 23.392 [2]), using IMS Data Channel capabilities (TS 23.228 [3]). It covers:

- DC Application management for PLMN operators offering MMTel with IMS Data Channel.
- DC Application downloading control to regulate app downloads on UE via data channel profiles.
- MMTel service usage, enabling application providers or verticals to use features such as calls between apps and DCMTSI clients, third-party calls, P2A, and A2P Data Channels.

The protocol aspect is specified in TS 24.392 [4] and the API aspect is specified in TS 29.392 [5]

References

List of related CRs: select "TSG Status = Approved" in:
https://portal.3gpp.org/ChangeRequests.aspx?q=1&workitem=1070036;1040073;1070008;1070037;1070038

[1] TS 22.261: "Service requirements for the 5G system".
[2] TS 23.392: "Application enablement aspects for MMTel".
[3] TS 23.228: "IP Multimedia Subsystem (IMS); Stage 2".
[4] TS 24.538: "Enabling MSGin5G Service; Protocol specification".
[5] TS 29.538: "Enabling MSGin5G Service; Application Programming Interfaces (API) specification; Stage 3".

12 Rel-19 XR (eXtended Reality), Augmented Reality (AR), Metaverse, Edge Computing

12.1 Localized Mobile Metaverse Services

1020056	**Localized Mobile Metaverse Services**	Metaverse		SP-230509	Guttman, Erik, Samsung
950005	*Study on Localized Mobile Metaverse Services*	FS_Metaverse	S1	SP-220353	Guttman, Erik, Samsung
1000028	**Stage 1 of Localized Mobile Metaverse Services**	Metaverse	S1	SP-230509	Guttman, Erik, Samsung
1040076	**Application enablement for mobile metaverse services**	Metaverse_App	S6	SP-241390	Shah, Sapan, Samsung
1020052	*Study on application enablement for Localized Mobile Metaverse Services*	FS_Metaverse_App	S6	SP-231806	Shah, Sapan, Samsung
1030039	*Study on security aspects of 5G Mobile Metaverse services*	FS_Metaverse_Sec	S3	SP-240508	Rohini Rajendran,
1050025	**CT aspects of Application enablement for mobile metaverse services**	Metaverse_App		CP-251132	Vijay Sangameshwara, Samsung
1050060	CT1 aspects of Application enablement for mobile metaverse services	Metaverse_App	C1	CP-251132	Vijay Sangameshwara, Samsung
1050061	CT3 aspects of Application enablement for mobile metaverse services	Metaverse_App	C3	CP-251132	Vijay Sangameshwara, Samsung
1060060	**Security aspects of 5G Mobile Metaverse services**	Metaverse_Sec	S3	SP-250075	Rohini Rajendran, r.rohini@samsung.com

Summary based on the input provided by Samsung by e-mail received Fri 10/04/2026 18:02, file "Summary_1020056".

The term metaverse has been used in various ways to refer to the broader implications of AR and VR media. This feature introduces specific capabilities related to advanced multimedia use cases; these are specifically supported by the 3GPP standard.

The Localized Mobile Metaverse Services feature comprises a number of service enablers related to delivery of XR-based services, including some new entirely services.

General capabilities were provided, to enable service delivery that has the expected performance characteristics, as well as targeted energy savings, as required by TS 22.156 [1].

- The ability to adjust energy usage of mobile metaverse services on the basis of subscription policies is supported through the EnergySys feature. [2]

- The ability to expose network performance information associated with one or more users of an authorized third party mobile metaverse application is supported through QoS Monitoring. [3]

- The ability to maintain consistent user experience, for a given UE, when XR media from different mobile metaverse services have different communication performance is supported through multi-modal QoS control. [4]

Localized mobile metaverse services are immersive and integrated into a user's ordinary experiences. Such service experiences are location-related and can include presentation of AR, MR media.

Localized experiences are effectively present in the user's environment, so that the mobile metaverse media provided for a given mobile metaverse service is both appropriate to and integrated with both the physical world and with mobile metaverse media content displayed. Localized mobile metaverse services can be associated with specific places (3D locations in the physical world). The association between these places and service information is termed a spatial anchor.

Spatial anchors enable mobile metaverse services to be discovered and accessed, if the user is authorized. For example, the service information can convey the mobile metaverse server access information. When the user's application accesses the mobile metaverse service, the media associated with the service can be obtained by the user.

Spatial anchors can be defined by third parties, e.g., service providers, to offer relevant localized services, e.g., associated with specific items or features in their place of business. This information and its associated authorization

information, determining who can discover the spatial anchor, can be managed - created, deleted and modified. These capabilities are supported by TS 23.437. [5]

Users' localization is important in order to discover spatial anchors. The 5G system offers a spatial localization service to determine this information. Using sensor data related to the user's location, the 5G system can identify where the user is. This is possible by means of processing the sensor data as well as a spatial map. The result, precise user location and orientation in space, can be exposed to authorized third parties.

The spatial map is created using processed sensor data. The 5G system supports a spatial mapping service to customers that, for example, want to offer mobile metaverse services associated with spatial anchors on their premises. Creation of a spatial map for a location makes localization there possible, as well as assignment of spatial anchors in that location. These capabilities are supported by TS 23.437. [5]

Services can benefit from common information, such as avatar parameters and configuration information, so that a user's digital representation is consistent across different applications. Users can benefit from the support of associating their digital assets with different User Identities, to flexibly control their stored information. In Release 19, digital asset support has been added, for adding, removing, controlling authorization for use (including by third parties) and associating the avatar with user identities. This is specified in TS 23.438. [6]

A user can take part in mobile metaverse services that provide digital representations of several other users simultaneously, for example, to support a Conference using XR media. As these services are interactive and immersive, the 5G system provides a means so that the experience of each user of the same service is compatible and consistent. Users can participate together in this way, whether some of those in the Conference service are located in the same place: these users can experience remote users as AR or MR media. This media is conversational - meaning that all parties can participate, and real time - meaning that all users perceive each other's actions effectively simultaneously.

Avatars are employed in mobile metaverse services mainly to represent a user. To improve the user experience of this representation, the 5G system makes it possible to associate the user and the avatar representation and to expose this information to authorized third parties. This information could be valuable, as avatars can represent organizations, individual users, even groups of users.

The 5G system supports a means by which user's pose, gestures and expressions are captured as input for the conversational mobile metaverse service. Devices (e.g., UEs) can capture this information in a standardized form, which is used for the creation of an animated digital representation of the user (e.g., 3D avatar) that can be presented to other users. The system supports communication of this animated digital representation as "avatar media" as well as audio and other media as needed. The 5G system supports the creation of avatar media. Privacy and user consent is needed for sensor data used to capture a user's pose, gestures and (facial) expressions is sent from the UE to the network encoded as avatar media to be rendered.

These capabilities are captured in TS 26.264 [7]

For each of the features above, aspects were specified in TS 33.434 [8] to satisfy the security, authorization and privacy requirements in TS 22.156. [1]

Charging aspects of this feature are covered by CAPIF charging support. [9]

References

List of related CRs: select "TSG Status = Approved" in:
https://portal.3gpp.org/ChangeRequests.aspx?q=1&workitem=1020056,950005,1000028,1040076,1020052,1030039,1050025,1050060,1050061

[1] TS 22.156: "Mobile Metaverse Services".
[2] Energy Efficiency as a service criteria (SP-230520)
[3] QoS Monitoring (SP-180141), including QoS Monitoring Enhancements (SP-240961)
[4] TS 23.501: "System architecture for the 5G System (5GS)", 5.37.2
[5] TS 23.437: "Service Enabler Architecture Layer for Verticals (SEAL); Spatial map and Spatial anchors".
[6] TS 23.438: "Service Enabler Architecture Layer for Verticals (SEAL); Digital assets".
[7] TS 26.264: "IMS-based AR Real-Time Communication".
[8] TS 33.434: "Security aspects of Service Enabler Architecture Layer (SEAL) for verticals".
[9] Charging aspects of CAPIF (SP-250820)

12.2 Extended Reality and Media

1060034	**Extended Reality and Media**				Georgios Gkellas (Nokia),
990052	**Supporting UE Mobility for XR Services**	XRMobility	S1	SP-230233	Xiaonan Shi, ChinaMobile
1010032	*Study on Extended Reality and Media service (XRM) Phase 2*	FS_XRM_Ph2	S2	SP-231671	Primary Devaki Chandramouli (Nokia)
1040032	**Extended Reality and Media service (XRM) Phase 2**	XRM_Ph2	S2	SP-241385	Georgios Gkellas (Nokia),
1060027	**CT aspects of Extended Reality and Media service (XRM) Phase 2**	XRM_Ph2		CP-252058	Ravindran, Parthasarathi, Nokia
1060035	CT1 aspects of Extended Reality and Media service (XRM) Phase 2	XRM_Ph2	C1	CP-252058	Ravindran, Parthasarathi, Nokia
1060036	CT3 aspects of Extended Reality and Media service (XRM) Phase 2	XRM_Ph2	C3	CP-252058	Ravindran, Parthasarathi, Nokia
1060037	CT4 aspects of Extended Reality and Media service (XRM) Phase 2	XRM_Ph2	C4	CP-252058	Ravindran, Parthasarathi, Nokia
1090012	**Charging Aspects for XRM phase 2**	XRM_PH2-CH	S5	SP-251214	Xiaoqian Jia (China Telecom)
1020051	*Study on Application enabler for XR Services*	FS_XRApp	S6	SP-2315739	Zheng, Shaowen, CMCC
1040074	**Application enablement for XRM Services Phase 2**	XRM_Ph2_App	S6	SP-241207	Zheng, Shaowen, CMCC
1060024	**CT aspects of Application enablement for XRM Services Phase 2**	XRM_Ph2_App		CP-243072	Zhenning, Huang, China Mobile
1060038	CT1 aspects of Application enablement for XRM Services Phase 2	XRM_Ph2_App	C1	CP-243072	Zhenning, Huang, China Mobile
1060039	CT3 aspects of Application enablement for XRM Services Phase 2	XRM_Ph2_App	C3	CP-243072	Zhenning, Huang, China Mobile
1070021	**Protection of XRM Media related information**	XRM_Ph2-SEC	S3	SP-250361	Daniel Cho
1080016	**OAM support for Extended Reality and Media service (XRM) phase 2**	XRM_Ph2-OAM	S5	SP-250880	Xie Pengxiang (ZTE),

Summary based on the input provided by Nokia in SP-260035, then including SA3 aspects provided by Ericsson by e-mail received Wed 21/01/2026 09:33, file "XRM_Ph2_summary". OAM aspects covered separately in SP-260152 from ZTE.

In Rel-18, the Extended Reality and Media services (XRM) work item introduced the support for power saving optimizations based on dynamic End of Data Burst indication, in addition to more static periodicity and burst size information.

The Rel-19 enhancements improve the XRM traffic handling, particularly for adaptive and encrypted applications. These enhancements enable more dynamic, fine-grained control of traffic by allowing the network to better interpret and act on application-level characteristics such as burst size and timing, leading to improved power efficiency and resource utilization.

Release 19 extends PDU Set–based handling to support more advanced QoS mechanisms, including alternative QoS profiles, enhanced notification control, and more accurate traffic differentiation through PDU Set identification and marking (e.g., DSCP). This allows more effective handling of multiplexed and encrypted data flows, where traditional QoS methods were previously insufficient. Additional packet filtering capabilities enable finer detection and mapping of different media types within a single transport connection.

The enhancements also introduce support for dynamic updates of traffic characteristics via the user plane, enabling the network to adapt in real time to changing application behavior. Application-driven mechanisms such as data boosting are supported, alongside improved exposure of network conditions (e.g., available bitrate and rate limitations) to assist application-layer optimization.

Support is expanded to cover multiple access types, including non-3GPP and wireline access, while also incorporating low-latency transport capabilities. Security is strengthened through protection of media-related information exchanged between network functions, ensuring that enhanced traffic handling can be achieved without compromising end-to-end encryption of application data.

Overall, these improvements enable more precise QoS enforcement, better handling of encrypted and multiplexed XR traffic, and more adaptive interaction between applications and the network, resulting in enhanced service quality and more efficient use of network resources.

More precisely, the following aspects are specified:

- *New indicator over NGAP to indicate support for PDU Set Information marking and UPF performing the PDU Set Information marking based on information from NG RAN.*
- *Enhancements to support PDU Set in Alternative QoS Profiles, Notification control.*
- *Enhancements to support QoS control and PDU Set identification for End-to-End encrypted XR traffic.*
- *Enhancements for PDU Set based DSCP marking.*
- *Support for additional packet filter for traffic detection and QoS Flow mapping for different media types multiplexed data flows within a single end-to-end transport connection.*
- *Support dynamic change (via user plane) in traffic characteristics (i.e., burst size, Time To Next Burst) provided by the application in the DN.*
- *Support data boosting triggered by the AF.*
- *Support L4S in untrusted/trusted (N3IWF, TNGF), wireline access (W-AGF, 5G-RG).*
- *Support PDU Set QoS in untrusted/trusted (N3IWF, TNGF), wireline access (W-AGF, 5G-RG).*
- *Support exposure of available bitrate, data rate limitation information towards the AF.*
- *Security support for protection of XRM media related information associated with end-to-end encrypted XRM traffic.*

As for OAM aspects to support XRM service, both the Network Resource Model (NRM) and the Performance Management (PM) enhancements are specified:

NRM aspects include enhancements on PreDefinedPccRule NRM for: ECN marking; for exposure of information related to XRM service based on QoSMonitoring and for PDU set based handling; QoSData NRM to support PDU Set QoS Parameters; QoS Monitoring Control NRM for exposing additional information related to XRM; Dynamic Traffic Characteristics Update; Deliver Media Related Information for Encrypted Traffic Using On-path N6 Signalling; PreDefinedPccRule to: Support Multi-modality Flows; and for UE Power Saving for XRM Services. The NRM enhancements are specified in TS 28.540 [5] and TS 28.541 [6], and the corresponding changes on YAML are updated on Forge.

The PM enhancements for the 3GPP management system include distribution of DL PDU set delay between NG-RAN and PSA UPF; Data Rate Measurements for XRM and N6 Measurements for XRM. The PM enhancement are specified in TS 28.552[7].

References

List of related CRs: select "TSG Status = Approved" in:
https://portal.3gpp.org/ChangeRequests.aspx?q=1&workitem=1060034,990052,1010032,1040032,1060027,1060035,1060036,1060037,1090012,1020051,1040074,1060024,1060038,1060039,1070021,1080016

[1] TS 23.501: " System architecture for the 5G System (5GS)"
[2] TS 23.502: " Procedures for the 5G System (5GS)"
[3] TS 23.503: " Policy and charging control framework for the 5G System (5GS); Stage 2"
[4] TS 33.501: "Security architecture and procedures for 5G System"
[5] TS 28.540: "Management and orchestration; 5G Network Resource Model (NRM); Stage 1"
[6] TS 28.541: "Management and orchestration; 5G Network Resource Model (NRM); Stage 2 and stage 3"
[7] TS 28.552: “Management and orchestration; 5G performance measurements”

12.3 XR (eXtended Reality) for NR Phase 3

1020098	XR (eXtended Reality) for NR Phase 3	NR_XR_Ph3		RP-250107	Nokia
1021098	**Core part: XR (eXtended Reality) for NR Phase 3**	NR_XR_Ph3-Core	R2	RP-250107	Benoist Sebire, Nokia
1022098	**Perf. part: XR (eXtended Reality) for NR Phase 3**	NR_XR_Ph3-Perf	R4	RP-250107	Benoist Sebire, Nokia

Summary based on the input provided by Nokia and Qualcomm in RP-252356.

This WI is a continuation of the Rel-18 work done to enhance the support of XR services in the RAN and complements the work done in SA groups. It provides enhancements addressing system capacity, efficient and effective mechanisms to meet QoS requirements and lower device power consumption, in the context of the demanding scenarios and traffic characteristics requirements of XR.

To enable the transmission and reception during some of the measurements gaps configured for RRM measurements, a measurement gap occasion may be cancelled via DCI. Once a measurement gap occasion is cancelled, the network can schedule the UE during the cancelled measurement gap occasion, and as a result, the measurement delay related to measurement objects associated with that gap is extended accordingly. In addition, the UE may provide, via UE Assistance information, a recommended gap occasion cancellation ratio per measurement gap configuration.

In uplink, the granularity of delay status reports is increased to depict the UEs buffers more accurately to the gNB, and buffered PDCP PDUs having a time left before discard crossing a threshold can be prioritised in logical channel prioritization at MAC. Furthermore, to enable faster adaptation of the uplink source rate (e.g. to handle to uplink congestion), an uplink physical-layer bit rate available to a QoS flow can be suggested by the gNB, and the UE can also request a desired one. Enhancements are also introduced in RLC to avoid unnecessary retransmissions and ensure timely retransmissions.

To enable faster adaptation of the source rate to uplink congestion, in downlink, the gNB can suggest an uplink physical-layer bit rate available to a QoS flow to the UE, based on the information indicating that the QoS Flow allows rate adaptation from the 5GC. This information is further provided from CU to DU, or from source gNB to the target gNB during the Xn-based handover preparation procedure.

Network interfaces are enhanced for the downlink support of PDCP discard, and timely RLC retransmissions. PDU Set based handling, ECN marking, Burst Arrival Time reporting and PSI Discard are all extended to NR-NR dual connectivity scenarios on the network side. Furthermore, to assist the gNB for radio resource management and scheduling in downlink, the UPF provides Data Burst Size and Time To Next Burst to the gNB. The 5GC may provide the Multi-modal Service ID (MMSID) to NG-RAN, as part of the QoS parameters of the QoS flow, when establishing and/or updating the corresponding QoS Flows, and the support for alternative PDU Set QoS parameters PSDB and PSER is added

For QoS monitoring, the gNB can inform the 5GC when the QoS is not met in DL or UL, and the available bit rate (referring to the bitrate that NG-RAN can provide) for a QoS flow can be reported for the UL and/or DL by the gNB to 5GC.

References

List of related CRs: select "TSG Status = Approved" in:
https://portal.3gpp.org/ChangeRequests.aspx?q=1&workitem=1020098,1021098,1022098

12.4 Avatar Communications in AR Calls

1070056	**Avatar Communications in AR Calls**	AvCall-MED	S4	SP-250264	Bouazizi, Imed, Qualcomm,

Summary based on the input provided by Qualcomm Inc. in SP-251147.

The work item specifies the Avatar call functionality as an extension to IMS-based AR Real-Time Communication in TS26.264 [1].

AvCall-MED adds the stage-3 pieces needed to enable avatar calls. It adopts ISO/IEC 23090-39 (ARF) for base avatars; specifies support for both 3D and 2D avatars; and defines data channel-based animation messages with support for facial, joint, and landmark animation.

The changes introduce SDP and data channel-based avatar negotiation to add avatars to an IMS call. The signalling allows for activation of the animation and avatar rendering on the sender side, in the network, or on the receiver side.

An Mbar_Management service for the Base Avatar Repository (BAR) is also defined to enable the creation and management of the user's base avatar, as well as the selection and sharing of subset of the avatar assets to be used in an Avatar call.

References

List of related CRs: select "TSG Status = Approved" in:
https://portal.3gpp.org/ChangeRequests.aspx?q=1&workitem=1070056

[1] TS26.264, IMS-based AR Real-Time Communication

12.5 Split rendering over IMS

1030003	**Split rendering over IMS**	SR_IMS	S4	SP-240492	He, Shane, Nokia Corporation

Summary based on the input provided by Nokia in SP-250982.

This Work Item has generated TS 26.567 [1], which specifies functional entities, reference points and protocols for IMS-based split rendering.

Key use cases enabled by this specification include XR services incorporating real-time and non-real-time media in industrial (e.g., for monitoring, maintenance, collaboration, and tele-operation), enterprise and educational environments; entertainment use-cases, including cloud-gaming, and shared and collaborative entertainment and productivity XR services over IMS.

TS 26.567 also specifies media codecs for delivery of split-rendered media and metadata for split-rendered content for both uplink and downlink. Split rendering metrics are specified to enable QoE metrics collection and reporting for split-rendered content. TS 26.567 also specifies procedures for split rendering session establishment, session management, adaptation, as well as other necessary procedures to support split rendering process based on network support functions.

In summary, TS 26.567 includes:
1. Functional entities and reference points for IMS-based split rendering
2. Media codecs, configuration and data transport
3. Split rendering metrics including metrics formats, metrics configuration and metrics reporting
4. Detailed procedures from split rendering session establishment, modification, adaptation, etc.
5. Metadata formats and message types.

References

List of related CRs: select "TSG Status = Approved" in:
https://portal.3gpp.org/ChangeRequests.aspx?q=1&workitem=1030003

[1] TS 26.567: “Split Rendering over IMS”.

12.6 Edge Computing Considering the Operational Needs of Service Hosting Environment

1010022	**Edge Computing Considering the Operational Needs of Service Hosting Environment**	EdgeOpNeeds	S1	SP-231037	Yuying Zhang, China Telecom

Summary based on the input provided by China Telecom in SP-260262.

This work item updates the edge computing requirements in TS 22.261 [1] by introducing support for selection and reselection of user-plane paths based on Service Hosting Environment usage information.

The stage 2 work is covered under the name “eEDGE_5GC_Ph3”.

References

List of related CRs: select "TSG Status = Approved" in:
https://portal.3gpp.org/ChangeRequests.aspx?q=1&workitem=1010022

[1] TS 22.261: "Service requirements for the 5G system; Stage 1".

12.7 Enhancement of support for Edge Computing in 5G Core network - Phase 3

1040036	**Enhancement of support for Edge Computing in 5G Core network - Phase 3**	eEDGE_5GC_Ph3		SP-241284	Changhong Shan (Intel)
1020004	*Study on Enhancement of support for Edge Computing in 5G Core network - Phase 3*	FS_eEDGE_5GC_Ph3	S2	SP-240962	Changhong Shan (Intel)
1030036	*Study on Security Aspects of Enhancement of Support for Edge Computing in 5GC phase 3*	FS_EDGE_Ph3	S3	SP-240510	Ge Yao,

1060064	**Security Aspects of Enhancement of Support for Edge Computing in 5GC phase 3**	EDGE_Ph3	S3	SP-241959	TBD
1050050	**Stage 2 of Enhancement of support for Edge Computing in 5G Core network - Phase 3**	eEDGE_5GC_Ph3	S2	SP-241505	Meghashree Dattatri Kedalagudde (Intel);
1050006	**CT aspects of Enhancement of support for Edge Computing in 5G Core network - Phase 3**	eEDGE_5GC_Ph3		CP-250016	Landais, Bruno, Nokia
1050051	CT3 aspects of Enhancement of support for Edge Computing in 5G Core network - Phase 3	eEDGE_5GC_Ph3	C3	CP-250016	Landais, Bruno, Nokia
1050052	CT4 aspects of Enhancement of support for Edge Computing in 5G Core network - Phase 3	eEDGE_5GC_Ph3	C4	CP-250016	Landais, Bruno, Nokia

Summary based on the input provided by Intel by e-mail received Fri 13/03/2026 17:13, file "Summary_1040036".

Edge Computing has been supported in the 5GS since Rel-15 [2]. While further enhancements were developed in Rel-17 and Rel-18, Rel-19 addresses remaining limitations related to deployment impact and latency-aware edge resource selection.

Rel-19 introduces enhancements that reduce the impact of Edge Computing support on central 5GC Control Plane functions. In particular, lightweight mechanisms are specified to avoid the need for extensive storage of edge network and application information in centralized functions, improving deployability while preserving existing capabilities. For the Distributed Anchor Point and Multiple PDU Session models, previously specified mechanisms continue to apply.

For session breakout models, Rel-19 enables enhanced local offloading management. The Intermediate Session Management Function (I-SMF) selection is performed by the Access and Mobility Management Function (AMF) and enhanced to support local offloading based on subscription data. Edge Application Server (EAS) discovery and selection are performed by the I-SMF, which can retrieve and manage Edge Application Server Deployment Information (EDI), be preconfigured with EAS deployment data, and select local User Plane Functions (UPFs) based on policies indicating allowed IP range(s) and/or Fully Qualified Domain Name (FQDN) patterns. Traffic influence information provided by an Application Function (AF) may also be made available to the I-SMF.

In addition, Rel-19 enables N6 delay-aware local PDU Session Anchor (PSA) UPF selection. The Session Management Function (SMF) can collect N6 delay measurements between candidate UPFs and application-side endpoints (e.g. EAS or Designated IP ranges) using existing measurement mechanisms. Taking N6 delay into account allows the selection of more suitable local UPFs and, optionally, EASs, reducing end-to-end latency for Edge Computing services.

References

List of related CRs: select "TSG Status = Approved" in:
https://portal.3gpp.org/ChangeRequests.aspx?q=1&workitem=1040036;1020004;1030036;1060064;1050050;1050006;1050051;1050052

[1] TR 23.700-49: “Study on enhancement of support for Edge Computing in 5G Core Network”
[2] TS 23.548: “5G System Enhancements for Edge Computing”

12.8 Edge Computing for Industrial Scenarios

990050	**Edge Computing for Industrial Scenarios**	EDGINDUS	S1	SP-230229	Philippe Lottin, Orange

Summary based on the input provided by Orange by e-mail received Tue 17/03/2026 12:36, file "Edge Computing for Industrial Scenarios".

5G-ACIA produced an internal report on edge computing use cases and requirements for industrial 5G networks and is currently working on a whitepaper regarding edge deployment options and architectural considerations for edge components. The report aimed to identify and understand industrial edge use-cases and their requirements concerning edge computing infrastructure and related communication requirements. The 5G-ACIA also sent a liaison statement to 3GPP to address gaps in performance KPIs and use cases and propose new service requirements for edge computing in industrial manufacturing in 3GPP specifications.

This Work Item updates TS 22.104 "Service requirements for cyber-physical control applications in vertical domains" as to:

- Add additional clarifications on privacy and security protection regarding Mobile Robots and AGV: text is added in additional information section of Mobile Robots (A 2.2.3) to clarify that having an edge cloud system enables lower latency for Mobile robots control for critical operations

• Add a new use case on "digital twin in a production quality inspection". This use case deploys telemetry data/video processing locally for immediate inspection and corrective action if needed. A new section is added.

References

List of related CRs: select "TSG Status = Approved" in:
https://portal.3gpp.org/ChangeRequests.aspx?q=1&workitem=990050

12.9 Enabling Edge Applications Phase 3

1040081	**Architecture for enabling Edge Applications Phase 3**	EDGEAPP_Ph3		SP-231160	Hyesung Kim, Samsung
1010007	**Stage 2 of enabling Edge Applications Phase 3**	EDGEAPP_Ph3	S6	SP-231160	Hyesung Kim, Samsung
1040008	**CT aspects for Enabling Edge Applications Phase 3**	EDGEAPP_Ph3		CP-241074	Narendranath Durga Tangudu, Samsung
1040058	CT1 aspects for Enabling Edge Applications Phase 3	EDGEAPP_Ph3	C1	CP-241074	Narendranath Durga Tangudu, Samsung
1040059	CT3 aspects for Enabling Edge Applications Phase 3	EDGEAPP_Ph3	C3	CP-241074	Narendranath Durga Tangudu, Samsung

Summary based on the input provided by Samsung by e-mail received Mon 30/03/2026 07:56, file "Summary_1040081".

This WI is a continuation of the Rel-18 (EDGEAPP_Ph2) to support the following aspects:

a) Enhancements to Edge Enabler Layer (EEL) to support additional scenarios for edge services via a common Edge Application Server (EAS), including:
   i) Common EAS discovery and selection: To support this, EAS service KPI and latency are considered to select a common EAS.
   ii) Service continuity when service is being provided by common EAS (e.g., mobility of some users using a common EAS): For S-EAS detected case, the S-EAS includes list of UEs for which Application Context Relocation (ACR) is required in the ACR request to indicate S-EES to perform ACR for UEs in the group as the S-EAS is overloaded and may not meet to KPIs for common EAS. The S-EES sends the ACR information notification (ACR complete) message to EEC for the identified UEs. For S-EES executed case, S-EES may detect common EAS overload situation based on notification from ADAES. S-EES identifies all UEs connected to the common EAS, and discovers T-EAS. S-EES identifies list of UEs for which T-EAS can act as common EAS. S-EES needs to inform S-EAS to initiate Application Context Transfer. ACR can be supported only some application group members by including 'list of UE IDs' in EAS discovery request for S-EAS or S-EES triggered EAS discovery request
   iii) Discovery and selection of a common EAS from federated partners
   iv) Discovery, selection and service continuity of common EAS bundle
b) Discovery, selection and service continuity for bundled EASs in federation and roaming scenarios
c) Enhancements to EEL to support additional functionalities for ENS scenarios
   i) EEL operation to support Dynamic EAS instantiation by ECSP management system in ENS scenarios
   ii) Service continuity for ENS scenarios: To support this, some EEC request (over EDGE-1) or EAS request (over EDGE-3) may be relayed between leading EES and partner EES (over EDGE-9) with necessary modification (e.g., along with the ECSP ID of the sending entity).

References

List of related CRs: select "TSG Status = Approved" in:
https://portal.3gpp.org/ChangeRequests.aspx?q=1&workitem=1040081;1010007;1040008;1040058;1040059

[1] TS 23.558: "Architecture for enabling Edge Applications"
[2] TS 23.436: "Functional architecture and information flows for Application Data Analytics Enablement Service"
[3] TS 24.558: "Enabling Edge Applications; Protocol specification"
[4] TS 29.558: "Enabling Edge Applications; Application Programming Interface (API) specification; Stage 3"
[5] TS 29.549: "Service Enabler Architecture Layer for Verticals (SEAL); Application Programming Interface (API) specification; Stage 3"

12.10 See also

"Management Data Analytics phase 3 (1060014, eMDAS_Ph3)" and "5G Advanced NRM features phase 3 (1020027, AdNRM_Ph3)"

13 Rel-19 High Power UEs (HPUE)

1040123	Rel-19 High power UE (power class 1.5 or 2) for NR intra-band CA or NR inter-band CA/DC band combinations with/without NR Supplementary Uplink (UL)	HPUE_NR_CADC_SUL_R19		RP-252540	China Telecom
1041123	**Core part: Rel-19 High power UE (power class 1.5 or 2) for NR intra-band CA or NR inter-band CA/DC band combinations with/without NR Supplementary Uplink (UL)**	HPUE_NR_CADC_SUL_R19-Core	R4	RP-252540	China Telecom
1040124	Rel-19 High power UE (power class 1.5 and 2) for NR FR1 TDD/FDD single band for handheld/FWA UEs, and high power UE operation (power class 1) for FWVM (fixed-wireless/vehicle-mounted) use cases in a single NR band	HPUE_NR_FR1_bands_R19		RP-252606	CMCC
1041124	**Core part: Rel-19 High power UE (power class 1.5 and 2) for NR FR1 TDD/FDD single band for handheld/FWA Ues, and high power UE operation (power class 2, 1.5 and 1) for FWVM (fixed-wireless/vehicle-mounted) use cases in a single NR band**	HPUE_NR_FR1_bands_R19-Core	R4	RP-252606	CMCC
1080076	**UE Conformance - Rel-19 High power UE (power class 1.5 and 2) for NR FR1 TDD/FDD single band for handheld/FWA UEs, and high power UE operation (power class 2, 1.5 and 1) for FWVM (fixed-wireless/vehicle-mounted) use cases in a single NR band**	HPUE_NR_FR1_bands_R19-UEConTest	R5	RP-252598	CMCC
1040122	Rel-19 High power UE (power class 1.5 or 2) for DC combinations of LTE band(s) and NR band(s)	HPUE_DC_LTE_NR_R19		RP-252336	China Unicom
1041122	**Core part: Rel-19 High power UE (power class 1.5 or 2) for DC combinations of LTE band(s) and NR band(s)**	HPUE_DC_LTE_NR_R19-Core	R4	RP-252336	China Unicom
1080081	**UE Conformance - Rel-19 High power UE (power class 1.5 or 2) for DC combinations of LTE band(s) and NR band(s)**	HPUE_DC_LTE_NR_R19-UEConTest	R5	RP-251512	China Unicom
1040116	Rel-19 High power UE (power class 2) and high power operation (power class 1) for fixed-wireless/vehicle-mounted use cases in a single LTE band	HPUE_LTE_bands_R19		RP-241681	Nokia
1041116	**Core part: Rel-19 High power UE (power class 2) and high power operation (power class 1) for fixed-wireless/vehicle-mounted use cases in a single LTE band**	HPUE_LTE_bands_R19-Core	R4	RP-242086	Nokia
1040128	Introduction of Power Class 2 and UE 40MHz Channel Bandwidth in NR band n28	NR_n28_PC2_40MHz_CBW		RP-243321	CMCC
1041128	**Core part: Introduction of Power Class 2 and UE 40MHz Channel Bandwidth in NR band n28**	NR_n28_PC2_40MHz_CBW-Core	R4	RP-243321	CMCC

1070062	**UE Conformance - Introduction of Power Class 2 and UE 40MHz Channel Bandwidth in NR band n28**	NR_n28_PC2_40MHz_CBW-UEConTest	R5	RP-250249	CMCC

All HPUE enhancements in this Release are band-related, i.e. no summaries provided.

These band enhanvements are:

- Rel-19 High power UE (power class 1.5 or 2) for NR intra-band CA or NR inter-band CA/DC band combinations with/without NR Supplementary Uplink (UL)
- Rel-19 High power UE (power class 1.5 and 2) for NR FR1 TDD/FDD single band for handheld/FWA UEs, and high power UE operation (power class 1) for FWVM (fixed-wireless/vehicle-mounted) use cases in a single NR band
- Introduction of Power Class 2 and UE 40MHz Channel Bandwidth in NR band n28
- Rel-19 High power UE (power class 1.5 or 2) for DC combinations of LTE band(s) and NR band(s)
- Rel-19 High power UE (power class 2) and high power operation (power class 1) for fixed-wireless/vehicle-mounted use cases in a single LTE band

14 Rel-19 RAN topology

14.1 5G NR Femto

Covered in 14.2.

14.2 Additional topological enhancements for NR

1060048	**5G NR Femto**	5G_Femto		SP-241327	Malla Reddy Sama
1020002	*Study on System aspects of 5G NR Femto*	FS_5G_Femto	S2	SP-231797	Malla Reddy Sama (NTT DOCOMO)
1040028	**System aspects of 5G NR Femto**	5G_Femto	S2	SP-241327	Malla Reddy Sama
1060002	**CT aspects of 5G NR Femto**	5G_Femto	C4	CP-250020	Dojiri, Shunsuke, NTT DOCOMO
1060049	CT3 aspects of 5G NR Femto	5G_Femto	C3	CP-250020	Dojiri, Shunsuke, NTT DOCOMO
1060050	CT4 aspects of 5G NR Femto	5G_Femto	C4	CP-250020	Dojiri, Shunsuke, NTT DOCOMO
1030028	*Study on security aspects of 5G NR Femto*	FS_5G_Femto_Sec	S3	SP-240505	Peilin LIU (ZTE)
1060062	**Security aspects of 5G NR Femto**	5G_Femto_Sec	S3	SP-241958	Liu,Peilin liu.peilin@zte.com.cn
1050125	**Additional topological enhancements for NR**	NR_WAB_5GFemto		RP-252479	NTT DOCOMO
1051125	**Core part: Additional topological enhancements for NR**	NR_WAB_5GFemto-Core	R3	RP-252479	NTT DOCOMO
1020082	*Study on additional topological enhancements for NR*	FS_NR_WAB_5GFemto	R3	RP-241264	Min, Tianyang, NTT DOCOMO

Summary based on the input provided by NTT DOCOMO, INC. and AT&T in RP-252480, then by e-mail from DoCoMo, received Tue 17/03/2026 09:02, file "Femto_Summary_SA2_CT".

This WI addresses both Wireless Access Backhaul (WAB) and 5G Femto enhancements.

5G Femto

The 5G NR Femto feature enables deployment of small 5G access nodes in customer premises (e.g., home, enterprise, campus) to improve indoor coverage, offload macro traffic, and enhance user experience. It supports plug-and-play deployment and allows access to the femto cell to be controlled based on subscriber authorization. For more details, see clause 5.50, TS 23.501.

The Network Exposure Function (NEF) northbound interface is enhanced to support Closed Access Group (CAG) information provisioning for NR Femto, with the corresponding data stored and maintained in the Unified Data Management (UDM) and Repository (UDR). For more details, see clauses 5.40, TS 29.522 and 5.6, TS 29.503.

Figure 14.2-1 depicts the deployment options of NR Femto with a locally deployed UPF close to the location of NR Femto node(s). The locally deployed UPF, which is a 5GC entity, has a direct N4/N9 interface with the 5GC and can be selected based on the edge computing functionality specified in clause 5.13, TS 23.501. The NR Femto GW may optionally be used as a concentrator for the N2 interface. The N3 interface may directly connect to a centrally deployed UPF (behind the NR Femto GW) or to a locally deployed UPF close to the NR Femto node location. The N3 interface may also traverse the NR Femto GW, in which case the NR Femto GW performs IP layer routing only. An NR Femto Security Gateway (SeGW) might also be used.

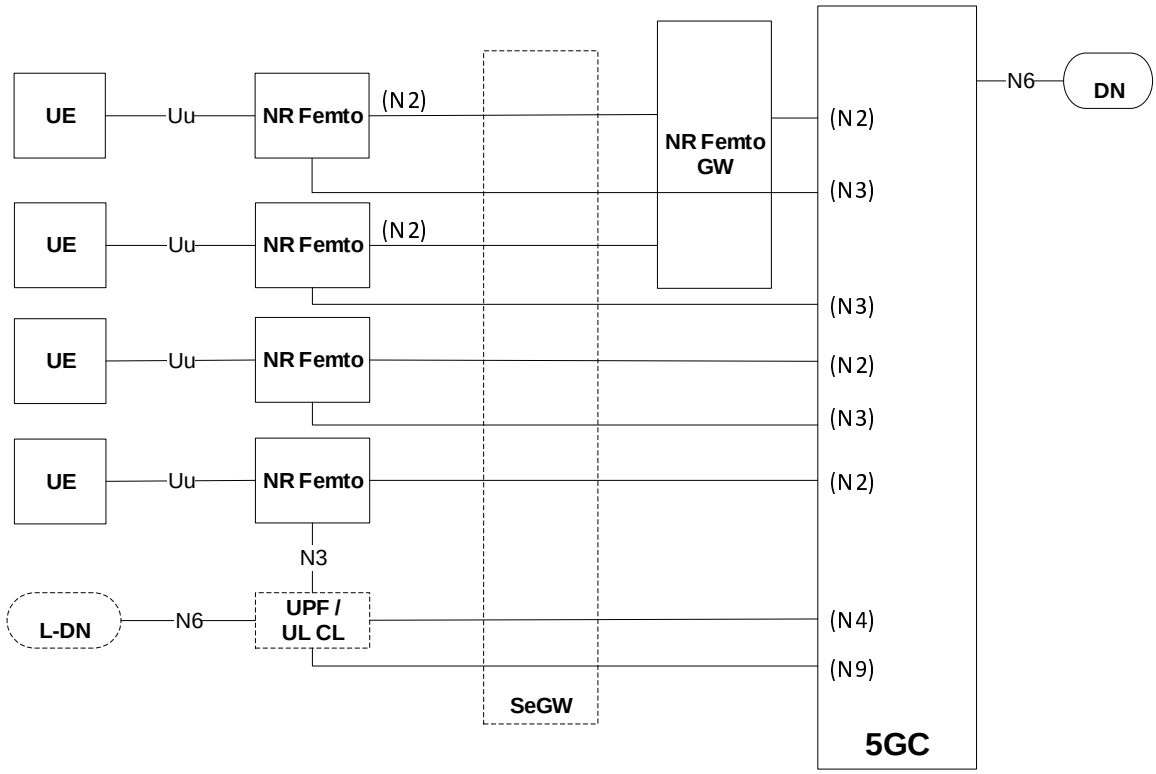


Figure 14.2-1: NR Femto deployment options for 5GS [Annex V.1, TS 23.501]

For 5G Femto, the following main functionalities have been defined:

- support NR Femto architecture with optional NR Femto GW for NG interface
- support access control for NR Femtos reusing existing CAG functionality

For WAB, corresponding stage-2 descriptions are specified in TS 38.300, TS 38.401. NG-AP related functionalities are specified in TS38.413. Xn-AP related functionalities are specified in TS38.423. Location services related functionalities are specified in TS 38.305 and TS 38.455.

For 5G Femto, corresponding stage-2 descriptions are specified in TS 38.300. NG related functionality is specified in TS 38.410. NG-AP related functionalities are specified in TS38.413.

More precisely:

- *Overall Architecture and Functional split:*

The NR Femto node may directly connect to the 5GC. The NG-RAN architecture may also deploy an NR Femto Gateway (NR Femto GW) to allow the concentration of the NG-C interface between the NR Femto node and the 5GC.

An NR Femto node hosts the same functions as a gNB, with the some additional specifics in case of connection to the NR Femto GW as specified in TS 38.300.

AMF hosts routing of handover messages and DOWNLINK RAN CONFIGURATION TRANSFER message towards NR Femto GWs based on the Selected TAI contained in these messages.

The NR Femto GW hosts relaying UE-associated NGAP messages between the AMF and the NR Femto node serving the UE and terminating non-UE associated NGAP procedures towards the NR Femto node and towards the AMF. The NR Femto GW supports TAC and PLMN ID used by the NR Femto node. The Femto GW Relays the PATH SWITCH REQUEST message towards the AMF indicated by the GUAMI of the source AMF received from the NR Femto node.

- *Access control:Cells served by an NR Femto node may be deployed as part of a PNI-NPN in order to restrict access to UEs according to the respective subscription.*

Wireless Access Backhaul (WAB)

For WAB, it defines support for a WAB node consisting of a WAB-gNB and a WAB-MT, and enables the backhauling of NG, Xn, and OAM traffic over the WAB-MT's PDU sessions. It specifies how WAB nodes are integrated into the network, how Xn connectivity is supported while avoiding unnecessary Xn connections between WAB-gNBs, and how WAB-gNB traffic is handled during mobility and changes in backhaul sessions. The work also covers authorization changes, AMF change handling for UEs served by WAB-gNBs, and UE location reporting reflecting the WAB node location. The corresponding architectural and protocol aspects are captured in the relevant stage-2 specifications.

For 5G Femto, it supports the NR Femto architecture, including the option of an NR Femto Gateway for the NG interface, and enables access control for NR Femtos by reusing existing Closed Access Group mechanisms. The associated stage-2 and NG/NG-AP specifications define the required functionality.

For WAB, the following main functionalities have been defined:

- Support of a WAB-node including a WAB-gNB and a WAB-MT.
- Support of backhauling of the WAB-gNB's NG, Xn and OAM traffic over the WAB-MT's PDU session(s).
- Support of Xn interface(s) by the WAB-gNB with the WAB-MTs serving BH RAN node and with other surrounding gNBs, including how to avoid setting up Xn between WAB-gNBs.
- Defining the behaviour of WAB-node in case the authorization status of WAB-MT and/or WAB-gNB changes.
- Network integration procedures for WAB nodes.
- Handling of WAB-gNB's traffic (including Xn, NG and OAM traffic) during WAB-node mobility, including the case where the WAB-MT's BH PDU session changes.
- Support the UE's AMF change for UEs connected to, or camped on, a WAB-gNB.
- UE's ULI that reflect the WAB node's location.

More precisely:

- *A WAB-node consists of a WAB-gNB and a WAB-MT.*
- *WAB node integration: WAB node integration procedure consists of WAB-MT setup, WAB-gNB setup, WAB-gNB initialization, NG connection setup and Xn connection setup.*
- *Configuration of WAB node:*

  When WAB node moves, the parameters that enable the WAB-gNB to select and connect to the AMF(s) that serve the UE(s), the parameters that enable the WAB-gNB to connect to, and communicate with, the OAM system, and the parameters that the WAB-gNB should broadcast, e.g., the TAC(s), the cell ID(s), the RANAC(s) may need to be updated.

  A WAB-MT obtains IP address(es) for the PDU sessions in the same manner as a legacy UE.

  The TAC/RANAC of WAB-gNB's cell is configured by the OAM. During the mobility of WAB-node, it can be reconfigured by the OAM, or it can remain unchanged in a certain geographical area.
- *NG connection management: Based on the OAM configuration, the WAB-gNB can set up NG interface with an AMF.*
- *WAB node authorization: WAB-node authorization includes authorization of the WAB-MT's backhaul support, and service authorization of the WAB-gNB, as defined in TS 23.501.*
- *User Location information for UEs served by a WAB-gNB:For UEs served by a WAB-gNB, in addition to the User Location Information (ULI), the WAB-gNB also provides the core network with Additional ULI, which includes a TAI and a NR CGI pertinent to the WAB-gNB's broadcasted PLMN/SNPN.*
- *PCI collision avoidance:In WAB-node deployments, the legacy mechanism for PCI collision avoidance can be reused. PCI space can be partitioned by allocating a range of PCIs to WAB-gNB cells.*
- *WAB-MT mobility:The WAB-MT reuses legacy mobility procedures defined for the UE.*
- *WAB-gNB mobility:Due to the WAB-gNB's mobility, the UEs moving together with the WAB-gNB may have to change AMF. For the AMF change, a new logical WAB-gNB is instantiated, which establishes NG connection(s) towards one or more new AMF(s). The new logical WAB-gNB may obtain from the OAM the configuration parameters needed to establish the connection(s) to the UE's new AMF(s), based on, e.g., WAB-node's location. The new logical WAB-gNB may shall activate one or more cell(s) with new TAC, cell ID, and PCI, which depend on the WAB-node's current location.*
- *Xn management:*

  A WAB-gNB can establish an Xn connection with the BH-gNB serving the co-located WAB-MT, and with the neighbouring gNBs. During the setup or update of its Xn connections, the WAB-gNB can include an ID of the WAB-MT, to indicate that it is a WAB-gNB. In case the peer gNB is the WAB-MT's BH-gNB, the WAB-MT ID makes the BH-gNB aware of the co-location of the WAB-MT and the WAB-gNB. The WAB-MT ID consists of the C-RNTI assigned to the WAB-MT by the BH-gNB, and the cell ID of BH-gNB´s cell serving the WAB-MT.

  Establishment of Xn connections between two WAB-gNBs can be avoided. To achieve this, the WAB-gNB may reject the Xn setup initiated by another WAB-gNB, e.g., based on the presence of the WAB-MT ID received in the XN SETUP REQUEST message.

References

List of related CRs: select "TSG Status = Approved" in:
https://portal.3gpp.org/ChangeRequests.aspx?q=1&workitem=1060048,1040028,1060002,1060049,1060050,1060062,1050125,1051125

14.3 Vehicle Mounted Relays Phase 2

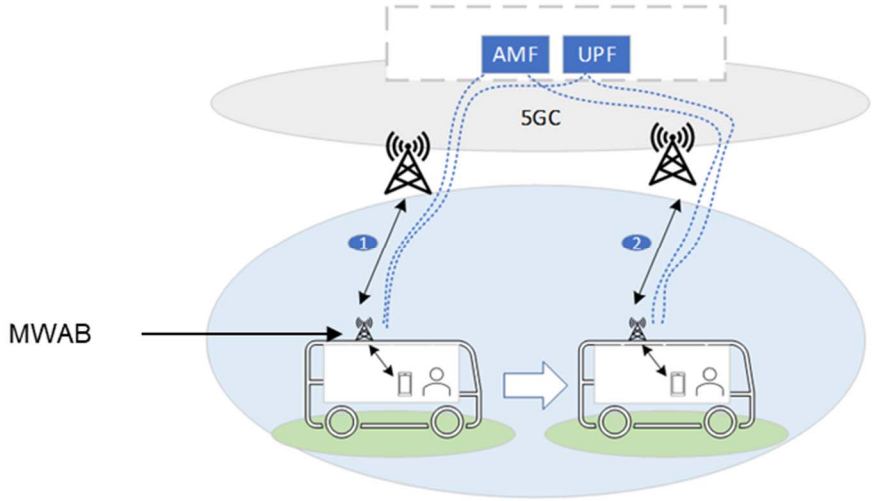
1050100	Vehicle Mounted Relays Phase 2	VMR_Ph2		SP-240989	Hong Cheng (Qualcomm)
1020067	*Study on Vehicle Mounted Relays Phase 2*	FS_VMR_Ph2	S2	SP-240493	Primary Hong Cheng (Qualcomm)

1040031	**Stage 2 of Vehicle Mounted Relays Phase 2**	VMR_Ph2	S2	SP-240989	Hong Cheng (Qualcomm)
1050026	**CT aspects of Vehicle Mounted Relays Phase 2**	VMR_Ph2		CP-251221	Kim, Sunghoon, Qualcomm Incorporated
1050101	CT1 aspects of Vehicle Mounted Relays Phase 2	VMR_Ph2	C1	CP-251221	Kim, Sunghoon, Qualcomm Incorporated
1050102	CT4 aspects of Vehicle Mounted Relays Phase 2	VMR_Ph2	C4	CP-251221	Kim, Sunghoon, Qualcomm Incorporated

Summary based on the input provided by Qualcomm Incorporated in SP-251360

Vehicle Mounted Relay Phase 2 (VMR_Ph2) specified a mobile relay solution that allows a Mobile gNB with Wireless Access Backhauling (MWAB) to provide NR access link to UEs in proximity and connects to the 5GC serving the UE through an IP connectivity provided by a Backhaul PDU session(s). The use cases and stage-1 service requirements are captured in TS 22.261 [1], and the corresponding RAN enhancements are introduced via the RAN Work Item on Additional Topological Enhancements for NR. The “MWAB” is equivalent to the term “WAB” used in RAN specifications.

UEs continuously served by a MWAB (e.g. inside the vehicle and/or in its vicinity), and this MWAB-gNB is moving around within a limited geographical area while keeping connecting with the same 5GC nodes (e.g. AMF and UPF) (TR 23.700-06 scenario A: mobility within the same 5GC node)



From system architecture perspective, the MWAB consists of a gNB component (MWAB-gNB) and a UE component (MWAB-UE). The MWAB-gNB is based on the gNB functionality specified in TS 38.300 [2] and TS 38.401 [3]. The MWAB may be mounted on a moving vehicle and may serve UEs inside or outside the vehicle.

The MWAB-UE establishes the IP connectivity for the backhaul links for the MWAB-gNB, via NR Uu, using the existing registration procedure and PDU session establishment procedure, specified in TS 23.501[4] and TS 23.502 [5]. The backhaul links are between the MWAB-gNB and entities of the network (e.g. AMF, UPF, other gNBs and OAM server) that a MWAB-gNB cell serves. The IP connectivity provided by the MWAB-UE may be either via the same PLMN/SNPN that the MWAB-gNB serves or a different PLMN/SNPN, depending on the MWAB-UE PLMN selection mechanism as specified in TS 23.122 [6]. A high level system architecture is shown in Figure 1.

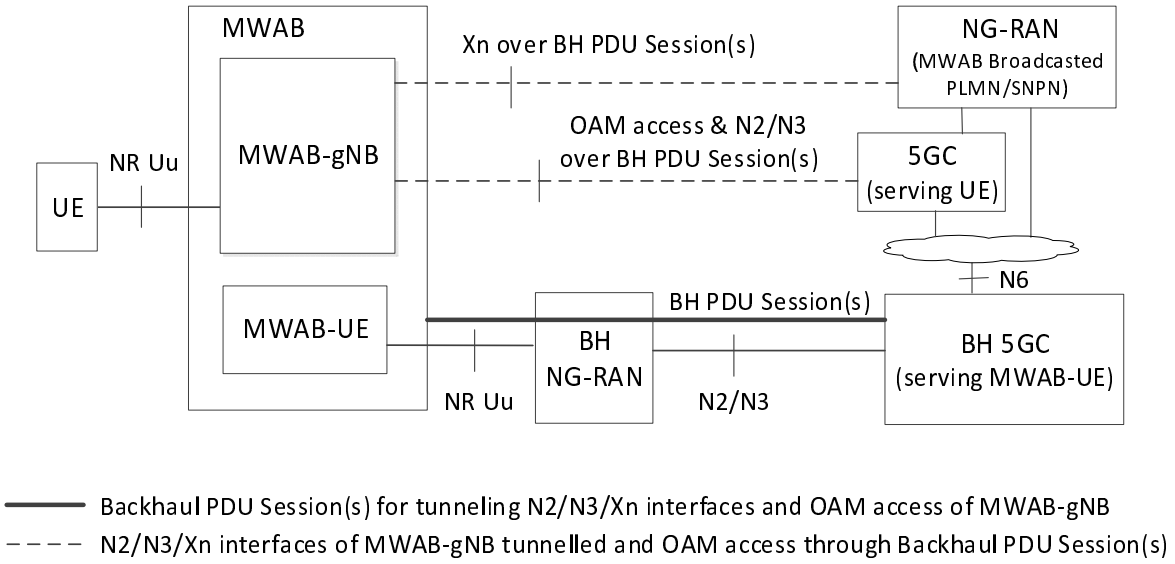


Figure 1: MWAB architecture for 5GS

MWAB operation supports both PLMN and SNPN cases. The UEs served by the MWAB are not aware of the network serving the MWAB-UE. MWAB-UE has a single NR Uu hop to the BH NG-RAN, using either TN or NTN access technology. The NR Uu access link between the MWAB-gNB and the served UE(s) does not use NTN access technology. 5G MOCN can be supported by the MWAB-gNB. Different BH PDU Sessions may be used for MWAB-gNB N2, N3, Xn interfaces and OAM access, based on configuration from OAM of the MWAB Broadcasted PLMN/SNPN.

Configuration of the MWAB

The configuration of the MWAB-UE can be UE Local Configuration, or URSP rules as defined in TS 23.503 [7], for the determination of the proper PDU session parameters, e.g. the DNN, S-NSSAI, the SSC Mode, etc. The URSP rules can be preconfigured or provisioned to the MWAB-UE by its HPLMN using UE Policy delivery as described in TS 23.502 [5].

The MWAB-gNB's configuration is managed by the OAM of the MWAB Broadcasted PLMN/SNPN and the address (or FQDN) of the OAM server per MWAB Broadcasted PLMN/SNPN may be configured at the MWAB-gNB. The OAM server of the MWAB Broadcasted PLMN/SNPN provides (re-)configuration parameters considering location of MWAB, the N2, N3 and Xn interface management, activating/deactivating the MWAB-gNB operation and to assist the MWAB-gNB providing traffic descriptor information used by MWAB-UE for the BH PDU Session(s) management via URSP processing.

Authorization of the MWAB

The MWAB-UE is authorized based on its subscription information which includes dedicated S-NSSAI(s) and DNN for MWAB operation. The MWAB-UE authorization supports time-based or location-based control. For time-based control, existing time-based network slice subscription control can be reused; and for location-based control, existing mechanism, e.g. service area restriction, LADN based control can be reused. The MWAB-UE may be connected to the BH PLMN/SNPN for other services, if allowed by the subscription data, even if the S-NSSAIs and DNNs for Backhaul PDU sessions are no longer authorized.

The authorization of the MWAB-gNB is managed by the OAM system of the MWAB-Broadcasted PLMN/SNPN as specified in TS 38.401 [3]. The MWAB-gNB establishes a secure and trusted connection to the OAM server using the IP connectivity provided by the MWAB-UE via the Backhaul PDU Session(s).

Support of Additional ULI

When a UE is served by a cell of MWAB-gNB, the MWAB-gNB shall also provide the Additional User Location Information (Additional ULI) that reflects the location of the MWAB-UE to the AMF when it sends the User Location Information (ULI) over N2 messages. In the Additional ULI, the MWAB-gNB provides the TAI/NR CGI of the MWAB Broadcasted PLMN/SNPN with which the UE is registered as specified in TS 38.401 [3]. The MWAB-gNB indicates the Additional ULI to AMF(s) it connects to by NG Setup and RAN Configuration Update procedures. When

MWAB Broadcasted PLMN/SNPN is different from the BH PLMN, or when the MWAB-UE is using the NTN access technology, MWAB-gNB formulates the Additional ULI as specified in TS 38.401 [3] based on MWABs geo-location.

If serving cell change report is required by AMF, the MWAB-gNB shall report Additional ULI (and ULI) for the UE even when only serving cell of the MWAB-UE changes. The AMF may consider the 'Additional ULI' when it determines UE location and manages the UE location related functions (e.g. Mobility Restrictions). Based on operator policy, the AMF may send the Additional ULI when the AMF provides user location information it has received over the N2 interface to a NF, e.g. the LMF.

Mobility support of MWAB

No MWAB-specific enhancement is needed for a UE mobility to and from a MWAB-gNB, irrespective of whether the MWAB is static or the MWAB is moving.

The mobility of the MWAB-UE reuses the exiting UE mobility procedures specified for normal UE. During the MWAB-UE mobility, a MWAB-gNB can obtain or apply a new configuration from the OAM server. If the MWAB-gNB is configured with a new AMF, the MWAB-gNB (i.e. the new logical gNB) sets up the N2 connection with the new AMF and the old logical gNB releases the N2 connection with any old AMF. The MWAB-UE may establish new or modify existing BH PDU Sessions for the corresponding N2, N3 and Xn interfaces. The MWAB-gNB updates the TNL association to use the new TNL information associated with the new BH PDU Session(s) as needed.

To prevent handover of a MWAB-UE towards a target MWAB-gNB, the target MWAB-gNB (i.e. during Xn handover or during N2 HO after target AMF slice control) fails the handover as specified in TS 38.401 [3] based on the dedicated slices support at the target MWAB-gNB.

Control of UE access to MWAB

If the MWAB-gNB is serving a PNI-NPN, the CAG mechanism is used. Time validity information can be provided to the UE together with the CAG identifier for accessing to MWAB in order to control that UE not accessing the MWAB cell outside of the time validity information. If the MWAB-gNB is serving an SNPN, the SNPN control mechanism is used. The MWAB-gNB and the 5GC can also reuse other existing mechanisms e.g. forbidden Tracking Area, UAC, to manage the UE's access to an MWAB-gNB cell, as specified in TS 24.501 [10].

Location service and regulatory service support

Location Service supports network-based positioning of a target UE when MWAB(s) are involved, as specified in TS 23.273 [8]. Based on the OAM configurations, an MWAB may support emergency services for UEs connected to the MWAB-gNB. The support of PWS functionality with MWAB is described in TS 23.041 [9].

References

List of related CRs: select "TSG Status = Approved" in:
https://portal.3gpp.org/ChangeRequests.aspx?q=1&workitem=1050100,1020067,1040031,1050026,1050101,1050102

[1] TS 22.261 “5G service requirements; Stage-1”
[2] TS 38.300 “NR; NR and NG-RAN Overall description; Stage-2”
[3] TS 38.401 "NG-RAN Architecture description".
[4] TS 23.501 “System architecture for the 5G System (5GS)”
[5] TS 23.502 “Procedures for the 5G System (5GS)
[6] TS 23.122 "Non-Access-Stratum (NAS) functions related to Mobile Station in idle mode".
[7] TS 23.503 "Policy and Charging Control Framework for the 5G System".
[8] TS 23.273 "5G System (5GS) Location Services (LCS); Stage 2".
[9] TS 23.041 "Public Warning System".
[10] TS 24.501 "Non-Access-Stratum (NAS) protocol for 5G System (5GS); Stage 3"

15 Rel-19 Sidelink, Proximity

15.1 NR sidelink multi-hop relay

1040114	NR sidelink multi-hop relay	NR_SL_relay_multihop		RP-250188	LG Electronics

1041114	**Core part: NR sidelink multi-hop relay**	NR_SL_relay_multihop-Core	R2	RP-250188	LG Electronics

Summary based on the input provided by LG Electronics Inc., InterDigital in RP-251942

NOTE: the document mentioned above has been significantly summarized to fit the format of this TR.

This WI extends the single-hop relay solutions defined in Releases 17 and 18 by specifying multi-hop Layer-2 UE-to-Network relay via sidelink relay UEs, enabling indirect network connectivity for remote UEs through relay paths involving up to two additional hops and covering aspects such as protocol architecture, relay discovery and selection, control-plane procedures, and service continuity.

Up to two additional relay hops are supported beyond the Rel-17 single-hop U2N relay, starting from one additional hop and extending to two, forming paths such as remote UE → first relay UE → second relay UE → last relay UE → gNB .

Sidelink relay enables 5G ProSe U2N Relay function for providing single or multi-hop connectivity to the network. Each unicast link is established between a Layer-2 U2N Relay UE and a U2N Remote UE or between relay UEs in multi-hop scenarios. Traffic to the NG-RAN for remote UEs via relay UEs is separated into different Uu RLC channels . An intermediate relay UE acts both as a relay and a remote UE and can simultaneously carry its own traffic .

Protocol Architecture

The user plane and control plane protocol stacks include the SRAP sublayer above RLC for both PC5 and Uu interfaces. The Uu SDAP, PDCP, and RRC terminate between the L2 U2N Remote UE and gNB, while SRAP, RLC, MAC, and PHY terminate at each hop. The SRAP sublayer over PC5 is used for bearer mapping but is not present for certain control messages like BCCH and PCCH on PC5 hops. SRAP headers are selectively present depending on message type and hop .

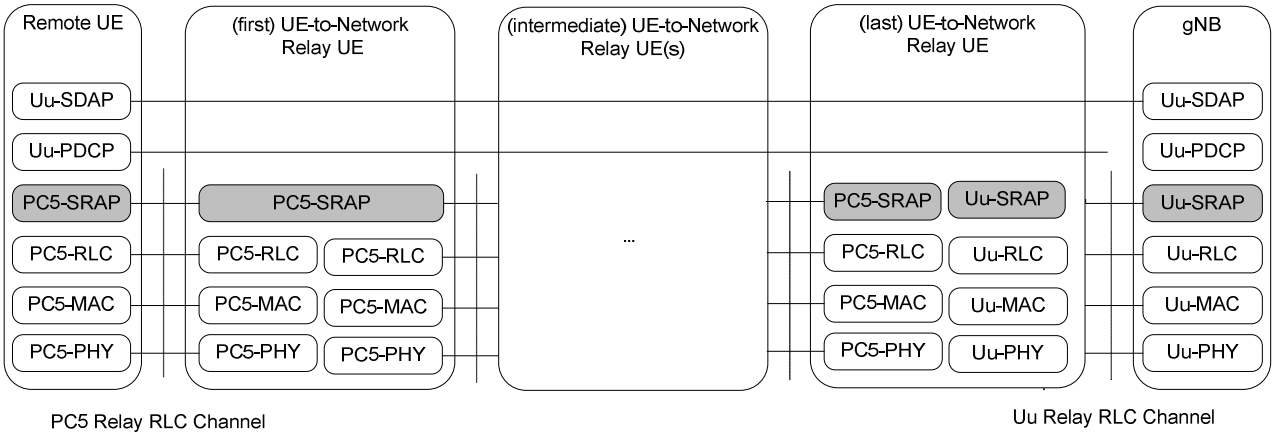


Figure 1: User plane protocol stack for multi-hop L2 UE-to-Network Relay

The Cotrol Plane has the same protocol stack except that "Ur-RRC" is used instead of "Uu-SDAP" shown above.

For uplink, the Uu/PC5 SRAP at the relay UE maps end-to-end Uu Radio Bearers of remote UEs to egress Uu/PC5 Relay RLC channels, multiplexing multiple bearers and UEs over the same channel. It includes identity information for correlating packets at the gNB. The PC5 SRAP at the remote UE supports similar bearer mapping .

For downlink, the gNB maps end-to-end Uu Radio Bearers into Uu Relay RLC channels. Relay UEs perform bearer mapping and multiplexing over Uu/PC5 interfaces and include identity information for correct bearer identification. The PC5 SRAP at the remote UE correlates received packets to the correct PDCP entity .

Relay Discovery: Each U2N Relay UE maintains a PC5 connection with its child UEs, and each intermediate or remote UE maintains a single PC5 connection with its parent UE. Relay discovery configuration can be provided by the network via broadcast or dedicated signaling, or by pre-configuration. Candidate intermediate relay UEs may forward discovery messages based on coverage and connection status using pre-configuration or SIB12/dedicated configuration. Remote UEs connected via single or multi-hop relays use resource allocation mode 2 for discovery message transmission .

Relay Selection/Reselection: Remote UEs perform PC5 radio measurements to select or reselect relays, using SD-RSRP measurements when no unicast PC5 connection exists. Relay UEs in RRCIDLE or RRCINACTIVE use PC5-

RRC messages to inform connected child UEs of cell changes or failures, triggering potential link release or re-establishment. Different discovery models (A and B) define how intermediate relays forward or respond to discovery messages based on PC5 link quality thresholds .

RRC Connection Management: The multi-hop connection establishment involves discovery and PC5-RRC connection setup between remote and relay UEs. The remote UE sends its initial RRC setup request via the first relay UE, which may establish its own relay connection if not already connected. This procedure cascades up to the last relay UE and then to the gNB. The gNB configures SRB0 relaying PC5 Relay RLC channels, and messages are forwarded through the relay chain. Security procedures and bearer setups follow, with relaying channels established over Uu and PC5 interfaces.

Radio Link Failure: Relay UEs declare Uu/PC5 radio link failure per existing criteria and may send PC5-RRC messages to connected child UEs to trigger RRC re-establishment or PC5 link release. Remote UEs may also trigger RRC re-establishment upon detecting PC5 RLF or link release .

System Information: Intermediate relay UEs acquire requested SIBs for child UEs or themselves, possibly over Uu if in the same cell as the last relay UE. Remote and intermediate UEs can receive system information from parent UEs after PC5 connection. On-demand SIB acquisition is supported via relay UEs using PC5-RRC messages .

Paging: Paging monitoring by intermediate relay UEs in RRCIDLE/INACTIVE is possible if they share the cell with the last relay UE. Relay UEs in RRCCONNECTED can monitor paging occasions for connected remote or downstream UEs or receive dedicated paging messages from the gNB containing remote UE IDs. Remote UEs provide identifiers and DRX cycle information to relay UEs for paging purposes. Relay UEs notify the gNB of downstream UE information and deliver relevant paging records to remote or child UEs .

Service Continuity for Multi-hop Relay

The WI specifies intra-gNB service continuity scenarios for multi-hop relay, including: Multi-hop indirect to direct path switching; Multi-hop indirect to single-hop indirect path switching; Direct to multi-hop indirect path switching; Single-hop indirect to multi-hop indirect path switching

Target relay UEs in multi-hop paths are assumed to be in RRC_CONNECTED, while single-hop targets may be in any RRCstate .

Switching from Multi-hop Indirect to Direct Path: The procedure involves measurement reporting by the remote UE, gNB decision and RRC reconfiguration to switch to a direct Uu path. The remote UE synchronizes and performs random access to the gNB directly, followed by completion messages and reconfiguration of relay UEs. PC5 unicast link release timing is implementation-dependent. Data path switching and PDCP re-establishment ensure lossless delivery during the switch .

Switching from Direct to Indirect Path: The remote UE reports candidate relay UEs and measurements. The gNB selects a target relay UE and sends reconfiguration messages to both the target relay and remote UEs. The remote UE establishes PC5-RRC with the target relay UE and completes the path switch, switching data transmission from direct to indirect path .

Switching Between Multi/single-hop Indirect Paths: The gNB can select relay UEs in any RRC state for path switching between multi-hop and single-hop indirect paths. The remote UE reports candidate relays and sidelink measurements. The gNB sends reconfiguration messages to the target relay UE and the remote UE, which establishes PC5-RRC with the target relay and completes the path switch. The gNB reconfigures the source relay UEs and link release timing is UE-implementation dependent. Data path switching occurs after reconfiguration. If the target relay UE is in RRCIDLE or RRCINACTIVE, the remote UE establishes the PC5 link and triggers the relay UE to enter RRC_CONNECTED.

References

List of related CRs: select "TSG Status = Approved" in:
https://portal.3gpp.org/ChangeRequests.aspx?q=1&workitem=1040114,1041114

[1] RP-250188, Revised WID on NR sidelink multi-hop relay
[2] RP-251941, Status report for WI on NR sidelink multi-hop relay

15.2 UE-to-UE multi-hop relay

1000025	UE-to-UE multi-hop relay	UEMHopRelay	S1	SP-230521	Lipford, Mark, FirstNet Authority,

Summary based on the input provided by FirstNet in SP-251512

This Work Item updates the current mission critical requirements to support "off-network device-to-device communications" using multiple relays to extend the coverage, resulting .in one CRs for TS22.261 and another one for TS22.280

References

List of related CRs: select "TSG Status = Approved" in:
https://portal.3gpp.org/ChangeRequests.aspx?q=1&workitem=1000025

[1] TS 22.261: “Service Requirements for the 5Gsystem”
[2] TS22.280:”Mission Critical Services Common Requirements (MCCoRe)”

15.3 NR Sidelink: Intra-band Carrier Aggregation in ITS band

1030084	NR Sidelink: Intra-band Carrier Aggregation in ITS band	NR_SL_intraB_CA_ITS		RP-250997	OPPO
1031084	Core part: NR Sidelink: Intra-band Carrier Aggregation in ITS band	NR_SL_intraB_CA_ITS-Core	R4	RP-250997	OPPO

Summary based on the input provided by Oppo in RP-250999.

This WI introduces UE RF requirements for sidelink intra-band carrier aggregation in ITS spectrum, covering both intra-band contiguous and non-contiguous CA, by defining power class 2 and power class 3 requirements: while Rel-18 already specified PC3 intra-band contiguous CA, Rel-19 further enhances the specifications to include PC2 intra-band contiguous CA as well as PC2 and PC3 intra-band non-contiguous CA, in response to industry requests.

Intra-band non-contiguous CA in ITS band:

From Rel-16 to Rel-18, RAN4 has defined single carrier requirement, NR/SL con-current operation requirements, and PC3 intra-band contiguous CA requirements in RAN4. Intra-band non-contiguous sidelink CA is also an important scenario, but requirements were not defined.

Based on the EU regulation, 10MHz is defined for each channel when operating in the ITS spectrum. Hence larger channel bandwidth for single carrier as 20/30/40MHz which have been defined in previous releases cannot be used. To account for car companies’ request for higher data rate, intra-band non-contiguous CA with two 10MHz carriers can be utilized to increase transmission bandwidth while still comply with the EU regulation. Besides, 10+20MHz is also introduced to account for other regions’ demand.

When defining requirements for intra-band non-contiguous CA, the TX and RX requirements have been checked one by one based on the NR intra-band non-contiguous CA requirements as well as the sidelink single carrier requirements. Based on the TR38.787, most of the NR CA requirements can be reused. The newly defined MPR and A-MPR requirements for power class 2 and power class 3 are based on simulation results considering different PA and LO architectures. The simulation results are captured in TR38.787.

Intra-band contiguous CA in ITS band:

The power class 3 intra-band contiguous CA requirements have already been defined in Rel-18. Higher power class as power class 2 means larger uplink coverage which is a basic demand from industry. The power class 2 requirements are fully based on the power class 3 requirements for intra-band contiguous CA. The MPR and A-MPR for power class 2 have been simulated based on both 2PA and 1PA architecture and the simulation results are captured in TR38.787. The aggregated channel bandwidth has been agreed as 70MHz to fully utilize the sidelink carriers.

Both Tx and Rx requirements were specified for this feature in this WI.

Tx requirements: Since the PC3 intra-band contiguous CA requirements has been introduced with suffix A under the sidelink requirement with suffix E, further PC2 intra-band CA and intra-band non-contiguous CA TX requirements are captured in the same sub-clause. A new NS value NS_33 has been introduced for sidelink CA and corresponding requirements are specified.

Rx requirements: Similarly, with the PC3 intra-band contiguous CA requirements has been introduced with suffix A under the sidelink requirement with suffix E, further PC2 intra-band CA and intra-band non-contiguous CA RX requirements are captured in the same sub-clause

References

List of related CRs: select "TSG Status = Approved" in:
https://portal.3gpp.org/ChangeRequests.aspx?q=1&workitem=1030084,1031084

[1] RP-240840, New WID on NR Sidelink: Intra-band Carrier Aggregation in ITS band, Moderator (Apple), RAN#103, Mar 2024.
[2] R4-2507365, Big CR for SL intra-band CA, LG Electronics, Oppo, RAN4#115, May

15.4 Charging Aspects of Ranging and Sidelink Positioning

1040013	**Charging Aspects of Ranging and Sidelink Positioning**	Ranging_SL_CH	S5	SP-241002	Zhiwei Mo, China Telecom,

Summary based on the input provided by China Telecom by e-mail received Thu 12/03/2026 10:28, file "Summary_for_Ranging_SL_CH".

This Work Item specifies the charging aspects of Ranging and Sidelink Positioning. TR 28.845 [1] has identified the business roles, potential charging scenarios and requirements, and developed and evaluated the potential solutions. This Work Item is the normative work Rel-19 based on the recommendations.

Based on the conclusions documented in the TR 28.845 [1], the charging aspects of Ranging and Sidelink Positioning is specified in TS 32.240 [2], TS 32.271 [3], TS 32.290 [4], TS 32.254 [5], TS 32.291 [6] and TS 32.298 [7] with the following aspects:

- Support of architecture for converged charging for Ranging and Sidelink Positioning
- Support of converged charging for UE positioning assisted by Sidelink Positioning and involving 5GC
- Support of converged charging for Ranging/SL Positioning service exposure.

References

List of related CRs: select "TSG Status = Approved" in:
https://portal.3gpp.org/ChangeRequests.aspx?q=1&workitem=1040013

[1] TS 28.845: "Study on Charging Aspects of Ranging and Sidelink Positioning".
[2] TS 32.240: "Telecommunication management; Charging management; Charging Architecture and Principles".
[3] TS 32.271: "Telecommunication management; Charging management; Location Services (LCS) charging".
[4] TS 32.290: "Telecommunication management; Charging management; 5G system; Services, operations and procedures of charging using Service Based Interface (SBI)".
[5] TS 32.254: "Telecommunication management; Charging management; Exposure function Northbound Application Program Interfaces (APIs) charging".
[6] TS 32.291: "Telecommunication management; Charging management; 5G system; Charging service, stage 3".
[7] TS 32.298: "Telecommunication management; Charging management; Charging Data Record (CDR) parameter description".

15.5 Proximity-based Services in 5GS Phase 3

1050115	**Proximity-based Services in 5GS Phase 3**	5G_ProSe_Ph3		SP-231037	Hugh Shieh (AT&T)
1020066	*Study on Proximity-based Services in 5GS Phase 3*	FS_5G_ProSe_Ph3	S2	SP-231798	Hugh Shieh (AT&T)
1040030	**Proximity-based Services in 5GS Phase 3**	5G_ProSe_Ph3	S2	SP-240988	Nassima Toumi (KPN N.V.)
1050023	**CT aspects of Proximity-based Services in 5GS Phase 3**	5G_ProSe_Ph3		CP-243266	Zhao, Xiaoxue (CATT)

1050038	CT1 aspects of Proximity-based Services in 5GS Phase 3	5G_ProSe_Ph3	C1	CP-243266	Zhao, Xiaoxue (CATT)
1050116	CT3 aspects of Proximity-based Services in 5GS Phase 3	5G_ProSe_Ph3	C3	CP-243266	Zhao, Xiaoxue (CATT)
1050039	CT4 aspects of Proximity-based Services in 5GS Phase 3	5G_ProSe_Ph3	C4	CP-243266	Zhao, Xiaoxue (CATT)
1050040	CT6 aspects of Proximity-based Services in 5GS Phase 3	5G_ProSe_Ph3	C6	CP-243266	Zhao, Xiaoxue (CATT)
1060057	**Security Aspects of Proximity Based Services in 5GS Phase 3**	5G_ProSe_Sec_Ph3	S3	SP-241956	Primary Ao Lei (Huawei)
1050030	**Charging for 5G ProSe Ph3**	5G_ProSe_Ph3-CH	S5	SP-241377	Xiaoqian Jia (China Telecom)
1030034	*Study on Security Aspects of Enhancement for Proximity Based Services in 5GS Phase 3*	FS_5G_ProSe_Ph3_SEC	S3	SP-240513	Lei, Ao (Huawei)

Summary based on the input provided by AT&T by e-mail received Thu 19/03/2026 22:59, file "Summary_1050115" for the first paragraph, and by KPN/TNO (e-mail received Tue 31/03/2026 16:05) for all the rest of the section.

Phase 3 of Proximity-based Services (ProSe) in the 5G System introduces new requirements, captured in TS 22.278 and TS 22.261, including **multi-hop support** for Layer-2 and Layer-3 UE-to-UE and UE-to-Network relays, as well as multi-path transmission via different UE-to-Network relays. To address these requirements, Release-19 introduces targeted architectural enhancements that strengthen Layer-2 and Layer-3 multi-hop relay support and enable the corresponding ProSe service capabilities in 5GS. The necessary system-level enhancements and dependencies across the Radio Access Network, security, core network, and charging aspects are incorporated through the relevant specifications.

In previous releases, 5G systems were enhanced to support Proximity Services (ProSe), where Release 17 introduced single-hop UE-to-Network Relay, and Release 18 added support for single-hop UE-to-UE Relay. In Release 19, further enhancements introduced support for Multi-hop relay, which covers scenarios where the path includes more than one relay. Study and normative work outcomes for ProSe_Ph3 were captured in TR 23.700-03 [1] and TS 23.304 [2] respectively, where the work was concluded by adding solutions to support Multi-hop relay for Layer-3 UE-to-Network and Layer-3 UE-to-UE, leaving Layer-2 UE-to-Network support to be added based on alignment with RAN work.

Security aspects for Layer 3 multi-hop UE-to-UE Relays and for multi-hop UE-to-Network Relays have also been addressed in TS 33.503 [3]. Charging mechanisms to support Layer 3 multi-hop UE-to-UE Relays and for multi-hop UE-to-Network Relays have been defined in TS 32.277 [4], while the corresponding Stage 3 is specified in TS 24.554 [5], TS 24.555 [6], and TS 29.559 [7].

Multi-hop UE-to-Network Relay Support

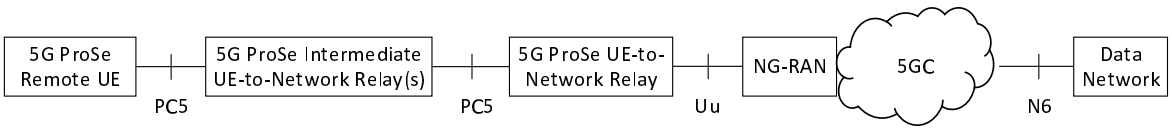


Figure 1: Reference architecture for 5G ProSe Multi-hop Layer-3 UE-to-Network Relay

For discovery and link setup and management, different approaches are supported for Model A and Model B discovery. In Model A, the Remote UE selects a directly connected Intermediate Relay based on discovery announcements, with each Intermediate Relay subsequently selecting its parent relay until the UE-to-Network Relay is reached and the full path is established. In Model B, the Remote UE selects both the UE-to-Network Relay and the complete end to end relay path, which is conveyed via Relay Discovery Response messages containing an ordered list of relay identifiers where each relay includes its own ID. Relay path selection may consider PC5 signal strength, hop count, the maximum hop limit for the RSC, or cumulative QoS metrics.

For multi-hop UE-to-Network relay operation, the 5G ProSe Direct Communication procedure defined in TS 23.304 is reused to establish Layer 2 links between each pair of UEs along the relay path. IP address allocation is performed by the UE to Network relay, where the Intermediate relays act as DHCP proxies or transparently relay IP allocation signalling between the Remote UE and the UE to Network relay.

End to end QoS management also follows the principles defined for single hop Layer 3 UE to Network relays, with enhancements to support QoS splitting across multiple PC5 hops based on hop count. Each relay derives per-hop QoS

constraints from the end-to-end QoS requirements and forwards the relevant QoS information along the path. QoS flow setup may be initiated by the network or by the 5G ProSe Layer 3 Remote UE during link establishment or modification.

In case of relay re-selection, in Model A, intermediate relays transparently repair the path unless a new root relay is needed, while in Model B the Remote UE is informed and reselects the entire end-to-end path through a new discovery procedure. If the same UE-to-Network Relay is retained, the Remote UE's IP address can be reused; otherwise, links and associations are gradually released.

Layer-3 multi-hop UE-to-UE Relays

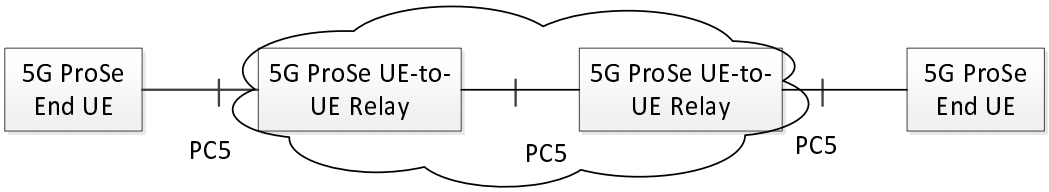


Figure 2: Reference architecture for Layer-3 5G ProSe Multi-hop UE-to-UE Relay

For UE-to-UE relay, it was agreed to use the MANET protocol for PDU sessions of IP type, while for PDU session type Ethernet and Unstructured, the support for UE-to-UE relay is based on the extension of Release 18 methods.

For IP PDU type, multi-hop UE-to-UE Relay discovery only needs to discover the presence of the 5G ProSe Multi-hop UE-to-UE Relay. The discovery of the target 5G ProSe End UEs are performed via DNS queries after establishing a link with the 5G ProSe Multi-hop UE-to-UE Relay(s). This procedure is also used by Multi-hop UE-to-UE Relay(s) among themselves in order to join or create the MANET network.

For Ethernet and unstructured PDU sessions, similar to multi-hop UE-to-Network Relay discovery with Model B, UE-to-UE relay discovery announcements or solicitations include relay IDs to build loop-free end-to-end paths, where each UE-to-UE relay extends the path information in the discovery message with its own information before forwarding it to the next UE to enable end-to-end path selection.

QoS support for non-IP PDU sessions is specified, where each relay splits and propagates the remaining QoS budget to the next hop while ensuring the previous hop's requirements are satisfied, allowing accumulated QoS information to be considered during link establishment or modification to ensure end-to-end QoS.

Relay re-selection involves discovering a new multi-hop path and exchanging link modification messages carrying the updated path information, after which traffic is switched to the new path while reusing existing identifiers such as IP or MAC addresses.

References

List of related CRs: select "TSG Status = Approved" in:
https://portal.3gpp.org/ChangeRequests.aspx?q=1&workitem=1050115;1020066;1040030;1050023;1050038;1050116;1050039;1050040;1060057;1050030

[1] TR 23.700-03: "Study on system enhancement for Proximity based Services (ProSe) in the 5G System (5GS); Phase 3".
[2] TS 23.304: "Proximity based Services (ProSe) in the 5G System (5GS)".
[3] TS 33.503: "Security Aspects of Proximity based Services (ProSe) in the 5G System (5GS)".
[4] TS 32.277: "Telecommunication management; Charging management; Proximity-based Services (ProSe) charging".
[5] TS 24.554: "Proximity based services (ProSe) in 5G system (5GS) protocol aspects; Stage 3".
[6] TS 24.555: "Proximity based services (ProSe) in 5G system (5GS); User Equipment (UE) policies; Stage 3".
[7] TS 29.559: "5G System; 5G ProSe Key Management Services; Stage 3".

15.6 See also

ProSe support in NPN; Multi-path relay (no summary)

16 NR and LTE Dual Connectivity (DC)

16.1 UE RF enhancements for NR FR1/FR2 and EN-DC, Phase 4

1030077	UE RF enhancements for NR FR1/FR2 and EN-DC, Phase 4	NR_ENDC_RF_Ph4		RP-251816	Huawei
1031077	**Core part: UE RF enhancements for NR FR1/FR2 and EN-DC, Phase 4**	NR_ENDC_RF_Ph4-Core	R4	RP-251816	Huawei
1032077	**Perf. part: UE RF enhancements for NR FR1/FR2 and EN-DC, Phase 4**	NR_ENDC_RF_Ph4-Perf	R4	RP-251816	Huawei

Summary based on the input provided by Huawei, HiSilicon and AT&T in RP-252732.

This Work Item develops enhancements for UE RF operation in FR1 and FR2, as well as EN-DC, introducing new RF requirements together with corresponding feature groups and functionalities. Two main aspects are covered: mostly power handling but also spatial aspects (MIMO).

For **High Power UE (HPUE)** operation in carrier aggregation and dual connectivity within terrestrial networks, requirements are specified for Power Class 1.5 UEs supporting NR TDD intra-band uplink contiguous and non-contiguous CA with two transmit chains. In intra-band contiguous CA, uplink MIMO with two transmitters is supported based on a TxD architecture, with an upper bound of 29 dBm for maximum output power. The defined **Maximum Power Reduction (MPR)** requirements are carrier-power based and Power Spectral Density (PSD)-agnostic, derived assuming equal PSD with margin to accommodate imbalance, and are specified separately for handheld and FWA devices. In intra-band non-contiguous CA, uplink MIMO is not supported due to the assumed dual-PA architecture; the maximum carrier power is limited to 26 dBm per component carrier, while the overall maximum power remains 29 dBm, with similarly defined PSD-agnostic MPR requirements and separate treatment for handheld and FWA UEs.

The WI also enables **higher-power operation for NR CA and EN-DC** with two or three transmit chains using the existing higherPowerLimit-R17 and higherPowerLimitMRDC-R17 capabilities, without introducing new power classes. These capabilities apply to any specified inter-band combination. No additional MSD requirements are introduced beyond those of the legacy power classes, no new SAR duty-cycle mechanisms are defined, and P-MPR may be used for SAR mitigation. Certain configurations, such as three-uplink-carrier CA or DC scenarios with three transmit chains, are excluded.

Inter-band CA HPUE Maximum SAR Duty cycle (MSD) requirements are simplified using look-up tables for configurations with one uplink band and one uplink carrier, and for configurations with two uplink bands and one uplink carrier per band, as documented in the agreed working and discussion papers. After implementation of the corresponding change request following RAN#109, existing Power Class 2 and Power Class 1.5 MSD tables remain unchanged in TS 38.101-1 and TS 38.101-3, with deviations from the look-up tables expected to be rare. Handling of Rel-19 baskets remains unchanged for the default power class, where MSD continues to be studied and captured as applicable.

A duty-cycle-based **Specific Absorption Rate (SAR)** solution is supported for inter-band uplink CA with two or three transmit chains, and support is provided for inter-band EN-DC with two and three transmit chains.

The WI further includes **power boosting and MPR** mechanisms. For FR1, MPR reduction is achieved by converting outer to inner resource block allocations using an extended channel bandwidth method, with extension location and ratio signaled by the network based on UE capability. Additional specification updates cover single-component-carrier activation in non-contiguous CA, MPR applicability for intra-band contiguous CA with a single active carrier, and emission requirements for Power Class 2 and 3 in such cases. For FR2, MPR is determined based on activated aggregated uplink bandwidth rather than configured bandwidth, and single-carrier uplink MPR requirements apply when combined with intra-band downlink CA.

Finally, support is introduced for **UEs equipped with six receive chains** for both handheld and FWA devices. New RF requirements include a six-antenna reference sensitivity allowance and associated requirements for defined SRS antenna switching patterns, without introducing new test cases. **Downlink MIMO** performance requirements support up to four layers for both handheld and FWA UEs, while six-layer operation is limited to FWA devices and explicitly captured in RAN4 specifications. Optional UE capability signaling is added to indicate support for up to six downlink MIMO layers for PDSCH. No solution is introduced in Rel-19 for SRS inter-layer imbalance.

References

List of related CRs: select "TSG Status = Approved" in:
https://portal.3gpp.org/ChangeRequests.aspx?q=1&workitem=1030077,1031077,1032077

[1] RP-251816 Revised WID UE RF enhancements for NR FR1 FR2 and EN-DC, Phase 4
[2] RP-252731, SR on NR_ENDC_RF_Ph4, RAN#109

16.2 Support of intra-band non-collocated EN-DC/NR-CA deployment Phase2: new receiver type(s)

1030082	**Support of intra-band non-collocated EN-DC/NR-CA deployment Phase2: new receiver type(s)**	NonCol_intraB_ENDC_NR_CA_Ph2		RP-242083	KDDI
1031082	**Core part: Support of intra-band non-collocated EN-DC/NR-CA deployment Phase2: new receiver type(s)**	NonCol_intraB_ENDC_NR_CA_Ph2-Core	R4	RP-242083	KDDI
1032082	**Perf. part: Support of intra-band non-collocated EN-DC/NR-CA deployment Phase2: new receiver type(s)**	NonCol_intraB_ENDC_NR_CA_Ph2-Perf	R4	RP-242083	KDDI

Summary based on the input provided by KDDI Corporationin RP-252837.

This Work Item defines the necessary RAN4 UE RF and RRM requirements together with the corresponding RAN2 signalling to **support Type 4 Fixed Wireless Access (FWA) UEs**. A Type 4 FWA UE is designed to operate in intra-band non-collocated EN-DC and NR-CA deployments by using separate receive chains for carriers transmitted from different geographic locations. In other words, a Type 4 FWA UE can reliably receive signals from multiple network sites at the same time thanks to its separate internal receivers, for signals coming from different places.

The need for intra-band non-collocated EN-DC and NR-CA deployments was identified during Rel-18, as such deployments are key to extending EN-DC and NR-CA coverage areas. While Rel-18 specified RF, RRM, and demodulation requirements for FR1 inter-band non-contiguous EN-DC with overlapping or partially overlapping bands and for FR1 intra-band non-contiguous NR-CA in the two-layer (Type 2 UE) case, requirements for Type 3 and Type 4 UEs were not addressed. To better support these deployment scenarios, Rel-19 introduces explicit requirements for Type 4 FWA UEs.

RAN4 has specified these requirements under the assumption that Type 4 FWA UEs employ separate receive chains to independently handle carriers transmitted from different locations. The specified requirements apply to both EN-DC and NR-CA scenarios and are limited to the relevant band combinations DC_42_n77, DC_42_n78, CA_n77, and CA_n78. They include UE RF requirements addressing receive power imbalance, as well as RRM requirements covering maximum uplink transmission timing difference and maximum relative transmission timing difference.

In addition, RAN2 has defined new base-station signalling and UE capability indications to support this feature. These include signalling to indicate whether Type 1 UE or Type 4 FWA UE requirements apply for EN-DC and NR-CA operation, along with UE capabilities indicating support for TDD-TDD inter-band EN-DC with overlapping or partially overlapping downlink bands and for TDD-TDD intra-band non-collocated NR-CA, both based on the Type 4 FWA UE requirements.

References

List of related CRs: select "TSG Status = Approved" in:
https://portal.3gpp.org/ChangeRequests.aspx?q=1&workitem=1030082,1031082,1032082

16.3 Rel-19 downlink interruption for NR and EN-DC band combinations at dynamic Tx Switching in Uplink

No summary provided (band-related).

16.4 Rel-19 DC of x LTE band(s), y NR band(s) (1<=x<6, 1<=y<6, x+y<=6) and single or two NR Supplementary Uplink (SUL) bands

No summary provided (band-related).

16.5 Simultaneous Rx/Tx band combinations for NR CA/DC, NR SUL and LTE/NR DC in Rel-19

No summary provided (band-related).

16.6 UE Conformance - Rel-19 NR CA and DC; and NR and LTE DC Configurations

No summary provided (band-related).

17 Rel-19 Other NR and LTE Radio

17.1 Adding channel bandwidth(s) support to existing NR bands and CA/ENDC combinations in REL-19

No summary provided (band-related).

17.2 Data collection for SON (Self-Organising Networks)/MDT (Minimization of Drive Tests) in NR standalone and MR-DC (Multi-Radio Dual Connectivity) Phase 4

1020092	**Data collection for SON (Self-Organising Networks)/MDT (Minimization of Drive Tests) in NR standalone and MR-DC (Multi-Radio Dual Connectivity) Phase 4**	NR_ENDC_SON_MDT_Ph4		RP-252557	China Unicom
1021092	**Core part: Data collection for SON (Self-Organising Networks)/MDT (Minimization of Drive Tests) in NR standalone and MR-DC (Multi-Radio Dual Connectivity) Phase 4**	NR_ENDC_SON_MDT_Ph4-Core	R3	RP-252557	Xufei Dong, China Unicom

Summary based on the input provided by China Unicom and CMCC in RP-252926.

Note: this is a summary of RP-252926. RP-252926 contains more information and can be retrieved from the 3GPP Server.

This Work Item specifies further NR SON/MDT enhancements, covering **Mobility Robustness Optimisation (MRO)** for intra-CU lower-layer triggered mobility (LTM), **conditional handover (CHO)** with candidate SCGs, subsequent CPAC, MR-DC SCG failures, as well as SON/MDT **support for intra-NTN mobility** and network slicing. It also completes remaining Release-18 items, including **RACH optimization for SDT** and MHI enhancements for SCG activation and deactivation.

For intra-CU LTM, the work item enhances **MRO** to detect cell switch failures caused by too-early or too-late triggering or wrong target cell selection, with RLF reports forwarded via the gNB-CU to the relevant gNB-DU. Additional LTM-specific MRO use cases over the F1 interface are supported, including beam failure recovery after LTM, RACH-less LTM failures due to beam issues, and failures caused by outdated timing advance, enabling

optimization of beam selection and TA parameters. The gNB-CU also detects **LTM ping-pong** scenarios and provides UE history information to assist resolution at the gNB-DU.

MRO enhancements for **CHO with candidate SCGs** address failures due to too-early or too-late handover execution or wrong target selection, with failure information exchanged over Xn or NG interfaces to optimize mobility configurations. SCG failures under CHO are analyzed by the serving node, which forwards SCG failure information to the responsible MN or SN as appropriate.

For subsequent CPAC, the work item enhances **MRO** to handle failures caused by timing issues or wrong PSCell selection, with root-cause analysis and optimization performed by the MN or SN depending on whether the procedure is MN- or SN-initiated, and with SCG failure information reporting when needed.

SON/MDT enhancements for **intra-NTN mobility** allow MDT configuration to include up to eight geographical areas, ensuring measurement collection is limited to the configured regions for both logged and immediate MDT. Network slicing support in MDT is enhanced by introducing a Network Slice Area Scope, restricting measurements to cells where the configured slices are available.

Finally, Release-18 leftovers are addressed by enabling **RACH optimization for SDT** through SDT-specific reporting, enhancing MHI and UHI to include SCG activation and deactivation time for improved dual-connectivity optimization, and improving MRO for MR-DC SCG failures by introducing new SCG failure reporting and transfer procedures for EN-DC and NG-EN-DC scenarios.

References

List of related CRs: select "TSG Status = Approved" in:
https://portal.3gpp.org/ChangeRequests.aspx?q=1&workitem=1020092,1021092

[1] RP-234038, New WID: Data collection for SON (Self-Organising Networks)/MDT (Minimization of Drive Tests) in NR standalone and MR-DC (Multi-Radio Dual Connectivity) Phase 4
[2] RP-252558, SR for NR_ENDC_SON_MDT_phase 4

18 Rel-19 NR Radio

18.1 NR mobility enhancements Phase 4

1020091	NR mobility enhancements Phase 4	NR_Mob_Ph4		RP-252111	Apple
1021091	**Core part: NR mobility enhancements Phase 4**	NR_Mob_Ph4-Core	R2	RP-252111	Naveen Palle, Apple Inc.
1022091	**Perf. part: NR mobility enhancements Phase 4**	NR_Mob_Ph4-Perf	R4	RP-252111	Naveen Palle, Apple Inc.
1040023	**Security aspects of NR mobility enhancement Phase 4**	NR_Mob_Ph4_Sec	S3	SP-240652	Rajavelsamy R

Summary based on the input provided by Apple Inc and China Telecom in RP-252884.

Note: This is a summary of the document mentioned above. RP-252884 contains more information, in a bulleted format.

This Work Item builds on the Rel-18 "Further NR Mobility Enhancements" and addresses **four key objectives: enabling L1/L2-based inter-CU[1] inter-cell mobility, enhancing measurements for LTM[2], introducing conditional**

[1] Central Unit (CU) is the logical part of a gNB that hosts higher-layer radio functions, such as RRC, PDCP, and parts of SDAP, while the lower-layer functions are handled by Distributed Units (DUs). The CU can control one or multiple DUs.

[2] L1/L2-Triggered Mobility (LTM) is a mobility mechanism where handover or cell switching decisions are triggered directly by Layer 1 and Layer 2 measurements—such as signal quality or beam measurements—rather than relying solely on higher-layer (RRC) procedures. This allows faster and more responsive mobility, particularly for high-mobility scenarios and beam-based operation, and forms the basis for features such as inter-CU mobility and conditional mobility introduced in recent NR releases.

LTM for intra-CU scenarios, and defining RRM core requirements supporting inter-CU LTM, conditional LTM, and LTM measurement enhancements.

Stage-2 aspects are specified in TS 38.300 and TS 37.340 for NR-DC, with physical-layer procedures defined in TS 38.312, TS 38.213, and TS 38.214. MAC and RRC signalling are specified in TS 38.321 and TS 38.331, while NG-RAN architecture and protocols are covered by TS 38.401, TS 38.420, TS 38.423, TS 37.483, TS 38.470, and TS 38.473. Measurement requirements and UE capabilities are specified in TS 38.133 and TS 38.306.

For inter-CU L1/L2-based mobility, the work **enhances L1/L2-triggered mobility** to support handovers between gNB-CUs by extending RRC configuration for candidate cells across multiple CUs, enabling security context provisioning via MAC CE, and introducing inter-CU signalling to generate the required LTM configurations. Measurement enhancements enable CSI-RS-based and beam-based LTM through new CSI-RS configurations, early CSI acquisition, and beam-triggered measurement events with reporting via MAC CE.

The work item also introduces **conditional LTM**, allowing the UE to trigger mobility based on configured conditions, supported by early timing-advance acquisition and enabling RACH-less execution when a valid timing-advance value is available, with additional F1AP signalling for intra-CU operation. Finally, RRM core requirements are defined for event-triggered L1 RSRP reporting, CSI-RS-based measurements on candidate cells, and performance requirements for conditional cell-switch delay and interruption.

References

List of related CRs: select "TSG Status = Approved" in:
https://portal.3gpp.org/ChangeRequests.aspx?q=1&workitem=1020091,1021091,1022091,1040023

18.2 Evolution of NR duplex operation: Sub-band full duplex (SBFD)

1020090	**Evolution of NR duplex operation: Sub-band full duplex (SBFD)**	NR_duplex_evo		RP-251874	Huawei
1021090	**Core part: Evolution of NR duplex operation: Sub-band full duplex (SBFD)**	NR_duplex_evo-Core	R1	RP-251874	Xinghua Song, He (Jackson) Wang, Huawei, Samsung
1022090	**Perf. part: Evolution of NR duplex operation: Sub-band full duplex (SBFD)**	NR_duplex_evo-Perf	R4	RP-251874	Xinghua Song, He (Jackson) Wang, Huawei, Samsung

Summary based on the input provided by Huawei in RP-252365

TDD is widely used in commercial NR deployments, typically with DL dominant TDD UL/DL configurations. The limited uplink transmission duration in TDD may result in reduced coverage, increased latency, and reduced capacity. In Rel-18, a study was conducted on the feasibility and benefits of Sub-Band non-overlapping Full Duplex (SBFD) operation at gNB, which enables simultaneous downlink and uplink on separate sub-bands within a TDD carrier. The study outcome is documented in [1].

Based on this, this WI specifies the following key functionalities

SBFD sub-bands configuration in time and frequency: To enable SBFD operation, the time and frequency resources of the SBFD sub-bands shall be provided to the UE. The configuration of cell-specific SBFD sub-bands is conveyed through SIB1 or dedicated signalling. SBFD sub-bands can be configured in both downlink symbols and flexible symbols. Within a TDD UL/DL pattern period, SBFD symbols are configured consecutively and may start or end at any symbol within a slot. The frequency location of an SBFD UL sub-band and SBFD DL sub-band(s) is defined by a starting RB and a bandwidth. At most one UL sub-band and up to two DL sub-bands can be configured for SBFD operation within an SBFD symbol in a TDD carrier.

UE transmission, reception and measurement in SBFD symbols and/or non-SBFD symbols: To facilitate SBFD operation at the gNB, the UE behaviour and physical layer procedures of transmission, reception and measurement in SBFD symbols and/or non-SBFD symbols are specified. In an SBFD symbol, except for cross-link interference measurements, a UE transmits or receives, only in UL PRBs that are both in the active UL BWP and in the UL sub-band, and, DL PRBs that are both in the active DL BWP and in the DL sub-band, respectively. A UE can be configured to transmit or receive in either non-SBFD symbols or SBFD symbols only, or across both SBFD symbols and non-SBFD symbols (each transmission/reception occasion restricted to either SBFD or non-SBFD symbols). For CSI

reporting, a UE may be configured to derive CSI based on CSI-RS transmission occasions restricted either to SBFD symbols or to non-SBFD symbols. Separate UL power control for PUSCH/PUCCH/SRS in SBFD and non-SBFD symbols are supported. A UE can be configured with separate SRS resource sets, PUCCH resources in SBFD and non-SBFD symbols.

Random access procedure in SBFD symbols: To support random access in SBFD symbols, a UE can be configured with an additional set of RACH occasions (additional-ROs) in addition to legacy RACH occasions (legacy-ROs). Two RACH configuration options are defined: a single RACH configuration and a dual RACH configuration, with only one option enabled at the gNB. For additional-ROs, separate SSB-to-RO mapping, association period, and association pattern period are specified. For contention-based random access (CBRA), the RO type can be selected based on explicit indication or configured SSB-RSRP threshold-related parameters. For contention-free random access (CFRA) triggered by PDCCH order, the UE is explicitly indicated in the PDCCH order whether to use a PRACH occasion associated with legacy-ROs or with additional-ROs. Switching between legacy-ROs and additional-ROs is supported when the number of PRACH transmission attempts exceeds a configured threshold, with only one fallback allowed.

UL resource muting for PUSCH: To mitigate gNB-to-gNB CLI, UL resource muting for PUSCH is introduced. UL resource muting can be applied for both CP-OFDM and DFT-S-OFDM waveform. A UE is semi-statically configured with up to two UL muting symbols within a slot for a PUSCH and a comb offset. For PUSCH scheduled or activated by a DCI, the TDRA field in the DCI indicates whether muting is applied to the corresponding PUSCH transmission. When UL resource muting is applied in a symbol, either even or odd sub-carriers of the frequency resource of the PUSCH are available, and the other sub-carriers are not used for the PUSCH transmission. A UE configured with SBFD symbols can be further configured to apply the resource muting in SBFD symbols and non-SBFD symbols.

Information exchange among gNBs: To facilitate gNB-to-gNB CLI handling and UE-to-UE CLI handling, information exchange among gNBs is specified. This includes SBFD sub-band configuration, NZP CSI-RS and/or SSB resource configuration, strongest DL beam information, CLI-mitigation request and one or more SRS resources configurations.

UE-to-UE CLI measurement and reporting: To mitigate UE-to-UE CLI, L1-based UE-to-UE CLI measurement and reporting is introduced based on the existing CSI framework. Two new reporting quantities are defined: SRS-RSRP and CLI-RSSI. The CSI resource configuration is extended by introducing two CLI measurement resource set lists, one for SRS-RSRP measurement and one for CLI-RSSI measurement. CLI-RSSI measurement may be configured in either a DL sub-band or a UL sub-band, while SRS-RSRP measurement is configured only in a UL sub-band. For SRS-RSRP, only aperiodic CSI reporting is supported, whereas for CLI-RSSI, both periodic and aperiodic CSI reporting are supported.

References

List of related CRs: select "TSG Status = Approved" in:
https://portal.3gpp.org/ChangeRequests.aspx?q=1&workitem=1020090,1021090,1022090

[1] TR 38.858, Study on evolution of NR duplex operation
[2] RP-251874, Revised WID: Evolution of NR duplex operation: Sub-band full duplex (SBFD)
[3] RP-252364, Status report for WI: Evolution of NR duplex operation: Sub-band full duplex (SBFD); rapporteur: Huawei

18.3 NR Radio Resource Management (RRM) Phase 5

1030079	NR Radio Resource Management (RRM) Phase 5	NR_RRM_Ph5		RP-252494	Apple
1031079	**Core part: NR Radio Resource Management (RRM) Phase 5**	NR_RRM_Ph5-Core	R4	RP-252494	Apple
1032079	**Perf. part: NR Radio Resource Management (RRM) Phase 5**	NR_RRM_Ph5-Perf	R4	RP-252494	Apple

Summary based on the input provided by Apple and CATT in RP-252495.

This WI defines RRM requirements for the three following UE features:

Reducing FR2-1 L3 measurement delay by optimizing Rx beam sweeping factor
RAN4 has specified the RRM requirements for L3 fast beam sweeping in TS38.133:
- Following UE feature was introduced in this enhancement:
o Feature 1: Fast Rx beam sweeping factor for FR2-1 L3 measurement delay reduction
- RAN4 specified the fast Rx beam sweeping in the following requirements:

o SSB based Intra-frequency measurement without MG
o SSB based Intra-frequency measurement with MG
o SSB based Inter-frequency measurement with MG
o Handover
o RRC Re-establishment
o RRC Connection Release with Redirection.
- RAN2 has specified the corresponding signalling for L3 fast beam sweeping in TS38.331.

Reducing FR2-1 L3 measurement delay by optimizing CSSF outside gap in CA/DC scenarios
RAN4 has specified the RRM requirements for L3 CSSF enhancement in TS38.133:
- Following UE features were introduced in this enhancement:
o Feature 1: L3 serving cell and neighbor cells measurement and report on one serving carrier per-band for intra-frequency measurements without measurement gap
o Feature 2: Simultaneous L3 measurement on three carriers for measurements without measurement gap under CA/DC operation.
- RAN4 specified the CSSF enhancements in following requirements:
o Requirements for FR2 in section 9.2.3.2 of TS38.133
o Intra-frequency cell identification requirements in section 9.2.5.1 of TS38.133
o Intra-frequency cell identification requirements in section 9.2.6.2 of TS38.133
o Intra-frequency measurements with NCSG requirements in section 9.2.7 of TS38.133
o Inter-frequency measurement with measurement gaps requirements in section 9.3.4 of TS38.133
o Inter-frequency measurement with NCSG requirements in section 9.3.10 of TS38.133
- RAN2 has specified the corresponding signalling for L3 CSSF enhancement in TS38.331.

Fast SCell activation for UE supporting Rel-18 EMR
RAN4 has specified the RRM requirements for fast SCell activation in TS38.133:
- Following UE feature was introduced in this enhancement:
o Feature 1: R19 SCell activation based on EMR
- RAN4 specified:
o SCell activation delay requirements based on R18 EMR in section 8.3.2A of TS38.133.

References

List of related CRs: select "TSG Status = Approved" in:
https://portal.3gpp.org/ChangeRequests.aspx?q=1&workitem=1030079,1031079,1032079

[1] RP-242380, Revised WID of NR RRM Phase 5, RAN #105, Apple, CATT
[2] RP-252493, SR of NR Radio Resource Management (RRM) Phase 5, RAN#109, Apple, CATT
[3] R4-2512348, Big CR for NR RRM Phase 5, RAN#109, Apple, CATT

18.4 Multi-carrier enhancements for NR Phase 3

1050120	**Multi-carrier enhancements for NR Phase 3**	NR_MC_enh2		RP-252564	Lenovo
1051120	**Core part: Multi-carrier enhancements for NR Phase 3**	NR_MC_enh2-Core	R1	RP-252564	Lenovo

Summary based on the input provided by Lenovo in RP-252565.

The WI introduces two enhancements for multi-carrier operation: use a single DCI to schedule multiple cells with different subcarrier spacing values or carrier types; and use a single DCI to schedule multiple cells with multiple PDSCH/PUSCH transmissions on each cell. These enhancements are not mutually exclusive.

Multi-carrier operation is very important for 5G commercial networks by aggregating various spectrum resources for providing high data rate and low latency communication.

Due to quite limited TU for Rel-18 multi-carrier enhancements, some important use cases were excluded from Rel-18, e.g., different SCSs among co-scheduled cells, different carrier types among co-scheduled cells. Co-scheduled carriers with different SCSs have high commercial needs for operators, e.g., 3.5GHz TDD + Sub-3GHz FDD, FR1 + FR2, etc. Furthermore, two new DCI formats were introduced in Rel-18, DCI format 0_3 and 1_3. Each DCI format 0_3 or 1_3 can schedule up to 4 cells with limitation of a single PUSCH or PDSCH per scheduled cell. In Rel-17, for FR2 with high SCS, multi-PDSCH/PUSCH scheduling was introduced, i.e., up to 8 PUSCHs or PDSCHs on a single serving cell can be scheduled by a single DCI format 0_1 or 1_1, in order to save UE power consumption and reduce PDCCH

overhead. Hence, it is straightforward to combine multi-cell scheduling and multi-PDSCH/PUSCH scheduling in Rel-19 to fully exploit the gain of power saving and PDCCH overhead reduction so that one DCI format 0_3 or 1_3 can schedule multiple cells with one or multiple PUSCHs/PDSCHs per scheduled cell. This is especially useful when scheduling cell in FR1 with a lower SCS schedules multiple cells in FR2 with higher SCS.

The key functionalities introduced in this WI is multi-cell PDSCH/PUSCH scheduling with a single DCI, including scheduling of different SCS/carrier types and one or multiple PUSCHs/PDSCHs per scheduled cell as shown in Figure 1. In Rel-18 MC enhancement, same SCS/carrier types (licensed/unlicensed, FR1/FR2/FR2-2) among co-scheduled cells, and single PDSCH/PUSCH scheduled per scheduled cell were supported. In Rel.19 MC enhancement, same or different SCS/carrier types among co-scheduled cells and one or more PDSCH/PUSCH scheduled per scheduled cell were specified. Single-cell multi-PUSCH/PDSCH scheduling and multi-cell multi-PUSCH/PDSCH scheduling on the same or different cells within a same PUCCH group are not expected to be configured to UE simultaneously.

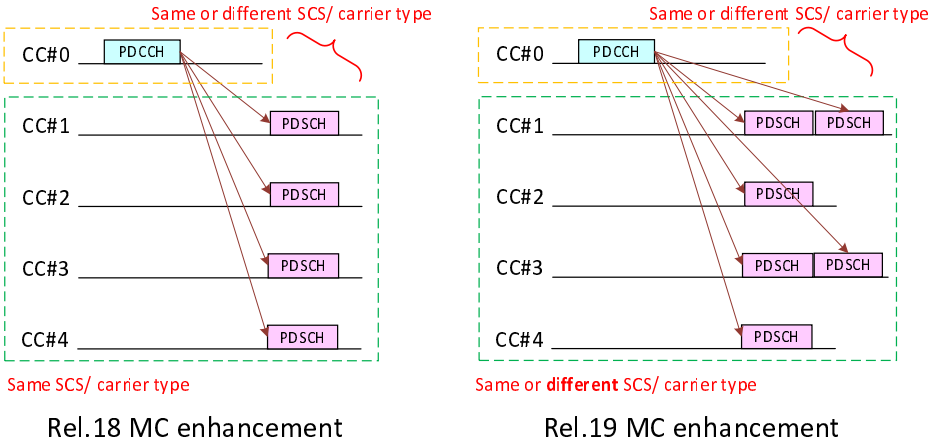


Figure 1 Rel.19 MC Enhancement Illustration

The existing DCI formats 1_3 and 0_3 are enhanced and optimized to support multi-cell PDSCH/PUSCH scheduling with up to 8 PUSCH/PDSCH for a scheduled cell by a single DCI with DCI payload size restriction (i.e. less than 140 bits). For some fields in DCI format 1_3/0_3, a single indication is commonly applied to all scheduled PUSCHs/PDSCHs and co-scheduled cell(s) so that the payload size could be compact while some other fields have separate indication for each PDSCH/PUSCH, e.g., for NDI/RV field, the number of bit for each block of the corresponding cell is equal to the maximum number of schedulable PUSCHs/PDSCHs by the DCI, and placed according to an ascending order of a serving cell index.

To support HARQ-ACK information feedback for multi-cell scheduling with multiple PDSCHs per scheduled cell by DCI format 1_3, Type-1 HARQ-ACK codebook was reused without enhancement, Type-2 HARQ-ACK codebook generation is enhanced (e.g., for the second sub-codebook, the number of HARQ-ACK information bits is generated based on all the configured cell set(s) in the PUCCH group for the UE). Furthermore, time-domain HARQ-ACK bundling configured per cell is supported.

References

List of related CRs: select "TSG Status = Approved" in:
https://portal.3gpp.org/ChangeRequests.aspx?q=1&workitem=1050120,1051120

[1] RP-242904 Revised WID: Multi-carrier enhancements for NR Phase 3
[2] R1-2504988 Introduction of Rel-19 Multi-carrier enhancements (TS38.212)
[3] R1-2504972 Introduction of multi-carrier enhancements for NR Phase 3 (TS38.213)
[4] R1-2504995 Introduction of Multi-carrier enhancements for NR Phase 3 (TS38.214)
[5] R2-2506252 Stage 2 CR for Rel-19 Multi-carrier enhancements (TS38.300)
[6] R2-2506253 Introduction of Rel-19 Multi-carrier enhancements (TS38.331)

18.5 NR demodulation performance Phase 5

No summary provided.

18.6 NR MIMO Phase 5

1020089	**NR MIMO Phase 5**	NR_MIMO_Ph5		RP-242394	Samsung
1021089	**Core part: NR MIMO Phase 5**	NR_MIMO_Ph5-Core	R1	RP-242394	Eko Onggosanusi, Samsung
1022089	**Perf. part: NR MIMO Phase 5**	NR_MIMO_Ph5-Perf	R4	RP-242394	Eko Onggosanusi, Samsung

Summary based on the input provided by Samsung in RP-252823.

Note: this is a summary of RP-252823, which provides more details.

This Work Item introduces enhanced specification support for key MIMO features where NR Releases 15 to 18 were found limited in terms of signaling latency, overhead, spectral efficiency, and coverage.

First, legacy NR relies on network-configured or network-triggered periodic, semi-persistent, or aperiodic beam reporting to identify preferred beams, which leads to significant uplink reporting and downlink control overhead as well as delayed adaptation. Since UEs have more accurate and timely awareness of beam quality variations, Rel-19 introduces **UE-initiated, event-driven beam reporting to enable faster and lower-overhead beam management**. Two UE-initiated reporting modes are supported: one based on a DCI-scheduled PUSCH following a one-bit UE-initiated report indication, and another based on a configured-grant PUSCH triggered by the same indication. Multiple events are defined to support use cases such as rapid degradation detection, timely beam switching, and accelerated TCI state updates, and cross-carrier UE-initiated beam reporting is also supported to facilitate multi-carrier deployments.

Second, Rel-19 addresses the limitation of PMI-based CSI reporting to 32 CSI-RS ports by extending **CSI support up to 128 ports** for both fully digital and hybrid MIMO architectures. This includes aggregation of multiple NZP CSI-RS resources, enhancements to Type-I and Type-II codebooks, and extensions to CRI-based reporting to allow multiple CSI hypotheses. To reduce UE complexity for higher-layer operation in TDD, SRS port grouping is introduced to support reciprocity-based CSI acquisition. In addition, to enable **coherent joint transmission (CJT)** with non-ideal backhaul, Rel-19 introduces aperiodic CSI reports to assist network **calibration** across multiple TRPs, covering inter-TRP delay, frequency, and DL/UL phase offsets.

Third, uplink support is extended to accommodate UEs with three transmit antennas. Rel-19 introduces enhancements for both codebook-based and non-codebook-based **3TX PUSCH transmission**, including corresponding SRS, DMRS, and PTRS updates, a new UE capability indicating a maximum PUSCH rank of three, and optimized DCI signaling. SRS antenna switching is extended to support 3T3R and 3T6R configurations by reusing existing multi-port SRS structures.

Finally, Rel-19 enhances support for **asymmetric multi-TRP scenarios** where downlink transmission and uplink reception are handled by different TRPs, as commonly seen in heterogeneous deployments. This includes configurable pathloss offsets for uplink channels and signals, support for two independent closed-loop power control adjustment states for SRS, and more flexible association of timing advance values with TCI states, removing previous restrictions tied to multi-DCI operation.

References

List of related CRs: select "TSG Status = Approved" in:
https://portal.3gpp.org/ChangeRequests.aspx?q=1&workitem=1020089,1021089,1022089

[1] RP-242394, Samsung (Moderator), WID revision: NR MIMO Phase 5

18.7 FR1 TRP, TRS and MIMO OTA testing enhancement Phase 3

1030085	**FR1 TRP, TRS and MIMO OTA testing enhancement Phase 3**	TRP_TRS_MIMO_OTA_Ph3		RP-252187	Vivo
1031085	**Core part: FR1 TRP (Total Radiated Power), TRS (Total Radiated Sensitivity) and MIMO OTA (Over the Air) testing enhancement Phase 3**	TRP_TRS_MIMO_OTA_Ph3-Core	R4	RP-252187	Vivo

1032085	**Perf. Part: FR1 TRP (Total Radiated Power), TRS (Total Radiated Sensitivity) and MIMO OTA (Over the Air) testing enhancement Phase 3**	TRP_TRS_MIMO_OTA_Ph3-Perf	R4	RP-252187	Vivo

Summary based on the input provided by vivo and CAICT in RP-252011

This WI defines the enhanced Over-the-Air (OTA) test methods for TRP/TRS, and dynamic MIMO OTA. The outcome of TRP/TRS test method is captured in technical report TR 38.870 [2], dynamic MIMO OTA method in a new technical report TR 38.762 [3], and few update are also reflected in TS 38.161 [4].

Rel-17 and Rel-18 OTA work specified the Antenna Coupling (AC) and Receive Correlation (RC) test method for TRP/TRS and multi-probe static MIMO OTA method.

This Rel-19 WI is a general WI covering both TRP/TRS and MIMO OTA, which specified enhanced AC test methods to support verification of radiated performance for XR and NTN, as well as advanced multi-probe MIMO OTA test method with dynamic channel model and link adaptions. The performance metric and usage scenarios for XR and NTN consider both full-sphere TRP/TRS and partial sphere performance metric. The performance metric and channel model of dynamic MIMO OTA are also different from those of the traditional static MIMO OTA methodology.

OTA (over the air) test methods and radiated performance verification (as considered in this WI) are important since if the above UE types (XR and NTN) for FR1 (frequency range 1) are verified by conducted methodology at temporary antenna ports, it remains unknown what the actual performance of the UE would be in realistic network conditions with the UE antenna included. For MIMO OTA, with the adoption of dynamic channel model in the chamber and link adaptive UE configurations, the test methodology is more close to the real usage scenarios of multi-antenna UE in the real network.

No matter TRP/TRS or MIMO OTA, the end-to-end radiated performance verification based on OTA methodology is one of the most important characteristics to present the entire UE performance under conditions more closely resembling the end user's interaction with the device and usage scenarios in the network.

The purpose of this WI is to develop test method for FR1 TRP TRS and dynamic MIMO OTA. Some key outcomes of this WI can be summarized as following:

The enhanced TRP TRS test methods support: XR devices, including glasses and head mounted display; NTN devices, including both NR-NTN and IoT-NTN.

The following specific Usage scenarios for NTN were defined:
- *Usage scenario 1: both head+hand mode and hand only mode*
- *Usage scenario 2: hand only mode targeted for specific UE orientation usage*
- *Usage scenario 3: hand only mode targeted for arbitrary UE orientation usage*
- *Applicability of NTN usage scenarios and corresponding test procedure*

New Usage scenario for XR was defined: Using XR Head phantom for headworn XR devices
The UE positioning guidelines for NTN were defined: Hand phantom only for NTN scenario 2, e.g., different elevation tilt angle.
New performance metric specified:Partial Radiated Metrics (PRP/PRS) for AC
Test methodology enhancement:
- *QZ and min measurement distance for 50cm UE*
- *Test procedure*
- *MU assessment*

Alternative methodologies to resolve battery issue
- *Alternate TRS Test procedure Using Low UL Power*
- *Split TRP/TRS measurement grids method*

The enhanced MIMO OTA test methods support: smartphone and other UE types
New DUT positioning guidelines:
- *Four device orientations, i.e., DMP, DML (both left and right tilt), and DMSU*
- *Four rotations of device (per orientation), i.e., 0°, 90°, 180°, 270°*

The newly defined dynamic Channel models: dynamic channel models with full path loss model, UMidyn and UMadyn; Channel model parameters
New performance metric specified:CDF of the Total Measured Throughput (CTMT)
New configuration of gNB and UE:gNodeB emulator settings; Dynamic link adaptation for MCS, rank and MIMO precoding
Test methodology enhancement:
- *Environmental conditions*
- *Minimum range length of multi-probe support 30cm QZ*
- *Test procedure, includes e.g., calibration, ripple test, throughput testing*
- *MU assessment*

Dynamic Channel model validation procedure and target values:
- *Path Loss (PL)*
- *Power Delay Profile (PDP)*
- *Doppler/Temporal correlation (TCF)*
- *Spatial Correlation (SCF)*

o Cross-polarization (XPO)
Channel model validation pass/fail limits:Pass/fail limits of each validation aspects

References

List of related CRs: select "TSG Status = Approved" in:
https://portal.3gpp.org/ChangeRequests.aspx?q=1&workitem=1030085,1031085,1032085

[1] RP-250197, WID: WI on FR1 TRP (Total Radiated Power), TRS (Total Radiated Sensitivity) and MIMO OTA (Over the Air) testing enhancement Phase 3, vivo, CAICT.
[2] TR 38.870, v19.2.0
[3] TR 38.762 v0.5.0
[4] TS 38.161, v19.1.0

18.8 Rel-19 NR CA/DC for x bands DL with y bands UL (x<7, y<3) and SUL/CA band combinations with a single SUL or two SUL cells

1040120	**Rel-19 NR CA/DC for x bands DL with y bands UL (x<7, y<3) and SUL/CA band combinations with a single SUL or two SUL cells**	NR_CADC_SUL_R19	R4	RP-252440	Ericsson

No summary for band-related items.

18.9 Low band carrier aggregation via switching

1060082	**Low band carrier aggregation via switching**	NR_LBCA_Sw		RP-252694	Telus
1061082	**Core part: Low band carrier aggregation via switching**	NR_LBCA_Sw-Core	R4	RP-252694	Telus
1062082	**Perf. part: Low band carrier aggregation via switching**	NR_LBCA_Sw-Perf	R4	RP-252694	Telus

Summary based on the input provided by TELUS in RP-252696, revised to RP-260682.

Note: All the editing work performed on RP-252696 has been lost when producing RP-260682. Consequently, this section is simply a copy-paste of RP-260682.

This Work Item (WI) enables low-band NR carrier aggregation (CA) via a time-domain switching mechanism, allowing devices to aggregate two low-band carriers without requiring simultaneous receptions. Before this WI, inter-band carrier aggregation between two low-bands (<1 GHz) was generally not supported by most devices due to the extreme complexity of the radio frequency (RF) front-end, which required bulky and expensive components like triplexers or quadplexers.

This enhancement was needed to solve the problem of underutilized Supplemental Downlink (SDL) spectrum (e.g., band n29) in deep indoor areas where higher-frequency bands cannot reach. By overcoming these hardware and antenna limitations, the WI allows operators to leverage the coverage of two low-band carriers simultaneously to provide enhanced capacity to deep indoors and/or rural and remote users.

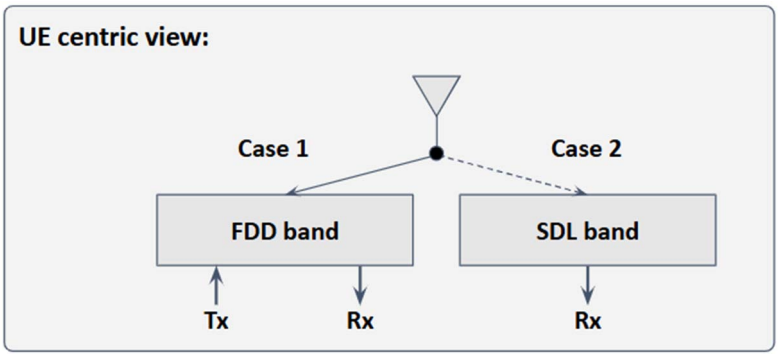
The primary objective of this Work Item was to enable Low-Band + Low-Band Carrier Aggregation (LB-LB CA) on devices where physical space and hardware costs make traditional simultaneous receptions impractical. By implementing a switching-based architecture, the UE can benefit from the coverage of two low-band carriers without the need for the heavy, expensive, and complex radio frequency (RF) components typically required for concurrent operation in conventional CA mode of operation.

Key functionalities of the work item and their purpose include:

- **Enabling SDL in low-band only scenarios:** A Supplemental Downlink (SDL) band (e.g., n29) is downlink-only and requires a paired uplink (UL) on a different band to function. Previously, most devices could not support LB-LB CA due to the complexity of the radio front-end. Consequently, in deep indoor areas where only low bands are available, the SDL band could not be used because of its lack of an UL. This WI introduces

a switching mechanism that allows the UE to provide the necessary UL signaling on the primary band and to receive data on the SDL band.

- **Time-Division Rx switching:** The UE does not receive two low-band frequencies simultaneously. Instead, it switches its internal radio resources between the primary FDD band and the SDL band (Case 1 and Case 2, as illustrated below). This allows the device to aggregate data from both carriers using a single receiver chain and a restricted number of antennas.



- **Front-End simplification (avoiding triplexers or quadplexers):** Simultaneous LB+LB (including SDL) operation requires a triplexer, a specialized, bulky component that merges three filters (Primary Tx, Primary Rx, and SDL Rx) into one. These components are expensive and may cause high insertion loss, which degrades the receiver's performance. The switching mechanism introduced via this WI allows the use of standard, high-efficiency duplexers, preserving signal sensitivity, and reducing device cost.
- **Rx antenna sharing:** This WI defines how a device with a restricted number of receive antennas (e.g., 2Rx) can logically share those antennas between two low-bands. This ensures the device meets performance requirements without the prohibitive cost and space of adding more physical antennas to the device.
- **Network-controlled switching pattern:** To keep the UE and the base station (gNB) synchronized, the WI defines signaling for the switching pattern and switching gaps. These gaps allow the hardware to retune. The gNB knows exactly when these gaps occur and schedules data transmissions only when the UE is listening to the correct band.

Physical layer procedures

The core mechanism introduces a semi-static switching pattern between a **PCell** (paired UL/DL) and an **SCell** (DL-only/SDL). Key procedures include:

- **Switching Pattern:** A UE is provided with switchingPattern-19, which is a **bitmap** of continuous slots. A bit value of '0' indicates the UE can transmit/receive on the PCell, while '1' indicates the UE can receive on the SCell.
- **Defined Guard Gaps:** Two specific switching durations are introduced:
  - gapDurationPCelltoSCell-r19: For switching from PCell to SCell.
  - gapDurationSCelltoPCell-r19: For switching from SCell back to PCell.
- **Transmission Restrictions:** The UE is prohibited from transmitting PUSCH, PUCCH, SRS, and PRACH overlapping with guard durations or in any slot where the bitmap value is '1'.
- **Reception Restrictions:** The UE is prohibited from receiving SS/PBCH block, PDSCH, or PDCCH, or CSI-RS, or PRS overlapping with guard durations or on PCell in any slot where the bitmap value is '1' or on SCell in any slot where the bitmap value is '0'.

RRM requirements

RAN4 has established several performance and maintenance requirements:

- **SCell Activation/Deactivation:** The switching pattern is applied during the SCell activation procedure.
- **Activation Delay:** Reference times for $T_{activation_time}$ are adjusted based on the switching pattern.
- **Radio link monitoring**: Evaluation period $T_{Evaluate_out_SSB}$ / $T_{Evaluate_out_CSI\text{-}RS}$ and $T_{Evaluate_in_SSB}$ / $T_{Evaluate_in_CSI\text{-}RS}$ are adjusted based on the switching pattern.

- **Beam failure detection:** Evaluation period $T_{Evaluate_BFD_SSB}$ and $T_{Evaluate_BFD_CSI\text{-}RS}$ are adjusted based on the switching pattern.
- **Candidate beam detection:** Evaluation period $T_{Evaluate_CBD_SSB}$ and $T_{Evaluate_CBD_CSI\text{-}RS}$ are adjusted based on the switching pattern.
- **Intra-frequency cell identification:** Time period for PSS/SSS detection T_{PSS/SSS_sync_intra} are adjusted based on the switching pattern.
- **L1-RSRP measurement:** Measurement period $T_{L1\text{-}RSRP_Measurement_Period_SSB}$ and $T_{L1\text{-}RSRP_Measurement_Period_CSI\text{-}RS}$ are adjusted based on the switching pattern.
- **L1-SINR reporting with CSI-RS based CMR and no dedicated IMR configured:** Measurement period $T_{L1\text{-}SINR_Measurement_Period_CSI\text{-}RS_CMR_Only}$ are adjusted based on the switching pattern.

UE Capabilities introduced

The following UE capabilities were introduced to support these procedures:

- ***featureSetCombinationLowBandSwitching-r19*:** Reported by the UE to indicate support for switching operation between PCell and SCell.
- ***switchingPeriodForFDD-SDL-r19***: Indicates the band pair that the UE supports of LB-LB carrier aggregation via switching between an FDD band (Case 1) and SDL band (Case 2)
- ***directSCellActivation-r19***: Indicates whether the UE supports RRC based direct SCell activation at SCell addition in NR LB-CA via switching

References

List of related CRs: select "TSG Status = Approved" in:
https://portal.3gpp.org/ChangeRequests.aspx?q=1&workitem=1060082,1061082,1062082

[1] RP-251315, Last approved WID: Low NR band carrier aggregation via switching, TELUS, 3GPP RAN #108, June 2025
[2] TR 38.768, NR; Low NR Band Carrier Aggregation via Switching
[3] TS 38.213, NR; Physical layer procedures for control
[4] TS 38.214, NR; Physical layer procedures for data
[5] TS 38.300, NR and NG-RAN Overall Description; Stage 2
[6] TS 38.331, NR; Radio Resource Control (RRC) protocol specification
[7] TS 38.101-1, NR; User Equipment (UE) radio transmission and reception; Part 1: Range 1 Standalone
[8] TS 38.133, NR; Requirements for support of radio resource management
[9] RP-252695, Status Report: Low NR band carrier aggregation via switching, TELUS, 3GPP RAN #109, September 2025

18.10 NR channel BW less than 5MHz for FR1 Phase 2

1030081	**NR channel BW less than 5MHz for FR1 Phase 2**	NR_FR1_lessthan_5MHz_BW_Ph2		RP-251670	Intel
1031081	**Core part: NR channel BW less than 5MHz for FR1 Phase 2**	NR_FR1_lessthan_5MHz_BW_Ph2-Core	R4	RP-251670	Intel
1032081	**Perf. part: NR channel BW less than 5MHz for FR1 Phase 2**	NR_FR1_lessthan_5MHz_BW_Ph2-Perf	R4	RP-251670	Intel
1080077	**UE Conformance - NR channel BW less than 5MHz for FR1 Phase 2**	NR_FR1_lessthan_5MHz_BW_Ph2-UEConTest	R5	RP-252552	China Telecom

Summary based on the input provided by Intel Corporation in RP-250464

In this work item, common co-located and non-co-located inter-band NR CA/DC UE RF requirements with 3MHz Bandwidth (BW) in the one band and 5MHz or 10MHz BW in the other band are defined. RRM requirements for inter-band CA and DC for combinations introduced are defined.

In Release-18 work item NR_FR1_lessthan_5MHz_BW, RAN4 specified requirements to support deploying NR in spectrum allocations from approximately 3 MHz up to below 5 MHz, in bands including n100, n106, n26, n28, n31, n72

and n85. This Work Item in Release-19 further specifies Carrier Aggregation and Dual Connectivity of a 3MHz carrier with one or more 3MHz or wider bandwidth carriers.

Carier Aggregation of a 3MHz carrier with one or more 3MHz or wider bandwidth carriers will increase the total available network capacity and hence the value of the 3MHz carrier. In fact, it in some cases enables the use of spectrum blocks that otherwise would not be used at all for 5G. Dual Connectivity of a 3MHz carrier with one or more 3MHz or wider bandwidth carriers will increase radio coverage and cell hand-over performance. For railways the latter is particularly relevant for handovers for example between tunnel entries or exits, and the open field network coverage areas.

The below UE RF requirements are specified to enable the CA/DC operations for example band combination CA/DC_n100-n101 in TS 38.101-1:
- *CA configurations and bandwidth combinations for band combination CA_n100A-n101A*
- *GSCN parameters for SCell with 12 and 15 PRB transmission bandwidth*
- *ΔTIB,c requirements for inter-band CA band combination CA_n100A-n101A*
- *Spurious emission requirements for band combination CA_n100-n101*
- *UE maximum output power for inter-band CA band combination CA_n100A-n101A*
- *DC configurations for band combination DC_n100A-n101A*
- *UE maximum output power for inter-band DC band combination DC_n100A-n101A*

The below RRM requirements are specified in TS 38.133:
- *IDLE mode measurements for inter-frequency CA/DC candidate cells with 12PRB SSB*
- *Conditional handover with PSCell-PSCell change delay requirements with 12PRB SSB PSCell*
- *PSCell conditional change delay requirements with 12PRB SSB target PSCell*
- *SCell activation delay requirements with 12PRB SSB target SCell*
- *PSCell addition delay requirements with 12PRB SSB target PSCell*
- *Conditional PSCell addition delay requirements with 12PRB SSB target PSCell*
- *Subsequent Conditional PSCell addition delay requirements with 12PRB SSB target PSCell*
- *PSCell change delay requirements with 12PRB SSB target PSCell*
- *Conditional PSCell change delay requirements with 12PRB SSB target PSCell*
- *Subsequent Conditional PSCell change delay requirements with 12PRB SSB target PSCell*
- *SCG activation delay requirements with 12PRB SSB PSCell*
- *Deactivated SCell measurement period requirements with 12PRB SSB target SCell*
- *Deactivated PSCell measurement period requirements with 12PRB SSB target PSCell*

References

List of related CRs: select "TSG Status = Approved" in:
https://portal.3gpp.org/ChangeRequests.aspx?q=1&workitem=1030081,1031081,1032081,1080077

[1] RP-243309, Revised WID: NR channel BW less than 5MHz for FR1 Phase 2, Intel Corporation
[2] RP-250462, Status report to TSG RAN #107 for NR_FR1_lessthan_5MHz_BW_Ph2, Intel Corporation
[3] R4-2502608, Big CR for less than 5MHz RRM requirements in Rel-19, Intel Corporation
[4] R4-2502262, Big CR for less than 5MHz UE RF requirements in Rel-19, Intel Corporation

18.11 mmWave in NR: UE spurious emissions and EESS (Earth Exploration Satellite Service) protection

18.12 NR base station (BS) RF requirement evolution for FR1/FR2 and testing

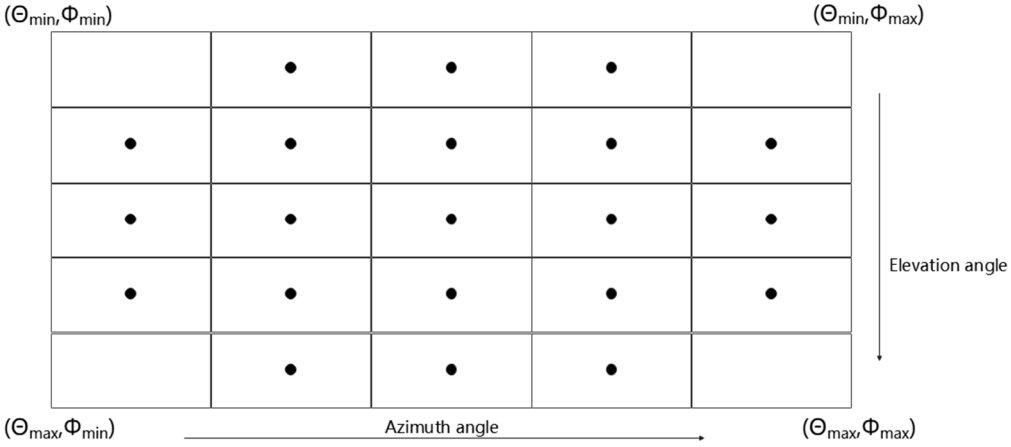
1030078	**NR base station (BS) RF requirement evolution for FR1/FR2 and testing**	NR_BS_RF_req_evo		RP-252578	ZTE
1031078	**Core part: NR base station (BS) RF requirement evolution for FR1/FR2 and testing**	NR_BS_RF_req_evo-Core	R4	RP-252578	ZTE
1032078	**Perf. part: NR base station (BS) RF requirement evolution for FR1/FR2 and testing**	NR_BS_RF_req_evo-Perf	R4	RP-252578	ZTE

Summary based on the input provided by ZTE in RP-253518, revised to RP-260625.

This Work Item develops several evolution aspects for NR base station (BS) RF requirement evolution for FR1/FR and testing, which are:

Expected EIRP mask for upper 6GHz

- The following EEIRP mask is specified for band n104 to protect the Earth-to-space fixed GSO satellite services (FSS) in the frequency range 6425 – 7075 MHz according to ITU-R regulatory requirement. In the conformance testing specification e.g.TS38.141-2 and TS38.115-2, RAN4 also defined the corresponding manufacture declaration for beam steering range, test beam direction set as illustrated as following, conformance test requirement and also test procedure and methodology to calculate the EEIRP in the discrete manner.



OTA test enhancement: OTA testing scope reduction and TRP simplification:

- in order to reduce the test burden for TX IMD requirement for FR1 BS, the worst case for different requirements for TX IMD was examined. The following reduction for the TX IMD conformance testing is specified:
  - For OTA ACLR requirements in clause 6.7.3, n=1.
  - For OTA OBUE in clause 6.7.4, n=3 for $BW_{Channel}$ of interfering signal is less than or equal to 5 MHz, otherwise n=1.
  - For OTA spurious emission in clause 6.7.5, n=3.

- in order to reduce the test burden for OOBB requirement for FR1 BS, RAN4 agreed to use the 1/3 of minimum supported bandwidth similar as FR2 BS OOBB requirement as the general principle and the detailed measurement step size for FR1 BS are agreed as following:

Minimum supported *BS channel bandwidth* (MHz)	Measurement step size (MHz)
<6	1
≥6 and <10	2
10, 15	3
20, 25	6
30, 35, 40	10
45, 50	15
60, 70, 80	20
90,100	30

- the following the TRP measurement updates are added:
  - The definition of anechoic chamber in TR 37.941;
  - The preferred method for measuring TRP for both in-band and out-of-band emissions as the “Spherical equal angle grid” and relevant clarification for other TRP measurement methods;
  - Update the preferred method in each of TRP based requirement in the test procedure section;
  - Capture the guidance can be used for selection of TRP measurement method, based on ERC Recommendation 74-01, Table 6 in Annex 2;
  - Some text improvement for Pre-scan method for the OTA spurious emission measurement.

References

List of related CRs: select "TSG Status = Approved" in:
https://portal.3gpp.org/ChangeRequests.aspx?q=1&workitem=1030078,1031078,1032078

[1] RP-253516 Status report for NR base station (BS) RF requirement evolution for FR1/FR2 and testing ZTE Corporation

18.13 UE Conformance - New Rel-19 NR licensed bands and extension of existing NR bands

18.14 Other band-related items

Note: no summary is provided for band-related items.

18.14.1 7MHz Channel Bandwidth for n26 and n5

1060083	**7MHz Channel Bandwidth for n26 and n5**	NR_FR1_7MHz_BW	R4	RP-251453	China Telecom

No summary expected for band-related items.

18.14.2 Introduction of the NR FDD 1.4 GHz band

1040127	**Introduction of the NR FDD 1.4 GHz band**	NR_FDD_1400MHz	R4	RP-250760	MidWave Wireless

No summary expected for band-related items.

18.14.3 Introduction of NR bands n87 and n88

1040131	**Introduction of NR bands n87 and n88**	NR_bands_n87_n88	R4	RP-242501	Nokia

No summary expected for band-related items.

18.14.4 Introduction of NR band n68

1040132	**Introduction of NR band n68**	NR_band_n68	R4	RP-250763	Ericsson

No summary expected for band-related items.

18.14.5 Additional NR bands for NR features in Rel-19

1040121	**Additional NR bands for NR features in Rel-19**	NR_bands_xFeature_R19	R4	RP-252463	ZTE

No summary expected for band-related items.

18.15 Study on spatial channel model for demodulation performance requirements for NR

1040100	*Study on spatial channel model for demodulation performance requirements for NR*	FS_NR_demod_SCM	R4	RP-251040	Nokia

Summary based on the input provided by Nokia in RP-252728

Note: Studies are not expected to appear in this document. Since the summary had been written anyway, it is included here.

Short summary: The study concludes that spatially agnostic TDL-based channel models are insufficient for accurate NR MIMO performance requirements, and that introducing aligned spatial channel models (rCDL-C1 and xTDL-C1) enables more realistic, repeatable, and normatively usable demodulation and CSI performance evaluation.

Limitations of the existing method for channel modelling for RAN4 performance requirements using the spatially agnostic Tapped Delay Line (TDL) or spatially deficient TDL + antenna correlation approaches have been discussed in RAN1 and in RAN4. These limitations may be attributed to the fact that TDL channels on their own are spatially agnostic, i.e., any direction to transmit signals through the channel is as good as any other, which impacts performance of different MIMO layers matched to differing large scale environments. Such shortcomings are, for example, visible in 2-codeword requirements in Rel-18, where both codewords have the same performance requirements at the same time.

Channel models with spatial property have been discussed since the beginning of NR. In Rel-14, two study items were progressed: the first one analyzing how to define a channel model representing the channel for frequencies up to 100 GHz [2], whose outcome was documented in TR 38.900 [3], and the second one aiming to develop an NR access technology to meet a broad range of use cases including enhanced mobile broadband, massive MTC, critical MTC [4], including the channel model definition that was finally documented in TR 38.901 [5]. RAN1 are using geometric based SCM(E) defined in 38.901 for system level simulation since this channel model is more realistic. The 38.901 CDL channel model is derived from SCM(E) with randomness reduction on some parameters, and the 38.901 TDL model is further derived from 38.901 CDL without spatial property [4]. In Rel-15, RAN4 decided to use simplified TDL model [6] for NR FR1 demodulation tests [7] and simplified TDL model for NR FR2 demodulation tests [8] since a majority of companies felt that TDL was enough for demodulation performance requirements, and it was hard to align simulation results using 38.901 CDL at that stage. A new SID [9] on the radiated metrics and test methodology for the verification of multi-antenna reception performance of NR UEs was approved in June 2018, from which TR 38.827 [10] was approved .

The present study item introduces Spatial Channel Models, one CDL based and one TDL based, for the purpose of demodulation requirements and CSI reporting requirements as agreed in TR 38.753, and as agreed in SID [1].

This SI has studied candidate spatial channel models for NR demodulation performance requirements in FR1, considering SU-MIMO scenarios, and evaluating CDL-based and Multi cluster TDL-based modelling approaches. The study has:
- Investigated methodology to generate repeatable spatial channel effects with manageable test complexity
- Down selected two stable SCM candidates for future requirements
- Compared performance outcomes across candidate models against agreed test cases
- Collected alignment results from multiple contributors to determine the span and average for key performance metrics.
- Found alignment for the SCM candidates in specified test cases

Regarding SCM Candidates the following observations were drawn:

Regarding rCDL-C1 the following observations can be drawn:
- The study has worked on alignment of rCDL-C1 implementations and has verified that there is a reasonable level of alignment.
- This level of alignment is expected to allow RAN4 to work on normative requirements.
- If the group considers the application of rCDL to settings that involve time-domain prediction it is necessary to further review the deterministic behaviour of rCDL and, if found necessary, identify countermeasures.

Regarding xTDL-C1 the following observations can be drawn:
- The study has worked on alignment of xTDL-C1 implementations and has verified that there is a reasonable level of alignment, among a lower number of contributors than other SCM candidates.
- This level of alignment likely allows RAN4 to work on normative requirements.

Regarding all channel models:
- Applicability of all channel models for band agnostic requirements to be evaluated for specific use case.
- All spatial channel models in this study have fixed angular properties and have non-uniform PMI statistics, hence a fixed precoder may perform better than a channel adaptive precoder found by PMI selection algorithm. Countermeasures can be identified if necessary, including but not limited to, e.g. the legacy beam steering approach defined in LTE and NR specifications [TS 38.101-4 section B.2.3.2.3].

References

List of related CRs: select "TSG Status = Approved" in:
https://portal.3gpp.org/ChangeRequests.aspx?q=1&workitem=1040100

[1] RP-250577, Revised SID: Study on spatial channel model for demodulation performance requirements, RAN#107 meeting, Nokia, BT Plc
[2] RP-160210, SID: Study on channel model for frequency spectrum above 6 GHz, RAN#71, Samsung, Nokia, March 2016
[3] TR 38.900, Study on channel model for frequency spectrum above 6 GHz
[4] RP-170379, SID: Study on New Radio Access Technology, NTT DOCOMO INC., RAN#75, March 2017
[5] TR 38.901, Study on channel model for frequencies from 0.5 to 100 GHz
[6] TS 38.101-4, User Equipment (UE) radio transmission and reception; Part 4: Performance requirements
[7] R4-1808400, NR Test Methods SI ad-hoc meeting notes, RAN4#87, Intel Corporation
[8] R4-1812001, RAN4#88 Meeting report, RAN4#88
[9] RP-181325, New SID on Study on radiated metrics and test methodology for the verification of multi-antenna reception performance of NR UEs, RAN#80, CATR, OPPO, Samsung
[10] TR 38.827, Study on radiated metrics and test methodology for the verification of multi-antenna reception performance of NR User Equipment (UE)
[11] TS 38.104, Base Station (BS) radio transmission and reception
[12] R4-2017678, “Way Forward on PMI Reporting Requirement for NR eMIMO”, RAN4#97-e
[13] R4-2214397, “WF on UE demodulation and CSI requirements for feMIMO”, RAN4#104b-e
[14] R4-2210669, “WF on CSI requirement for Rel-17 feMIMO”, RAN4#103-e
[15] R4-2402277, Discussion on 8Rx general demodulation aspects and spatial channel models, RAN4#110, Nokia, Nokia Shanghai Bell, BT

18.16 See also

Most of the Rel-19 improvements impact the NR (e.g. “Rel-19 Energy Efficiency, Energy Saving”; “Rel-19 Satellite (5GSAT), NTN, UAS, Aerial”; “HPUE”; “IoT”; “Ambient IoT”; etc.).

19 Rel-19 LTE Radio

19.1 LTE-based 5G Broadcast Phase 2

1060081	LTE-based 5G Broadcast Phase 2	LTE_terr_bcast_Ph2		RP-250794	EBU
1061081	Core part: LTE-based 5G Broadcast Phase 2	LTE_terr_bcast_Ph2-Core	R1	RP-250794	EBU
1062081	Perf. part: LTE-based 5G Broadcast Phase 2	LTE_terr_bcast_Ph2-Perf	R4	RP-250794	EBU

Summary based on the input provided by EBU in RP-252829.

For MBMS-dedicated cells, this work item specifies time-frequency interleavers (specified by RAN1) and corresponding signalling (specified by RAN2 and RAN3).

Time and frequency interleavers are crucial components for enhancing transmission robustness in time-varying (e.g., mobile reception) and frequency-selective (e.g., Single Frequency Network (SFN) deployments) channels. Physical-layer time interleaving, in particular, offers significant performance benefits, especially in high-mobility scenarios.

Following on the design principles of NR TBoMS[3], the basic approach of the proposed Time-Frequency Interleaver is to reuse the building blocks behind HARQ retransmissions and combining, which allows to spread the coded bits of a Transport Block (TB) across multiple non-consecutive slots/subframes.The HARQ chip memory remains unused for dedicated MBMS carriers (i.e. no unicast transmissions) and can be repurposed to perform PMCH time-interleaving, i.e. no additional in-chip memory may be required.

[3] Transport-Block-over-Multiple-Slots (TBoMS) is a physical-layer transmission concept introduced for NR broadcast/multicast, where the coded bits of a single transport block are spread across multiple non-consecutive slots or subframes rather than being transmitted in one slot. This improves robustness against mobility-induced time variation and SFN-induced frequency selectivity, while allowing reuse of existing HARQ mechanisms (e.g. redundancy versions and LLR combining) without introducing additional UE memory requirements, since HARQ feedback is not used for broadcast services.

In Figure 1, this mechanism is depicted: n 'redundancy versions' (RV) of each of m TBs are generated and then interleaved across a time unit of m× n subframes. To maintain the same overall throughput with non-interleaved transmission, the Transport Block Size (TBS) is scaled according to n. Hence, the same HARQ memory can be reused to store the Log-Likelihood Ratios (LLRs) corresponding to each redundancy version and decode after all n redundancy versions for a TB have been received.

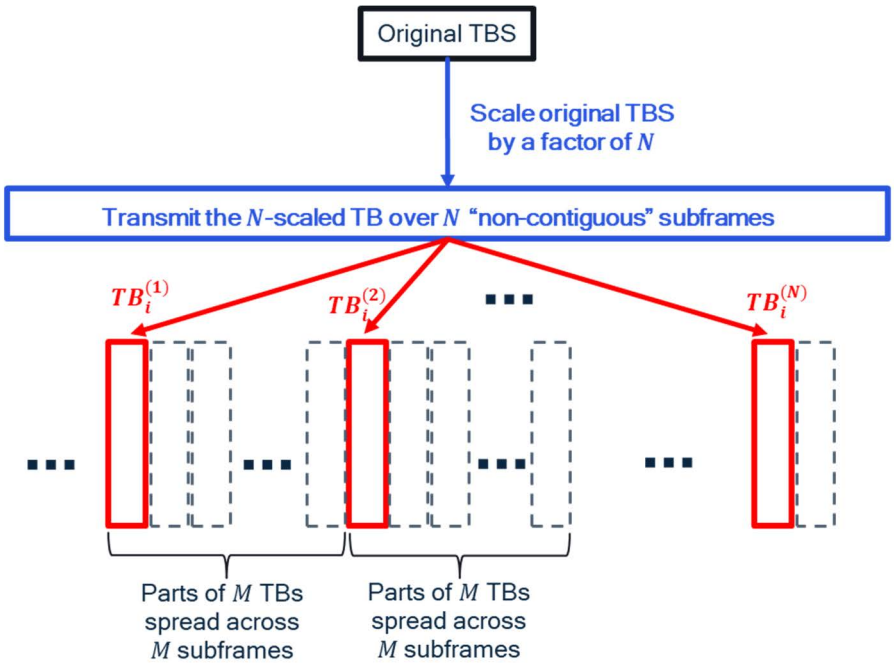


Figure 1: Subframe mapping of TBs for time interleaved PMCH transmission.

Frequency interleaving is key to counteract the erasure of OFDM subcarriers due to periodic frequency selectivity. This is a well-known effect of Single Frequency Network (SFN) scenarios, where frequency selectivity can erase entire codeblocks. Spreading the information of each codeblock across the entire system bandwidth would optimally mitigate the issue.

The main principles behind the proposed frequency interleaver are:

- The tones corresponding to the codeblocks are written column-wise.
- Pseudo random shifts are applied to the columns. This prevents periodic fades in the frequency domain (e.g., from two strong taps spaced apart in the delay domain) from affecting specific codeblocks negatively.
- Data symbols in an OFDM symbol are read row-wise.

References

List of related CRs: select "TSG Status = Approved" in:
https://portal.3gpp.org/ChangeRequests.aspx?q=1&workitem=1060081,1061081,1062081

[1] RP-250794, “Revised WID on LTE-based 5G Broadcast Phase 2”, TSG RAN Meeting #107, Incheon, Korea, March 2025;
[2] RP-251975, Status Report for “New WID on LTE-based 5G Broadcast Phase 2”

19.2 Rel-19 LTE-Advanced Carrier Aggregation for x bands (1<=x<= 6) DL with y bands (y=1, 2) UL

No summary provided for band-related items.

19.3 Band-related items

19.3.1 New bands for LTE based 5G terrestrial broadcast for early deployments

No summary provided for band-related items.

19.3.2 Introduction of LTE FDD band in 1800–1830 MHz for Canada

No summary provided for band-related items.

20 Rel-19 Mission Critical, eCall, Emergency

20.1 Enhanced Mission Critical Architecture

1000039	**Enhanced Mission Critical Architecture**	enhMC	S6	SP-241691	Negalaguli, Harish, Motorola Solutions,

Summary based on the input provided by Motorola Solutions by e-mail received Wed 25/03/2026 05:17, file "Summery_1000039".

This Work Item enhances the Mission Critical (MC) architecture by addressing identified gaps and continuing enhancements from previous Releases, while ensuring full backward compatibility.

Enhancements to location management are specified, including support for default configurations, trigger override mechanisms, and improved reporting for UEs associated with multiple devices.

The Common Location Service (CLS) is specified as an independent common service, decoupled from the MCPTT framework. Corresponding architecture, procedures, and functional aspects are defined to enable dedicated location management clients and servers to interact via standardized interfaces.

The protocol is enhanced through the use of HTTP-based mechanisms, aligned with HTTP/3, to support efficient retrieval and notification of location information for MC users.

References

List of related CRs: select "TSG Status = Approved" in:
https://portal.3gpp.org/ChangeRequests.aspx?q=1&workitem=1000039

[1] TS 23.379: " Functional architecture and information flows to support Mission Critical Push To Talk (MCPTT); Stage 2".
[2] TS 23.379: "Functional architecture and information flows to support Mission Critical Push To Talk (MCPTT); Stage 2".
[3] TS 23.281: "Functional architecture and information flows for mission critical video; Stage 2".
[4] TS 23.282: "Functional model and information flows for Mission Critical Data".
[5] TS 24.379: "Mission Critical Push to Talk (MCPTT) call control Protocol specification".
[6] TS 23.283: "Mission Critical Communication Interworking with Land Mobile Radio Systems; Stage 2".
[7] TS 24.482: " Mission Critical Services (MCS) identity management; Protocol specification".
[8] TS 24.483: "Mission Critical Services (MCS) Management Object (MO)".
[9] TS 24.484: "Mission Critical Services (MCS) configuration management Protocol specification".
[10] TS 24.281 Mission Critical Video (MCVideo) signalling control; Protocol specification;
[11] TS 24.282 Mission Critical Data (MCData) signalling control; Protocol specification;

20.2 Enhanced Mission Critical Location Management

1040045	**Enhanced Mission Critical Location Management**	enhMCLoc	C1	CP-241288	Magnus Tränk, Ericsson

Summary based on the input provided by Ericsson by e-mail received Thu 19/03/2026 15:17, file "CP-26xxxx enhMCLoc WI summary".

The enhMCLoc work item enhances the location management functionality of Mission Critical (MC) servicesby introducing a new protocol for the interface between the Location Management Client and Location Management Server (reference point CSC-14), as well as between the MC Service Server and Location Management Server (reference point CSC-15). This new HTTP-based protocol is specified in TS 24.283 [1].

Prior to this enhancement, location management functionality was limited to location configuration and reporting between each MC service client and its corresponding MC service server. With the introduction of the new reference points, location management traffic can be handled via a common interface shared across all three MC services. This reduces the need to distribute location information individually to each MC service server and enables a more centralized and efficient handling of location data.

The newly specified protocol provides the same capabilities as the existing location management functionality between MC service clients and MC service servers, while also introducing improved procedures for managing location subscriptions. These enhancements allow for subscriptions to location information from MC clients using MC service IDs, Functional Alias IDs, or MC service group IDs, supporting a more adaptable and scalable approach to distributing location information.

References

List of related CRs: select "TSG Status = Approved" in:
https://portal.3gpp.org/ChangeRequests.aspx?q=1&workitem=1040045

[1] TS 24.283: "Mission Critical Location Management (MCLoc); Protocol specification".

20.3 Alignment of eCall over IMS with CEN

1050027	Alignment of eCall over IMS with CEN	eCallCEN	C1	CP-251223	Osama Lotfallah, Qualcomm Incorporated
1060075	UE Conformance - Alignment of eCall over IMS with CEN	eCallCEN-UEConTest	R5	RP-251963	Qualcomm

Summary based on the input provided by Qualcomm in CP-260142/SP-260255.

3GPP specifications define support for eCall over IMS for Europe using LTE and NR access since Release 16. CEN specifications, which are largely based on 3GPP specifications, define High Level Application Protocol (HLAP) support for eCall over IMS for Europe using LTE and NR access. While most requirements in 3GPP specifications align with those in CEN, there are a small number of UE (IVS) requirements that are different.

The European Commission has issued new regulations mandating support of eCall over IMS by: PSAPs starting on 1 January 2026; by new vehicles without a preexisting type approval starting on 1 January 2026; and by new vehicles with a preexisting type approval starting on 1 January 2027. The new regulations require eCall over IMS compliance with CEN specifications and not directly with ETSI specifications. Although CEN specifications allow for 3GPP compliance as an option where there are differences in UE requirements, an evaluation by CEN TC 278 WG 15 showed that for some of the different requirements, the CEN requirements would result in more reliable eCall over IMS service. CEN TC 278 WG 15 has therefore asked 3GPP to consider aligning 3GPP specifications to CEN specifications for these requirements.

The following changes were made to the 3GPP specifications for eCall over IMS to align with CEN:
1) Support for transfer of an updated Minimum Set of Data (MSD) during a PSAP callback was added.
2) Support for transfer of an MSD during a test eCall, and support of an updated MSD during a callback following a test eCall, were added.
3) It was clarified that receipt of the MSD using audio media stream encoded as described in TS 26.267 [3] is optional for a PSAP that supports eCall over IMS according to CEN specifications.
4) Upon sending an INVITE for an eCall over IMS while not in limited service state, the UE shall treat a SIP 4xx or 6xx response including a positive acknowledgement of the MSD as being equivalent to a successful eCall that is now released.
5) If the UE receives a SIP 200 OK response to an INVITE for an eCall over IMS including a negative acknowledgement of the MSD, the UE shall consider the MSD transmission as unsuccessful but shall not consider it as a trigger to send the MSD using audio media stream encoded as described in TS 26.267 [3].
6) If the UE receives a SIP 200 OK response to an INVITE for an eCall over IMS not including any acknowledgement of the MSD, the UE shall send the MSD using audio media stream encoded as described in TS 26.267 [3] if the PSAP requests it.

References

List of related CRs: select "TSG Status = Approved" in:
https://portal.3gpp.org/ChangeRequests.aspx?q=1&workitem=1050027;1060075

[1] TS 24.229: "IP multimedia call control protocol based on Session Initiation Protocol (SIP) and Session Description Protocol (SDP); Stage 3".
[2] TS 23.167: "IP Multimedia Subsystem (IMS) emergency sessions".
[3] TS 26.267: "eCall Data Transfer; In-band modem solution; General description".
[4] CEN EN 17184:2024: “Intelligent transport systems — eSafety — eCall High level application Protocols (HLAP) using IP Multimedia Subsystem (IMS) over packet switched network”.

20.4 UE Conformance - Alignment of eCall over IMS with CEN

No summary provided for testing-only items.

20.5 Multiple Location Procedure for Emergency LCS Routing

1030017	**Multiple Location Procedure for Emergency LCS Routing**	TEI19_MLR4RTR	S2	SP-240123	Simon Cai, Nokia Shanghai-Bell

Summary based on the input provided by Nokia in SP-260038.

This WI introduces support for providing device location information to multiple emergency call routing entities during a single emergency session, in order to meet regulatory requirements for location-based emergency call routing without delaying normal call setup.

To achieve this, the LCS in the operator network is enhanced to provide location information to multiple LCS clients (such as emergency routing entities) for the same device, while meeting the required QoS and latency constraints at each routing hop.

As emergency calls are routed hop by hop, each routing entity can receive increasingly accurate location information as additional measurements become available. Initial routing may use coarse, cell-based location, while later routing decisions can rely on more precise estimates, which is especially important near PSAP service area boundaries.

The existing multiple location procedure defined in TS 23.273 currently supports only a single LCS client. This WI extends that procedure to allow multiple emergency LCS clients to receive a sequence of location updates within a single LCS session associated with an emergency session.

This is implemented as an extension of the 5GC-MT-LR multiple location procedure without UDM query, where the GMLC manages successive requests from emergency routing entities and responds with the latest or most accurate location information received from the LMF.

References

List of related CRs: select "TSG Status = Approved" in:
https://portal.3gpp.org/ChangeRequests.aspx?q=1&workitem=1030017

[1] TS 23.273: "5G System (5GS) Location Services (LCS); Stage 2)"

20.6 Multimedia Priority Service (MPS) for Messaging services

1040083	**MPS for Messaging services**	MPS4msg		SP-220939	Singh, Ray, Peraton Labs
970042	**(stage 1 of) MPS for Messaging services**	MPS4msg	S1	SP-220939	Singh, Ray, Peraton Labs
1040029	**(stage 2 of) MPS for IMS Messaging and SMS services**	MPS4msg	S2	SP-240987	Robert Streijl (Peraton Labs),
1010031	*Study on MPS for IMS Messaging and SMS services*	FS_MPS4msg	S2	SP-231197	Robert Streijl (Peraton Labs)

1050007	**CT aspects of MPS for IMS Messaging and SMS services**	MPS4msg		CP-243047	Monnes, Peter (Peraton Labs)
1050063	CT1 aspects of MPS for IMS Messaging and SMS services	MPS4msg	C1	CP-243047	Monnes, Peter (Peraton Labs)
1050064	CT3 aspects of MPS for IMS Messaging and SMS services	MPS4msg	C3	CP-243047	Monnes, Peter (Peraton Labs)
1050065	CT4 aspects of MPS for IMS Messaging and SMS services	MPS4msg	C4	CP-243047	Monnes, Peter (Peraton Labs)

Summary based on the input provided by Peraton Labs by e-mail received Wed 01/04/2026 03:00, file "Summary_MPS4msg_Rel19features_03-31-2026".

MPS for Messaging provides priority for messaging services in periods of network congestion during which normal commercial messaging services are degraded.

Stage 1 summary

Normative stage 1 requirements were added in TS 22.153 [1] Release 19 for the system to support MPS priority for messaging services for an authorized Service User using a UE with a subscription for MPS. This includes MPS priority for messaging services supported using IMS Messaging, SMS and/or MMS, or MSGin5G.

Stage 2 summary

Normative stage 2 solutions were added in Release 19 based on the study in TR 23.700-75 [2] which addressed how MPS for messaging services is requested and authorized using a UE with a subscription for MPS, gaps for MPS priority treatment for messaging services using TS 23.228 [3] specified SIP MESSAGE and SIP sessions over the EPC and 5GC, and gaps for MPS priority treatment in the EPC and 5GC for messaging services using MO/MT SMS over IP, and MO/MT SMS over NAS.

The following is a summary of key SA2 updates for EPS specifications and interworking with SMS:

- In TS 23.401 [4], text was added to clause 4.3.18.1 to indicate that an MPS subscription may include an optional MPS for Messaging indication which sets(enables)/clears(disables) the MPS priority treatment of messaging service (i.e. SMS over NAS, SMS over IP and messaging over IMS are all controlled by one indication). The MPS for Messaging indication is used in EPC entities.
- In TS 23.203 [5], the text on dynamic invocation in clause 6.1.11.1 was expanded to include MPS for Messaging, text was added to extend PCC rules and QoS rule generation to MPS for Messaging in clause 6.1.11.2, and a new clause was added on IMS Signalling Bearer priority support for MPS for Messaging, and the use of the Rx interface to trigger priority treatment.
- In TS 23.682 [6], text in clause 4.5.21 was updated to introduce AS triggered MPS for Messaging indication parameter provisioning in the HSS to invoke/revoke MPS for Messaging, and the clause 5.18 procedure is updated for the SCS/AS, SCEF and HSS to support the new parameter.
- In TS 23.204 [7], updates were included in clause 5.3.1.1 to add Resource-Priority handling to the list of IP-SM-GW functions, clause 5.3.2 to add storing of the MPS for Messaging indication to the list of HLR/HSS functions, clauses 6.7 and 6.13 for a mobile originated Instant Message to support P-CSCF check that MPS for Messaging is enabled for the UE in the HSS, and for the IP-SM-GW to convert it (at the originating or terminating side) to an SMS and include a transport level priority marking, and clause 6.14 to support priority for the case a mobile terminated priority SMS, where the SMS-GMSC includes a transport level priority marking when MPS for messaging is enabled for the UE, and the SMS is converted by the IP-SM-GW to an Instant message and marked with MPS appropriate Resource-Priority information for delivery with priority to the UE.
- In TS 23.272 [8], clauses C.1, C.4.2, C.4.3, C.7.1, C.7.2, C.8.1 were updated for MPS support for MO and MT SMS delivery via MME. Updates include the MPS for Messaging indicator in the HSS and specify its distribution to the MME, and add a new trigger for paging priority in the MME.

The following is a summary of key SA2 updates for 5GS specifications:

- In TS 23.501 [9], text was added in clause 5.16.5 to indicate that an MPS subscription supports setting (enabling)/clearing (disabling) priority treatment of the Messaging service (i.e. SMS over NAS, SMS over IP and messaging over IMS are all controlled by one indication) for an MPS-subscribed UE. Text was added in clause 5.22.2 to specify usage of the MPS for Messaging indication in the UDM subscription data, and to allow it to be provisioned by an AF via a NEF. The MPS for Messaging indication is used in 5GC entities.
- In TS 23.502 [10], text was added in clause 4.15.6.3a for the AF triggered MPS for Messaging Indication parameter provisioning, and the NEF and UDM service operations were updated to include the new parameter.

The procedures in clauses 4.2.2.2.2, 4.13.3.1, 4.13.3.3, 4.13.3.6 and 5.2.3.3.1 were updated to support MPS for Messaging priority for SMS. Changes include download of the MPS for Messaging indication from the UDM to the concerned FEs, the use of an MPS-appropriate Message Priority header or DSCP value in the steps pertaining to delivery of an MPS SMS, and a new trigger for paging priority in the AMF.

- In TS 23.503 [11], new text for MPS for Messaging was added in clause 6.1.3.11, including support for dynamic invocation via a request to the PCF over N5, and upgrade of the IMS Signalling Bearer priority.

The following is a summary of key SA2 updates for IMS specification:

- In TS 23.228 [3], updates were included in clause 3.1 to add a definition of MPS for Messaging, clause 4.6.1 to add Resource-Priority information handling and MPS for Messaging to the list of P-CSCF functions, clause 5.2.2.3 to add MPS for Messaging to the information that the S-CSCF makes available to the P-CSCF, clause 5.16.1.1.1 for the Immediate messaging case to add a P-CSCF check of the MPS for Messaging indication and paging priority support at the MME/AMF for MPS for Messaging, clauses 5.16.2.2.1, 5.16.2.2.2, and 5.16.2.2.3 for the Session-based cases to add a P-CSCF check of the MPS for Messaging indication and add Resource-Priority information to the INVITE before forwarding, and clause 5.21 to add general information about all MPS for IMS messaging features.

The following is a summary of the key CT4 stage 2 specification update for 5GS:

- In TS 23.540 [12], the call flows are extended to include the “MPS for Messaging indication” to support MPS for Messaging priority for Mobile Originated (clause 5.2.2) and Mobile Terminated (clauses 5.1.2, 5.1.3, 5.1.4) service-based SMS over NAS. The MPS for Messaging indication may be obtained from the UDM using the SM1 (UDM to SMS-GSMSC), SM4 (UDM to SMS Router), or SM3 (UDM to IP-SM-GW) interfaces. The solution also permits priority for the MT case to be triggered by an out-of-scope priority indication provided in the SMS received by the SC.

Stage 3 summary

Normative stage 3 solutions in Release 19 are specified for support of MPS for Messaging services including MO/MT SMS over NAS, MO/MT SMS over IP, and MO/MT messaging over IMS.

The following is a summary of key CT1 specification updates:

- In TS 24.229 [13], updates for the P-CSCF to include an MPS-appropriate RPH in MO SIP MESSAGEs and INVITE requests and to handle these requests with priority when the MPS for Messaging indication is set, and to specify the MPS for Messaging indication is conveyed in the regEvent from the S-CSCF to the P-CSCF.
- In TS 24.247 [14], updates for the list server or conferencing application server to add an MPS-appropriate RPH in support of MPS for Messaging when a text is sent to a list of recipients or to a conference number.
- In TS 24.301 [15], updates provide paging priority for MPS for Messaging when the MPS for Messaging indication for the UE received from the HSS is set.
- In TS 24.501 [16], updates provide paging priority for MPS for Messaging when the MPS for Messaging indication for the UE received from the UDM is set.

The following is a summary of CT3 specification updates:

- In TS 29.122 [17], updates provide the means for an SCS/AS to employ the T8 interface to a SCEF to provision the HSS with the MPS for Messaging indication.
- In TS 29.213 [18], call flow and mapping table updates add support for invocation of MPS for Messaging on the Rx interface.
- In TS 29.214 [19], procedural updates are added for the PCRF/PCF receiving the MPS for Messaging indication on the Rx interface and controlling the MPS priority of the IMS signalling bearer/QoS flow.
- In TS 29.513 [20], call flow and mapping table updates add support for invocation of MPS for Messaging on the N5 interface.
- In TS 29.514 [21], procedural updates are added for the PCF receiving the MPS for Messaging indication on the N5 interface and controlling the MPS priority of the IMS signalling bearer/QoS flow.
- In TS 29.522 [22], updates provide the means for an AF to employ the N33 interface to a NEF to provision the UDM with the MPS for Messaging indication.

The following is a summary of CT4 specification updates:

- In TS 29.228 [23], updates convey the MPS for Messaging indication on the Cx (HSS to S-CSCF) interface at times of S-CSCF registration and HSS initiated update of User Information.

- In TS 29.272 [24], updates convey the MPS for Messaging indication on the S6a (HSS to MME) interface within the MPS-Priority AVP as part of the Update Location and Insert Subscriber Data procedures.
- In TS 29.336 [25], updates convey the MPS for Messaging indication on the S6t (SCEF to HSS) within the MPS-Priority AVP as part of the Configuration Information on S6t procedure.
- In TS 29.338 [26], updates convey MPS for Messaging indication in the MPS-Priority AVP on the S6c (HSS/UDM to SMS-GMSC / SMS Router) interface, and a DRMP AVP with a value appropriate for MPS Priority. For the SGd (MME/SMSF/IP-SM-GW to SMS-GMSC/SMS-IWMSC/SMS Router) interface the MPS-Priority AVP is included from the SMS-GMSC with a MT Short Message Request, and the DRMP AVP is set with MPS priority for participating FEs on the SGd interface.
- In TS 29.503 [27], updates for MPS for Messaging indication in the UDM, the distribution of the indication to the AMF and SMS-GMSC, and to permit an on-demand service, a means to set the indication through a NEF. Towards the SMS-GMSC, the UDM sets the SBI-Message-Priority header to a value appropriate for MPS.
- In TS 29.540 [28], updates for MPS for Messaging to include a priority transport layer marking on a MT SMS sent towards the SMSF. Priority markings are included either based on MPS for Messaging indication in the UDM or a priority indication (out of 3GPP scope) contained in the SMS received from the SC.
- In TS 29.562 [29], updates convey the MPS for Messaging indication on the Nhss (HSS to S-CSCF) interface using a new attribute in the PriorityLevels type.
- In TS 29.577 [30], updates allow the SMS-GMSC to set the Sbi-Message-Priority for MPS for Messaging on the SM8 (IP-SM-GW to SMS-GMSC) and SM6 (SMS Router to SMS-GMSC) interfaces, and the UDM to set the Sbi-Message-Priority for MPS for Messaging on the SM3 (UDM to IP-SM-GW) and SM4 (UDM to SMS Router) interfaces.

References

List of related CRs: select "TSG Status = Approved" in:
https://portal.3gpp.org/ChangeRequests.aspx?q=1&workitem=1040083;970042;1040029;1010031;1050007;1050063;1050064;1050065

[1] TS 22.153: "Multimedia priority service".
[2] TR 23.700-75: "Study on Multimedia Priority Service (MPS) for IMS Messaging and SMS services".
[3] TS 23.228: "IP Multimedia Subsystem (IMS); Stage 2".
[4] TS 23.401: "General Packet Radio Service (GPRS) enhancements for Evolved Universal Terrestrial Radio Access Network (E-UTRAN) access".
[5] TS 23.203: "Policy and charging control architecture".
[6] TS 23.682: "Architecture enhancements to facilitate communications with packet data networks and applications".
[7] TS 23.204: "Support of Short Message Service (SMS) over generic 3GPP Internet Protocol (IP) access; Stage 2".
[8] TS 23.272: "Circuit Switched (CS) fallback in Evolved Packet System (EPS); Stage 2".
[9] TS 23.501: "System architecture for the 5G System (5GS)".
[10] TS 23.502: "Procedures for the 5G System (5GS)".
[11] TS 23.503: "Policy and charging control framework for the 5G System (5GS); Stage 2".
[12] TS 23.540: "5G System: Technical realization of Service Based Short Message Service; Stage 2".
[13] TS 24.229: "IP multimedia call control protocol based on Session Initiation Protocol (SIP) and Session Description Protocol (SDP); Stage 3".
[14] TS 24.247: "Messaging service using the IP Multimedia (IM) Core Network (CN) subsystem; Stage 3".
[15] TS 24.301: "Non-Access-Stratum (NAS) protocol for Evolved Packet System (EPS); Stage 3".
[16] TS 24.501: "Non-Access-Stratum (NAS) protocol for 5G System (5GS); Stage 3".
[17] TS 29.122: "T8 reference point for Northbound APIs".
[18] TS 29.213: "Policy and charging control signalling flows and Quality of Service (QoS) parameter mapping".
[19] TS 29.214: "Policy and charging control over Rx reference point".
[20] TS 29.513: "5G System; Policy and Charging Control signalling flows and QoS parameter mapping; Stage 3".
[21] TS 29.514: "5G System; Policy Authorization Service; Stage 3".
[22] TS 29.522: "5G System; Network Exposure Function Northbound APIs; Stage 3".
[23] TS 29.228: "IP Multimedia (IM) Subsystem Cx and Dx Interfaces; Signalling flows and message contents".
[24] TS 29.272: "Evolved Packet System (EPS); Mobility Management Entity (MME) and Serving GPRS Support Node (SGSN) related interfaces based on Diameter protocol".

[25] TS 29.336: "Home Subscriber Server (HSS) diameter interfaces for interworking with packet data networks and applications".
[26] TS 29.338: "Diameter based protocols to support Short Message Service (SMS) capable Mobile Management Entities (MMEs)".
[27] TS 29.503: "5G System; Unified Data Management Services; Stage 3".
[28] TS 29.540: “5G System; SMS Services; Stage 3”.
[29] TS 29.562: "5G System; Home Subscriber Server (HSS) services; Stage 3".
[30] TS 29.577: "5G System; IP Short Message Gateway and SMS Router for Short Message Services; Stage 3".

20.7 Mission Critical (MC) services for generic support on Isolated Operation for Public Safety (IOPS) mode of operation

1050036	MC services for generic support on IOPS mode of operation	Generic_IOPS	S6	SP-241382	FirstNet, Mark Lipford

Summary based on the input provided by FirstNet in SP-251513.

The work item transforms the LTE IOPS functionality into an access-agnostic service capable of operating over LTE, 5G NR, non-3GPP access, and future 3GPP technologies. No new features were introduced, and the resulting specification has no dependencies on other 3GPP work items.

The updated IOPS specification ensures continuity of mission-critical capabilities as networks transition from 4G to 5G. The previous LTE-specific design was re-evaluated and revised to remove access-dependent elements, enabling a common, access-independent service architecture suitable for modern and future deployments. Security considerations raised in the specification were resolved following exchanges with SA3.

As defined in Release 17, IOPS provides operational capability in scenarios where base station–to–core network connectivity is lost due to catastrophic events. With the updated access-agnostic design, users equipped with IOPS-capable devices can continue to access limited MCPTT and MCData services regardless of the underlying access technology.

References

List of related CRs: select "TSG Status = Approved" in:
https://portal.3gpp.org/ChangeRequests.aspx?q=1&workitem=1050036

[1] TS 23.180: “Mission critical services in the Isolated Operations for Public Safety (IOPS) mode of operations”

20.8 Sharing of administrative configuration between interconnected MC service systems

1000036	Sharing of administrative configuration between interconnected MC service systems	MCShAC	S6	SP-230692	andreas.flander@bdbos.bund.de

Summary based on the input provided by BDBOS by e-mail received Mon 30/03/2026 14:01, file "Summary_1000036".

This Work Item complements the mission critical Common Functional Architecture defined in TS 23.280 [1] through optional functional entities which are required for sharing administrative configuration between interconnected Mission Critical (MC) service systems.

The new functional entities such as Administration Configuration Management Client (ACMC) and Server (ACMS) provide means to exchange securely administrative configuration data between interconnected MC service systems. New information flows and procedures permit creating and managing interconnected MC service groups. MC service users can be enabled or disabled for migration into a Partner MC system. The MC service UE initial configuration can be updated for migrated MC service users. ACM clients, which trigger such exchanges and activities, are tight to profiles managed by MC system administrators who assign authorizations based on interconnected MC service system’s agreed policy.

References

List of related CRs: select "TSG Status = Approved" in:
https://portal.3gpp.org/ChangeRequests.aspx?q=1&workitem=1000036

[1] TS 23.280: "Common functional architecture to support mission critical services; Stage 2".

20.9 Future Railway Mobile Communication System (FRMCS) Phase 5

1000031	**FRMCS Phase 5**	FRMCS_Ph5	S1	SP-230512	Nikolopoulou, Vassiliki, UIC
1000038	**Stage 2 of Railways specific Enhancements to Mission Critical Services**	FRMCS_Ph5	S6	SP-230780	Oettl, Martin, Nokia,
1050018	**CT aspects of Railways specific Enhancements to Mission Critical Services**	FRMCS_Ph5		CP-242102	Won, Sung Hwan, Nokia
1050103	CT1 aspects of Railways specific Enhancements to Mission Critical Services	FRMCS_Ph5	C1	CP-242102	Won, Sung Hwan, Nokia
850044	*Study on Supporting of Railway Smart Station Services*	FS_RAILSS	S1	SP-190838	Han, Andrew Min-gyu, Hansung University
950007	*Study on FRMCS Phase 5*	FS_FRMCS_Ph5	S1	SP-241693	Patrick WIMART, UIC

Summary based on the input provided by UIC by e-mail received Mon 02/03/2026 16:47 , file "Summary_1000031".

Rel-19 mainly focuses on the enhancement of Ad hoc Group communications procedures and functionalities to support any identified gaps with the operational procedures (e.g., modifying criteria during an active communication, interconnection and migration, interworking with GSM-R).

Moreover, to support some smart station services, location-based services were enhanced (e.g., Mission-critical mobility service by leveraging sophisticated controllable location information along with high accuracy)

The work was based on Rel-18 achievements and aims to complete and enhance several aspects which could not be finished within Rel-18 timeframe.

The Rel-19 enhancements for railways were mainly updates and additions aiming to:
- enable modifying the criteria for determining the participants during an ongoing ad hoc group emergency alert within single or multiple MC systems
- enable modification of Ad hoc Group call criteria by an authorized user involving single or multiple MC systems
- enable release of an Ad hoc Group call by an authorized user
- conditions of termination of Ad hoc Group calls
- enable floor revoke by the authorized user
- delivery of initiator's location information to the target users' application for Ad hoc Group calls
- the usage of functional alias in the participants list and notifications of Ad hoc Group calls
- distribute the Application Priority to all participants of Ad hoc Group calls
- specify the architecture and procedures to support interworking with GSM-R for MCPTT private calls, MCPTT Ad hoc Group calls, MCData SDS and automatic subsequent MCPTT Ad hoc Group calls following an Ad hoc Group emergency alert.
- apply multi-talker features to Ad hoc Group calls
- update user profile configuration data and service configuration data
- enhance the architecture to allow the separation of the signaling and the media paths within an MC system.

The following technical specifications on MCPTT (Mission Critical Push-To-Talk), MCVideo, MCData have been updated:
Stage 1: TS 22.179 [1], TS22.280 [2]
Stage 2: TS 23.280 [3], TS 23.281 [4], TS 23.282 [5], TS 23.283 [6], TS23.289 [7], TS 23.379 [8],
Stage 3: TS 24.281 [9], TS 24.282 [10], TS 24.379 [11], TS 24.380 [12], TS 24.481 [13], TS 24.482 [14], TS 24.484 [15], TS 29.379 [16]

References

List of related CRs: select "TSG Status = Approved" in:
https://portal.3gpp.org/ChangeRequests.aspx?q=1&workitem=1000031;1000038;1050018;1050103;850044

[1] TS 22.179: "Mission Critical Push to Talk (MCPTT); Stage 1"
[2] TS 22.280: "Mission Critical Services Common Requirements (MCCoRe); Stage 1"
[3] TS 23.280: "Common functional architecture to support mission critical services; Stage 2 "
[4] TS 23.281: "Functional architecture and information flows to support Mission Critical Video (MCVideo); Stage 2".
[5] TS 23.282: "Functional architecture and information flows to support Mission Critical Data (MCData); Stage 2".
[6] TS 23.283: "Mission Critical Communication Interworking with Land Mobile Radio Systems "
[7] TS 23.289: Mission Critical services over 5G System; Stage 2
[8] TS 23.379: " Functional architecture and information flows to support Mission Critical Push To Talk (MCPTT); Stage 2".
[9] TS 24.281: "Mission Critical Video (MCVideo) signalling control; Protocol specification".
[10] TS 24.282: "Mission Critical Data (MCData) signalling control; Protocol specification".
[11] TS 24.379: "Mission Critical Push To Talk (MCPTT) call control; Protocol specification".
[12] TS 24.380: "Mission Critical Push To Talk (MCPTT) media plane control; Protocol specification"
[13] TS 24.481: "Mission Critical Services (MCS) group management; Protocol specification"
[14] TS 24.482: "Mission Critical Services (MCS) identity management; Protocol specification"
[15] TS 24.484: "Mission Critical Services (MCS) configuration management; Protocol specification"
[16] TS 29.379: "Mission Critical Push To Talk (MCPTT) call control interworking with Land Mobile Radio (LMR) systems; Stage-3"

20.10 Mission critical security enhancements for release 19

1020035	Mission critical security enhancements for release 19	MCXSec4	S3	SP-231783	Woodward, Tim, Motorola Solutions, Inc.

Summary based on the input provided by Motorola Solutions in SP-251565.

MCXSec4 enhances the well-defined security architecture, procedures and information flows to protect the mission critical service (MCX). The security architecture, messages and procedures include mechanisms to protect the Common Functional Architecture for mission critical applications. This includes Push-To-Talk (MCPTT), Video (MCVideo) and Data (MCData). Additionally, security mechanisms relating to on-network use, off-network use, roaming, migration, interconnection, interworking, Isolated E-UTRAN Operation for Public Safety (IOPS), and multiple security domains are covered.

It complements the mission critical Common Functional Architecture defined in TS 23.280 [1], the functional architecture for MCPTT defined in TS 23.379 [2], the functional architecture for MCVideo defined in TS 23.281 [3], the functional architecture for MCData defined in TS 23.282 [4], Interworking of 3GPP mission critical systems with Land Mobile Systems defined in TS 23.283 [5], and Mission critical services support in the Isolated Operation for Public Safety (IOPS) mode of operation in TS 23.180 [6].

The MC Service can be used for public safety applications and also for general commercial applications e.g. utility companies and railways. As the security model is based on the public safety environment, some MC security features may not be applicable for commercial purposes.

References

List of related CRs: select "TSG Status = Approved" in:
https://portal.3gpp.org/ChangeRequests.aspx?q=1&workitem=1020035

[1] TS 23.280: "Common functional architecture to support mission critical services; Stage 2".
[2] TS 23.379: "Functional architecture and information flows to support Mission Critical Push To Talk (MCPTT); Stage 2".
[3] TS 23.281: "Functional architecture and information flows for mission critical video; Stage 2".
[4] TS 23.282: "Functional model and information flows for Mission Critical Data".
[5] TS 23.283: "Mission Critical Communication Interworking with Land Mobile Radio Systems; Stage 2".
[6] TS 23.180: "Mission critical services support in the Isolated Operation for Public Safety (IOPS) mode of operation; Functional architecture and information flows."TS 22.261: “Service Requirements for the 5Gsystem”
[2] TS22.280:”Mission Critical Services Common Requirements (MCCoRe)”

20.11 Protocol enhancements for Mission Critical Services

No summary provided for protocol enhancements.

20.12 See also:

Support for PWS in Satellite E-UTRAN and Satellite NG-RAN; NTN for IoT Phase 3 (for LTE) (1020096, IoT_NTN_Ph3); slide on Sidelink/UE-to-UE

21 Rel-19 Network Slicing

21.1 Network Controlled Network Slice Selection

1070041	**Network Controlled Network Slice Selection**	TEI19_SliceSel		SP-241502	Saito, Koji, KDDI
1060030	**Stage 2 of Network Controlled Network Slice Selection**	TEI19_SliceSel	S2	SP-241502	Saito, Koji, KDDI
1070003	**CT aspects of Network Controlled Network Slice Selection**	TEI19_SliceSel		CP-251133	Xiaojian Yan, ZTE
1070042	CT3 aspects of Network Controlled Network Slice Selection	TEI19_SliceSel	C3	CP-251133	Xiaojian Yan, ZTE
1070043	CT4 aspects of Network Controlled Network Slice Selection	TEI19_SliceSel	C4	CP-251133	Xiaojian Yan, ZTE

Summary based on the input provided by KDDI by e-mail received Thu 19/03/2026 09:22, file "SP-26xxxx_Summary_of_TEI19_SliceSel".

This Work Item extends the existing Network Slice Replacement mechanism by allowing Application Functions to replace a certain S-NSSAI (Replaced S-NSSAI) for a UE with another S-NSSAI (Alternative S-NSSAI) which is part of the UE subscription.

In non-roaming cases, this feature enables an authorized AF or an AF subject to authorization via the NEF to request modification of the Set of Network Slice(s) for a UE by replacing a certain S-NSSAI (Replaced S-NSSAI) with another S-NSSAI (Alternative S-NSSAI) which is part of the UE subscription. The AF may also indicate a policy duration and may subscribe to Notification on the outcome of the AF requested Network Slice Replacement. The request is sent to the PCF for the UE handling the impacted subscriber, and only a single AF can request the modification of the Set of Network Slice(s) for one Replaced S-NSSAI.

Based on operator policies, the PCF considers the AF request and, if accepted, triggers the Network Slice Replacement at the AMF, by sending Slice replacement management information in the access and mobility management policies. The AMF then checks whether the Alternative S-NSSAI is within the UE subscription and supported (or partially supported) in the UE Registration Area, updates Allowed NSSAI / Partially Allowed NSSAI and Rejected S-NSSAI via UE Configuration Update and the AMF and UE release PDU sessions using the Replaced S-NSSAI. The AMF stores the related access and mobility management policy information in the UE context and reports the outcome of the network slice replacement to the PCF, which further notifies the AF.

These enhancements have been specified in TS 23.501 [1] , TS 23.502 [2], TS 23.503 [3], TS 29.522[4], TS 29.534[5], TS 29.519[6], TS 29.507[7], TS 29.523[8], TS 29.502[9], TS 29.504[10] and TS29.518[11].

References

List of related CRs: select "TSG Status = Approved" in:
https://portal.3gpp.org/ChangeRequests.aspx?q=1&workitem=1070041;1060030;1070003;1070042;1070043

[1] TS 23.501: "System architecture for the 5G System (5GS) "
[2] TS 23.502: "Procedures for the 5G System (5GS)"
[3] TS 23.503: "Policy and charging control framework for the 5G System (5GS); Stage 2"
[4] TS 29.522: "5G System; Network Exposure Function Northbound APIs; Stage 3"
[5] TS 29.534: "5G System; Access and Mobility Policy Authorization Service; Stage 3"

[6] TS 29.519: "5G System; Usage of the Unified Data Repository Service for Policy Data, Application Data and Structured Data for Exposure; Stage 3"
[7] TS 29.507: "5G System; Access and Mobility Policy Control Service; Stage 3"
[8] TS 29.523: "5G System; Policy Control Event Exposure Service; Stage 3"
[9] TS 29.502: "5G System; Session Management Services; Stage 3"
[10] TS 29.504: "5G System; Unified Data Repository Services; Stage 3"
[11] TS29.518: "5G System; Access and Mobility Management Services; Stage 3"

21.2 See also

- Advanced Media Delivery (1030006, AMD)
- Data collection for SON (Self-Organising Networks)/MDT (Minimization of Drive Tests) in NR standalone and MR-DC (Multi-Radio DC) Phase 4 (1020092, NR_ENDC_SON_MDT_Ph4)
- Enhancements for AI/ML for NG-RAN (1050124, NR_AIML_NGRAN_enh)
- 5.3 Energy Efficiency as Service Criteria;
- 9.6 Enhancements for Artificial Intelligence (AI)/Machine Learning (ML) for NG-RAN;
- 14.3 Vehicle Mounted Relays Phase 2

22 Rel-19 Service-Based Architecture (SBA)

22.1 UPF enhancement for Exposure And SBA Phase 2

1050097	**UPF enhancement for Exposure And SBA Phase 2**	UPEAS_Ph2	S2	SP-240995	Zhu Jinguo, ZTE
1020003	*Study on UPF enhancement for Exposure And SBA Phase 2*	FS_UPEAS_Ph2	S2	SP-231795	Zhu Jinguo, ZTE
1040035	**Stage 2 of UPF enhancement for Exposure And SBA Phase 2**	UPEAS_Ph2	S2	SP-240995	Zhu Jinguo, ZTE
1050001	**CT aspects of UPF enhancement for Exposure And SBA Phase 2**	UPEAS_Ph2	C4	CP-250018	Shuang Liang, ZTE
1050098	CT3 aspects of UPF enhancement for Exposure And SBA Phase 2	UPEAS_Ph2	C3	CP-250018	Shuang Liang, ZTE
1050099	CT4 aspects of UPF enhancement for Exposure And SBA Phase 2	UPEAS_Ph2	C4	CP-250018	Shuang Liang, ZTE

Summary based on the input provided by ZTE, Vodafone in SP-251560.

UPEAS_Ph2 specifies the system enhancements on UPF event exposure as concluded within TR 23.700-63 for the following aspects:

For key issue #1
- Add the following UPF functionalities in the N4 capabilities and UPF NF profile stored in NRF:
- The functionality of NAT information exposure;
- Packet Inspection functionality (to differentiate between IP or MAC filter based packet detection, and the packet detection based on other means, e.g. layer 7 DPI)
- Define operator configurable parameter in the N4 capabilities and UPF NF profile which can be used for non-standard or partially supported features and configured by operator to extend the baseline UPF capabilities.
- Define a subscription information in UDM/UDR that can be used for UPF discovery and selection, indicating functionalities that are required and/or preferred for the PDU session.

For key issue #2
- Enhance UPF exposure service to allow NF consumer to obtain one NATed UE public IP address and Port for a particular PDU Session from the SMF and UPF, based on the private UE IP address allocated by 5GC and remote end IP address.
- Allow the NEF/NWDAF to use IP address to discover the PSA UPF(i.e. not local PSA UPFs) and subscribe directly to PSA UPF for data collection.
- Enhance the SMF/UPF event exposure service with a new Remaining Data reporting indication to report any data collected since the last notification from the UPF when the N4 session is released.

- Enhance the UPF event exposure service with a new Subscription termination reporting indication to report that the subscription to the UPF event(s) has been terminated and a cause code from the UPF.

For key issue #3
Introduce a new optional feature: Handling of Payload Headers that allows exchange of information in-band in the user plane packets between the application (in the UE or in the application server) and UPF in the 5GS by supporting insertion, replacement, or removal of payload headers of PDUs. Details include the following:
- Enhance Nnef_TrafficInfluence service to accommodate requests from AFs for handling of Headers/Tags.
- Enhance the procedure in clause 4.3.6 of TS 23.502 to provision the handling information to the SMF/UPF
- A new data subset in the Application data set is provisioned in the UDR.
- NEF, PCF, SMF and UPF are enhanced to support the handling of Headers/Tags
- UPF profile in NRF is enhanced with the new UPF capability to support the corresponding UPF selection and discovery
- Enhancing Nupf_EventExposure for Header/Tag reporting.
- Enhancing N4 PFCP rules to instruct reporting of headers/tags handling upon AF request.
- See more details in TS 23.501[1]

References

List of related CRs: select "TSG Status = Approved" in:
https://portal.3gpp.org/ChangeRequests.aspx?q=1&workitem=1050097,1020003,1040035,1050001,1050098,1050099

[1] TS 23.501: “System architecture for the 5G System (5GS)

22.2 Automatic Certificate Management Environment (ACME) for the Service Based Architecture (SBA)

1060066	**Automatic Certificate Management Environment (ACME) for the Service Based Architecture (SBA)**	ACME_SBA	S3	SP-241960	Charles Eckel (Cisco)

Summary based on the input provided by Cisco Systems by e-mail received Wed 11/03/2026 04:10, file "SP-26-ACME-summary".

Automated Certification Management for the Service Based Architecture (ACM for SBA) was specified in Release 18. Support was defined using Certificate Management Protocol version 2 (CMPv2) [1]. In Release 19, the Automatic Certificate Management Environment (ACME) for the Service Based Architecture (SBA) feature enables the use of ACME as an optional protocol for ACM for SBA.

SBA is secured using X.509 certificates across a large number of SBA components and corresponding Network Functions (NFs). Virtualization and increased modularity of NFs have resulted in multi-vendor environments becoming more prevalent. It is now common for NFs to come from different vendors and for the cloud native environment in which they run to come from yet another vendor and for all of these to be independent of the Certificate Authority that is authoritative for the certificates used to secure communications. In such deployments, it is impractical to manage certificates manually.

ACME [2] was defined specifically for automated certificate management and is particularly well suited for some scenarios. Infrastructure deployment such as NFs deployed on cloud native platforms often have built-in support for ACME, so it is a natural fit. Another important benefit of ACME is automated validation of authority to represent an identifier (i.e., to be authoritative for the resource for which the certificate is issued). This is particularly helpful for multi-vendor environments and in cross-carrier scenarios.

The Release 19 ACME for SBA feature enables the use of ACME as an optional protocol for ACM for SBA.

References

List of related CRs: select "TSG Status = Approved" in:
https://portal.3gpp.org/ChangeRequests.aspx?q=1&workitem=1060066

[1] IETF RFC 6712: "Internet X.509 Public Key Infrastructure -- HTTP Transfer for the Certificate Management Protocol (CMP)".
[2] IETF RFC 8555: "Automatic Certificate Management Environment (ACME)".

22.3 Reducing Information Exposure over SBI

Unique ID	Title	Acronym	Working Group	Work/Study Item	Rapporteur
1060055	Reducing Information Exposure over SBI	RedInfExp_SBI	C4	CP-243308	Nivedya Parambath Sasi
1060003	Reducing Information Exposure over SBI	RedInfExp_SBI	C4	CP-243308	Nivedya Parambath Sasi

Provided by Samsung by e-mail, file "CP-26xxxx pCR TR29.919 Summary of Rel-19 Redinfoexp".

APIs defined by SBI currently does not limit the amount of information exposed to different NF-Service Consumers. The result of this is that NF receives excessive data, even if it does not require it and there is no mechanism to reduce or avoid communication of excessive data that is not required by the consumer. Therefore, a mechanism is proposed to strike a balance between data exposure aspects and efficiency/performance between the NFs. Study has performed for scenarios where excessive data may be exposed over SBI when an NF-Consumer accesses an API exposed by an NF-Producer, and studies potential solutions to limit the same.

Below is a brief description of the functionalities introduced in the Rel-19 Reducing Information Exposure over SBI work item:

Selective Parameter Omission: New rules in TS 29.500 and NRF Spec (TS 29.510) allows NFs to omit non-essential attributes in responses based on consumer NF type and Access scope.

Enhanced Access Scopes: Expanded 3gpp-Sbi-Access-Scope header usage across interfaces like Nnrf and Namf to limit exposed fields, enabling SCP-delegated filtering without full data revelation. ApplicableResourceContentFilters and consumerRcfInfo attributes have been added to type AccessTokenClaims.

Default/Null Value mechanism: Support for default or null values in NF profiles, subscriptions, and contexts (e.g., in NRF discovery responses) to prevent unnecessary info sharing while maintaining interoperability.

References

List of related CRs: select "TSG Status = Approved" in:
https://portal.3gpp.org/ChangeRequests.aspx?q=1&workitem=1060055,1060003

[1] TS 29.510: " Network function repository services "

22.4 See also

“UPF enhancement for Exposure And SBA, Phase 2 (1050097, UPEAS_Ph2)”; "Service Based Interface Protocol Improvements Release 19" (no summary since considered as protocol enhancements)

23 Rel-19 QoS and Policy

23.1 Rel-19 Enhancements of UE Policy

1060025	Rel-19 Enhancements of UE Policy	UEP19	C3	CP-243073	Susana Fernández, Ericsson

Summary based on the input provided by Ericsson in CP-243174.

The UE Policy Control service is used as part of the provisioning of UE policies to the UE and as part of the provisioning of N2 PC5 policy (for the applicable UE policies) to the NG-RAN.

During development and deployment of this service, it has been identified that there is a need to apply technical improvements and enhancements, specifically in the following areas:

- UE Policy Provisioning completion.
- Completion of failure scenarios.
- Clarification and completion of functionality defined in previous releases.

This Work Item enhances the procedures for UE policy provisioning functionality (e.g. inclusion of connectivity group ID, establishment of URSP rule precedence, URSP derivation for BDT scenarios) improves specification clarity for the handling of UE policies and completes failure scenarios, considering partial success of UE policy delivery and PCF service authorization scenarios.

The functionality introduced in 3GPP Work Item for Release 19 regarding UE policies handling enhances the way UE policy provisioning and failure management is performed.

Firstly, the Policy Control Function (PCF) now assigns a proper URSP (UE Route Selection Policy) rule precedence so that UE evaluates VPLMN-specific URSP rules before non-VPLMN ones, ensuring that the serving PLMN's URSP rules take priority. Additionally, the traffic descriptor within URSP rules has been modified by including the Connectivity Group ID, along with the corresponding updates to service parameter data provided by the AF and stored in the UDR.

Provisioning of URSP rules for Background Data Transfer (BDT) is now fully specified, incorporating how information is derived from the Unified Data Repository (UDR) and defining criteria for when the PCF should revalidate and provision these URSP rules.

Also, the PCF will be able to remove URSP rules when corresponding policy data is removed from the UDR.

Regarding handling of failure scenarios, the PCF now supports notifications for cases when UE policy delivery outcome is unsuccessful or only partially successful. Furthermore, the PCF can report authorization failures to the Network Exposure Function (NEF) and then to the Application Function (AF), specifically when the PCF detects that the AF provided service parameter data is not authorized.

Lastly, the Work Item clarifies and completes functionality related to UE policy provisioning and reporting procedures, such as defining the applicability of features, specifying the derivation of PLMN IDs in scenarios where the V-AF initiates service parameter provisioning or supporting the unconditional report of URSP rule enforcement in the PCF and AF.

References

List of related CRs: select "TSG Status = Approved" in:
https://portal.3gpp.org/ChangeRequests.aspx?q=1&workitem=1060025

[1] TS 29.525: "5G System; UE Policy Control Service; Stage 3".
[2] TS 29.522: "5G System; Network Exposure Function Northbound APIs; Stage 3".
[3] TS 29.513: "5G System; Policy and Charging Control signalling flows and QoS parameter mapping; Stage 3".
[4] TS 29.519: "5G System; Usage of the Unified Data Repository service for Policy Control Data, Application Data and Structured Data for Exposure; Stage 3".
[5] TS 29.523: "5G System; Policy Control Event Exposure Service; Stage 3".
[6] TS 29.512: "5G System; Session Management Policy Control Service; Stage 3".
[7] TS 29.514: "5G System; Policy Authorization Service; Stage 3".

23.2 Rel-19 Enhancements of Session Management (SM) Policy

1070005	Rel-19 Enhancements of SM Policy	SMPC19	C3	CP-250075	Chi Zhang, Huawei

Summary based on the input provided by Huawei in CP-250075.

The Session Management Policy Control services and service procedures have been specified during the previous 3GPP releases, e.g., Stage2 requirement defined in TS 23.503, TS 23.316, or Stage 3 requirement. During development of the above services, it has been identified that there is a need to apply technical improvements and enhancements (e.g. support the missed requirement in the previous releases, improve the signaling and processing efficiency, increase the flexibility, enhance the reliability, improve specification clarity, etc.).

There is a need to apply the technical improvements and enhancements in Release 19 (e.g. complete the missing non-critical stage-2 requirements, enhancements on the definition of the API in stage-3 level, etc.) to these Session

Management Policy Control related services, as such enhancements may not be covered by the other dedicated Rel-19 work items.

This Work Item specified the corresponding stage 3 procedures related to the technical improvements identified for the following areas:

- Specified the presence condition of handling the precedence.

- Specified the handing of default QoS for GERAN/UTRAN interworking scenarios.

- Complete for wireless and wireline convergence with missing feature (e.g., PCRT) limitation for already specified functionality.

- Corrected the metering method used for SM policy.

- Corrections on policy control for DDN events.

- Corrections on the SM policy termination by PCF.

- Corrected and updated descriptions of session management policy control missed in the previous 3GPP Releases, which are not covered by the other dedicated Rel-19 work items.

References

List of related CRs: select "TSG Status = Approved" in:
https://portal.3gpp.org/ChangeRequests.aspx?q=1&workitem=1070005

[1] TS 29.512: "5G System; Session Management Policy Control Service; Stage 3".
[2] TS 29.513: "5G System; Policy and Charging Control signalling flows and QoS parameter mapping; Stage 3".

23.3 Minimize the Number of Policy Associations

1030012	Minimize the Number of Policy Associations	TEI19_MINPA		SP-240118	Ericsson, Belen Pancorbo
1040078	**Stage 2 of Minimize the Number of Policy Associations**	TEI19_MINPA	S2	SP-240118	Ericsson, Belen Pancorbo
1040004	**CT Aspects on Minimize the Number of Policy Associations**	TEI19_MINPA		CP-241295	Jones Lu, Ericsson
1040052	CT3 Aspects on Minimize the Number of Policy Associations	TEI19_MINPA	C3	CP-241295	Jones Lu, Ericsson
1040053	CT4 Aspects on Minimize the Number of Policy Associations	TEI19_MINPA	C4	CP-241295	Jones Lu, Ericsson

Summary based on the input provided by Ericsson in SP-251568.

Two enhancements have been specified by this WID:

1. Establish UE Policy Association only when there are UE Policies to be sent to the UE.

2. Establish AM Policy Association only when there are AM Policies to be provided to the AMF.

The following enhancements are defined for the AMF to be able to determine whether an establishment of the UE Policy Association is required. The Access and Mobility Subscription Data Type in UDM is extended with an indication of whether the UE Policy Association is enabled, disabled or no control of UE Policies. At initial registration or EPC to 5GS mobility, the AMF triggers the UE Policy Association Establishment to the PCF using both the reception of UE Policy Container and the value of the new indication received from UDM.

Similar enhancements are defined for the AMF to be able to determine whether an establishment of the AM Policy Association is required. The Access and Mobility Subscription Data Type in UDM is extended with an indication of whether the AM Policy Association is enabled or disabled. At initial registration or EPC to 5GS mobility, the AMF triggers the AM Policy Association Establishment to the PCF using both the local configuration and the new indication received from UDM.

These enhancements have been specified in TS 23.502 [1] and TS 23.503 [2].

References

List of related CRs: select "TSG Status = Approved" in:
https://portal.3gpp.org/ChangeRequests.aspx?q=1&workitem=1030012,1040078,1040004,1040052,1040053

[1] TS 23.502: "Procedures for the 5G System (5GS)"
[2] TS 23.503: "Policy and charging control framework for the 5G System (5GS)

23.4 Spending Limits for UE Policies in Roaming scenario

1050089	**Spending Limits for UE Policies in Roaming scenario**	TEI19_SLUPiR		SP-240486	uri baniel, oracle
1030014	**Stage 2 of Spending Limits for UE Policies in Roaming scenario**	TEI19_SLUPiR	S2	SP-240486	uri baniel, oracle
1050011	**CT aspects of Spending Limits for UE Policies in Roaming scenario**	TEI19_SLUPiR		CP-242060	SUN, Yue, China Telecom
1050090	CT3 aspects of Spending Limits for UE Policies in Roaming scenario	TEI19_SLUPiR	C3	CP-242060	SUN, Yue, China Telecom

Summary based on the input provided by Oracle in SP-260233.

This work item extends spending-limit–based policy control to UE policies in roaming scenarios, enabling consistent enforcement of UE Route Selection Policies (URSP) based on charging information. The “spending limit” is a subscriber-specific threshold on chargeable usage, maintained by the Charging Function (CHF) and exposed to policy control so that the network can adapt its behavior when that threshold is reached or approaches.

Policy decisions based on spending limits allow the Policy Control Function (PCF) to take actions according to the status of policy counters maintained in the Charging Function (CHF). In Release 17, this capability is limited to session-management–related policy control, as specified in TS 23.503 [1] (version 17.4, clause 6.1.3).

In Release 18, the scope was extended to also cover Access and Mobility (AM) policy and UE policy (URSP only); however, this enhancement was limited to non-roaming scenarios. As a reminder, URSP (UE Route Selection Policy) is a set of policy rules provisioned by the 5G CN to the UE that determines how application traffic is routed, i.e. which PDU session, Data Network (DNN), and network slice (S-NSSAI) are used for different types of traffic.

The TEI19_SLUPiR work item (Spending Limits for UE Policies in Roaming Scenarios) further extends spending-limit support for UE policies (URSP only) to roaming scenarios.

In addition to the PCF handling PDU sessions, the PCF associated with a UE may use the Nchf_SpendingLimitControl service to obtain the status of policy counters from the CHF. A change in the status of these counters may trigger the UE-level PCF to dynamically push updated AM policy decisions to the AMF and/or updated UE policy decisions to the UE via the AMF.

From Release 19 onward, the scope of this work item for UE policies applies to both non-roaming and roaming scenarios.

As part of this feature, the PCF for a UE provides the address of the CHF to the AMF to ensure consistent use of the same CHF. Starting from Release 19 onward, in roaming scenarios, this CHF address corresponds to the Home CHF (H-CHF).

References

List of related CRs: select "TSG Status = Approved" in:
https://portal.3gpp.org/ChangeRequests.aspx?q=1&workitem=1050089;1030014;1050011;1050090

[1] TS 23.501: “System architecture for the 5G System (5GS)”
[2] TS 23.502: “Technical Specification Group Services and System Aspects; Procedures for the 5G System (5GS)”
[3] TS 23.503: “Policy and charging control framework for the 5G System (5GS)”
[4] TS 29.513: “5G System; Policy and Charging Control signalling flows and QoS parameter mapping”
[5] TS 29.525: “5G System; UE Policy Control Service; Stage 3”
[6] TS 29.594: “5G System; Spending Limit Control Service; Stage 3”

23.5 Enhancing Parameter Provisioning with static UE IP address and UP security policy

1030015	**Enhancing Parameter Provisioning with static UE IP address and UP security policy**	TEI19_IP_SP_EXP		SP-240121	Stefan Rommer, Ericsson
1040079	**Stage 2 of Enhancing Parameter Provisioning with static UE IP address and UP security policy**	TEI19_IP_SP_EXP	S2	SP-240121	Stefan Rommer, Ericsson
1040007	**CT aspects of Enhancing Parameter Provisioning with static UE IP address and UP security policy**	TEI19_IP_SP_EXP		CP-243076	Maria Tianmei, Liang, Ericsson
1040056	CT3 aspects of Enhancing Parameter Provisioning with static UE IP address and UP security policy	TEI19_IP_SP_EXP	C3	CP-243076	Maria Tianmei, Liang, Ericsson
1040057	CT4 aspects of Enhancing Parameter Provisioning with static UE IP address and UP security policy	TEI19_IP_SP_EXP	C4	CP-243076	Maria Tianmei, Liang, Ericsson

Summary based on the input provided by Ericsson in SP-251569..

Two enhancements have been specified by this WID:

1. Allow an AF to configure static UE IP address for a UE via NEF exposure interface.

2. Allow an AF to provide User Plane Security Policy for a 5G VN group via NEF exposure interface.

To allow UE IP address information to be provided by an AF via NEF exposure interface, the NEF exposure service has been enhanced. Corresponding updates have been done to the UDM Parameter Provisioning service. This allows organizations running e.g. NPN networks to assign static UE IP addresses for some or all their UEs.

For 5G VNs, all 5G VN group members need to have the same User Plane Security Policy [3]. The NEF 5G VN exposure service has therefore been enhanced to allow an AF to also provision the User Plane Security Policy together with other 5G VN Group Data. Corresponding enhancements have been made for the 5G VN group data handled by UDM.

References

List of related CRs: select "TSG Status = Approved" in:
https://portal.3gpp.org/ChangeRequests.aspx?q=1&workitem=1030015,1040079,1040007,1040056,1040057

[1] TS 23.501: "System architecture for the 5G System (5GS)"
[2] TS 23.502: "Procedures for the 5G System (5GS)"
[3] TS 33.501: "Security architecture and procedures for 5G System"
[4] TS 29.503: "5G System; Unified Data Management Services; Stage 3"
[5] TS 29.522: "5G System; Network Exposure Function Northbound APIs; Stage 3"

23.6 Providing per-subscriber VLAN instructions from UDM and DN-AAA

1030022	**Providing per-subscriber VLAN instructions from UDM and DN-AAA**	TEI19_VLANSUB		SP-240128	Stefan Rommer, Ericsson
1040080	**Stage 2 of Providing per-subscriber VLAN instructions from UDM and DN-AAA**	TEI19_VLANSUB	S2	SP-240128	Stefan Rommer, Ericsson
1040006	**CT aspects of Providing per-subscriber VLAN instructions from UDM and DN-AAA**	TEI19_VLANSUB		CP-250078	Maria Tianmei, Liang, Ericsson
1040054	CT3 aspects of Providing per-subscriber VLAN instructions from UDM and DN-AAA	TEI19_VLANSUB	C3	CP-250078	Maria Tianmei, Liang, Ericsson
1040055	CT4 aspects of Providing per-subscriber VLAN instructions from UDM and DN-AAA	TEI19_VLANSUB	C4	CP-250078	Maria Tianmei, Liang, Ericsson

Summary based on the input provided by Ericsson in SP-251570.

In Rel-18, it is possible to configure the SMF with VLAN handling instructions, for instructing UPF to add or remove VLAN tags for Ethernet PDU Session. This WI enhances this capability by allowing per-subscriber VLAN handling instructions to be provided from UDM and/or DN-AAA server to SMF.

SM subscription data in UDM is enhanced to include VLAN handling instructions (C-tag and/or S-tag to be inserted/removed) for Ethernet PDU Sessions. Similarly, DN-AAA can also be configured with such information. At PDU Session establishment, this information is provided to SMF. The SMF in turn provides the information to UPF, instructing the UPF for how to handle VLAN-tags. In case overlapping information is received from DN-AAA and UDM, the information from DN-AAA takes precedence.

References

List of related CRs: select "TSG Status = Approved" in:
https://portal.3gpp.org/ChangeRequests.aspx?q=1&workitem=1030022,1040080,1040006,1040054,1040055

[1] TS 23.501: "System architecture for the 5G System (5GS)"
[2] TS 23.502: "Procedures for the 5G System (5GS)"
[3] TS 29.503: "5G System; Unified Data Management Services; Stage 3"
[4] TS 29.561: "5G System; Interworking between 5G Network and external Data Networks; Stage 3"
[5] TS 29.230: "Diameter applications; 3GPP specific codes and identifiers"
[6] TS 29.244: "Interface between the Control Plane and the User Plane nodes"

23.7 QoS monitoring enhancement

1050084	QoS monitoring enhancement	TEI19_QME		SP-240961	Aihua Li, China Mobile
1030020	**Stage 2 of QoS monitoring enhancement**	TEI19_QME	S2	SP-240961	Aihua Li, China Mobile
1050012	**CT aspects of QoS monitoring enhancement**	TEI19_QME		CP-243312	Huang, Zhenning, China Mobile
1050085	CT3 aspects of QoS monitoring enhancement	TEI19_QME	C3	CP-243312	Huang, Zhenning, China Mobile
1050086	CT4 aspects of QoS monitoring enhancement	TEI19_QME	C4	CP-243312	Huang, Zhenning, China Mobile

Summary based on the input provided by China Mobile by e-mail received Tue 31/03/2026 14:17, file "21919-110-revision".

This work item introduces a mechanism for configuring and transferring QoS monitoring capabilities within the 5GC.

Measurement of QoS parameters and reporting of the results must be supported by the RAN and UPF. However, not all RANs and UPFs can be assumed to support QoS monitoring. Therefore, the SMF needs to determine whether the RAN supports QoS monitoring.

References

List of related CRs: select "TSG Status = Approved" in:
https://portal.3gpp.org/ChangeRequests.aspx?q=1&workitem=1050084;1030020;1050012;1050085;1050086

[1] TS 23.501: “System architecture for the 5G System (5GS)
[2] TS 23.502: “Procedures for the 5G System (5GS)”
[3] TS 23.503: “Policy and charging control framework for the 5G System (5GS); Stage 2”
[4] TS 29.122: "T8 reference point for Northbound APIs"
[5] TS 29.244: "Interface between the Control Plane and the User Plane nodes"
[6] TS 29.502: “Session Management Services; Stage 3”
[7] TS 29.512: “Session Management Policy Control Service; Stage 3”
[8] TS 29.514: "5G System; Policy Authorization Service; Stage 3".
[9] TS 29.522: "5G System; Network Exposure Function Northbound APIs; Stage 3"
[10] TS 29.565: "5G System; Time Sensitive Communication and Time Synchronization Function Services; Stage 3"

24 Rel-19 multi-access

24.1 Void

24.2 Multi-Access (ATSSS_Ph4)

1050056	**Multi-Access (ATSSS_Ph4)**	MASSS		SP-240994	Krisztian Kiss (Apple),
1020070	*Study on Multi-Access (DualSteer and ATSSS_Ph4)*	FS_MASSS	S2	SP-240467	Krisztian Kiss (Apple)
1030037	*Study on security aspects for Multi-Access (DualSteer + ATSSS Ph-4)*	FS_MASSS_Sec	S3	SP-240511	Kolekar, Abhijeet (Intel)
1040034	**Stage 2 of Multi-Access (ATSSS_Ph4)**	MASSS	S2	SP-241331	Krisztian Kiss (Apple),
1050020	**CT aspects of Multi-Access (ATSSS_Ph4)**	MASSS		CP-251226	Gupta, Vivek, Apple
1050057	CT1 aspects of Multi-Access (ATSSS_Ph4)	MASSS	C1	CP-251226	Gupta, Vivek, Apple
1050058	CT3 aspects of Multi-Access (ATSSS_Ph4)	MASSS	C3	CP-251226	Gupta, Vivek, Apple
1050059	CT4 aspects of Multi-Access (ATSSS_Ph4)	MASSS	C4	CP-251226	Gupta, Vivek, Apple

Summary based on the input provided by Apple by e-mail received Sat 28/03/2026 02:23, file "Summary_1050056".

The Access Traffic Steering, Switching and Splitting Phase-4 (ATSSS_Ph4) work item builds upon the efforts completed in Release 18, which encompassed ATSSS_Ph3. In Release 18, a high-layer steering functionality—MPQUIC steering using UDP proxying over HTTP—was specified, enabling the steering, switching, and splitting of UDP traffic based on IETF protocols. With Release 19, ATSSS_Ph4 expands this capability, allowing MPQUIC steering functionality to also manage non-UDP traffic, including both IP and Ethernet traffic.

To support this expansion, two additional steering functionalities and new procedures have been incorporated, facilitating MPQUIC operations via CONNECT-IP and CONNECT-Ethernet, as defined by the IETF. The MPQUIC-IP steering functionality can be utilized to steer, switch, and split IP traffic identified within the ATSSS/N4 rules, while the MPQUIC-E steering functionality applies to Ethernet traffic similarly identified.

MPQUIC-IP operation is grounded in RFC 9484 "Proxying IP in HTTP," detailing how IP traffic may be transmitted between a client (UE) and a proxy (UPF) utilizing the RFC 9114 HTTP/3 protocol. This functionality is available for MA PDU Sessions of type IPv4, IPv6, or IPv4v6, contingent on support from both the UE and the network. It is not enabled when the MA PDU Session is of Ethernet type.

Conversely, MPQUIC-E operation is based on the IETF draft-ietf-masque-connect-ethernet "Proxying Ethernet in HTTP," specifying Ethernet traffic transfer via the HTTP/3 protocol. MPQUIC-E functionality may be activated when the MA PDU Session is of Ethernet type.

Additionally, this work item refines the negotiation process for steering functionalities between UE and the network, incorporating previously established ATSSS-LL, MPTCP (Release 16), and MPQUIC-UDP (Release 18) functionalities. The prior negotiation approach, used before Release 19, was determined to be insufficiently scalable and difficult to extend with new steering functionalities introduced in Release 19. Accordingly, the procedures for handling UE capabilities during PDU Session establishment have been revised, and new UE capabilities for MPQUIC-IP and MPQUIC-E support have been added.

References

List of related CRs: select "TSG Status = Approved" in:
https://portal.3gpp.org/ChangeRequests.aspx?q=1&workitem=1050056;1020070;1030037;1040034;1050020;1050057;1050058;1050059

[1] TS 23.003 “Numbering, addressing and identification”
[2] TS 23.501: “System architecture for the 5G System (5GS)”
[3] TS 23.502: “Procedures for the 5G System (5GS)”
[4] TS 23.503: “Policy and charging control framework for the 5G System (5GS); Stage 2”
[5] TS 24.193: “Access Traffic Steering, Switching and Splitting (ATSSS); Stage 3”
[6] TS 24.302: “Access to the 3GPP Evolved Packet Core (EPC) via non-3GPP access networks; Stage 3”
[7] TS 24.501: “Non-Access-Stratum (NAS) protocol for 5G System (5GS); Stage 3”

[8] TS 27.007: “AT command set for User Equipment (UE)”
[9] TS 29.244: “Interface between the Control Plane and the User Plane Nodes; Stage 3”
[10] TS 29.512: “5G System; Session Management Policy Control Service; Stage 3”
[11] TS 29.571: “5G System; Common Data Types for Service Based Interfaces; Stage 3”

24.3 ATSSS Rule Provisioning via 3GPP access connected to EPC

1080039	**ATSSS Rule Provisioning via 3GPP access connected to EPC**	TEI19_ARP3E	S2	SP-241500	Chunhui Zhu (NEC)
1060029	**Stage 2 of ATSSS Rule Provisioning via 3GPP access connected to EPC**	TEI19_ARP3E	S2	SP-241500	Chunhui Zhu (NEC)
1080020	**CT aspects of ATSSS Rule Provisioning via 3GPP access connected to EPC**	TEI19_ARP3E-CT	C1	CP-251217	Chunhui Zhu (NEC)
1080040	CT1 aspects of ATSSS Rule Provisioning via 3GPP access connected to EPC	TEI19_ARP3E-CT	C1	CP-251217	Chunhui Zhu (NEC)

Summary based on the input provided by NEC in SP-260151.

This work item enables ATSSS rules to be delivered to the UE even when the UE is connected via EPS (EPC-based 3GPP access), ensuring that ATSSS can be fully supported in mixed EPC/5GC deployments. ATSSS applies to MA (Multi-Access) PDU Sessions, which allow traffic to be distributed across 3GPP and non-3GPP accesses.

Previously, ATSSS rules were only provided to the UE after an MA PDU Session was established via 5GC signaling under the control of the SMF (Session Management Function). As a result, when the UE was connected through 3GPP access anchored in EPC, ATSSS rules could not be delivered over that access, creating a gap in scenarios where 3GPP access is served by EPC while non-3GPP access is served by 5GC.

With this feature, the UE can indicate—during PDN Connection Establishment over EPS—that it supports ATSSS rule provisioning via 3GPP access connected to EPC. When this capability is indicated, the PGW-C combined with SMF functionality is allowed to provide ATSSS rules to the UE over the 3GPP access, completing ATSSS rule provisioning for MA PDU Sessions in EPC–5GC interworking scenarios.

These enhancements have been specified in TS 23.502 [1] and TS 24.193 [2].

References

List of related CRs: select "TSG Status = Approved" in:
https://portal.3gpp.org/ChangeRequests.aspx?q=1&workitem=1080039;1060029;1080020;1080040

[1] TS 23.502: "Procedures for the 5G System (5GS)"
[2] TS 24.193: "5G System; Access Traffic Steering, Switching and Splitting (ATSSS); Stage 3"

24.4 Local traffic routing for multi-access UE

25 Other topics

25.1 Network Sharing

25.1.1 Indirect Network Sharing

1030045	**Indirect Network Sharing**	TEI19_NetShare			Xing, Tianqi, China Unicom,
950006	*Study on Network Sharing Aspects*	FS_NetShare	S1	SP-220087	Wei, Qun, China Unicom

1000029	**(Stage 1 of) Indirect Network Sharing**	NetShare	S1	SP-230511	Wei, Qun, China Unicom
1030011	**(Stage 2 of) Indirect Network Sharing**	TEI19_NetShare	S2	SP-240488	Xing, Tianqi, China Unicom,
1040019	**Charging enhancement for indirect network sharing**	NetShare_CH	S5	SP-241004	Robert Törnkvist, Ericsson,
1050004	**CT aspects of indirect network sharing**	TEI19_NetShare		CP-250019	Xing Tianqi, China Unicom
1050074	CT1 aspects of indirect network sharing	TEI19_NetShare	C1	CP-250019	Xing Tianqi, China Unicom
1070048	CT3 aspects of indirect network sharing	TEI19_NetShare	C3	CP-250019	Xing Tianqi, China Unicom
1050075	CT4 aspects of indirect network sharing	TEI19_NetShare	C4	CP-250019	Xing Tianqi, China Unicom

Summary based on the input provided by China Unicom in SP-260034.

Indirect Network Sharing is designed to allow multiple participating operators to share resources of a single shared network according to agreed allocation schemes, without mandatory direct connectivity of the shared NG-RAN to the core networks of the participating operators.

It can significantly improve 5G network resource utilization, 5G network coverage in remote areas, 5G services user experience, and promote the high-quality development of the communication infrastructure.

The 5G System can support Indirect Network Sharing deployment between hosting operator (i.e. shared network operator) and participating operator, in which the NG-RAN is shared. The communication between the shared NG-RAN and the core network of the participating operator is routed through the core network of the hosting operator that connects to the shared NG-RAN. The architecture is illustrated in Figure 1:

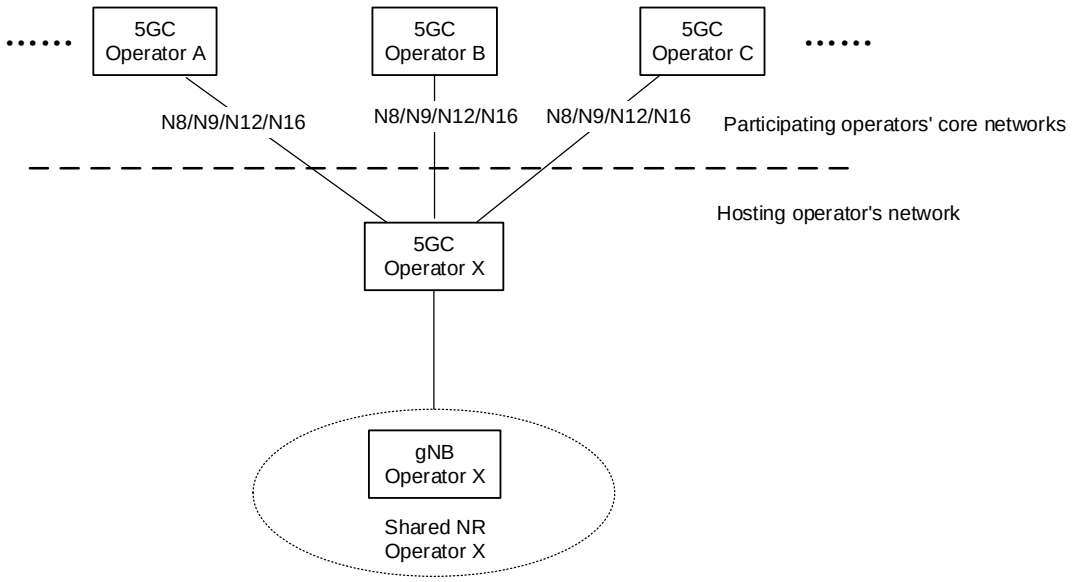


Figure 1: Indirect Network Sharing in which multiple participating operators' CNs connect to hosting operator's CN to share NR

The architecture of Indirect Network Sharing is based on the basic principle of home routed roaming architecture. Not all interfaces between the hosting operator and the participating operator are depicted in the Figure 1 for simplicity.

Enhancements for the Indirect Network Sharing:
o The definition and architecture of Indirect Network Sharing in TS 22.261 [1] and TS 23.501 [2] accompanied by signaling/procedures updates in TS 23.502 [3];
o NF discovery in the case of Indirect Network Sharing considering the UE location information as specified in TS 23.501 [2], TS 23.502 [3], TS 29.510 [9];
o Network slice handling considering the different PLMN IDs broadcast by shared NG-RAN as specified in TS 23.501 [2], TS 23.502 [3], TS 24.501 [4], TS 29.502 [5], TS 29.531 [11], TS 29.536 [12].

o Serving PLMN ID setting for interaction between the Network Functions in the hosting operator network and the participating operator network as specified in TS 23.501 [2], TS 29.502 [5], TS 29.503 [6], TS 29.507 [7], TS 29.509 [8], TS 29.525 [10];
o Charging aspect for Indirect Network Sharing as specified in TS 32.240 [13]..

References

List of related CRs: select "TSG Status = Approved" in:
https://portal.3gpp.org/ChangeRequests.aspx?q=1&workitem=1030045;950006;1000029;1030011;1040019;1050004;1050074;1070048;1050075

[1] TS 22.261: "Service requirements for the 5G system; Stage 1".
[2] TS 23.501: “System architecture for the 5G System (5GS); Stage 2”.
[3] TS 23.502: "Procedures for the 5G System; Stage 2".
[4] TS 24.501: "Non-Access-Stratum (NAS) protocol for 5G System (5GS); Stage 3".
[5] TS 29.502: "5G System: Session Management Services: Stage 3".
[6] TS 29.503: "5G System; Unified Data Management Services; Stage 3".
[7] TS 29.507: "Access and Mobility Policy Control Service; Stage 3".
[8] TS 29.509: "5G System; Authentication Server Services; Stage 3".
[9] TS 29.510: "5G System: Network function repository services; Stage 3".
[10] TS 29.525: "5G System; UE Policy Control Service; Stage 3".
[11] TS 29.531: "Network Slice Selection Services; Stage 3".
[12] TS 29.536: "5G System; Network Slice Admission Control Services; Stage 3".
[13] TS 32.240: "Charging management; Charging architecture and principles".

25.1.2 Management of Network Sharing Phase 3

1060017	**Management of Network Sharing Phase 3**	NetShare_OAM_Ph3	S5	SP-241950	ZhaoNing Wang (China Unicom)
1020015	*Study on Management of Network Sharing Phase3*	FS_NetShare_OAM_Ph3	S5	SP-240966	ZhaoNing Wang (China Unicom)

Summary based on the input provided by China Unicom in SP-260232.

Network sharing is a key approach for operators to reduce the capital expenditure (CAPEX) associated with 5G deployments. The increasing density of access nodes required to meet future performance targets creates significant challenges, particularly with respect to deployment complexity and the acquisition of spectrum and antenna sites. RAN sharing is therefore considered a technical solution to address these challenges.

This study and work item focus on management enhancements for network sharing scenarios, specifically Multiple Operator Core Network (MOCN) and Indirect Network Sharing (INS). The scope includes enhancements related to trace job management and trace data collection for Participating Operators (POPs), as well as the identification of management issues such as improvements to service-based management architecture (SBMA) support in MOCN scenarios, access control for CRUD operations on MOCN management data, and potential management enhancements for Indirect Network Sharing scenarios.

For MOCN, the management system of the Master Operator (MOP) is enhanced to allow authorized POP users to access, read, and download configuration data, performance metrics (including performance measurements and KPIs), and alarm information. In addition, the service-based management architecture is enhanced for MOCN operation, as illustrated in the corresponding figure in TS 32.130.

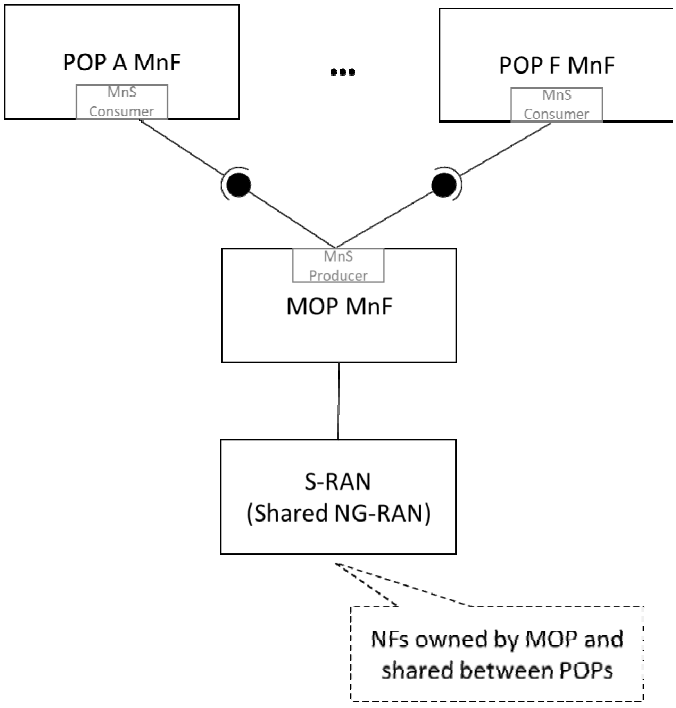


Figure 1: enhancement of the service-based management architecture for MOCN

For Indirect Network Sharing (INS), each POP may have requirements to manage the shared RAN (S-RAN). In this context, the hosting operator, acting as the MOP, is required to provide capabilities that allow authorized consumers to manage the S-RAN. RAN and core network configuration support is enhanced to enable Indirect Network Sharing in TS 28.541, and performance measurements at PLMN granularity for core network functions are enhanced in TS 28.552.

References

List of related CRs: select "TSG Status = Approved" in:
https://portal.3gpp.org/ChangeRequests.aspx?q=1&workitem=1060017

[1] TS 32.130: "Telecommunication management;Network sharing;Concepts and requirements".
[2] TS 28.541: “Management and orchestration; 5G Network Resource Model (NRM); Stage 2 and stage 3”
[3] TS 28.552: “Management and orchestration; 5G performance measurements”.

25.1.3 Charging aspects for Multi-Operator Core Network (MOCN) Network Sharing

1070020	**Charging aspects for Multi-Operator Core Network (MOCN) Network Sharing**	CH_MOCN_NetShare	S5	SP-250819	Robert Törnkvist, Ericsson,

Summary based on the input provided by Ericsson by e-mail received Thu 12/03/2026 12:48, file "Summary for CH_MOCN_NetShare".

The feature introduces the possibility to gather charging information for 5G Multi-Operator Core Network (5G MOCN) network sharing as specified in TS 23.501 [6]. For 5G MOCN network sharing, the communication between the shared RAN and the core network of the participating operator is routed directly to the shared RAN.

The introduction of this feature allows the hosting operator to use charging information from the performance measurements (TS 32.130 [7]) to charge participating operators, as specified in Network slice performance and analytics charging TS 28.201 [2].

References

List of related CRs: select "TSG Status = Approved" in:
https://portal.3gpp.org/ChangeRequests.aspx?q=1&workitem=1070020

[1] TS 32.240: “Charging architecture and principles”

[2] TS 28.201: “Network slice performance and analytics charging in the 5G System (5GS); Stage 2”
[3] TS 32.255: “5G data connectivity domain charging; Stage 2”
[4] TS 32.291: “5G system, charging service; Stage 3”
[5] TS 32.298: “Charging Data Record (CDR) parameter description”
[6] TS 23.501: “System architecture for the 5G System (5GS)”
[7] TS 32.130: “Network sharing; Concepts and requirements”

25.2 Rel-19 IMS

25.2.1 PS Data Off for IMS Data Channel Service

990049	PS Data Off for IMS Data Channel Service	IMSDCDataOff	S1	SP-230227	Yue Hu, China Mobile

Summary based on the input provided by China Mobile by e-mail received Tue 31/03/2026 10:31, file "24.3.1_Summary for PS Data Off for IMS Data Channel Service".

This Work Item adds a new stage 1 requirement: adding "IMS Data Channel" to the list of "3GPP PS Data Off Exempt Services", so that operators can configure whether the "IMS Data Channel" will still be enabled when the user sets the UE to "PS Data Off".

The corresponding architecture impacts are included in the NG_RTC_Ph2 work item and documented in TS 23.228 [1]. The stage 3 protocol aspect is specified in TS 24.186 [2].

References

List of related CRs: select "TSG Status = Approved" in:
https://portal.3gpp.org/ChangeRequests.aspx?q=1&workitem=990049

[1] TS 23.228: "IP Multimedia Subsystem (IMS); Stage 2".
[2] TS 24.186: "IMS Data Channel applications; Protocol specification".

25.2.2 IMS Disaster Prevention and Restoration Enhancement

1080001	IMS Disaster Prevention and Restoration Enhancement	IMS_RES-CT		CP-251016	Liu, Liu, China Telecom
1090034	**CT1 aspects of IMS Disaster Prevention and Restoration Enhancement**	IMS_RES-CT	C1	CP-251016	Liu, Liu, China Telecom
1080035	**CT3 aspects of IMS Disaster Prevention and Restoration Enhancement**	IMS_RES-CT	C3	CP-251016	Liu, Liu, China Telecom
1080036	**CT4 aspects of IMS Disaster Prevention and Restoration Enhancement**	IMS_RES-CT	C4	CP-251016	Liu, Liu, China Telecom

Summary based on the input provided by China Telecom by e-mail received Fri 13/03/2026 04:08, file "Summary for IMS Disaster Prevention and Restoration Enhancement-Liu Liu".

This Work Item specifies the aspect of IMS Disaster Prevention and Restoration Enhancement. TR 29.866 [1] evaluates 10 solutions for a more stable IMS network. Stage 2 specifications are in TS 23.380 [2], with stage 3 covered by TS 23.334 [3] and related documents.

Enhancements include:

- Enabling PCRF/PCF bypass if all instances fail.
- Accelerating restoration of terminating sessions when EPC/5GC NF fails and restoration procedures begin.
- Speeding up restoration when CN nodes (PGW-C/SMF, PGW-U/UPF) fail, using HSS-based P-CSCF Restoration per TS 23.380.
- Allowing dynamic extension of registration expiration timers via Registration Management.

References

List of related CRs: select "TSG Status = Approved" in:
https://portal.3gpp.org/ChangeRequests.aspx?q=1&workitem=1080001;1090034;1080035;1080036

[1] TR 29.866: “Study on IMS Disaster Prevention and Restoration Enhancement”
[2] TS 23.380: “IMS Restoration Procedures”
[3] TS 23.334: “IP Multimedia Subsystem (IMS) Application Level Gateway (IMS-ALG) - IMS Access Gateway (IMS-AGW) interface: Procedures descriptions”
[4] TS 29.244: “Interface between the Control Plane and the User Plane nodes”
[5] TS 29.500: “5G System; Technical Realization of Service Based Architecture; Stage 3”
[6] TS 29.562: “5G System; Home Subscriber Server (HSS) services; Stage 3”
[7] TS 29.213: “Policy and charging control signalling flows and Quality of Service (QoS) parameter mapping”
[8] TS 29.513: “5G System; Policy and Charging Control signalling flows and QoS parameter mapping; Stage 3”
[9] TS 24.229: “IP multimedia call control protocol based on Session Initiation Protocol (SIP) and Session Description Protocol (SDP); Stage 3”

25.2.3 IMS Stage-3 IETF Protocol Alignment

25.3 Roaming and Inter-PLMN enhancements

25.3.1 Roaming Value-Added Services

1030046	**Roaming Value-Added Services**	TEI19_RVAS			
960015	*Study on roaming value added services*	FS_RVAS	S1	SP-220442	Peter Bleckert, Ericsson
990051	**Roaming Value-Added Services**	RVAS	S1	SP-230231	Peter Bleckert, Ericsson
1030013	**Architecture support of roaming value-added services**	TEI19_RVAS	S2	SP-240119	Qian Chen, Ericsson,
1050002	**CT aspects of Architecture support of roaming value-added services**	TEI19_RVAS		CP-242022	Frank Yong, Yang, Ericsson
1050087	CT3 aspects of Architecture support of roaming value-added services	TEI19_RVAS	C3	CP-242022	Frank Yong, Yang, Ericsson
1050088	CT4 aspects of Architecture support of roaming value-added services	TEI19_RVAS	C4	CP-242022	Frank Yong, Yang, Ericsson

Summary based on the input provided by Ericsson by e-mail received Mon 16/03/2026 17:13, file "RVAS Rel19 Summary_Ericsson".

Roaming Value Added Services (RVAS) form part of the roaming services ecosystem and these three RVAS aspects have been studied in SA1 in report TR 22.877 “Study on roaming value-added services”, and been standardised as part of this release in TS 22.261:

- The "Welcome SMS" service sends a SMS to a roaming subscriber’s UE when the UE is registered in a new network for the first time
- The "Steering of Roaming (SoR) during the registration procedure" service makes the home operator able to steer a user to a certain network during the registration procedure when the user tries to register to a new (non-preferred) network.
- The "Subscription-based routing to a particular core network" service forwards the traffic from the HPLMN to a target PLMN.

About the control of sending the welcome SMS when UE registers the first time in a new network, "Roaming Status event exposure from 5GC" is enhanced to include the PEI/GPSI information.

About the steering of roaming during registration procedure, it is considered that existing SoR function and the PRINS support in SA3 for roaming intermediary handling can fulfil the requirements.

About the support of subscription-based routing to a particular core network, the following aspects are specfied:

a. Interactions among NRFs in different PLMNs, e.g. VPLMN, HPLMN and it’s partner/target PLMN, during NF discovery are clarified to facilitate the NF producer selection in partner/target PLMN.

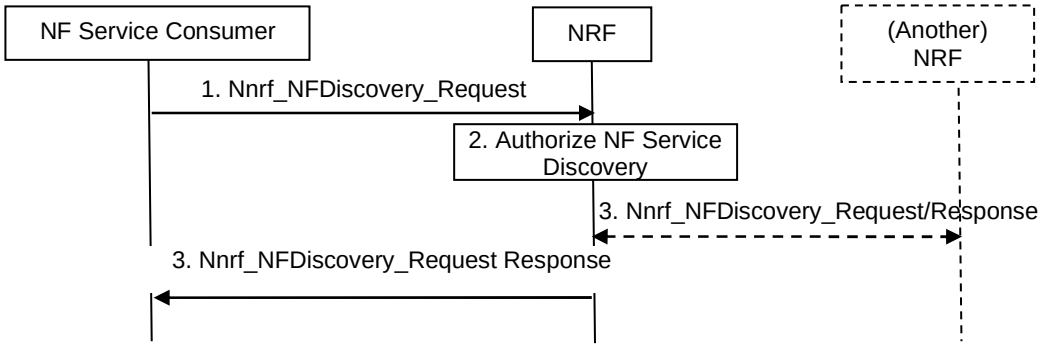


Figure 1: NF/NF service discovery in the same PLMN

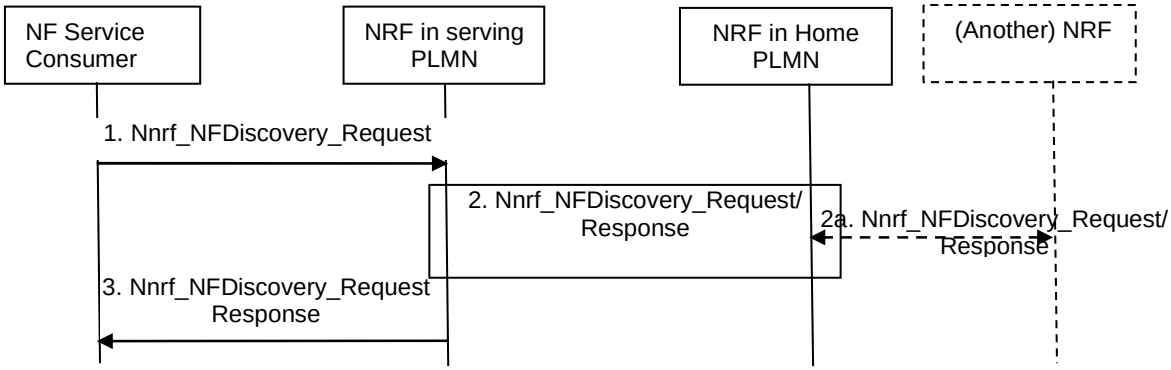


Figure 2: NF/NF service discovery across PLMNs

b. Subscription data is enhanced to support the possibility for the HPLMN to control the formulation of NRF query parameter (i.e. the targeted PLMN) for the SMF selection during the PDU session establishment in non-roaming scenario or roaming with home routed PDU session scenario.

References

List of related CRs: select "TSG Status = Approved" in:
https://portal.3gpp.org/ChangeRequests.aspx?q=1&workitem=1030046;960015;990051;1030013;1050002;1050087;1050088

[1] TR 22.877: “Study on roaming value added services”
[2] TS 22.261: “Service requirements for the 5G system”
[2] TS 23.501: “System architecture for the 5G System (5GS)”
[3] TS 23.502: “Procedures for the 5G System (5GS)”
[4] TS 29.122: "T8 reference point for Northbound APIs"
[5] TS 29.502: “Session Management Services; Stage 3”
[6] TS 29.503: "5G System; Unified Data Management Services; Stage 3"
[7] TS 29.510: “5G System; Network function repository services; Stage 3”
[8] TS 29. 522: “Network Exposure Function Northbound APIs; Stage 3”

25.3.2 Roaming traffic offloading via session breakout in HPLMN

1030018	Roaming traffic offloading via session breakout in HPLMN	TEI19_HSBO	S2	SP-240491	Hyesung Kim, Samsung

Summary based on the input provided by Samsung by e-mail received Mon 30/03/2026 07:56, file "Summary_1030018".

In the home routed roaming scenario, the H-UPF operated by HPLMN operator can be deployed nearby VPLMN. The home routed roaming traffic offloading via session breakout in HPLMN is a type of session breakout to support that roaming traffic of an application is offloaded at the H-UPF, logically belonging to HPLMN but deployed in the visited region (e.g., deployed on the public cloud of which data center is located at the visited region). The HPLMN utilizes a forward-deployed H-UPF to offer a better user experience to the HR roaming users via a shortened transmission path for an application server providing global service (i.e., having its servers distributed in both visited and home countries). If the traffic offloading at the H-UPF deployed in the visited region is not supported, the global service application traffic of an HR roaming UE is routed towards its corresponding server in the home country.

To support this, the following are some considerations:

1. The H-SMF decides to support roaming traffic offloading at the H-UPF for HR PDU Session based on the configured operator policy.

2. When performing H-UPF selection, UE location information (e.g. MNC and MCC of the VPLMN) can be considered in order to select a H-UPF nearby VPLMN for a given HR PDU Session, based on the HPLMN operator policy.

3. Based on the operator policy, the selected H-UPF deployed nearby VPLMN can provide the H-SMF with the DNS server information, if it is locally configured.

Also, for the deployment scenario where the H-UPF of HR PDU Session is deployed nearby VPLMN, if the H-SMF receives from H-UPF the UP Path failure report for the path between the H-UPF (PSA) and V-UPF, the H-SMF may decide to delete the PDU session contexts associated with the path in failure, the H-SMF can trigger release of the HR PDU Session and include an existing cause value indicating that a PDU Session re-establishment is requested. During the new HR PDU Session re-establishment, a new H-UPF is selected based on operator policy.

References

List of related CRs: select "TSG Status = Approved" in:
https://portal.3gpp.org/ChangeRequests.aspx?q=1&workitem=1030018

[1] TS 23.501: “System architecture for the 5G System (5GS)”

25.3.3 NF discovery and selection by target PLMN

1050076	NF discovery and selection by target PLMN	TEI19_NFsel_by_tPLMN		SP-240490	Dr. Thomas Belling, Nokia,
1030023	**Stage 2 of NF discovery and selection by target PLMN**	TEI19_NFsel_by_tPLMN	S2	SP-240490	Dr. Thomas Belling, Nokia,
1050005	**CT aspects of NF discovery and selection by target PLMN**	TEI19_NFsel_by_tPLMN		CP-242025	Landais, Bruno, Nokia
1050077	CT4 aspects of NF discovery and selection by target PLMN	TEI19_NFsel_by_tPLMN	C4	CP-242025	Landais, Bruno, Nokia

Summary based on the input provided by Nokia in SP-260037.

This WI refines service-based communication between 5G core networks in roaming scenarios by enabling the target PLMN to control NF producer discovery and selection when the NF consumer and producer serving a roaming subscriber are in different PLMNs.

Specifically, it allows the target PLMN to select the appropriate NF producer according to its own policies using indirect communication, based on selection criteria provided by the NF consumer in the source PLMN and applied by an SCP in the target PLMN.

References

List of related CRs: select "TSG Status = Approved" in:
https://portal.3gpp.org/ChangeRequests.aspx?q=1&workitem=1050076;1030023;1050005;1050077

[1] TS 23.501, “System architecture for the 5G System (5GS)”, Clause 6.3.1
[2] TS 23.502, “Procedures for the 5G System (5GS)”, Clause 4.17.10a
[3] TS 29.510, “5G System; Network function repository services; Stage 3”
[4] TS 29.500, “5G System; Technical Realization of Service Based Architecture; Stage 3”
[5] TS 23.527, “5G System; Restoration procedures”
[6] TS 29.502, “5G System; Session Management Services; Stage 3”

25.4 Service Continuity and Network Resilience

25.4.1 Minimization of Service Interruption During Core Network Failure Phase 2

1050104	**Minimization of Service Interruption During Core Network Failure Phase 2**	MINT_Ph2	S1	SP-220992	Yinglin Chen, China Telecom
1050029	*Study on MINT support in EPS for 5G-only national roaming UE*	FS_MINT_Ph2	C1	CP-251227	Li Mingxue, China Telecom
970041	**Stage 2 of Minimization of Service Interruption During Core Network Failure Phase 2**	MINT_Ph2	S1	SP-220992	Yinglin Chen, China Telecom
1080022	**CT aspects of Minimization of Service Interruption During Core Network Failure Phase 2**	MINT_Ph2	C1	CP-252193	Li Mingxue, China Telecom
1080037	CT1 aspects of Minimization of Service Interruption During Core Network Failure Phase 2	MINT_Ph2	C1	CP-252193	Li Mingxue, China Telecom
1090035	CT4 aspects of Minimization of Service Interruption During Core Network Failure Phase 2	MINT_Ph2	C4	CP-252193	Li Mingxue, China Telecom
1080038	CT6 aspects of Minimization of Service Interruption During Core Network Failure Phase 2	MINT_Ph2	C6	CP-252193	Li Mingxue, China Telecom
1090014	**Charging aspects of MINT**	MINT-CH	S5	SP-251262	Robert Törnkvist, Ericsson

Summary based on the input provided by China Telecom in SP-260228.

This work item enables 5G-only national roaming UEs to access Disaster Roaming services via EPS within the same VPLMN when the VPLMN's 5GS is unable to provide service due to a disaster condition. While Release 17 MINT (SP-190814) introduced Disaster Roaming support for 5GS, this WI extends that capability to EPS..

Based on the service requirements specified in TS 22.261 [1], support for Disaster Roaming in EPS with minimized service interruption is specified in TS 23.401 [2]. This includes defining:

- when the UE attempts Disaster Roaming via EPS;
- what information the UE is configured with;
- how the UE determines that a disaster condition applies;
- how the UE attaches to EPS for Disaster Roaming service;
- how the UE behavior is handled once the disaster condition no longer applies; and
- how signalling overload related to disaster conditions and Disaster Roaming is prevented.

References

List of related CRs: select "TSG Status = Approved" in:
https://portal.3gpp.org/ChangeRequests.aspx?q=1&workitem=1050104;1050029;970041;1080022;1080037;1090035;1080038;1090014

[1] TS 22.261: "Service requirements for the 5G system; Stage 1".
[2] TS 23.401: "General Packet Radio Service (GPRS) enhancements for Evolved Universal Terrestrial Radio Access Network (E-UTRAN) access".

25.4.2 See also:

"IMS Disaster Prevention and Restoration Enhancement"

As summarised in a corresponding section above.

25.5 Broadcast, Multicast and Media Integration

25.5.1 Interworking of Non-3GPP Digital Terrestrial Broadcast Networks with 5GS Multicast Broadcast Services

No summary provided.

25.6 Location Services enhancements

25.6.1 Enhanced application layer support for location services

1050053	**Enhanced application layer support for location services**	eLSAPP		SP-241005	Liping Wu, CATT
1000035	*Study on enhanced application layer support for location services*	FS_eLSAPP	S6	SP-230778	Liping Wu, CATT
1040072	**Stage 2 of Enhanced application layer support for location services**	eLSAPP	S6	SP-241005	Liping Wu, CATT
1050015	**CT aspects of Enhanced application layer support for location services**	eLSAPP		CP-242064	Wang, Baixiao (CATT)
1050054	CT1 aspects of Enhanced application layer support for location services	eLSAPP	C1	CP-242064	Wang, Baixiao (CATT)
1050055	CT3 aspects of Enhanced application layer support for location services	eLSAPP	C3	CP-242064	Wang, Baixiao (CATT)

Summary based on the input provided by CATT in SP-260218.

To enhance the 5G location services in the application enabled layer , this work item specifies the set of functions and mechanisms shown in the next paragraphs (Geofencing, etc).

A preliminary dedicated study (FS_eLSAPP) was performed in TR 23.700-72[1]. The (normative) Stage 2 is covered by TS 23.434 [2]. Stage 3 is covered by TS 24.545 [3] and TS 29.549 [4].

Support of Geofencing
To specify how the SEAL location-related architecture can be enhanced to support Geofencing at the application enabled layer.
Geofencing is a type of location-based marketing and advertising and known as a virtual geographical boundary.
The current SEAL-LM enabler is enhanced to support the Geofencing and dynamic Geofencing services. The Geofencing subscription/unsubscribe procedure can reuse the one as defined in clause 9.3.12 of TS 23.434 [2] with some clarification and enhancements. The Geofencing UE(s) information request/response procedure can reuse the procedure defined in clause 9.3.10 of TS 23.434 [2] with some clarification and enhancement.

Support of history tracing request or playback
To specify how the SEAL location-related architecture can be enhanced to support history tracing request at the application enabled layer.
In the current positioning system for personnel safety management, the historical track function is one of the more important functions besides real-time positioning.
The current SEAL-LM enabler can be enhanced to enable the location history tracing query and provide the value-added location services to the VAL servers via introducing the procedure of location tracing configuration request, location history request, and location tracing configuration cancel, etc.

Location QoS improvement
To further improve location QoS, especially for location accuracy and latency (service response time) in application layer, the existing application enabler consider utilizing positioning methods (e.g. Ranging) to meet the strict LCS QoS requirements from vertical applications.
The current SEAL-LM enabler can be enhanced to provide more accurate UE location exposure by additionally collecting target UE location via other UEs close to the target UE. Or it can improve the LCS QoS by reducing the service response time.

Enhancement to support location information exposure
To support to expose value-added location information to the consumers, the following aspects are introduced:

- Location Management Server could statistic and calculate the received UE location data per the temporal/spatial granularity and expose the value-added UE location data with different granularity (e.g., frequency) as well as additional information (e.g., velocity) to the VAL server.
- Introduce a new ADAE analytics on location-related UE Group for location services, including time duration prediction/statistics for UE staying in an area, UE density prediction/statistics in an area, UE group route prediction/statistics, and UE group member deviation prediction/statistics, etc.
- When the VAL server application trigger the subscription for verification service, the Location Management Server will support the verification of UE provided location by the Location Management Client and notify the VAL server for the verification result of the UE locations.

Support for ranging/sidelink positioning services
Ranging-based services are the applications utilizing the distance between two UEs and/or the direction of one UE from the other one.
To support to expose value-added location information to the consumers, the following aspects are introduced:
- The current SEAL-LM enabler support an SL ranging/positioning management procedure which enables selection and configuration of reference UEs and provisioning of UEs with SL ranging/positioning configuration information.
- The ADAE Server support exposing ranging information to 3rd party consumers and application layer analytics on collision detection among UEs based on ranging/sidelink positioning information.
Support of location services for multiple USIMs/UEs sharing the same location
To specify how the SEAL location-related architecture can be enhanced to optimize the location service operations when multiple USIMs/UEs sharing the same location.
The Location Management Server supports to associates several UEs via the associated ID and verifies if the associated UEs share the same location. And it also could select the proper UE among multiple UEs in case they are sharing the same location.

References

List of related CRs: select "TSG Status = Approved" in:
https://portal.3gpp.org/ChangeRequests.aspx?q=1&workitem=1050053;1040072;1050015;1050054;1050055

[1] TR 23.700-72: "Study on enhanced application layer support for location services ".
[2] TS 23.434: "Service Enabler Architecture Layer for Verticals (SEAL) ".
[3] TS 24.545: "Location Management - Service Enabler Architecture Layer for Verticals (SEAL)".
[4] TS 29.549: "Service Enabler Architecture Layer for Verticals (SEAL); Application Programming Interface (API) specification; Stage 3".

25.6.2 Deferred 5GC-MT-LR Procedure for Periodic Location Events based NRPPa Periodic Measurement Reports

1060043	**Deferred 5GC-MT-LR Procedure for Periodic Location Events based NRPPa Periodic Measurement Reports**	TEI19_DLPMR		SP-240125	Richard, Batorfi, Ericsson,
1030019	**Deferred 5GC-MT-LR Procedure for Periodic Location Events based NRPPa Periodic Measurement Reports**	TEI19_DLPMR	S2	SP-240125	Richard, Batorfi, Ericsson,
1060004	**CT aspects of Deferred 5GC-MT-LR Procedure for Periodic Location Events based NRPPa Periodic Measurement Reports**	TEI19_DLPMR	C4	CP-243048	Jones Lu, Ericsson
1060044	CT4 aspects of Deferred 5GC-MT-LR Procedure for Periodic Location Events based NRPPa Periodic Measurement Reports	TEI19_DLPMR	C4	CP-243048	Jones Lu, Ericsson

Summary based on the input provided by Ericsson in SP-251571.

LMF may decide to execute different methods to serve Deferred 5GC-MT-LR Procedure for Periodic Location Events. LMF either executes UE based deferred location standardized in Rel-16 or the RAN based method using NRPPa Periodic Measurements Reports introduced in Rel-19. This feature impacts core network only since RAN already supports NRPPa periodic report from Rel-16.

More specifically, the following impacts have been introduced:

On the LMF: LMF selects what method is executed to serve Deferred 5GC-MT-LR Procedure for Periodic Location Events; and if LMF selects to execute NRPPa Periodic Measurement Reports, then LMF provides NRPPa Periodic Indication to GMLC and may subscribe to NG-RAN change.

On the AMF: in case of receiving NRPPa Periodic Indication from GMLC in case of cancellation, AMF does not send cancellation message to UE.

On the GMLC: receiving NRPPa Periodic Indication from LMF; and provides NRPPa Periodic Indication to AMF in case of cancellation triggered by clients.

These enhancements have been specified in TS 23.273 [1], TS 29.515 [2], TS 29.518 [3], TS 29.572 [4].

References

List of related CRs: select "TSG Status = Approved" in:
https://portal.3gpp.org/ChangeRequests.aspx?q=1&workitem=1060043,1030019,1060004,1060044

[1] TS 23.273: ”5G System (5GS) Location Services (LCS); Stage 2”
[2] TS 29.515: “5G System; Gateway Mobile Location Services; Stage 3”
[3] TS 29.518: “5G System; Access and Mobility Management Services; Stage 3”
[4] TS 29.572: “5G System; Location Management Services; Stage 3”

25.7 Data Analytics and Automation

25.7.1 Rel-19 Application Data Analytics Enablement Service

1070044	**Rel-19 Application Data Analytics Enablement Service**	TEI19_ADAES		SP-241695	Yue, Jing, Ericsson
1060068	**Stage 2 of Rel-19 Application Data Analytics Enablement Service**	TEI19_ADAES	S6	SP-241695	Yue, Jing, Ericsson
1070004	**CT aspects of Rel-19 Application Data Analytics Enablement Service**	TEI19_ADAES		CP-250074	Pastushok, Igor, Ericsson
1070045	CT3 aspects of Rel-19 Application Data Analytics Enablement Service	TEI19_ADAES	C3	CP-250074	Pastushok, Igor, Ericsson

Summary based on the input provided by Ericsson by e-mail received Tue 07/04/2026 09:09, file "draft Summary for TEI19_ADAES".

This Work Item introduces enhancements to application data analytics enablement (ADAE) services.

ADAE is a logical framework introduced in Rel-18 to support application data analytics in the enablement layer. It contains logical entities such as the ADAES (Application Data Analytics Enabler Server), ADAEC (Client), A-DCCF (Application layer - Data Collection and Coordination Function), and A-ADRF (Analytical Data Repository Function). These entities and the related functionalities are defined in TS 23.436 [1], TS 24.559 [2], and TS 29.549 [3].

This Rel-19 ADAE enhancements introduced here are the clarification of the metrics of analytics inputs/outputs, some enhancements to the Information Elements in the request/response of data collection from Data Producer and/or A-ADRF (or via A-DCCF), and the completion of the IEs definition in the data/analytics storage subscription request to A-ADRF.

References

List of related CRs: select "TSG Status = Approved" in:
https://portal.3gpp.org/ChangeRequests.aspx?q=1&workitem=1070044;1060068;1070004;1070045

[1] TS 23.436: “Functional architecture and information flows for Application Data Analytics Enablement Service”
[2] TS 24.559: “Application Data Analytics Enablement Service (ADAES); Stage 3”
[3] TS 29.549: “Service Enabler Architecture Layer for Verticals (SEAL); Application Programming Interface (API) specification; Stage 3”

25.7.2 Rel-19 Enhancements of Network Automation Enablers

1050010	Rel-19 Enhancements of Network Automation Enablers	eNetAE19	C3	CP-243078	Xuefei Zhang (Huawei)

Summary based on the input provided by Huawei in CP-260174.

The network data analytics related services (e.g. Nnwdaf services defined in TS 29.520, Ndccf services defined in TS 29.574, Nadrf services defined in TS 29.575, Nmfaf services defined in TS 29.576, etc.) are specified in Rel-15, Rel-16, Rel-17 and R18 in order to efficiently collect the data, store the data and analytics, provide the analytics to the consumer, and monitor the accuracy of the analytics and the ML models.

There is a need to apply the technical improvements and enhancements in Release 19 (e.g. complete the missing non-critical stage-2 requirements, enhancements on the definition of the API in stage-3 level, etc.) to these network data analytics related services, as such enhancements may not be covered by the other dedicated Rel-19 work items.

This work item specifies the technical improvements and enhancements to the network data analytics related services in Release 19 stage 3 level, mainly (but not exhaustively) including:
1. Support of Processing Instructions in MFAF and ADRF APIs.
2. Completion of the analytics transfer procedure.
- Enhance the Nnwdaf_MLModelMonitor_Register service to include the Subscription Correlation ID and the source NF ID of the NWDAF containing AnLF, as well as the Nnwdaf_MLModelMontior_Deregister service to include the ML Model accuracy provisioning termination information.
- Enhance the Nnwdaf_MLModelMonitor_Subscribe service operation to include the Analytics ID, Target of Analytics Reporting and Analytics filter.
3. Completion of the analytics aggregation.
- Enhance the AnalyticsMetadataInfo data structure to include the Data source(s) of the data used for the generation of the output analytics and the Data Formatting and Processing instructions.
4. Clarifications for input data information in ML model training procedure.
- Add clarification for the attributes contained in the InputDataInfo data type defined in TS 29.520 because it may be provided by the consumer to indicate the various settings of the data that are expected to be used by AnLF during inferences, and may also be provided by the MTLF to indicate the various settings of the data that have been used by MTLF during training.
5. Enhancements of Collective Behaviour of UEs in AF, e.g., to include the average moving speed of the UE.
6. Enhancements of Nnwdaf_MLModelTraining_Notify service operation to provide the the Global ML Model Accuracy information.
- Support of the Global ML Model Accuracy information if the Model Accuracy Check Flag is present in the Nnwdaf_MLModelTraining_Subscribe.

References

List of related CRs: select "TSG Status = Approved" in:
https://portal.3gpp.org/ChangeRequests.aspx?q=1&workitem=1050010

[1] TS 23.288: "Architecture enhancements for 5G System (5GS) to support network data analytics services".
[2] TS 29.520: "5G System; Network Data Analytics Services; Stage 3".
[3] TS 29.574: "5G System; Data Collection Coordination Services; Stage 3".
[4] TS 29.575: "5G System; Analytics Data Repository Services; Stage 3".
[5] TS 29.576: "5G System; Messaging Framework Adaptor Services; Stage 3".
[6] TS 29.517: "5G System; Application Function (AF) event exposure service".
[7] TS 29.552: "5G System; Network Data Analytics signalling flows; Stage 3".
[8] TS 29.508: "5G System; Session Management Event Exposure Service; Stage 3".
[9] TS 29.551: "5G System; Packet Flow Description Management Service; Stage 3".
[10] TS 29.522: "5G System; Network Exposure Function Northbound APIs; Stage 3".

25.8 Device and Connectivity Enhancements

25.8.1 Identifying non-3GPP Devices Connecting behind a UE or 5G-RG

1050093	**Identifying non-3GPP Devices Connecting behind a UE or 5G-RG**	UIA_ARC		SP-240971	Michael Starsinic, Interdigital
1020063	*Study on User Identities and Authentication Architecture*	FS_UIA_ARC	S2	SP-231804	Michael Starsinic, Interdigital
1040085	**Stage 2 of Identifying non-3GPP Devices Connecting behind a UE or 5G-RG**	UIA_ARC	S2	SP-240971	Michael Starsinic, Interdigital
1050009	**CT aspects of Identifying non-3GPP Devices Connecting behind a UE or 5G-RG**	UIA_ARC		CP-242029	Sethi, Anuj, InterDigital
1050094	CT1 aspects of Identifying non-3GPP Devices Connecting behind a UE or 5G-RG	UIA_ARC	C1	CP-242029	Sethi, Anuj, InterDigital
1050095	CT3 aspects of Identifying non-3GPP Devices Connecting behind a UE or 5G-RG	UIA_ARC	C3	CP-242029	Sethi, Anuj, InterDigital
1050096	CT4 aspects of Identifying non-3GPP Devices Connecting behind a UE or 5G-RG	UIA_ARC	C4	CP-242029	Sethi, Anuj, InterDigital
1030032	*Study on security aspects of User Identities and Authentication*	FS_UIA_Sec	S3	SP-240507	Samir Ferdi,

Summary based on the input provided by InterDigital Inc. in SP-251376.

The User Identities and Authentication (UIA_ARC) work item covers the case of non-3GPP devices that connect to 3GPP data network through a UE or 5G Residential Gateway (5G-RG). These enhancements allow the UE or 5G-RG to request or suspend QoS differentiation for traffic originating from or destined to such non-3GPP devices.

The Session Management (NAS-SM) procedures are updated so that the UE or 5G-RG can indicate, within a PDU Session Modification Request, that traffic from a non-3GPP device is linked to a specific non-3GPP Device Identifier. This association is conveyed by including the device's addressing information (e.g., IP address and/or port numbers) in the NAS-SM message. The non-3GPP Device Identifier is provided as a string.

Subscription information for the UE or 5G-RG is extended to include the non-3GPP Device Identifier(s) associated with it. For each identifier, the subscription may also contain the corresponding QoS Reference or individual QoS parameters, which can be provisioned by an AF through the NEF.

During a PDU Session Modification, the SMF can forward the non-3GPP Device Identifier(s) and related addressing information to the PCF. Using these identifiers, the PCF retrieves the associated QoS Reference or QoS parameters from the UDR and applies them to determine the PCC Rules. In line with existing procedures, these PCC Rules are then used to derive updated QoS Rules.

References

List of related CRs: select "TSG Status = Approved" in:
https://portal.3gpp.org/ChangeRequests.aspx?q=1&workitem=1050093,1020063,1040085,1050009,1050094,1050095,1050096,1030032

[1] TR 23.700-32: "Study on User Identities and Authentication Architecture"
[2] TR 33.700-32: "Study on the security aspects for usage of user identifiers in the 5G system"
[3] TS 23.501: "System architecture for the 5G System (5GS)"
[4] TS 23.502: "Procedures for the 5G System (5GS)"
[5] TS 23.503: "Policy and charging control framework for the 5G System (5GS); Stage 2"
[6] TS 23.316: "Wireless and wireline convergence access support for the 5G System (5GS)"
[7] TS 24.501: "Non-Access-Stratum (NAS) protocol for 5G System (5GS); Stage 3"
[8] TS 29.512: "Session Management Policy Control Service; Stage 3"
[9] TS 29.513: "Policy and Charging Control signalling flows and QoS parameter mapping; Stage 3"
[10] TS 29.519: "Usage of the Unified Data Repository Service for Policy Data, Application Data and Structured Data for Exposure; Stage 3"
[11] TS 29.522: "Network Exposure Function Northbound APIs; Stage 3"
[12] TS 29.502: "Session Management Services; Stage 3"
[13] TS 29.504: "Unified Data Repository Services; Stage 3"
[14] TS 23.003: "Numbering, addressing and identification"

25.8.2 Lower Selection-priority for PLMN Selection

1090031	**Lower Selection-priority for PLMN Selection**	LoSePLMN			Cristina Romaguera, Vodafone
1090040	**Stage 1 for Lower Selection-priority for PLMN Selection**	LoSePLMN-REQ	S1	SP-251194	Cristina Romaguera, Vodafone
1090009	**CT aspects of Lower Selection-priority for PLMN Selection**	LoSePLMN-CT		CP-252194	Yang Lu; Vodafone
1090032	CT aspects of Lower Selection-priority for PLMN Selection	LoSePLMN-CT	C1	CP-252194	Yang Lu; Vodafone
1090033	CT aspects of Lower Selection-priority for PLMN Selection	LoSePLMN-CT	C6	CP-252194	Yang Lu; Vodafone

Summary based on the input provided by Vodafone by e-mail received Thu 12/03/2026 23:06, file "Summary_PLMN_Selection".

Public Land Mobile Network (PLMN) selection enables a User Equipment (UE) to identify and register with an appropriate network. In automatic PLMN selection, the UE evaluates available networks based on predefined priorities and supported services, while in manual PLMN selection, the subscriber explicitly chooses the preferred network from the list of available options.

Some PLMNs provide limited capacity or restricted service capabilities. Treating such networks with the same priority as fully capable PLMNs may result in repeated registration attempts on sub optimal networks, leading to inefficient network selection behavior.

This Work Item introduces an operator controlled lower-selection priority category for PLMN selection and enhances the UE network selection logic. Its primary objective is to ensure that UEs deprioritize networks with limited capacity or restricted service capabilities, such as satellite networks that reuse terrestrial radio access technologies but offer only constrained resources.

By assigning these networks a lower-selection priority, unnecessary registration attempts are avoided when higher capacity PLMNs are available. This enhancement applies to both manual and automatic PLMN selection and requires extensions to UE priority handling mechanisms, along with corresponding updates to mobility management functions.

Overall, this enhancement improves the efficiency and robustness of network selection by enabling operator-controlled prioritization based on service capability, thereby guiding UEs toward the most appropriate available network.

References

List of related CRs: select "TSG Status = Approved" in:
https://portal.3gpp.org/ChangeRequests.aspx?q=1&workitem=1090031,1090040,1090009,1090032,1090033

[1] TS 22.011: ”Service accessibility”
[2] TS 23.122: ”Non-Access-Stratum (NAS) functions related to Mobile Station (MS) in idle mode”
[3] TS 24.368: ”Non-Access Stratum (NAS) configuration Management Object (MO)”
[4] TS 24.501: ”Non-Access-Stratum (NAS) protocol for 5G System (5GS); Stage 3”
[5] TS 31.102: ”Characteristics of the Universal Subscriber Identity Module (USIM) application”

25.8.3 Enhancement of controlling RAT utilization

1040042	**Enhancement of controlling RAT utilization**	ECRATU		CP-251220	Yang Lu, Vodafone
1040067	**CT1 aspects of Enhancement of controlling RAT utilization**	ECRATU	C1	CP-251220	Yang Lu, Vodafone
1040068	**CT4 aspects of Enhancement of controlling RAT utilization**	ECRATU	C4	CP-251220	Yang Lu, Vodafone

Summary based on the input provided by Vodafone by e-mail received Tue 17/03/2026 09:20, file "Summary_ECRATU".

National roaming services are vital for ensuring seamless connectivity, especially in areas where the primary network operator lacks coverage. Allowing unrestricted use of access technologies for national roamers can result in technical challenges including interoperability issues, quality of service concerns and network congestion concerns.

The current mechanism to limit the UEs utilization of certain access technology employed by the network operators involves rejecting the UE's attach/TAU request or the registration request with CC#15 or #27 upon the UE attempts to attach/register via a specific access technology. However, this existing mechanism resulted in:
a) higher signaling loads within the network;
b) service outage until the UE selects another access technology; and
c) UE keeping re-attempting on the same PLMN/access technology upon the UE re-enabling the corresponding access technology.

In light of the above considerations, the following enhancements have been done:

CT1:
- Enhancements of EPS and 5GS NAS to the delivery of access technology utilization restriction information to the UE, and specifying the corresponding UE behaviour.
- Enhancements of PLMN selection procedure.

CT4:
- Access Restriction Data update in the HSS and the UDM/UDR to enable limitation of access technology utilization performed at the MME and the AMF level.

RAN2:
Description associated with RAT utilization restriction information including the following aspects:
- The usage of the list of "PLMNs with associated RAT restrictions" during cell selection and cell reselection.
- The impact on the suitable cell defination. (NOTE: The restrictions are not applicable if the UE is selecting a cell for limited service)
- Update on rules for inter-RAT and intra-RAT cell reselction to accommodate the changes needed to support "RAT restriction" in targeted cell.

References

List of related CRs: select "TSG Status = Approved" in:
https://portal.3gpp.org/ChangeRequests.aspx?q=1&workitem=1040042;1040067;1040068

[1] TS 23.122: "Non-Access-Stratum (NAS) functions related to Mobile Station (MS) in idle mode"
[2] TS 24.301: "Non-Access-Stratum (NAS) protocol for Evolved Packet System (EPS)"
[3] TS 24.501: "Non-Access-Stratum (NAS) protocol for 5G System (5GS)"
[4] TS 29.272: "Evolved Packet System (EPS); Mobility Management Entity (MME) and Serving GPRS Support Node (SGSN) related interfaces based on Diameter protocol"
[5] TS 29.503: "5G System; Unified Data Management Services; Stage 3"
[6] TS 36.304: "LTE; Evolved Universal Terrestrial Radio Access (E-UTRA); User Equipment (UE) procedures in idle mode"
[7] TS 38.304: "3rd Generation Partnership Project;Technical Specification Group Radio Access Network; NR;User Equipment (UE) procedures in Idle mode and RRC Inactive state"
[8] TS 36.300: "3rd Generation Partnership Project; Technical Specification Group Radio Access Network; Evolved Universal Terrestrial Radio Access (E-UTRA) and Evolved Universal Terrestrial Radio Access Network (E-UTRAN); Overall description; Stage 2"
[9] TS 38.300: "3rd Generation Partnership Project; Technical Specification Group Radio Access Network; NR; NR and NG-RAN Overall Description; Stage 2"

25.9 Core Network and Protocol Evolution

25.9.1 Stage-3 5GS NAS protocol development 18

No summary expected for protocol enhancements.

25.9.2 Stage-3 SAE Protocol Development

No summary expected for protocol enhancements.

25.9.3 CT Aspects for IP Domain usage

1070006	CT Aspects for IP Domain usage	IPD	C3	CP-250248	Zhenning Huang, China Mobile

Summary based on the input provided by China Mobile in CP-260158.

This work item specifies the IP domain use cases and related data definition references for typical usage and update TS 29.513 and TS 29.213 in informative annex .

The IP domain ID has been introduced to indicate different IP address domains for IPv4 addresses out of the same private address range to Use. There is a need to clear the use cases when the IP domain address can be used to indicate the PCEF/SMF/UPF or the domain name for the IPv4 addresses. And it is also needed to indicate that how to use the IP domain address when it is an identification for public IP Address or directly indicate the IP addresses.

References

List of related CRs: select "TSG Status = Approved" in:
https://portal.3gpp.org/ChangeRequests.aspx?q=1&workitem=1070006

[1] TS 29.213: "Policy and charging control signalling flows and Quality of Service (QoS) parameter mapping"
[2] TS 29.513: "5G System; Policy and Charging Control signalling flows and QoS parameter mapping; Stage 3"

25.9.4 MSISDN verification operation support to Nnef_UEId Service

1070046	MSISDN verification operation support to Nnef_UEId Service	TEI19_MVOSNS		SP-241498	Rohit Abhishek (AT&T)
1060028	**Stage 2 of MSISDN verification operation support to Nnef_UEId Service**	TEI19_MVOSNS	S2	SP-241498	Rohit Abhishek (AT&T)
1070007	**CT aspects of MSISDN verification operation support to Nnef_UEId Service**	TEI19_MVOSNS		CP-250077	Shahram Mohajeri, AT&T
1070047	CT3 aspects of MSISDN verification operation support to Nnef_UEId Service	TEI19_MVOSNS	C3	CP-250077	Shahram Mohajeri, AT&T

Summary based on the input provided by ATT by e-mail received Tue 17/03/2026 21:08, file "Summary_1070046".

The Nnef_UEId Service in Release-18 supported MSISDN retrieval operation (i.e. GetMsisdn) where the MSISDN is retrieved from the UDM and exposed to an authorized AF. However, there are many use cases in the industry where the operator would need to support MSISDN verification use cases in which the authorized AF provides the MSISDN in the request and asks the operator to verify whether the UE's MSISDN as recorded in 5GC matches the provided MSISDN in the AF's request, as an alternative to the MSISDN retrieval operation.

In order to support such a use case, the Nnef_UEId Service in Release-19 is enhanced with a new operation (i.e. VerifyMsisdn) to allow NEF to perform verification between the MSISDN it retrieves from the UDM with the one it receives from the AF (in the request) and responds with the verification result accordingly (i.e. true/false depending on whether the two MSISDNs match or not)..

References

List of related CRs: select "TSG Status = Approved" in:
https://portal.3gpp.org/ChangeRequests.aspx?q=1&workitem=1070046;1060028;1070007;1070047

25.9.5 Subscription control for reference time distribution in EPS

1060045	Subscription control for reference time distribution in EPS	TEI19_TIME_SUB_EPS		SP-240127	Sebastian Speicher, Qualcomm
1030021	**Stage 2 of Subscription control for reference time distribution in EPS**	TEI19_TIME_SUB_EPS	S2	SP-240127	Sebastian Speicher, Qualcomm
1060001	**CT aspects of Subscription control for reference time distribution in EPS**	TEI19_TIME_SUB_EPS		CP-243017	Hanna Lim, Qualcomm Incorporated

1060046	CT4 aspects of Subscription control for reference time distribution in EPS	TEI19_TIME_SUB_EPS	C4	CP-243017	Hanna Lim, Qualcomm Incorporated

Summary based on the input provided by Qualcomm in SP-251361.

This Work Item introduces support for reference time distribution based on subscription for WB-EUTRA.

"Timing as a service" refers to providing timing information to UEs as a monetizable operator service. Rel-18 enabled timing as a service for NR by introducing subscription control for timing distribution to individual UEs. This allowed operators to control whether timing information is to be delivered to a particular UE.

Timing as a service is only supported for NR. This is an issue since NR coverage is still quite limited and may not be available in many areas for quite some time. This means that operators cannot offer timing as a service nation-wide but only in areas where NR connected to 5GC has been deployed, which limits opportunities for operators.

Therefore, this work item specifies support for controlling reference time distribution to individual UEs based on subscription for WB-EUTRA.

WB-EUTRA RRC already supported delivering reference time to individual UEs (see TS 36.331 [1] clause 6.2.2). Building on that, this work item specifies the following [2]:

- EPS Subscription enhancement: Indication whether UE is subscribed for reference time distribution;

- Based on the subscription indication for reference time distribution, MME to indicate to eNB to distribute reference time to the UE;

- eNB to deliver reference time using existing unicast RRC to a UE based on indication from MME.

References

List of related CRs: select "TSG Status = Approved" in:
https://portal.3gpp.org/ChangeRequests.aspx?q=1&workitem=1060045,1030021,1060001,1060046

[1] TS 36.331: “Evolved Universal Terrestrial Radio Access (E-UTRA); Radio Resource Control (RRC); Protocol specification”
[2] TS 23.401: “General Packet Radio Service (GPRS) enhancements for Evolved Universal Terrestrial Radio Access Network (E-UTRAN) access”

25.10 Network Monitoring and Measurement

25.10.1 Monitoring of signalling traffic in 5G

1050114	**Monitoring of signalling traffic in 5G**	MonSTra	S1	SP-240793	Cristina Romaguera
1040082	**Stage 1 for MonSta**	MonSTra	S1	SP-240793	Cristina Romaguera
1050028	**Security for MonStra**	MonStra-Sec	S3	SP-241376	Vodafone, Susana Sabater
1060012	**Management for MonStra**	Monstra-OAM	S5	SP-241945	Susana Sabater (Vodafone)

Summary based on the input provided by Vodafone by e-mail received Thu 12/03/2026 23:06, file "Summary_Monstra".

External monitoring systems are widely used by mobile network operators for network surveillance, troubleshooting, diagnosis, and fault analysis, typically at the signalling level and under full operator control. With the introduction of encryption for signalling exchanged between network functions in the 5G Core, standardized and secure means to share signalling information with such monitoring systems were no longer available.

This Work Item defines capabilities that enable signalling monitoring in 5G networks, while minimising impact on real-time traffic performance and ensuring appropriate protection of signalling information exposed to external monitoring systems.

The protection of signalling data exchanged between the 5G Core and external monitoring systems is achieved through robust security mechanisms that ensure confidentiality, integrity, and trust. Security credentials remain strictly within

the 5G Core, while any information shared for monitoring purposes is protected to prevent disclosure of sensitive security-related information. This enables effective monitoring while preserving the overall security of the 5G system.

The solution provides the capability for network elements to securely export copies of both incoming and outgoing signalling traffic to an external monitoring system. It supports flexible operator control of monitoring, including the ability to enable or disable monitoring per network element and to select the network elements and types of signalling traffic to be monitored. And it is designed to operate across different deployment environments.

References

List of related CRs: select "TSG Status = Approved" in:
https://portal.3gpp.org/ChangeRequests.aspx?q=1&workitem=1050114,1040082,1050028,1060012

[1] TS 22.261: ” Service requirements for the 5G system; Stage 1”
[2] TS 33.501: ”Security architecture and procedures for 5G system”
[3] TS 28.560: ”Management and orchestration; Signalling traffic monitoring management (Stage 1, stage 2, and stage 3)”

25.10.2 Measurement Data Collection

980108	**Measurement Data Collection**	MeasureData	S1	SP-221263	Yingjun Zhou, ZTE Corporation

Summary based on the input provided by ZTE by e-mail received Thu 19/03/2026 07:31, file "SP-26xxxx Summary of R19 Measuredata".

This Work Item proposes new QoS monitoring requirements for TS22.261 [1], which are:
- QoS monitoring and reporting with per packet granularity.
- Activation and deactivation QoS monitoring to report on data packets not meeting the required QoS level.
- Clarification of the existing refresh rate requirement for QoS monitoring.

It applies only to direct network connection scenarios in non-public networks, and there is no stage 2 and stage 3 related work item.

References

List of related CRs: select "TSG Status = Approved" in:
https://portal.3gpp.org/ChangeRequests.aspx?q=1&workitem=980108

[1] TS 22.261: “Service requirements for the 5G system; Stage 1

25.11 Testing and Interoperability

25.11.1 Harmonization of test case definitions for cross-RAT usability

No summary provided.

25.12 Generic Enablers and Miscellaneous Enhancements

25.12.1 PRU Usage Extension supported by Core Network

1070039	**PRU Usage Extension supported by Core Network**	TEI19_PRUE		SP-241504	Hou, Yunjing, CATT
1060031	**Stage 2 of PRU Usage Extension supported by Core Network**	TEI19_PRUE	S2	SP-241504	Hou, Yunjing, CATT
1070001	**CT aspects of PRU Usage Extension supported by Core Network**	TEI19_PRUE		CP-250017	Wang, Baixiao, CATT
1070040	CT4 aspects of PRU Usage Extension supported by Core Network	TEI19_PRUE	C4	CP-250017	Wang, Baixiao, CATT

Summary based on the input provided by CATT by e-mail received Thu 19/03/2026 04:24, file "Summary_1070039".

This Work Item specifies the new functions and mechanisms for Positioning Reference Unit (PRU). A PRU is a network-controlled radio node that transmits reference signals specifically for positioning purposes, rather than for user data (it acts as a terrestrial positioning anchor, similar in concept to a GNSS satellite, but operating within the 3GPP cellular network). The 3 new functions are descripbed in the next 3 paragraphs.

Support the network assisted positioning based on target UE measurements and PRU measurements when the serving LMF of target UE and the serving LMF of PRU are different: the serving LMF of target UE provides the positioning method, time window(s) and the identifier of TRPs for the measurements to the serving LMF of PRU. The serving LMF of PRU triggers network assisted positioning based on the received information and returns the PRU location measurements to the serving LMF of target UE.

Support UE assisted positioning based on target UE measurements and PRU measurements when the serving LMF of target UE and the serving LMF of PRU are different: the serving LMF of target UE provides the positioning method and time window(s) to the serving LMF of PRU. The serving LMF of PRU triggers UE assisted positioning based on the received information and returns the PRU location measurements to the serving LMF of target UE.

Support UE-based positioning using PRU measurements when the serving LMF of target UE and the serving LMF of PRU are different: the serving LMF of target UE provides the assistance data including location measurements from the PRU, the associated timestamp and the PRU location to the target UE which is responsible to calculate the location based on the assistance data.

References

List of related CRs: select "TSG Status = Approved" in:
https://portal.3gpp.org/ChangeRequests.aspx?q=1&workitem=1070039,1060031,1070001,1070040

26 Rel-19 miscellaneous Security

26.1 Security Assurance Specification for maintenance of 5G features

1030029	Security Assurance Specification for maintenance of 5G features	SCAS_5G_Maint	S3	SP-240515	Rong Wu (Huawei)

Summary based on the input provided by Huawei by e-mail received Thu 26/03/2026 09:07, file "Summary_1030029 for SA3 SCAS_5G_Maint".

This Work Item is a routine work to maintain and enhance the assurance specification that is used by NESAS/SCAS scheme jointly developed by GSMA and 3GPP. This scheme has drawn eyes from European Commission to Regulators, and from inside 3GPP to other SDOs for security assurance for telecommunication network equipment.

It ensures up-to-date and complete SCAS specifications for the 3GPP systems. This covers more than 20 SCAS specifications. During the course of the release, the scope of the scheme has considered the input from different stakeholders with adding new test cases and enhance the existing content based on the improvement comments from GSMA NESASG. The work done in this work item includes at least: revision of several specifications based on requests and feedback from GSMA, additions of new test cases covering the new features of Rel-19.

References

List of related CRs: select "TSG Status = Approved" in:
https://portal.3gpp.org/ChangeRequests.aspx?q=1&workitem=1030029

26.2 5G Security Assurance Specification (SCAS) for the Unified Data Repository (UDR)

1010010	5G Security Assurance Specification (SCAS) for the Unified Data Repository (UDR)	SCAS_5G_UDR	S3	SP-230864	Andreas, Joerg, BSI Germany

Summary based on the input provided by BSI Germany in SP-251312.

The Unified Data Repository (UDR) is a critical component of 5G core networks, responsible for the storage of key material in close connection with the UDM functionality. To ensure the security of this key exchange functionality, it is necessary to extend the security assurance specifications by the UDR SCAS TS 33.530 (SCAS_5G_UDR). This TS fills a gap regarding the availability of Security Assurance Specifications (SCAS) for 3GPP specified 5G core network functions. It is used to evaluate the security requirements of the UDR network function via GSMA NESAS and BSI NESAS certification schemes.

Further on, TS 33.530 is the test specification for the UDR core network function. As a SCAS the TS has no impact on system behaviour or other network functions. The UDR SCAS enhances a list of test specifications for the 3GPP 5G core network functions. From architecture side the UDR is defined in [3] as SBI core network function. A specific test case content was contributed as pCR to TS 33.530. The current assembly of test cases is specific to UDR network function with further references to the catalogue of general security assurance [1].

References

List of related CRs: select "TSG Status = Approved" in:
https://portal.3gpp.org/ChangeRequests.aspx?q=1&workitem=1010010

[1] TS 33.117: "Catalogue of general security assurance requirements"
[2] TS 33.926: "Security Assurance Specification (SCAS) threats and critical assets in 3GPP network product classes"
[3] TS 23.501 "System architecture for the 5G System (5GS)"

26.3 5G Security Assurance Specification (SCAS) for the Short Message Service Function (SMSF)

No summary provided.

26.4 Addition of 256-bit security Algorithms

1010012	**Addition of 256-bit security Algorithms**	256_Algo	S3	SP-240476	Orkopoulos, Stawros
1020041	*Study on enabling a cryptographic algorithm transition to 256-bits*	FS_CAT256	S3	SP-231788	Yuto Nakano

Summary based on the input provided by Nokia by e-mail received Tue 24/03/2026 15:45, file "Summary_1010012".

This Work Item introduces new 256-bit security algorithms (SNOW 5G, AES, and ZUC) by defining a dedicated set of 35-series specifications (TS 35.240 to 35.248). Each algorithm is documented in a full specification set that includes the algorithm description, implementers' test data, and conformance test data.

Because the detailed algorithm specifications were initially confidential and under review by the French Government, two versions of the documents were produced to manage different levels of information disclosure. A redacted version, meaning a version where sensitive technical details have been deliberately removed or hidden for public release, was published for general use. The non-redacted (full) version was prepared for restricted distribution and shared only with the French Government for evaluation.

After approval was granted, the non-redacted versions were authorized for publication and incorporated into the specifications via change requests. This resulted in fully detailed and publicly available 256-bit algorithm specifications, with the feature approved in Rel-18 and completed in Rel-20.

References

List of related CRs: select "TSG Status = Approved" in:
https://portal.3gpp.org/ChangeRequests.aspx?q=1&workitem=1010012

[1] TS 35.240: "Specification of the Snow 5G based 256-bits algorithm set: specification of the 256-NEA4 encryption, the 256-NIA4 integrity, and the 256-NCA4 authenticated encryption algorithm for 5G; Document 1: algorithm specification"
[2] TS 35.241: "Specification of the Snow 5G based 256-bits algorithm set: Specification of the 256-NEA4 encryption, the 256-NIA4 integrity algorithm, and the 256-NCA4 authenticated encryption algorithm for 5G; Document 2: implementation test data"

[3] TS 35.242: “Specification of the Snow 5G based 256-bits algorithm set: Specification of the 256-NEA4 encryption, the 256-NIA4 integrity, and the 256-NCA4 authenticated encryption algorithm for 5G; Document 3: design conformance test data “

[4] TS 35.243: “Specification of the AES based 256-bits algorithm set: Specification of the 256-NEA5 encryption, the 256-NIA5 integrity, and the 256-NCA5 authenticated encryption algorithm for 5G; Document 1: algorithm specification”

[5] TS 35.244: “Specification of the AES based 256-bits algorithm set: Specification of the 256-NEA5 encryption, the 256-NIA5 integrity, and the 256-NCA5 authenticated encryption algorithm for 5G; Document 2: implementation test data”

[6] TS 35.245: “Specification of the AES based 256-bits algorithm set: Specification of the 256-NEA5 encryption, the 256-NIA5 integrity, and the 256-NCA5 authenticated encryption algorithm for 5G; Document 3: design conformance test data”

[7] TS 35.246: “Specification of the ZUC based 256-bits algorithm set: Specification of the 256-NEA6 encryption, the 256-NIA6 integrity, and the 256-NCA6 authenticated encryption algorithm for 5G; Document 1: algorithm specification”

[8] TS 35.247: “Specification of the ZUC based 256-bits algorithm set: Specification of the 256-NEA6 encryption, the 256-NIA6 integrity, and the 256-NCA6 authenticated encryption algorithm for 5G; Document 2: implementation test data”

[9] TS 35.248: “Specification of the ZUC based 256-bits algorithm set: Specification of the 256-NEA6 encryption, the 256-NIA6 integrity, and the 256-NCA6 authenticated encryption algorithm for 5G; Document 3: design conformance test data”

26.5 Addition of Milenage-256 algorithm

No summary provided.

26.6 Roaming and interconnect authorization aspects in indirect communication

1080043	Roaming and interconnect authorization aspects in indirect communication	SBA_RoamInt-SEC	S3	SP-250815	Anja Jerichow

Summary based on the input provided by Nokia in SP-260039.

This WI addresses security aspects for roaming and interconnect scenarios involving indirect communication, which were not explicitly covered in TS 33.501 in previous Releases. Previously, indirect communication relied on the same security handling as direct communication when NF selection was performed by the source PLMN. With the introduction of target-PLMN-based NF selection, additional security considerations needed to be specified.

The work item clarifies those aspects by adding a new clause on indirect communication in TS 33.501, aligned with CT4 work on security handling for all indirect communication cases, recognizing that these differ from direct communication scenarios where NF selection is controlled by the source PLMN.

Overall, the WI SBA_RoamInt-SEC introduces authorization aspects for roaming and interconnect scenarios covering both NF selection by the source PLMN (CR2133, TEI 19, SBA-related) and NF selection by the target PLMN (CR2135, TEI19_TPlmnNfSel, CT4-related).

References

List of related CRs: select "TSG Status = Approved" in:
https://portal.3gpp.org/ChangeRequests.aspx?q=1&workitem=1080043

[1] TS 33.501: “Security architecture and procedures for 5G System”

26.7 Public key distribution and Issuer claim verification of the Access Token

1080046	**Public key distribution and Issuer claim verification of the Access Token**	SBA_KDATV-SEC	S3	SP-250816	Mohsin Khan

Summary based on the input provided by Ericsson by e-mail received Thu 19/03/2026 09:12, file "Summary_1080046".

The work item introduces a mechanism for an NF Service Producer to fetch the right public key from the appropriate NRF to validate an access token presented to the NF Service Producer.

The feature is introduced in TS 33.501 [1]. To validate an access token, an NF Service Producer needs the public key identified by the header parameter. Previously, this public key has been configured in the NF service producer. According to this new feature, if the NF Service Producer does not have the public key, the NF Service Producer retrieves the key using the new service operation (Nnrf_AccessToken_RetrieveKey Service Operation) specified in clause 14.3.3 in TS 33.501 [1].

References

List of related CRs: select "TSG Status = Approved" in:
https://portal.3gpp.org/ChangeRequests.aspx?q=1&workitem=1080046

[1] TS 33.501: "Security architecture and procedures for 5G System"

26.8 3GPP profiles for cryptographic algorithms and security protocols

1020042	**3GPP profiles for cryptographic algorithms and security protocols**	CryptoSP	S3	SP-231793	Mohsin Khan A

Summary based on the input provided by Ericsson by e-mail received Thu 19/03/2026 09:12, file "Summary_1020042".

The work item updates of cryptographic algorithms and the 3GPP security protocol profiles for security algorithms from other standards bodies (e.g., IETF).

In TS 33.501 [1], issue with DTLS over SCTP (RFC 6083) is described, references to RFC 9191 and RFC 9247 are added, the procedure for EAP-TLS 1.3 is added as an example. In TS 33.210 [2], several new RFCs are added to the references, and profiles for ESP, IKEv2, TLS 1.2, and TLS 1.3 are updated. In TS 33.310 [3] too, several new RFCs are added to the references, and profiles for CRL and IKEv2 are updated. In TS 33.203, the use of "Security Mechanism Agreement for SIP Sessions" for security mode set-up has been updated with the use of hmac-sha2-256-128. Also, in TS 33.203 [4], the Key expansion functions for IPsec ESP has been updated on how to compute IKESP when the selected authentication algorithm is HMAC-SHA2-256-128.

References

List of related CRs: select "TSG Status = Approved" in:
https://portal.3gpp.org/ChangeRequests.aspx?q=1&workitem=1020042

[1] TS 33.501: "Security architecture and procedures for 5G System"
[2] TS 33.210: "Network Domain Security (NDS); IP network layer security"
[3] TS 33.310: "Network Domain Security (NDS); Authentication Framework (AF)"
[4] TS 33.203: "3G security; Access security for IP-based services"

26.9 Mobility over non-3GPP access to avoid full primary authentication

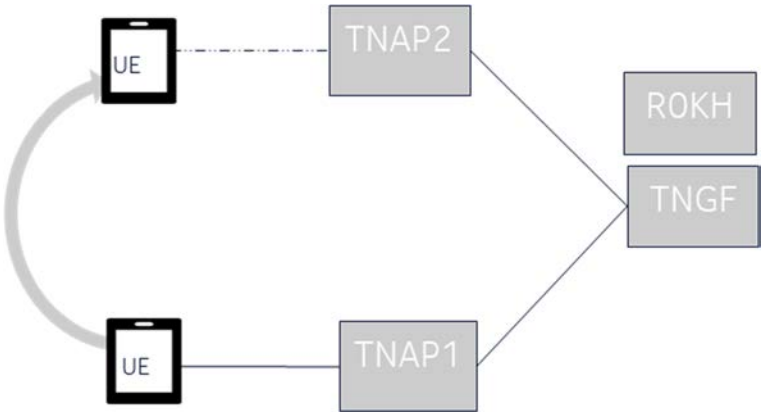
1060054	**Mobility over non-3GPP access to avoid full primary authentication**	Non3GPPMob_Sec		SP-241088	Khare, Saurabh
1020045	*Study on Security for mobility over non-3GPP access to avoid full primary authentication*	FS_Non3GPPMob_Sec	S3	SP-231791	Saurabh Khare, Nokia

1040024	**Security for mobility over non-3GPP access to avoid full primary authentication**	Non3GPPMob_Sec	S3	SP-241088	Khare, Saurabh
1060020	**CT aspects of security for mobility over non-3GPP access to avoid full primary authentication**	Non3GPPMob_Sec	C1	CP-250133	Nassar, Mohamed Amin, Nokia

Summary based on the input provided by in .

This feature focuses on enhancing security for mobility over non-3GPP access networks such as Wi-Fi and fixed broadband. Its goal is to avoid the need for full primary authentication when devices move between two non-3GPP accesses, thereby improving handover efficiency and user experience. The feature is designed to minimize authentication overhead similar to mechanisms seen in Wi-Fi fast BSS transition (FT).

UE mobility across TNAP

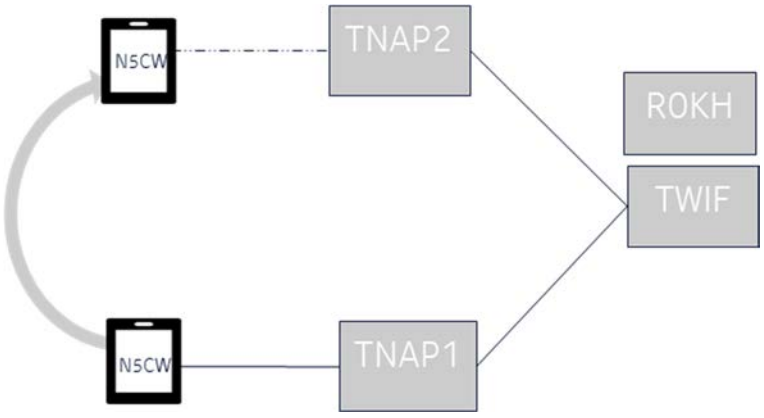


• If the Wi-Fi access point (TNAP) supports Fast Transition (FT), the gateway (TNGF) creates a new key for re-authentication using the FT process. If the user device (UE) and the access network (TNAN) both support FT, the device will use the FT procedure to re-authenticate and will generate a special FT key (KFT) from a key provided by the gateway (KTNGF).

• The device and the authentication server (R0KH) then use KFT as the master key to create the key hierarchy needed for FT, following the IEEE 802.11 standard. This lets the device switch between different Wi-Fi access points (TNAPs) in the same mobility domain without a full re-authentication each time.

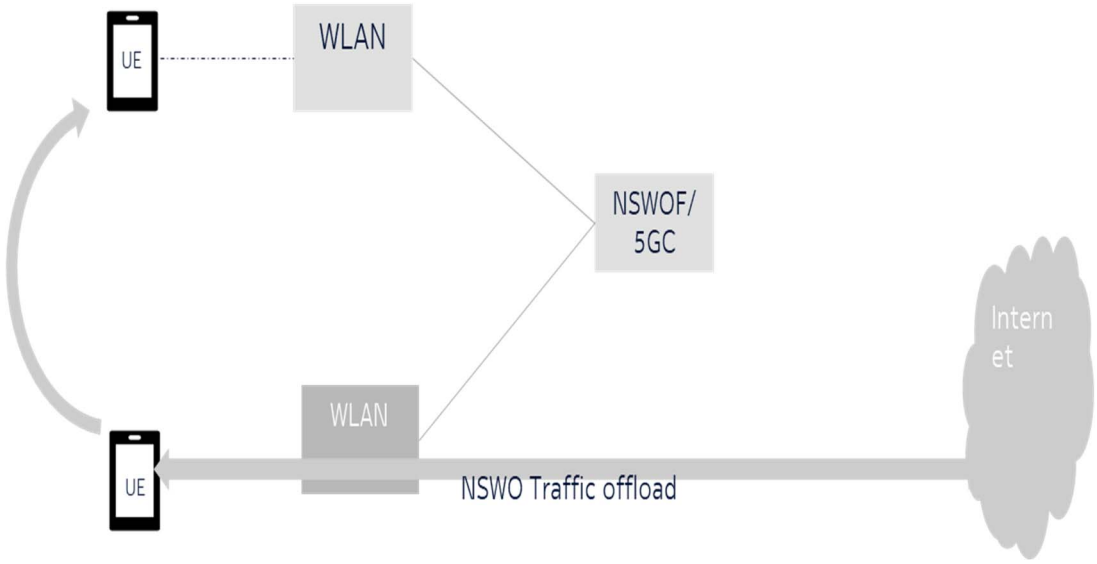
• Once the device has switched access points, if the MOBIKE protocol is supported, the device can request the restoration of its encrypted IPSec connection and all associated security associations as described in RFC 4555.

N5CW device mobility across TNAP



Similar concept is applicable to N5CW devices moving from TNAP1 to TNAP2 under the same TWIF.

NSWO related mobility



• NSWO (Non-Seamless WLAN Offload) in 3GPP allows a device to connect directly to Wi-Fi and offload selected data traffic without routing it through the mobile core network.

• IEEE 802-11 FT protocol based fast BSS transition already were supported in certain existing deployments. Therefore, note added. i.e., An alternative deployment when UE moves to a new WLAN AP and when both the UE and WLAN AN support FT procedure as specified in IEEE 802.11 is to use FT relying on MSK (available to the WLAN AN and UE) to generate the keys necessary.

References

List of related CRs: select "TSG Status = Approved" in:
https://portal.3gpp.org/ChangeRequests.aspx?q=1&workitem=1060054;1040024;1060020

[1] TS 33.501: “Security architecture and procedures for 5G System”
[2] TS 23.502: “Procedures for the 5G System (5GS); Stage 2”
[3] TS 24.502: “Access to the 3GPP 5G Core Network (5GCN) via Non-3GPP Access Networks (N3AN); Stage 3”

26.10 Other security features (no summary)

The following Features are marked in the Work Plan as Release 19, but no summary was proposed for this document:

- LI Handling of Protected Services
- Lawful Interception Rel-19
- Lawful Interception Guidance Rel-19
- Specification of example algorithm for alternative f5* (f5) function

27 Rel-19 Management and Orchestration - centric Features

27.1 Introduction

This section presents all the standalone management functionalities. Sub-clauses 27.2 – 27.15 contain work item summaries including relevant references for all these management functionalities in Rel-19.

In addition management aspects related to other features are reported in the relevant section as described below:
1. Management Aspects of NTN Phase 2, see 5.3.5

2. Management Aspects of RedCap features, see 6.2
3. AI/ML Management Phase 2, see 8.1.4
4. Enhanced OAM for management service exposure to external consumers through CAPIF, see 10.3
5. OAM support for Extended Reality and Media service (XRM) phase 2, see 12.2
6. Management for MonStra in “Monitoring of signalling traffic in 5G”

27.2 Service Based Management Architecture enhancement phase 3

1060006	**Service Based Management Architecture enhancement phase 3**	SBMA_Ph3	S5	SP-241939	Kai Zhang (Huawei)
1020011	*Study on Service Based Management Architecture enhancement phase 3*	FS_SBMA_Ph3	S5	SP-241569	Kai Zhang (Huawei)

Summary based on the input provided by Huawei in SP-251548.

SBMA_Ph3 provides the following :

1 Further elaboration on usage guidance of model driven service based management services. Enhance the SBMA definitions in TS 28.533 [3] to clarify the model driven approach.
2 Enhance the "5G specifications overview" in TS 28.533 [3] for PM TSs and NRM TSs, including
2.1 Augment the "5G specifications overview" in TS 28.533 [3] Annex E to include the performance components.
2.2 Augment the "5G specifications overview" in TS 28.533 [3] Annex E to include the NRM components.
3 Enhance TS 28.537 [3] and TS 28.622 [6] to improve the overview information for each generic management capability, to clearly describe the MnS registry and discovery capabilities and corresponding solutions for different deployment scenarios.
4 Enhance MnSInfo IOC in TS 28.622 [6] to provide capabilities allowing MnS consumers to retrieve the MnS information for MnS instances based on specified areas (geographical area or TAI), to support the capabilities allowing MnS consumers to retrieve the m5anagement capability information of MnS instance provided by MnS producer.
5 Clarify in TS 32.156 [8] that <<SupportIOC>> instances are not used when an MnS is designed based on a model driven approach using a NRM and CRUD operations.
6 Update the ProvMnS schema in TS 28.532 [2] to decouple the ProvMnS schema and supported feature schema.
7 Update the Annex F “Usage of CRUD operations and NRM fragments to support management capabilities in SBMA” in TS 28.533 [3] with the overview of management capabilities and corresponding Solution Sets.
8 Add generic MnS version handling, including the specific MnS versioning in TS 28.532 [2] and TS 28.111 [1] for each MnS.
9 Add definition of the common notification header to TS 28.532 [2] and reliable notification transfer in TS 28.532 [2], TS 28.111 [1], TS 28.622 [6] and TS 28.623 [7].
10 Add input parameters for filtering on alarm conditions in TS 28.111 [1].

References

List of related CRs: select "TSG Status = Approved" in:
https://portal.3gpp.org/ChangeRequests.aspx?q=1&workitem=1060006

[1] TS 28.111: "Fault Management"
[2] TS 28.532: "Management and orchestration; Generic management services"
[3] TS 28.533: "Management and orchestration; Architecture framework"
[4] TS 28.536: "Management and orchestration; Management services for communication service assurance; Stage 2 and stage 3"
[5] TS 28.537: "Management and orchestration; Management capabilities"
[6] TS 28.622: "Generic Network Resource Model (NRM) Integration Reference Point (IRP); Information Service (IS)"
[7] TS 28.623: "Generic Network Resource Model (NRM) Integration Reference Point (IRP); Solution Set (SS) definitions"
[8] TS 32.156: "Fixed Mobile Convergence (FMC) model repertoire"
[9] TS 32.158: "Design rules for REpresentational State Transfer (REST) Solution Sets (SS)"
[10] TS 32.160: "Management service template"
[11] TR 28.871 " Study on Service Based Management Architecture enhancement phase 3"

27.3 Management Data Analytics phase 3

1060014	**Management Data Analytics phase 3**	eMDAS_Ph3	S5	SP-241947	Brendan T Hassett (Huawei)
1020019	*Study on Management Data Analytics (MDA) – Phase 3*	FS_eMDAS_Ph3	S5	SP-231726	1020019

Summary based on the input provided by Huawei in SP-251551.

eMDAS_Ph3 adds Management Data Analytics (MDA) capabilities and enhances existing Management Data Analytics capabilities and data types. Management Data Analytics (MDA), as a key enabler of automation and intelligence, is considered a foundational capability for mobile networks and services management and orchestration.

The following new MDA capability have been added in TS 28.104 [1]:

- End-to-end performance analytics including edge computing domain: perform analysis and provide recommendations related to edge application deployment location; estimate latency between UE and EAS.
- Data correlation analytics. perform analysis and provide recommendations related to data correlation; - handover optimization analysis (enhanced).
- ATSSS performance analytics. traffic steering (added recommendations);
- UE throughput analytics. traffic congestion.
- Fault management related analytics. report related to threshold crossing events; recommendations related to service failure recovery; failure prediction (enhanced).
- Software upgrade validation. validating a software upgrade.
- Control plane congestion analytics: 5GC Control plane congestion analysis (enhanced)

References

List of related CRs: select "TSG Status = Approved" in:
https://portal.3gpp.org/ChangeRequests.aspx?q=1&workitem=1060014

[1] TS 28.104: "Management and orchestration; Management Data Analytics (MDA)"

27.4 Enhancement of Management Aspects related to NWDAF Phase 2

1050031	**Enhancement of Management Aspects Related of NWDAF Phase 2**	NWDAF_OAM_Ph2	S5	SP-241378	Song Zhao (China Telecom)
1020020	*Study on Enhancement of Management Aspects related to NWDAF Phase 2*	FS_NWDAF_OAM_Ph2	S5	SP-231724	Primary Song Zhao (China Telecom)

Summary based on the input provided by China Telecom in S5-252641.

This work item is the phase two work on the enhancement of management aspects related to NWDAF. It is based on TR 28.877 [1], which studies the potential management enhancements on two selected aspects related to the NWDAF deployed in the system.

In order to provide the CM on the roaming exchange capability of the NWDAF, the IOC of "NWDAFFunction" and the data types of "NwdafCapability" and "NWDAFInfo" are updated in TS 28.541[2]. With these enhancements, the operators can configure the behavior of the NWDAFs, i.e. the exchange of analytic exposure information and related data, towards the other PLMNs belonging to different MNOs.

And more specifically, when the NWDAF is configured by the roamingAnalytics attributes to support Nnwdaf_RoamingAnalytics services in the roaming scenario, a series of performance measurements are provided in TS 28.552, reflecting the number, the time consumption,and the failure rate of the Nnwdaf_RoamingAnalytics services invoked.

Th second aspect is management enhancement related to the NWDAF data collection. New performance measurements are provided in TS 28.552 [3]:

- Performance measurements reflect the amount of data collected by NWDAF. Two cases are considered: a) data is collected directly by NWDAF; b) data is collected via DCCF/NWDAF hosting DCCF.

- Performance measurements estimate the NWDAF data collection efficiency. Two factors are considered to estimate the efficiency: a) the transmission overhead; b) the time used for the transmission (the time used for data preparation is not included).

The NRM enhancements are specified in TS 28.541 [2], the PM enhancements are specified in TS 28.552 [3], and the corresponding changes on YAML are updated on Forge.

References

List of related CRs: select "TSG Status = Approved" in:
https://portal.3gpp.org/ChangeRequests.aspx?q=1&workitem=1050031

[1] TS 28.877: “Study on Enhancement of the management aspects related to Network Data Analytics Functions (NWDAF) Phase 2”
[2] TS 28.541: “TS 28.541: "Management and orchestration; 5G Network Resource Model (NRM); Stage 2 and stage 3”
[3] TS 28.552: “Management and orchestration; 5G performance measurements”

27.5 Intent driven management services for mobile network phase 3

1060013	**Intent driven management services for mobile network phase 3**	IDMS_MN_Ph3	S5	SP-241946	Ruiyue Xu (Huawei)
1020008	*Study on intent driven management services for mobile networks phase 3*	FS_IDMS_MN_Ph 3	S5	SP-231737	1020008

Summary based on the input provided by Huawei and Ericsson in SP-251546.

This Work Item was the continuation of Intent Driven Management Services for Mobile Network work items from Release 18. It focused on following main topics:

- Requirements and solutions for new scenarios for intent driven management for 3gpp networks and services.

- New capabilities and solutions for intent driven management services.

5G Advanced networks bring operational complexities due to the large number of devices and diversity of services. Network autonomy is one of the important topics for 5G and 5G Advanced networks. Network autonomy aims to reduce network operating expenditure (OPEX) through increased operational efficiency and to improve the service experience of various vertical industries (e.g. autonomous vehicle, smart city) via a variety of intelligence and automation mechanisms. As networks continue to evolve the level of complexity for different scenarios including scenarios for design/planning, deployment, maintenance and optimization can explode. The need for abstraction of the operational requirements, such as is supported by intents, becomes more apparent.

An intent is a set of expectations including requirements, goals and constraints given to a 3GPP system, without specifying how to achieve them. Intents allow customers to request networks and services without detailed knowledge of how they will be provided. Inherently, this assumes that the system can learn the behaviour of networks and services and use intelligence and automation mechanisms to fulfil the expectations expressed via intents. This not only relieves the consumer of the burden of having to know implementation details but also provides flexibility allowing the system to explore alternative options to find optimal solutions.

Based on the intent driven management solution defined in R17 and R18, this WI further specifies the new use case and management capabilities for intent driven management for mobile networks.

The following content is added in TS 28.312 [1]:

1. Enhance the Intent Driven Management Services use cases to support new scenarios, including:

- Enhance the radio service delivery and assurance use case to support the scenario of delivering a radio service in a scheduled time.
- Enhance the radio network performance assurance use case to support the following scenarios:

- Radio network traffic assurance for scheduled events
- Radio network support for UAV pre-flight preparation
- Radio Network support MOCN undifferentiated radio service

- Enhance the RAN energy savings use case to support scenarios with differentiated RAN UE throughput performance assurance for different frequencies or RATs.

- Add new network maintenance intent use case.

2. Add new generic capabilities for Intent Driven Management Service, including:

- Intent handling capability obtaining. This functionality allows an MnS consumer to find an intent handling function which has the necessary domain responsibilities and supports their required intent expectations.

- Intent exploration. This functionality allows MnS Consumer to explore potential results of requested intent expectations and/or the best results the intent handling function could achieve.

- Intent feasibility check. This functionality allows MnS consumer to verify or check the feasibility of whether the proposed intent expectation is feasible for an intent handling function.

- Intent fulfilment negotiation. This functionality enables the intent handling function and the MnS consumer to agree on their preferred way to fulfil an intent.

- Intent utility function. This functionality defines a method by which MnS consumers can express the relative value of an intent's expectations to assist an intent handling function in fulfilling their intents in the most acceptable manner.

3. Enhance the following definitions for Intent Driven Management Services to support the new use cases and generic capabilities identified above, including:

- Enhancement of common intent model framework to support generic capabilities for intent driven management, including IntentHandlingFunction model, Intent model and IntentReport model.

- Enhancement of scenario specific intents to support the intent driven scenarios above including RadioNetworkExpectation,

- RadioServiceExpectation and NetworkMaintenanceExpectation.

4. Add guidelines for using the Intent Generic Information Model to support new scenarios using vendor defined extensions to the generic ExpectationObject(s) and vendor defined ExpectationTarget(s).

5. Add the OpenAPI solution set for the enhancement of Intent Driven Management Services definition.

References

List of related CRs: select "TSG Status = Approved" in:
https://portal.3gpp.org/ChangeRequests.aspx?q=1&workitem=1060013

[1] TS 28.312: "Intent driven management services for mobile networks"

27.6 Management of planned configurations

1050032	**Management of planned configurations**	PlanM	S5	SP-241379	Olaf Pollakowski (Nokia)
1020033	*Study on Management of planned configurations*	*FS_PlanM*	*S5*	*SP-231721*	*Olaf Pollakowski (Nokia)*

Summary based on the input provided by Nokia by e-mail received Fri 13/03/2026 02:52, file "Rel-19 WI Summary for Management of planned configurations".

TS 28.572 [1] defines the information models and management services for creating, validating, scheduling, activating, and rolling back planned configurations across 5G/5G Advanced and future 6G networks, thus enabling safe, automated, and interoperable network configuration management.

As networks become more dynamic, their configurations need to be changed more frequently (deploy new RAN features, adjust slice resources, rebalance traffic, update NF parameters, perform energy-saving/green-RAN optimizations, introduce new QoS or policy rules). Configuration changes are initiated by the operator or automation systems. The existing standard supports only changing directly the current configuration (i.e. the configuration that is active in the managed system).

Planned configurations and a management framework for these planned configurations are introduced by TS 28.572 [1]. Planned configurations can be managed without impacting the current configuration. The current configuration is updated only with a planned configuration upon a dedicated request (activation). Besides the basic features for the life-cycle

management of planned configurations (creating, updating, deleting, and reading) the standard provides many advanced features. For example, rules can be configured to control the time when planned configurations are activated automatically. Planned configurations can also be validated before activation. Another feature allows the detection of conflicts between different planned configurations for the same managed system. Fallback to a previous current configuration is also provided. Planned configurations can be grouped into planned configuration groups.

Planned configurations are introduced for usage in numerous management scenarios. For example, two configurations can be provided, one for low traffic hours and one for high traffic hours. Switching between both happens automatically based on configured rules. Conflict detection can be used for detecting conflicts between configurations provided by two independent automation systems for the same network portion. Validation helps to detect violations of operational constraints or conflicts of the planned configuration with the current network configuration. This reduces the risk of outages caused by misconfiguration, which is one of the most common sources of operational failures in telecom networks.

Another feature allows to validate and activate a new configuration without prior creation of a planned configuration. This is particularly useful for bringing large configuration changes into the network, which are inherently asynchronous.

Planned configurations are therefore a key capability for enabling automation and autonomous network operations.

The framework for management of planned configurations includes a detailed model for planned configurations and the rules for automatic activation. Management services are provided for creating, updating, deleting and reading planned configurations and activation rules. Furthermore, the standard features management services for validation, activation, and fallback. Detailed error responses are provided for each management service. A special progress monitor helps the operator to observe the progress of the activation or validation process.

Management services are specified using OpenAPI. They follow the REST design paradigm. Special flavours for the format of planned configurations are provided for YAML and YANG defined network models.

References

List of related CRs: select "TSG Status = Approved" in:
https://portal.3gpp.org/ChangeRequests.aspx?q=1&workitem=1050032

[1] TS 28.572: "Management and orchestration; Management of planned configurations"

27.7 Management aspects of Network Digital Twins

1060005	**Management aspects of Network Digital Twins**	NDT	S5	SP-241938	Yushuang Hu (China Mobile)
1020018	*Study on Management aspect of Network Digital Twin*	FS_NDT	S5	SP-231727	1020018

Summary based on the input provided by China Mobile in SP-251501.

This Work Item defines key terms and concepts for the Network Digital Twin (NDT) and specifies the use cases, requirements, and solutions for applying NDT within the 3GPP management system. The documented solutions—both generic and use-case-specific—are based on a new Management Service and related network resource models.

As 5GA networks grow more complex and service expectations rise, efficient management becomes essential. NDTs address this need by offering deep insights and enabling proactive, scenario-based analysis. They allow operators to predict outcomes, optimize configurations, and prepare for high-risk operations without affecting the live network.

An NDT is a virtual representation of a mobile network, or part of it, capturing its characteristics and behaviour. It combines hardware, software, and data flows to provide a comprehensive view of the network's current and potential future states. NDTs may be implemented using simulation, emulation, AI-based modelling, or other techniques that replicate network behaviour.

NDTs provide a range of capabilities that enhance network management:

No.	Capabilities	Use cases
1	Usage of control and Life-cycle management of NDTs	1) Life cycle management of NDT instances 2) Control of NDT instances 3) Synchronization with network
2	NDT support for network automation	1) General capabilities on NDT support for network automation

		2) Support for evaluation of high-risk network operations 3) Support for evaluation of failure events including signaling storm 4) Network issue inducement
3	NDT support for verification	1) General verification 2) Verification of network response to events 3) Verification of network configurations 4) Verification of automation-function configurations
4	NDT support for data generation	1) General use case on NDT support for data generation 2) Using NDT to generate ML training data 3) Using NDT to generate user experience data
5	Advanced NDT capabilities	1) Collaboration between NDTs

The NDT in 3GPP is defined as a management service (MnS) to provide the digital twin capabilities, including definitions in stage 2 and stage 3 which allows operator to manage NDT in a multi-vendor environment.

For stage 2, the operations (e.g., createMOI operations) and notifications (e.g., notifyMOIcreation) of generic provisioning MnS can be used for NDT management, including NDT function management, NDT job management and NDT report management. The NDTFunction, NDTJob, NDTReport instances can be treated as Managed Object instances, integrating with existing network management systems and processes, ensuring seamless operation.

For stage 3, the NDT MnS is defined by standardized operations such as the use of RESTful APIs for HTTP-based solutions. The RESTful HTTP-based solution set for generic provisioning management service, corresponding className are NDTFunction, NDTJob and NDTReport.

References

List of related CRs: select "TSG Status = Approved" in:
https://portal.3gpp.org/ChangeRequests.aspx?q=1&workitem=1060005

[1] TS 28.561 - Management aspects of Network Digital Twins

27.8 Closed Control Loop Management

1060010	**Closed Control Loop Management**	CCLM	S5	SP-241943	Deepanshu Gautam (Samsung)
1020009	*Study on closed control loop management*	FS_CCLM	S5	SP-231735	Deepanshu Gautam (Samsung)

Summary based on the input provided by Samsung by e-mail received Thu 12/03/2026 09:08, file "SP-26xxxx pCR TR21.919 Summary of Rel-19 Closed Control Loop Management".

A Closed Control Loop (CCL) is a control mechanism that monitors and regulates a set of managed entities in order to meet specific requirements. A CCL can be logically decomposed into a number of stages or steps, each providing a distinct function. These steps operate together to achieve the intended objective.

Different CCLs may provide the same overall functionality while being implemented with a different number of steps. Even when two CCLs have the same number of steps and the same high-level functionality, the functions performed by the individual steps may differ.

The following CCL management functionalities are defined as part of CCLM [1].

- Dynamic control and composition of CCLs: CCLs may be dynamically realized in two ways: by instantiating a CCL from an existing template, or by dynamically composing a CCL from discrete components based on given requirements. Trigger mechanisms define when a CCL is instantiated and when its actions are executed. Information about previously instantiated and executed CCLs must be maintained as historical CCL information, which may be used to support the creation of new CCLs.

- CCL Performance Monitoring: Advanced monitoring functions can provide real-time insights into the performance and outcomes of a CCL. Monitoring results may trigger further actions during the operation phase, such as evaluation and adjustment of CCL settings to improve performance. Therefore, it is necessary to assess the performance of the

CCL itself. Performance metrics are essential for understanding and adapting CCL behavior and for improving its effectiveness in meeting the assigned requirements.

- CCL usage scenarios: CCLs can be applied to different use cases or scenarios. Example scenarios include fault management and network performance problem recovery.

- CCL Conflict management and coordination Capability: A CCL may experience direct conflicts on its requirements, targets, scopes, trigger time and execution time. The management system needs to support capabilities to avoid, detect and resolve the conflicts. Various conflict are defined as Scope, Trigger, Actions, metric-value.

- CCL decision escalation: Decisions made by a CCL may not always be sufficiently effective under certain network conditions or contexts. In such cases, the CCL may need to indicate a lack of confidence in its decision and request escalation to another entity. For example, a CCL optimizing energy savings may be unable to determine the order in which cells should be deactivated following cell failures, and may therefore escalate the situation to a problem recovery CCL.

References

List of related CRs: select "TSG Status = Approved" in:
https://portal.3gpp.org/ChangeRequests.aspx?q=1&workitem=1060010

[1] TS 28.567: "Management aspects of closed control loops "

27.9 Data management phase 2

1020025	Data management phase 2	MADCOL_Ph2	S5	SP-250116	Sreekumar Pothera Kalloor (Nokia)

Summary based on the input provided by Nokia in SP-260211.

Data management phase 2 introduces support for management of historical data and discovery, request and collection of external management data. Additionally, the feature introduces reporting format for management data collections. The feature also introduces condition-based data collection and external data collection based on geographic area and DNs.

Discovery and collection of external management data enables Management Service (MnS) Consumers to perform "discovery" of external management data categories based on name and type of data and supported reporting methods. An MnS is defined as a set of offered capabilities for management and orchestration of networks and services, provided through a standardized service interface by an MnS producer and consumed by an MnS consumer. An MnS Consumer, based on the information received, can "request" the data from MnS Producer which will do the "delivery" of the requested data. The diagram below shows the interaction between MnS Consumer and MnS Producer on this, where MnF stands for Management Function , a logical entity that plays the role of an MnS producer and/or an MnS consumer (this is the functional block that implements and uses MnSs).

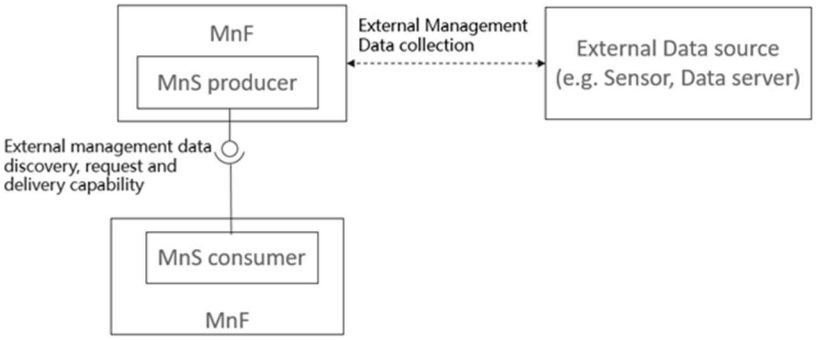


Figure 1: interaction between MnS Consumer and MnS Producer for Discovery and collection of external management data

Request management data based on conditions enables requests for management data based on certain conditions. The consumer can specify the conditions as part of the request and producer provides the data when the conditions specified are met. Examples of the condition can be a specific performance scenario.

Support for historic management data, external data scope, granularity period and reporting periods:

Management data collection introduces support for management of historic management data which enables discovery and retrieval of data from the past based on specified time period; External data scope support enables the retrieval of external management data based on specified geographic area and / or DNs; and Discovery of management data supports selection based on granularity periods and reporting periods..

References

List of related CRs: select "TSG Status = Approved" in:

https://portal.3gpp.org/ChangeRequests.aspx?q=1&workitem=1020025

[1] TS 28.537: "Management and orchestration; Management capabilities"
[2] TS 28.622: "Telecommunication management; Generic Network Resource Model (NRM) Integration Reference Point (IRP); Information Service (IS)"
[3] TS 28.623: "Telecommunication management; Generic Network Resource Model (NRM) Integration Reference Point (IRP); Solution Set (SS) definitions"

27.10 5G performance measurements and KPIs phase 4

1020026	5G performance measurements and KPIs phase 4	PM_KPI_5G_Ph4	S5	SP-250511	XiuMin Chen (China Telecom)

Summary based on the input provided by China Telecom in SP-251592.

Release-18's work on "5G performance measurements and KPIs" is extended in Release-19 to support advanced 5G features, enhanced UE-level measurements for data analytics and AI/ML applications, and further improvements to the PM streaming framework. The work enhances essential performance measurements and KPIs to address Rel-19 5G advanced capabilities, completes several Rel-18 measurements and KPIs that were previously left pending, and removes the limitation that restricted a PM stream to measurements from a single measured object.

The WID triggered several work in other WGs, including the introduction of a metric for UEs in the RRC_INACTIVE state with data transmission in TS 38.413 (RAN2) and maintenance updates to UE-level measurements in TS 29.244 (CT4).

New and enhanced 5G performance measurements for NG-RAN and 5GC are specified in TS 28.552, covering areas such as RAN availability and mobility, multicast and small data transmission management, RedCap and NTN support, PRB usage, QoS fulfilment notifications, ATSSS MA PDU sessions, PDU session establishment, user-plane delay between NG-RAN and PSA UPF, and NWDAF data collection and service subscriptions. New KPIs defined in TS 28.554 support monitoring of advanced 5G features, including extended DRX negotiation success, uplink and downlink time constraints, PDU session establishment accessibility, connected-mode RRM relaxation usage, MA PDU sessions, URLLC network slice end-to-end reliability, PRB-based high-load evaluation, and QoS flow maintainability.

The performance data streaming schema is enhanced through updates to the PDSU content in TS 28.550 and revisions to PM file extensions and JobID delimiters in TS 28.532. UE-level measurements are further specified in TS 28.558, including downlink packet loss with delay thresholds over Uu, round-trip packet delay between PSA UPF and NG-RAN, and correct data volume, together with aligned 5GC collection and reporting procedures, corrected managed object classes for UPF-related UE-level measurements, and enhanced examples in TS 32.423.

References

List of related CRs: select "TSG Status = Approved" in:
https://portal.3gpp.org/ChangeRequests.aspx?q=1&workitem=1020026

[1] TS 28.552: “Management and orchestration; 5G performance measurements”
[2] TS 28.554: "Management and orchestration; 5G end to end Key Performance Indicators (KPI)."
[3] TS 28.550: "Management and orchestration; Performance assurance".
[4] TS 28.532: "Management and orchestration; Generic management services".
[5] TS 28.558: "Management and orchestration; UE level measurements for 5G system".

[6] TS 32.423: "Telecommunication management; Subscriber and equipment trace; Trace data definition and management".

27.11 5G Advanced NRM features phase 3

1020027	5G Advanced NRM features phase 3	AdNRM_Ph3	S5	SP-250123	Sean Sun (Nokia) Shi Xiaoli (Huawei)

Summary based on the input provided by Huawei in SP-251547 and Nokia in SP-260036.

Note: The summaries mentioned above provide additional details; however, their structure, as a difficult-to-read list of items, does not align well with the style of this TR. They may be consulted for further information.

This work item continues the 5G Advanced NRM Phase 2 activities initiated in Releases 18, incorporating updates from SA2 and RAN2/3 while extending the scope to include management capabilities for edge applications. It further develops the 5G Network Resource Model (NRM), which provides the standardized management and service framework required to support new functionalities in NR and the 5G Core (5GC). The enhancements cover NG-RAN, 5GC, and Network Slice management, along with updates to the stage 3 Solution Set, ensuring consistency and alignment across specifications.

For NG-RAN and 5GC, the work introduces several functional and modeling improvements. On the NG-RAN side, it adds use cases and requirements for WAB-gNB management, support for Ambient IoT through new information object classes and related models, as well as enhancements for VMR Phase 2, cell coverage configuration, and ProSe multi-hop relay services. On the 5GC side, it extends key NRM components such as NFService and NFProfile, supports hierarchical NRF deployments, refines notification and IP address modeling, and aligns attribute definitions with TS 29.510. Additional enhancements include new attributes for various network functions and improved support for NTN management, LCS, and UPF capabilities. Across both domains, OpenAPI definitions are refined, in particular regarding multiplicity, default values, and attribute constraints.

In parallel, the work strengthens the Generic NRM and Edge NRM. For the Generic NRM, it introduces common data types, resolves alignment issues between stage 2 and stage 3, improves ENUM-based modeling, and refines schema definitions for SubNetwork and ManagedElement. For the Edge NRM, it enhances the EAS bundle and introduces managed services for edge entities, including improvements to ManagedEdgeNFService. Overall, these updates ensure a more consistent, extensible, and interoperable management framework across core, access, slicing, and edge domains.

References

List of related CRs: select "TSG Status = Approved" in:
https://portal.3gpp.org/ChangeRequests.aspx?q=1&workitem=1020027

[1] TS 28.540: "Management and orchestration; 5G Network Resource Model (NRM); Stage 1".
[2] TS 28.541: "Management and orchestration; 5G Network Resource Model (NRM); Stage 2 and stage 3"
[3] TS 28.622: "Telecommunication management; Generic Network Resource Model (NRM) Integration Reference Point (IRP); Information Service (IS)".
[4] TS 28.623: "Telecommunication management; Generic Network Resource Model (NRM) Integration Reference Point (IRP); Solution Set (SS) definitions".
[5] TR 28.538: "Management and orchestration; Edge Computing Management".

27.12 Subscriber and Equipment Trace and QoE collection management

1020028	Subscriber and Equipment Trace and QoE collection management	TraceQoE_OAM	S5	SP-250115	Zu Qiang (Ericsson)

Summary based on the input provided by Ericsson by e-mail received Thu 12/03/2026 16:59 (no file, plain e-mail text sent instead).

This Work Item delivers enhancements to Trace data collection and reporting mechanisms. The feature provides a completed solution for QoE in NR, enables RRC reporting collection, supports enhancements on Trace/MDT procedure and modelling (e.g. areaScope, UE consent, etc.), and allows the management based MDT to be measured continuously.

Enhancements to Trace data collection and reporting mechanisms introduced by this work item are specified in TS 32.421 [1] and TS 32.422 [2],TS 32.423 [3], TS 28.404 [4], TS 28.405 [5], TS 28.406 [6], TS 28.622 [7], and TS 28.623 [8].

References

List of related CRs: select "TSG Status = Approved" in:
https://portal.3gpp.org/ChangeRequests.aspx?q=1&workitem=1020028

[1] TS 32.421: “Management and orchestration; 5G performance measurements”
[2] TS 32.422: "Telecommunication management; Subscriber and equipment trace; Trace control and configuration management."
[3] TS 32.423: "Telecommunication management; Subscriber and equipment trace; Trace data definition and management".
[4] TS 28.404: "Telecommunication management; Performance Management (PM); Performance measurements; Definitions and template".
[5] TS 28.405: "Telecommunication management; Performance Management (PM); Performance measurements; Universal Terrestrial Radio Access Network (UTRAN)".
[6] TS 28.406: "Telecommunication management; Performance Management (PM); Performance measurements; Core Network (CN) Packet Switched (PS) domain".
[7] TS 28.622: "Generic Network Resource Model (NRM) Integration Reference Point (IRP); Information Service (IS)"
[8] TS 28.623: "Generic Network Resource Model (NRM) Integration Reference Point (IRP); Solution Set (SS) definitions"

27.13 Management of IAB nodes

1060015	**Management of IAB nodes**	NR_mobile_IAB_OAM	S5	SP-241948	JunFeng Wang (Ericsson)
1060014	Study on Management of IAB Nodes	FS_NR_mobile_IAB_OAM	S5	SP-231729	JunFeng Wang (Ericsson)

Summary based on the input provided by Ericsson by e-mail received Thu 12/03/2026 10:1, no attached file.

This Work Item adds management support to the following Integrated Access Backhauling (IAB) features: IAB-node connectivity to OAM system, using Plug and Connect (PnC) procedures; and location-based IAB-node configuration, using notification mechanisms.

References

List of related CRs: select "TSG Status = Approved" in:
https://portal.3gpp.org/ChangeRequests.aspx?q=1&workitem=1060015

27.14 Data management regarding subscriptions and reporting

1050033	**Data management regarding subscriptions and reporting**	Data_SREP	S5	SP-241380	Robert Petersen (Ericsson)
1020012	*Study on data management regarding subscriptions and reporting*	FS_Data_SREP	S5	SP-231732	Robert Petersen (Ericsson)

Summary based on the input provided by Ericsson by e-mail received Thu 12/03/2026 16:52, no file (summary provided in the e-mail).

This Work Item provides enhancements to data collection and reporting mechanisms, namely Trace Reference uniqueness for multi-consumer use cases, and supports Trace Admin notifications for different scenarios.

References

List of related CRs: select "TSG Status = Approved" in:
https://portal.3gpp.org/ChangeRequests.aspx?q=1&workitem=1050033

28 Rel-19 Charging-centric Features

28.1 Introduction

This section presents all the standalone charging functionalities. Charging Management aspects related to other features are reported in the relevant section. For instance, charging management for “Network Slicing" is reported in the section on " Network Slicing" .

In addition, chargin aspects related to other features are reported in the relevant section as described below:

1. Charging aspects for Multi-Operator Core Network (MOCN) Network Sharing, see 25.2

28.2 CHF Segmentation

1040012	**CHF Segmentation**	CHFSeg	S5	SP-250122	Cetinkaya Egemen Kemal,Verizon

Summary based on the input provided by Verizon by e-mail received Thu 26/03/2026 10:40, file "Summary_1040012".

The Charging Function (CHF) segmentation feature enables network deployments in which multiple groups of CHF instances are defined to support different categories of subscribers or services. For example, separate CHF groups may be deployed for enterprise customers and for non-enterprise consumers. Additionally, CHF groups may be organized to support subscribers associated with specific geographical regions. Such segmentation allows Network Function (NF) consumers (e.g., AMF, SMF, SMSF) to perform CHF discovery and selection, as specified in TS 32.255, TS 32.356, and TS 32.274.

The WID triggered several works in other WGs, including updates to the methods of CHF discovery and selection in the 3GPP 5G system as specified in TS 23.501, TS 23.502, and TS 23.503. Furthermore, charging characteristics IE is updated in TS 29.503, TS 29.510, and TS 29.518 (CT4).

References

List of related CRs: select "TSG Status = Approved" in:
https://portal.3gpp.org/ChangeRequests.aspx?q=1&workitem=1040012

[1] TS 32.255: "Telecommunication management; Charging management; 5G data connectivity domain charging; Stage 2"
[2] TS 32.256: "Charging management; 5G connection and mobility domain charging; Stage 2"
[3] TS 32.274: "Telecommunication management; Charging management; Short Message Service (SMS) charging"
[4] TS 32.290: "Telecommunication management; Charging management; 5G system; Services, operations and procedures of charging using Service Based Interface (SBI)"
[5] TS 32.291: "Telecommunication management; Charging management; 5G system, charging service; Stage 3"
[6] TS 23.501: "System architecture for the 5G System (5GS)"
[7] TS 23.502: "Procedures for the 5G System (5GS)"
[8] TS 23.503: "Policy and charging control framework for the 5G System (5GS); Stage 2"

29 No summary provided

29.1 No summary expected

No summaries are expected for studies, nor for work items dealing with protocol maintenance and frequency bands.

Studies

1040071	*Study on MC services support on IOPS mode of operation for 5G*	FS_Generic_IOPS	S6	SP-241017	FirstNet, Mark Lipford

980008	*Study on Diverse audio Capturing system for End-user Devices*	FS_DaCED	S4	SP-221330	Wang Bin, Xiaomi
1020014	*Study on management of IAB nodes*	FS_NR_mobile_IAB_OAM	S5	SP-231729	JunFeng Wang (Ericsson)
1020020	*Study on Enhancement of Management Aspects related to NWDAF Phase 2*	FS_NWDAF_OAM_Ph2	S5	SP-231724	Primary Song Zhao (China Telecom)
1000001	*Study on Reducing Information Exposure over SBI*	FS_RedInfExp_SBI	C4	CP-231350	Varini Gupta, Samsung
1020022	*Study on Enhanced OAM for management exposure to external consumers*	FS_MExpo	S5	SP-241570	Srilakshmi Srinivasaraju (Nokia)
1080002	*Study on IMS resiliency*	FS_IMSResil	C4	CP-251280	Kenichi Yamamoto, KDDI
1020034	*Study on enablers for Zero Trust Security*	FS_eZTS	S3	SP-231784	Sheeba Backia Mary Baskaran
1020040	*Study of ACME for Automated Certificate Management in SBA*	FS_ACME_SBA	S3	SP-231787	Charles Eckel
1020043	*Study on mitigations against bidding down attacks*	FS_MiBiDA	S3	SP-231789	Noamen Ben Henda
1060056	*Study on 3GPP Cryptographic Inventory*	FS_CryptoInv	S3	SP-241955	Stawros Orkopoulos
1080051	**LI Handling of Protected Services**	FS_LIHOPES	S3LI	SP-250813	Mats Näslund, NDRE
1040101	*Study on NR FR1 DL fragmented carriers*	FS_NR_FR1_DL_Frag_Carrier	R4	RP-243002	Mediatek
1020081	*Study on channel modelling enhancements for 7-24GHz for NR*	FS_NR_7_24GHz_CHmod	R1	RP-234018	LEE Daewon, Intel Corporation
780060	*Study on 6 GHz for LTE and NR*	FS_6GHz_LTE_NR	RP	RP-234033	Dominique Everaere, Ericsson
1030074	*Study on NR FR2 OTA (Over the Air) testing Phase 3*	FS_NR_FR2_OTA_Ph3	R4	RP-242833	Qualcomm
1030075	*Study on IMT parameters for NR for 4400 to 4800 MHz, 7125 to 8400 MHz and 14800 to 15350 MHz*	FS_NR_IMT_4400_7125_14800MHz	R4	RP-242944	Ericsson
1060078	*Study on Artificial Intelligence (AI)/Machine Learning (ML) for NR air interface Phase 2*	FS_NR_AIML_air_Ph2	R1	RP-250308	Qualcomm
950011	*Study on Artificial Intelligence (AI) and Machine Learning (ML) for Media*	FS_AI4Media	S4	SP-220328	Eric Yip, Samsung
950008	*Study on AI/ML Model Transfer Phase2*	FS_AIML_MT_Ph2	S1	SP-220439	Yang Xu, OPPO
1010003	*Study on IMS Disaster Prevention and Restoration Enhancement*	FS_IMS_RES	C4	CP-232281	Liu, Liu, China Telecom
1000016	*Study on Film Grain Synthesis*	FS_FGS	S4	SP-241389	Brian Lee (Dolby Laboratories)

Protocol maintenance

1040001	**Service Based Interface Protocol Improvements Release 19**	SBIProtoc19		CP-241021	Wang Rong, China Mobile
1040050	**CT3 aspects of Service Based Interface Protocol Improvements Release 19**	SBIProtoc19	C3	CP-241021	Wang Rong, China Mobile
1040051	**CT4 aspects of Service Based Interface Protocol Improvements Release 19**	SBIProtoc19	C4	CP-241021	Wang Rong, China Mobile
1040039	**IMS Stage-3 IETF Protocol Alignment**	IMSProtoc19		CP-251219	Won, Sung Hwan, Nokia
1040062	**CT1 aspects of IMS Stage-3 IETF Protocol Alignment**	IMSProtoc19	C1	CP-251219	Won, Sung Hwan, Nokia
1040063	**CT3 aspects of IMS Stage-3 IETF Protocol Alignment**	IMSProtoc19	C3	CP-251219	Won, Sung Hwan, Nokia

1040041	Protocol enhancements for Mission Critical Services	MCProtoc19		CP-241283	Magnus Tränk, Ericsson
1040064	**CT1 aspects of Protocol enhancements for Mission Critical Services**	MCProtoc19	C1	CP-241283	Magnus Tränk, Ericsson
1040065	**CT3 aspects of Protocol enhancements for Mission Critical Services**	MCProtoc19	C3	CP-241283	Magnus Tränk, Ericsson
1040066	**CT4 aspects of Protocol enhancements for Mission Critical Services**	MCProtoc19	C4	CP-241283	Magnus Tränk, Ericsson
1060025	Rel-19 Enhancements of UE Policy	UEP19	C3	CP-243073	Susana Fernández, Ericsson
1070005	Rel-19 Enhancements of SM Policy	SMPC19	C3	CP-250075	Chi Zhang, Huawei
1040044	Stage-3 5GS NAS protocol development 18	5GProtoc19		CP-241280	Sedlacek, Ivo, Ericsson,
1040046	**Stage-3 5GS NAS protocol development 18 general aspects**	5GProtoc19	C1	CP-241280	Sedlacek, Ivo, Ericsson,
1040047	**Stage-3 5GS NAS protocol development 18 non 3GPP aspects**	5GProtoc19-non3GPP	C1	CP-241280	Sedlacek, Ivo, Ericsson,
1040040	Stage-3 SAE Protocol Development	SAES19		CP-241284	Sethi, Anuj, InterDigital
1040048	**Stage-3 SAE Protocol Development general**	SAES19	C1	CP-241284	Sethi, Anuj, InterDigital
1040049	**Stage-3 SAE Protocol Development non 3GPP**	SAES19-non3GPP	C1	CP-241284	Sethi, Anuj, InterDigital
970002	Lawful Interception Rel-19	LI19	S3LI	SP-250817	Koen Jaspers, PIDS
1040010	Lawful Interception Guidance Rel-19	LI19_Guide	S3LI	SP-240562	Nag Rao, Nokia,

Frequency bands

0	Rel-19 Radio: NR and LTE	-		-	-
1040126	Adding channel bandwidth(s) support to existing NR bands and CA/ENDC combinations in REL-19	NR_bands_CA_ENDC_R19_BWs		RP-252020	Ericsson
1040125	Simultaneous Rx/Tx band combinations for NR CA/DC, NR SUL and LTE/NR DC in Rel-19	LTE_NR_R19_Simult_RxTx		RP-251536	Huawei
0	Rel-19 Radio: LTE	-		-	-
1060080	New bands for LTE based 5G terrestrial broadcast for early deployments	LTE_terr_bcast_bands_sub_108		RP-250793	EBU
1040117	Introduction of LTE FDD band in 1800–1830 MHz for Canada	LTE_FDD_1800_1830MHz_CAN		RP-242461	Novamint

29.2 Other items

No summary was deemed necessary for the following items:

1010011	5G Security Assurance Specification (SCAS) for the Short Message Service Function (SMSF)	SCAS_5G_SMSF	S3	SP-231158	Manjesh K. Hanawal, IIT Bombay
1020036	Addition of Milenage-256 algorithm	Milenage_256	S3	SP-241087	Mireille Pauliac
1070023	Specification of example algorithm for alternative f5* (f5) function	Crypto_f5	S3	SP-250360	Ferhat Karakoc
970043	Interworking of Non-3GPP Digital Terrestrial Broadcast Networks with 5GS Multicast Broadcast Services	DTT4MBS	S1	SP-220941	Saha, Anindya, Saankhya Labs
1060003	**Reducing Information Exposure over SBI**	RedInfExp_SBI	C4	CP-243308	Nivedya P S, Samsung
1040003	Subscriber Data Migration	SUBDMIG	C4	CP-251038	Jesus de Gregorio, Ericsson
1010025	Local traffic routing for multi-access UE	LTR_MA	S1	SP-231035	Yinglin Chen, China Telecom
970044	Multi-path relay	MultiRelay	S1	SP-220943	Yuying Zhang, China Telecom

0	**Rel-19 Radio: NR**	-		-	-
1040133	**NR power class 2 RedCap (Reduced Capability) UE in FR1**	NR_PC2_RedCap_UE		RP-243267	China Telecom
1050126	**mmWave in NR: UE spurious emissions and EESS (Earth Exploration Satellite Service) protection**	NR_mmWave_protect		RP-251811	China Unicom
1030080	**NR demodulation performance Phase 5**	NR_demod_Ph5		RP-252538	China Telecom
0	**Rel-19 Radio: NR and LTE**	-		-	-
1040118	**Rel-19 downlink interruption for NR and EN-DC band combinations at dynamic Tx Switching in Uplink**	DL_intrpt_combos_TxSW_R19		RP-250369	China Telecom
1040119	**Rel-19 DC of x LTE band(s), y NR band(s) (1<=x<6, 1<=y<6, x+y<=6) and single or two NR Supplementary Uplink (SUL) bands**	DC_R19_xBLTE_yBNR		RP-252016	Nokia
0	**Rel-19 Radio: LTE**	-		-	-
1040115	**Rel-19 LTE-Advanced Carrier Aggregation for x bands (1<=x<= 6) DL with y bands (y=1, 2) UL**	LTE_CA_R19_xBDL_yBUL		RP-252735	Huawei

Annex A:
Acronyms and short overview of the associated concepts

AI: Artificial Intelligence / ML: Machine Learning

API: Application Programming Interface. A set of standardized service interfaces (also called “Northbound APIs”) that expose the 3GPP network capabilities to external applications (operators, verticals, cloud services) in a secure and interoperable way, as to enable applications to consume the mobile network capabilities such location, QoS, or connectivity status.

BWP: Multi-Initial Bandwidth Part. A configured portion of the NR carrier bandwidth used by a UE at a given time for uplink or downlink operation.

CA: Carrier Aggregation. A radio feature that allows a UE to simultaneously transmit and/or receive data over multiple component carriers (CCs), which are combined to form a wider effective bandwidth and increase data throughput. The CCs can be contiguous or not, Intra-band or Inter-band.

CAG: Closed Access Group. An access-control mechanism: in NPN, only UEs that belong to the configured CAG are permitted to select and connect to those cells.

CJT: Coherent Joint Transmission. A downlink transmission technique (introduced in Rel-18) where multiple Transmission Reception Points (TRPs) transmit the same data stream to a UE with tight phase alignment, so that the signals combine coherently (constructively) at the receiver, instead of interfering with each other.

CMCD: Common Media Client Data. A standardized way for media players to report real-time playback information—such as buffer level, bitrate, latency, and session identifiers—to networks and CDNs. It gives operators and content providers better visibility into how streams are performing so they can optimize delivery and improve user experience.

CSI: Channel State Information. Information measured by the UE about the radio channel conditions and reported to the network (gNB) to enable efficient link adaptation and scheduling.

DC: Dual Connectivity. A 3GPP feature that allows a UE to be simultaneously connected to two different radio nodes, called a Master Node (MN) and a Secondary Node (SN), in order to aggregate radio resources and improve performance. It can be inter-RAT (LTE–NR) or intra-RAT (LTE–LTE or NR–NR)

DCI: Downlink Control Information. Control information sent by the eNB/gNB to a UE on the PDCCH to tell the UE how, when, and where to transmit or receive data.

DL: Downlink (Base Station to UE). See also Uplink (UE to Base station).

FDD (Frequency Division Duplexing). A duplexing method where uplink and downlink transmissions occur simultaneously on two separate, paired frequency bands, enabling full-duplex communication between the UE and the network (simultaneous UL and DL using different frequencies). Used in LTE and in NR. See also TDD.

FR1: NR Frequency Range 1 (410 MHz to 7.125 GHz, aka “sub-6 GHz”). Wider coverage as FR2.

FR2: NR Frequency Range 2 (24.25 GHz to 52.6 GHz, aka “mmWave”). Higher high data rates but shorter range than FR1.

FWA: Fixed-Wireless Access. A broadband access over LTE or NR for fixed locations. “Type 4 FWA” is the most stringent FWA variant, providing fully fixed, high-capacity, fiber-like broadband using cellular radio.

FWVM: Fixed-Wireless/Vehicle-Mounted or Frequency-Weighted Vector Magnitude

GNSS: Global Navigation Satellite System (or “A-GNSS” for Assisted-GNSS in the 3GPP context). The satellite-based positioning (GPS, Galileo, GLONASS, BeiDou, etc.) as integrated, assisted, and standardized within cellular systems (LTE and 5G).

IMS: Internet-Protocol (IP) Multimedia Subsystem. The set of CN entities that provides multimedia services in 4G and 5G (such as voice, video, and messaging) over IP networks using the IETF’s Session-Initiation Protocol (SIP).

INS: Indirect Network Sharing. It is a network-sharing architecture in which multiple operators share a common RAN, but the connectivity between the shared RAN and each participating operator’s CN is routed indirectly via the hosting operator’s CN, rather than through direct N2/N3 connections for each operator. See also MOCN.

IoT: Internet of Things. A family of cellular technologies standardized by 3GPP to connect massive numbers of low-cost, low-power devices over licensed mobile networks.

ITS: Intelligent Transport Systems. Refers to 3GPP support for V2X (Vehicle-to-Everything). The “ITS spectrum” or “ITS band” is region-specific (around 5.9 GHz)

LMF: Location Management Function. A 5G CN function that collects and manages positioning-related information.

LTE: “Long Term Evolution” – the name of 3GPP’s 4G radio technology (introduced in Rel-8).

LTM: L1/L2-Triggered Mobility. A 5G NR mobility enhancement (introduced in Rel-18) in which mobility decisions (cell switch / handover execution) are triggered using Layer-1 / Layer-2 measurements and procedures, rather than being driven solely by higher-layer (RRC) control, with the objective of reducing mobility latency and interruption time.

MIMO: Multiple-Input Multiple-Output. The use of multiple transmit and receive antennas to increase data rate, reliability, and spectral efficiency through spatial multiplexing, diversity, and beamforming.

MOCN: Multi-Operator Core Network. It is a network-sharing architecture in which multiple operators share the same RAN (radio equipment and radio resources) while each operator maintains its own independent core network. See also INS.

NAS: Non-Access Stratum. The control-plane signaling between the UE and the CN (independent and carried transparently through the radio access network). Used for mobility, registration/attach, authentication, security and session management.

NR: “New Radio” – the name of 3GPP’s 5G radio technology (introduced in Rel-15).

NTN: Non-Terrestrial Networks (satellite, UAV, etc.)

OTA: Over-The-Air. Refers to radiated transmission, reception, and performance testing of UEs or base stations using antennas, without direct RF cable connections.

RedCap: Reduced Capability. It refers to a specific NR device capability profile (also called NR-Light) with constrained bandwidth, antennas, and complexity, used by certain IoT-like use cases.

RF: Radio Frequency. Refers to the radio transmission, reception, and performance requirements of UEs and base stations, including spectrum, bands, power, emissions, and receiver characteristics.

RRM: Radio Resource Management. Refers to the functions and procedures that control and optimize the use of radio resources—such as power, spectrum, scheduling, mobility, and interference—to meet performance and QoS requirements.

RTK: Real-Time Kinematic. A high-precision positioning technique that improves standard GNSS accuracy from meter level to centimeter level by using real-time correction data.

SAN: Satellite Access Node. A 3GPP-defined node that provides radio access for Non-Terrestrial Networks (NTN), terminating user plane and control plane protocols towards satellite-capable UEs and connecting to the 5G Core via the NG interface.

SEAL: Service Enabler Architecture Layer. A common , modular enabler layer that provides reusable service enablers on top of the 3GPP system, allowing vertical applications and services to interact with the network in a consistent way.

SIB1: SystemInformationBlockType1. The broadcast system information block that tells a UE whether it is allowed to access a cell and provides the scheduling information for acquiring other system information blocks.

SSB: Synchronization Signal Block. The downlink block that enables a UE to find and synchronize to a cell, consisting of the PSS, SSS, and PBCH transmitted together.

TDD: Time Division Duplexing. A duplexing method where uplink and downlink transmissions use the same frequency band, but are separated in time (UL and DL share one frequency and alternate in time). Used in LTE and in NR. See also FDD.

TN: Terrestrial Networks. By opposition to NTN.

TRP: Total Radiated Power. The total transmit power radiated by a device, obtained by integrating the radiated power over all directions and polarizations in free space.

TRS: Total Radiated Sensitivity. A measure of a receiver's sensitivity averaged over all directions and polarizations, defined as the integrated minimum received power required to meet a specified performance criterion in free space.

UE: User Equipment. In the general case, this is the mobile phone together with its SIM card.

UL: Uplink (UE to Base Station). See also Downlink (Base station to UE).

URSP: UE Route Selection Policy. A set of policy rules provisioned by the 5G CN to the UE that determines how application traffic is routed, i.e. which PDU Session, Data Network, and network slice (S-NSSAI) should be used for different types of traffic. The rules are defined by the Policy Control Function (PCF) and delivered to the UE via the AMF.

USS: UAS Service Supplier. A service provider that supports UAV operations by offering functions such as flight planning, in-flight monitoring, regulatory compliance, and coordination with the mobile network and UTM (UAS Traffic Management) systems.

WAB: Wireless Access Backhaul. WAB allows to use 5G NR radio links not only to connect user equipment (UEs) but also to provide wireless connectivity between radio network nodes, avoiding the need for wired (e.g. fiber) backhaul at every site.

Annex B:
Change history

Change history							
Date	**Meeting**	**TDoc**	**CR**	**Rev**	**Cat**	**Subject/Comment**	**New version**
2025-12	TSG#110					Initital draft, call for contributions	0.1.0
2026-03	TSG#111					**Inclusion of summaries received at TSG#110, i.e.:** TDoc Title Source CP-253174 Summary for Rel-19 Enhancements of UE Policy Ericsson CP-253216 Work Item Summary for SMPC19 Huawei CP-253217 Summary for Rel-19 Enhancements of 3GPP Northbound Interfaces and Application Layer APIs Huawei RP-253362 WI summary for AI/ML for NR air interface Qualcomm Technologies RP-253518 WI summary: NR base station (BS) RF requirement evolution for FR1/FR2 and testing ZTE Corporation SP-251312 Summary for 5G Security Assurance Specification (SCAS) for the Unified Data Repository (UDR) BSI Germany SP-251360 Summary for Vehicle Mounted Relays Phase 2 (VMR_Ph2) Qualcomm SP-251361 Summary for WI on subscription control for reference time distribution in EPS Qualcomm SP-251375 Work Item Summary for EVS Codec Extension for Immersive Voice and Audio Services, Phase 2 (IVAS_Codec_Ph2) vivo Mobile Communication Co., Ltd. (Rapporteur) SP-251376 Summary for Identifying non-3GPP Devices Connecting behind a UE or 5G-RG InterDigital Inc. SP-251501 Summary for Management aspects of Network Digital Twins China Mobile SP-251503 Summary for WI Video Operating Points - Harmonization and Stereo MV-HEVC (VOPS) Apple Switzerland AG SP-251504 Summary for SEALDD_Ph2 Huawei Technologies SP-251511 WI Summary for Application enablement for AI/ML services for Rel-19 Lenovo SP-251512 Summary for UE-to-UE multi-hop relay FirstNet SP-251513 Summary for Generic IOPS FirstNet SP-251544 Summary for ATIAS_Ph2 HEAD acoustics GmbH SP-251546 Summary for Rel-19 Intent driven Management Service for Mobile Network phase 3 Huawei, Ericsson SP-251547 Summary for Rel-19 5G Advanced NRM features phase 3 Huawei Tech.(UK) Co.. Ltd SP-251548 Summary for Rel-19 Service based management architecture phase 3 Huawei Tech.(UK) Co.. Ltd SP-251551 Summary for Rel-19 Management Data Analytics phase 3 Huawei Tech.(UK) Co.. Ltd SP-251554 Summary for Rel-19 Charging for Ambient power-enabled Internet of Things Huawei Tech.(UK) Co.. Ltd SP-251555 Summary for Rel-19 Charging Aspects for Energy Efficiency of Network Slice Huawei Tech.(UK) Co.. Ltd SP-251559 Summary for WI 'Stage 3 for Advanced Media Delivery (AMD_PRO-MED)' Qualcomm, BBC (rapporteurs) SP-251560 UPEAS_Ph2 WID summary ZTE, Vodafone SP-251565 WI Summary for Mission Critical Security Enhancements for Rel-19 Motorola Solutions Germany SP-251568 Summary for WI Minimize the Number of Policy Associations (TEI19_MINPA) Ericsson SP-251569 Summary for WI on Enhancing Parameter Provisioning with static UE IP address and UP security policy (TEI19_IP_SP_EXP) Ericsson SP-251570 Summary for WI on Providing per-subscriber VLAN instructions from UDM and DN-AAA (TEI19_VLANSUB) Ericsson SP-251571 Summary for Deferred 5GC-MT-LR Procedure for Periodic Location Events based NRPPa Periodic Measurement Reports (TEI19_DLPMR) Ericsson SP-251592 Summary for 5G performance measurements and KPIs phase 4 China Telecom, ZTE (Rapporteurs) **Inclusion of summaries from RAN#106 to RAN#109, i.e.:**	1.0.0

						RP-250464 WI summary: NR channel BW less than 5MHz for FR1 Phase 2 Intel Corporation RP-250999 WI summary for NR Sidelink: Intra-band Carrier Aggregation in ITS band OPPO RP-251942 WI Summary for NR sidelink multi-hop relay LG Electronics Inc., InterDigital RP-252009 WI Summary: Low-power wake-up signal and receiver for NR (LP-WUS/WUR); rapporteurs: vivo, NTT DOCOMO vivo, NTT DOCOMO RP-252011 WI Summary: TRP (Total Radiated Power), TRS (Total Radiated Sensitivity) and MIMO OTA (Over the Air) testing enhancement Phase 3; rapporteur: vivo, CAICT vivo, CAICT RP-252075 WI summary for enhanced requirements and conductive test methodology for NR NTN and IoT NTN Samsung, Xiaomi RP-252107 Summary for WI: Introduction of BDS (BeiDou Navigation Satellite System) B2b Signal in A-GNSS for LTE and NR CAICT, CATT RP-252149 WI summary: Enhancements for Artificial Intelligence (AI)/Machine Learning (ML) for NG-RAN ZTE Corporation, NEC RP-252333 WI summary for enhancements of network energy savings for NR Ericsson, Apple RP-252356 Summary for WI on XR (eXtended Reality) for NR Phase 3 Nokia, Qualcomm RP-252365 WI summary: Evolution of NR duplex operation: Sub-band full duplex (SBFD)Huawei RP-252480 Summary for WI additional topological enhancements for NR NTT DOCOMO, INC., AT&T RP-252495 WI summary of R19 RRM phase 5 Apple, CATT RP-252505 WI Summary: Non-Terrestrial Networks (NTN) for Internet of Things (IoT) Phase 3MediaTek Inc. RP-252565 WID Summary for Multi-carrier enhancements Lenovo RP-252603 Summary for WI Enhancements for Air-to-ground network for NR CMCC RP-252696 Summary for WI: Low NR band carrier aggregation via switching TELUS RP-252728 WI Summary: Study on Spatial Channel Model for demodulation performance requirements for NR Nokia RP-252732 WI summary: UE RF enhancements for NR FR1/FR2 and EN-DC, Phase 4 Huawei, HiSilicon, AT&T RP-252823 WI summary for WI Core part: NR MIMO Phase 5 Samsung RP-252828 WI summary for WI Core part: Introduction of A-GNSS support for NavIC (Navigation with Indian Constellation) L1 SPS (Standard Positioning Service) in NR & LTE Reliance Jio RP-252829 WI summary for WI Core part: LTE-based 5G Broadcast Phase 2 EBU RP-252837 WI summary for WI Core part: Support of intra-band non-collocated EN-DC/NR-CA deployment Phase2: new receiver type(s) KDDI RP-252846 WI summary for WI Core part: Inter-RAT mode mobility support from E-UTRAN TN to NR NTN Thales RP-252872 WI summary for WI Core part: Introduction of IoT-NTN TDD mode Iridium RP-252884 Summary for WI NR Mobility Enhancements Phase 4 Apple Inc, China Telecom RP-252885 Summary for WI New LTE band for 5G broadcast for region 3 utilizing a geosynchronous satellite ASTRUM MOBILE PTE. LTD. RP-252886 WI summary for WI Core part: Non-Terrestrial Networks (NTN) for NR Phase 3 Thales, CATT RP-252926 WI Summary for Data collection for SON (Self-Organising Networks)/MDT (Minimization of Drive Tests) in NR standalone and MR-DC (Multi-Radio Dual Connectivity) Phase 4 China Unicom, CMCC RP-252928 WI summary for Rel-19 Ambient IoT (Internet of Things) in NR CMCC, Huawei, T-Mobile USA Inc.	
						Clause numberring corrected (their were two "9" at level 1 headings). **Inclusion of Rel-19 summaries from CT#107 to CT#109 and SA#107 to SA#109, i.e.:**	1.1.0

						Note: all summaries provided at CT and SA#107 were for Rel-18 Features. Rel-19 summaries, included here,are: SP-250982 Work Item Summary for 'Split Rendering over IMS' (SR_IMS) Nokia (Rapporteur) SP-251129 Summary for UAS_Ph3 LG Electronics SP-251144 Work Item Summary of '5G Real-time Media Transport Protocol Configurations, Phase 2 (5G_RTP_Ph2)' Nokia (5G_RTP_Ph2 Rapporteur) SP-251147 Summary for WI AvCall-MED Qualcomm SP-251148 Summary for WI 'Media Messaging Enhancements (MeME-MED)' Qualcomm **Summaries provided at TSG#111 – not yet included in this version:** CP-260142 Summary for Rel-19 Alignment of eCall over IMS with CEN Qualcomm Incorporated CP-260144 (CP-260174?) Summary for Rel-19 Enhancments of Network Automation Enablers Huawei CP-260158 Summary for Rel-19 IP Domain usage China Mobile CP-260159 Summary for Rel-19 Protocol for AI Data Collection from UPF China Mobile RP-260625 WI summary: NR base station (BS) RF requirement evolution for FR1/FR2 and testing ZTE Corporation RP-260682 Revised Summary for WI: Low NR band carrier aggregation via switching TELUS SP-260034 Summary for Indirect Network Sharing China Unicom SP-260035 Summary for Extended Reality and Media Service Phase 2 (XRM_Ph2) Nokia SP-260036 Summary for 5G Advanced NRM features phase 3 Nokia SP-260037 Summary for NF discovery and selection by target PLMN (TEI19_NFsel_by_tPLMN) Nokia SP-260038 Summary for Multiple Location Procedure for Emergency LCS Routing (TEI19_MLR4RTR) Nokia SP-260039 Summary for Roaming and interconnect authorization aspects in indirect communication (SBA_RoamInt-SEC) Nokia SP-260040 Summary for Mobility over non-3GPP access to avoid full primary authentication (Non3GPPMob_Sec) Nokia SP-260151 Summary for ATSSS Rule Provisioning via 3GPP access connected to EPC (TEI19_ARP3E) NEC SP-260152 Summary for OAM support for Extended Reality and Media service (XRM) phase 2 ZTE SP-260211 Summary for Data management phase 2 (MADCOL_Ph2) Nokia SP-260218 Summary for Rel-19 Enhanced application layer support for location CATT SP-260227 Summary for Security for PLMN hosting a NPN China Telecom SP-260228 Summary for MINT_Ph2 China Telecom SP-260232 Summary for Management of Network Sharing Phase 3 China Unicom SP-260233 Summary for TEI19_SLUPiR Oracle, China Telecom SP-260235 Summary for Management Aspects of RedCap features China Unicom SP-260238 Summary of Rel-19 Work Items NTN_OAM_Ph2 China Unicom SP-260239 Summary for Rel-19 Study on 3GPP AI/ML Consistency Alignment Deutsche Telekom AG, vivo SP-260255 Summary for Rel-19 Alignment of eCall over IMS with CEN Qualcomm SP-260256 Summary for CAPIF_Ph3 Nokia Solutions & Networks (I) SP-260262 Summary for EdgeOpNeeds China Telecom Corporation Ltd. SP-260281 Summary for ProSe support in NPN Philips International B.V. Identified that "Sensing" shall be moved to Rel-20 (only non-viable Stage 1 in Rel-19) **Summaries provided by e-mail after at TSG#111 included in this version:** 1070009; Support for PWS in Satellite E-UTRAN and Satellite NG-RAN; amerc@qti.qualcomm.com; e-mail 16/03/2026 21:54, file "Summary_1070009_1070049" 1060018; NAS layer overhead reduction for data transfer using CP CIoT; e-mail 16/03/2026 21:54, file "Summary_1060018". 1000036; Sharing of administrative configuration between interconnected MC service systems; e-mail 30/03/2026 14:01, file "Summary_1000036".	

						1070039; PRU Usage Extension supported by Core Networ (TEI19_PRUE); e-mail 19/03/2026 04:24, file "Summary_1070039". 1040012; CHF Segmentation; CHFSeg; e-mail 26/03/2026 10:40, file "Summary_1040012". 1050114; Monitoring of signalling traffic in 5G (MonSTra) ; Vodafone; e-mail 12/03/2026 23:06, file "Summary_Monstra". 1090031; Lower Selection-priority for PLMN Selection (LoSePLMN); Vodafone; e-mail 12/03/2026 23:06, file "Summary_PLMN_Selection". 1040032; Extended Reality and Media Phase 2; XRM_Ph2; Nokia in SP-260035, then Ericsson e-mail 21/01/2026 09:33, file "XRM_Ph2_summary". 1060010; Closed Control Loop Management (CCLM); Samsung; e-mail 12/03/2026 09:08, file "SP-26xxxx pCR TR21.919 Summary of Rel-19 Closed Control Loop Management". 1060048; 5G NR Femto (5G_Femto); NTT DOCOMO, e-mail 17/03/2026 09:02, file "Femto_Summary_SA2_CT". 1020056; Localized Mobile Metaverse Services; Metaverses; Samsung; e-mail 10/04/2026 18:02, file "Summary_1020056". 1020031; Upper layer traffic steering and switching over dual 3GPP access (DualSteer); Qualcomm e-mail 17/03/2026 17:47, file "Summary_1020031". => deleted since DualSteer Stage 1 moved to Rel-20 Rel-19 Application Data Analytics Enablement Service (1070044, TEI19_ADAES): Ericsson; e-mail 07/04/2026 09:09, file "draft Summary for TEI19_ADAES". Proximity-based Services in 5GS Phase 3 (1050115, 5G_ProSe_Ph3); AT&T e-mail 19/03/2026 22:59, and and by KPN/TNO (e-mail received Tue 31/03/2026 16:05: 2 files named "Summary_1050115". Multi-Access (ATSSS_Ph4) (1050056, MASSS); Apple e-mail 28/03/2026 02:23 , file "Summary_1050056". IMS Disaster Prevention and Restoration Enhancement (1080001, IMS_RES-CT); China Telecom ; e-mail 13/03/2026 04:08, file "Summary for IMS Disaster Prevention and Restoration Enhancement-Liu Liu". PS Data Off for IMS Data Channel Service (990049, IMSDCDataOff); China Mobile; e-mail ; 31/03/2026 10:31, file "24.3.1_Summary for PS Data Off for IMS Data Channel Service". 5G Message Services (5GMSG) Service phase 3 (1060051, 5GMARCH_Ph3); China Mobile e-mail 31/03/2026 10:31, file "10.4_Summary for 5GMARCH_Ph3". Application enablement aspects for MMTel(1070036, MMTel_App); China Mobile; e-mail 31/03/2026 10:31, file "10.9_Summary for Application enablement aspects for MMTel". Enhanced Mission Critical Location Management (1040045, enhMCLoc) ; Ericsson ; e-mail ; 19/03/2026 15:17, file "CP-26xxxx enhMCLoc WI summary". Data management regarding subscriptions and reporting (1050033, Data_SREP); Ericsson ; e-mail 12/03/2026 16:52, no file (summary provided in the e-mail). Satellite access Phase 3 (1020058, 5GSAT_Ph3); Novamint and CATT by e-mail, latest version received Fri 03/04/2026 04:10, file "Summary_Update_1020058_V2.1". Security Aspects (1060065, 5GSAT_Ph3_SEC): also in Novamint and CATT file Charging aspects (1070014, 5GSAT_Ph3-CH): CATT in SP-250365 MSISDN verification operation support to Nnef_UEId Service (1070046, TEI19_MVOSNS); Summary based on the input provided by ATT by e-mail received Tue 17/03/2026 21:08, file "Summary_1070046".	

						3GPP profiles for cryptographic algorithms and security protocols (1020042, CryptoSP); Summary based on the input provided by Ericsson by e-mail received Thu 19/03/2026 09:12, file "Summary_1020042". 3GPP profiles for cryptographic algorithms and security protocols (1020042, CryptoSP); Summary based on the input provided by Ericsson by e-mail received Thu 19/03/2026 09:12, file "Summary_1080046". Future Railway Mobile Communication System (FRMCS) Phase 5 (1000031, FRMCS_Ph5); Summary based on the input provided by UIC by e-mail received Mon 02/03/2026 16:47 , file "Summary_1000031". Enhancement of controlling RAT utilization (1040042, ECRATU); Summary based on the input provided by Vodafone by e-mail received Tue 17/03/2026 09:20, file "Summary_ECRATU". Edge Computing for Industrial Scenarios (990050, EDGINDUS) ; Summary based on the input provided by Orange by e-mail received Tue 17/03/2026 12:36, file "Edge Computing for Industrial Scenarios". Charging aspects for Multi-Operator Core Network (MOCN) Network Sharing (1070020, CH_MOCN_NetShare) ; Summary based on the input provided by Ericsson by e-mail received Thu 12/03/2026 12:48, file "Summary for CH_MOCN_NetShare". Roaming Value-Added Services (1030046, TEI19_RVAS) ; Summary based on the input provided by Ericsson by e-mail received Mon 16/03/2026 17:13, file "RVAS Rel19 Summary_Ericsson". Multimedia Priority Service (MPS) for Messaging services (1040083, MPS4msg); Summary based on the input provided by Peraton Labs by e-mail received Wed 01/04/2026 03:00, file "Summary_MPS4msg_Rel19features_03-31-2026". Energy Efficiency as Service Criteria (1020060, SP-250069); Summary based on the input provided by Samsung by e-mail received Mon 30/03/2026 07:56, file "Summary_1020060"; Charging aspects: from Huawei, SP-251555. OAM aspects: from Huawei and Samsung, e-mail from Huawei received Thu 26/03/2026 03:28, file "Summary_1060007 Energy efficiency and energy saving aspects of 5G networks and services_final".s Roaming traffic offloading via session breakout in HPLMN (1030018, TEI19_HSBO); Summary based on the input provided by Samsung by e-mail received Mon 30/03/2026 07:56, file "Summary_1030018". Xxxxx Enabling Edge Applications Phase 3 (1040081, EDGEAPP_Ph3); Summary based on the input provided by Samsung by e-mail received Mon 30/03/2026 07:56, file "Summary_1040081". Interconnect of SNPN (1030041, ISN); Summary based on the input provided by Novamint by e-mail received Tue 31/03/2026 22:08, file "Summary_1030041". Next Generation Real time Communication services Phase 2 (1050066, NG_RTC_Ph2) ; Summary based on the input provided by China Mobile by e-mail received Fri 27/03/2026 11:58, file "Summary_1050066", including Security Aspects provided by Ericsson. Security Assurance Specification (SCAS) for maintenance of 5G features (1030029, SCAS_5G_Maint); Summary based on the input provided by Huawei by e-mail received Thu 26/03/2026 09:07, file "Summary_1030029 for SA3 SCAS_5G_Maint". Core Network Enhanced Support for AIML (1050047, AIML_CN); Summary based on the input provided by Vivo by e-mail received Tue 31/03/2026 11:59, file "Summary_1050047 for Rel-19 Core Network Enhanced	

						Support for AIML_v0". Addition of 256-bit security Algorithms (1010012, 256_Algo); Summary based on the input provided by Nokia by e-mail received Tue 24/03/2026 15:45, file "Summary_1010012". Management of planned configurations (1050032, PlanM); Summary based on the input provided by Nokia by e-mail received Fri 13/03/2026 02:52, file "Rel-19 WI Summary for Management of planned configurations". Network Controlled Network Slice Selection (1070041, TEI19_SliceSel); Summary based on the input provided by KDDI by e-mail received Thu 19/03/2026 09:22, file "SP-26xxxx_Summary_of_TEI19_SliceSel". Charging Aspects of Ranging and Sidelink Positioning (1040013, Ranging_SL_CH); Summary based on the input provided by China Telecom by e-mail received Thu 12/03/2026 10:28, file "Summary_for_Ranging_SL_CH". Summary based on the input provided by ZTE by e-mail Measurement Data Collection (980108, MeasureData); received Thu 19/03/2026 07:31, file "SP-26xxxx Summary of R19 Measuredata". Summary based on the input provided by Oppo by e-mail received Thu 19/03/2026 09:17, file "Summary_1000030".	
						All expected summaries provided. Global clean-up made by the Rapporteur (MCC).	2.0.0
2026-06	TSG#112	CP-261267 / SP-260604 / RP-261487	-	-	-	Clarified that only Stage 1 for AIML_MT_Ph2 (UID 1000030) specified in Rel-19, so deleted from Rel-19 Summary for AI/ML Management Phase 2 (NEC in SP-260231) added Summary for Enhancement of Management Aspects related to NWDAF Phase 2 (China Telecom in S5-252641) added	2.1.0
2026-06	TSG#112					Approved by TSG#112. Final clean-up.	19.0.0

History

Version	Date	Status
V19.0.0	August 2026	Publication